

Primary Glomerular Disease

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CHAPTER OUTLINE

GENERAL DESCRIPTION OF GLOMERULAR SYNDROMES, 1008

GLOMERULAR DISEASES THAT CAUSE NEPHROTIC SYNDROME AND GLOMERULONEPHRITIS, 1011

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS AND CRESCENTIC GLOMERULONEPHRITIS, 1071

KEY POINTS

- Primary membranous nephropathy (PMN) is an autoimmune disease characterized by circulating IgG autoantibodies against two major podocyte antigens, muscle-type phospholipase A2 receptor (PLA₂R1), which has been detected in 70% of cases, and thrombospondin type 1 domain-containing 7A (THSD7A) in 5% to 10% of cases.
- A new classification scheme for membranoproliferative glomerulonephritis (MPGN) has been proposed that is based on glomerular immunofluorescence. It is classified as an immune complex-mediated MPGN when both immunoglobulin and complement are deposited. Those with predominant C3 staining are classified as complement-mediated MPGN or C3 glomerulopathy.
- Fibrillary glomerulonephritis is now postulated to be an autoimmune disease based on the observation of autoantibodies targeting a putative autoantigen, DNAJB9.
- Patients with immunotactoid glomerulonephritis should be evaluated for underlying monoclonal gammopathy, malignancy, and lymphoproliferative disorders.
- In a large cohort of subjects from the European validation study of the Oxford classification of IgAN (VALIGA), no significant differences were found in respect to proteinuria or decline in renal function in patients with or without tonsillectomy.
- In anti-GBM disease, glomeruli with crescents typically have fibrinoid necrosis in adjacent glomerular segments, whereas nonnecrotic segments may look entirely normal by light microscopy or may have slight infiltration by neutrophils or mononuclear leukocytes. This differs from crescentic immune complex glomerulonephritis and C3 glomerulopathy, which typically have capillary wall thickening and endocapillary hypercellularity in the intact glomeruli.
- Autoantibodies in sporadic anti-GBM disease are directed against the NC1 domain of the α_3 chain of type IV collagen. By contrast, the anti-GBM alloantibodies that cause posttransplantation nephritis in some patients with X-linked Alport syndrome are directed against conformational epitopes in the NC1 domain of α_5 (IV) collagen, which are normally expressed in the allograft but absent in the recipient.
- In ANCA-associated vasculitis, patients with anti-proteinase-3 (PR3) antibodies, lung involvement, or upper respiratory tract involvement are more likely to relapse than others.

Glomerular diseases may be categorized into those that primarily involve the kidney (primary glomerular diseases) and those in which kidney involvement is part of a systemic disorder (secondary glomerular diseases). This chapter focuses on primary glomerular diseases. Some forms of glomerular disease occur not only as a renal-limited (primary) disease in some patients, but also as a component of systemic disease in other patients—for example, anti-glomerular basement membrane (anti-GBM) glomerulonephritis, with and without pulmonary disease, and antineutrophil cytoplasmic autoantibody (ANCA) glomerulonephritis, with and without systemic vasculitis and granulomatosis. Chapter 32 concentrates on secondary glomerular diseases and Chapter 33 focuses on therapy for glomerular disease. The separation of glomerular disease into primary versus secondary is somewhat problematic because, in some cases, what are considered primary glomerular diseases are similar, if not identical, to secondary glomerular diseases. For example, immunoglobulin A (IgA) nephropathy, pauci-immune necrotizing and crescentic glomerulonephritis, anti-GBM glomerulonephritis, membranous nephropathy (MN), and membranoproliferative glomerulonephritis (MPGN) can occur as primary kidney diseases or as components of systemic diseases such as IgA vasculitis, pauci-immune small-vessel vasculitis, Goodpasture syndrome, systemic lupus erythematosus (SLE), and cryoglobulinemic vasculitis, respectively. Of note, there has been a successful shift, as in many settings, to phase out the nosologic use of eponyms and to substitute noneponymous terms that more accurately reflect pathophysiologic specificity in the nomenclature of vasculitides. This nomenclature of vasculitides was formally revised in 2012 at the second international Chapel Hill Consensus Conference, and the following vasculitides with new names remain relevant to this edition: IgA vasculitis (formerly designated Henoch-Schönlein purpura), granulomatosis with polyangiitis (GPA, formerly designated Wegener granulomatosis) and eosinophilic granulomatosis with polyangiitis (EGPA, formerly designated Churg-Strauss syndrome).¹

A temporal change has emerged over the past 30 years in the frequencies of glomerular disease for reasons that remain unclear but could reflect changing behavioral and/or environmental exposures. A cross-sectional observational study of 18 widely recognized glomerular diseases between 1986 and 2015, identified through diagnostic kidney biopsy specimens from the Glomerular Disease Collaborative Network spanning the southeastern United States, revealed a dramatic increase in the frequency of diabetic glomerulosclerosis.² The frequency of focal segmental glomerulosclerosis (FSGS) initially increased but then declined, and the frequencies of other common glomerular disease subtypes remained stable (IgA nephropathy and ANCA glomerulonephritis) or declined (minimal change glomerulopathy, MN, MPGN, and lupus nephritis).²

When a patient presents with glomerular disease, the clinician not only must evaluate the clinical signs and symptoms but also must be vigilant for evidence of a systemic process or disease that could be causing the kidney disease. Clinical evaluation includes assessment of proteinuria, hematuria, the presence or absence of renal insufficiency, and the presence or absence of hypertension. Some glomerular diseases cause isolated proteinuria or isolated hematuria, with no other signs or symptoms of disease. More severe glomerular

Table 31.1 Manifestations of Nephrotic and Nephritic Features by Glomerular Diseases

Disease	Nephrotic Features	Nephritic Features
Minimal change disease	++++	—
Membranous nephropathy	++++	+
Focal segmental glomerulosclerosis	+++	++
Fibrillary glomerulonephritis	+++	++
Mesangioproliferative glomerulopathy ^a	++	++
Membranoproliferative glomerulonephritis ^b	++	+++
Proliferative glomerulonephritis ^a	++	+++
Acute diffuse proliferative glomerulonephritis ^c	+	++++
Crescentic glomerulonephritis ^d	+	++++

^aMesangioproliferative and proliferative glomerulonephritis (focal or diffuse) are structural manifestations of a number of glomerulonephritides, including IgA nephropathy and lupus nephritis.

^bBoth mesangiocapillary (type I) and dense deposit disease (type II).

^cOften a structural manifestation of acute poststreptococcal glomerulonephritis.

^dCan be immune complex-mediated, anti-glomerular basement membrane antibody-mediated, or associated with antineutrophil cytoplasmic autoantibodies.

Modified from Jennette JC, Mandal AK: The nephrotic syndrome. In Mandal AK, Jennette JC, eds. *Diagnosis and Management of Renal Disease and Hypertension*. Durham, NC: Carolina Academic Press; 1994:235–272.

disease often results in the nephrotic syndrome or nephritic (glomerulonephritic) syndrome. Glomerular disease may have an indolent course or begin abruptly, leading to acute or rapidly progressive glomerulonephritis. Although some glomerular disorders consistently cause a specific syndrome (e.g., minimal change disease [MCD] results in the nephrotic syndrome), most disorders are capable of causing features of both nephrosis and nephritis (Table 31.1). This sharing and variability of clinical manifestations among different glomerular diseases may not allow an accurate diagnosis based on clinical features alone. Therefore, kidney biopsy has an important role in the evaluation of many patients and remains the gold standard for the definitive diagnosis of many glomerular diseases.

This chapter describes the clinical syndromes caused by glomerular diseases, including isolated proteinuria, isolated hematuria, and specific forms of primary glomerular disease that cause the nephrotic or nephritic syndrome and reviews their distinctive pathologic features.

GENERAL DESCRIPTION OF GLOMERULAR SYNDROMES

PROTEINURIA

Proteinuria can be caused by the systemic overproduction of proteins that cross glomerular capillary walls (e.g., multiple

myeloma with Bence Jones proteinuria), tubular dysfunction (e.g., Fanconi syndrome), or glomerular dysfunction. It is important to identify patients in whom proteinuria is a manifestation of substantial glomerular disease as opposed to patients who have benign functional, transient, postural (orthostatic), or intermittent proteinuria. (Proteinuria is discussed further in Chapters 23 and 30.)

Plasma proteins larger than 70 kDa cross the basement membrane in a manner normally restricted by both size-selective and charge-selective barriers.^{3,4} The functional characteristics of the glomerular capillary filter have been extensively studied by the evaluation of the fractional clearance of molecules of different size and charge.⁵ The size-selective barrier is most likely a consequence of functional pores within the glomerular basement membrane (GBM) that restrict the filtration of plasma proteins larger than 150 kDa. There is also a shape restriction of molecules that allows elongated molecules to cross the glomerular capillary wall more readily than other molecules of the same molecular weight, and there is a charge-selective nature of the barrier, largely a consequence of glycosaminoglycans arranged along the capillary wall. Loss of charge selectivity may be a defect in MCD, whereas a loss of size selectivity may be the cause of proteinuria in, for example, patients with MN.⁴

A number of factors have proven to be important in the disruption of the glomerular capillary wall as a consequence of tissue-degrading enzymes, complement components that assemble or deposit in it, and oxygen radicals that target both the GBM and the slit diaphragm. Heparinase and hyaluronidase alterations in the aminoglycan content of the glomerular capillary wall may play a role in increased protein excretion.^{6,7} Genetic studies have provided exciting clues about the specific components of the glomerular capillary wall, including mutations in the podocyte or proteins in the slit diaphragm, that result in proteinuria (reviewed by Tryggvason and colleagues⁸ and more recently by Garg and Rabelink⁹).

Another major mechanism resulting in proteinuria is impaired reabsorption of plasma proteins by proximal tubular epithelial cells. A number of low-molecular-weight proteins, including albumin and β_1 -, β_2 -, and α_1 -microglobulins, are filtered by the glomerulus and absorbed by tubular epithelial cells. When tubular epithelial cells are damaged, these proteins are excreted. Russo and colleagues have studied the critical importance of the tubular absorption of proteins.¹⁰ The glomerular capillary sieving coefficient for albumin was examined in normal and nephrotic rats by two-photon (laser) intravital microscopy. The glomerular capillary sieving coefficient for albumin was 3.4×10^{-2} rather than 6.2×10^{-4} , as found by earlier micropuncture studies in nonproteinuric rats.

Several important observations emanate from this study. First, there is a large amount of albumin filtered across the glomerular capillary bed daily in the normal rat. Second, investigators have found no evidence for a charge-based restriction to the passage of albumin through the glomerular filter. Third, in normal and nephrotic animals, the vast majority of the filtered albumin was “reclaimed” from the filtrate by a high-capacity transcytotic pathway in the proximal tubule, which returns intact (unaltered) albumin to the peritubular capillary circulation. This is an important concept, because most nephrologists view albuminuria as resulting

solely from enhanced glomerular permeability (reviewed by Vallon¹¹).

The term “isolated proteinuria” is used for several conditions, including mild transient proteinuria of less than 1 g protein/day that typically accompanies physiologically stressful conditions, such as fever in hospitalized patients, exercise, and congestive heart failure.¹² In other patients, transient proteinuria is a consequence of the overflow of proteins of low molecular weight due to the overproduction of light chains, heavy chains, or other fragments of immunoglobulins. Additional examples of proteinuria caused by increased circulating proteins are β_2 -microglobulinuria, myoglobinuria, and hemoglobinuria.

The term “orthostatic proteinuria” is defined by the absence of proteinuria while the patient is in a recumbent posture and its appearance during an upright posture, especially during ambulation or exercise.¹³ The total amount of protein excretion in a 24-hour period is generally less than 1 g, but may be as much as 2 g. Orthostatic proteinuria is more common in adolescents and is uncommon in individuals older than 30 years.^{13,14} About 2% to 5% of adolescents have orthostatic proteinuria. Among patients with orthostatic proteinuria who underwent kidney biopsy, 47% were found to have normal glomeruli by light microscopy, 45% to have minimal to moderate glomerular abnormalities of nonspecific nature, and the remainder to have evidence of a primary glomerular disease.¹⁵

Why is proteinuria increased during an upright posture in individuals with normal glomeruli by light microscopy? Although the answer to this question is not fully known, there are several likely possibilities. Orthostatic proteinuria may occur as a consequence of alterations in glomerular hemodynamics. It is possible that even in histologically “normal” glomeruli in which there are no specific lesions, subtle glomerular abnormalities exist, including abnormal basement membranes or focal changes of the mesangium.¹⁶ Alternatively, orthostatic proteinuria has been demonstrated with anatomic entrapment and subsequent obstruction of the left renal vein between the aorta and superior mesenteric artery (commonly referred to as the “renal nutcracker”).¹⁷ In addition, the observation that surgical correction of a kink in an allograft renal vein resulted in the disappearance of orthostatic proteinuria gives credence to venous entrapment as a cause for orthostatic proteinuria.¹⁶

There are several approaches to the diagnosis of orthostatic proteinuria. These include comparison of protein excretion in two 12-hour urine collections, one during recumbency and one during ambulation. Another approach is to compare protein levels in a split collection of 16 hours during ambulation and 8 hours of overnight collection. It is important that patients be recumbent for at least 2 hours before their ambulatory collection is completed to avoid the possibility of contamination of the recumbent collection by urine formed during ambulation. The diagnosis of orthostatic proteinuria requires that protein excretion during recumbency be less than 50 mg during those 8 hours. Few convincing data exist on the usefulness of comparing urinary protein to creatinine ratio measurements during recumbency versus ambulation as a diagnostic test for orthostatic proteinuria.

A 20-year follow-up of orthostatic proteinuria has suggested a benign long-term course.¹⁴ Orthostatic proteinuria resolves in most patients. It is present in 50% of patients after 10

years and in only 17% of patients after 20 years.¹⁴ In the absence of a kidney biopsy, an underlying glomerulopathy cannot be completely excluded, and an orthostatic component of proteinuria may be found in early glomerular disease. Thus, it is important to reassess patients after an interval of about 1 year to be certain that the degree or pattern of proteinuria has not changed.

Fixed proteinuria is present whether the patient is upright or recumbent. The proteinuria disappears in some patients, whereas others will have a more ominous glomerular lesion that portends an adverse long-term outcome. The prognosis depends on the persistence and severity of the proteinuria. If proteinuria disappears, it is less likely that the patient will develop hypertension or a reduced glomerular filtration rate (GFR). These patients must be evaluated periodically for as long as proteinuria persists.

RECURRENT OR PERSISTENT HEMATURIA

Hematuria is the presence of an excessive number of red blood cells in the urine and is categorized as microscopic (visible only with the aid of a microscope) or macroscopic (urine that is tea-colored or cola-colored, pink, or even red). Hematuria can result from injury to the kidney or to another site in the urinary tract (see Chapter 23).

Healthy individuals may excrete as many as 10^5 red cells in the urine in a 12-hour period. An acceptable definition of hematuria is more than two red cells per high-power field (HPF) in centrifuged urine.¹⁸ The approach to processing urine varies from laboratory to laboratory; therefore, the number of red cells/HPF that is an accurate indicator of hematuria may vary slightly among different laboratories. The urine dipstick test detects one to two red cells/HPF per and is a very sensitive test. A negative result on dipstick examination virtually excludes hematuria.¹⁹

Hematuria is present in about 5% to 6% of the general population²⁰ and 4% of schoolchildren. In most children, follow-up urinalyses are normal.²¹ In most people, hematuria emanates from the lower urinary tract, especially in conditions affecting the urethra, bladder, and prostate. Less than 10% of cases of hematuria are caused by glomerular bleeding.¹⁸ Persistent hematuria, especially in older individuals, should raise the possibility of malignancy. The incidence of malignancy, especially of the bladder, ranges from 5% in individuals with persistent microscopic hematuria to over 20% in individuals with gross hematuria.²² Other causes of nonglomerular hematuria include neoplasms, trauma, metabolic defects such as hypercalciuria, vascular diseases, including renal infarctions and renal vein thrombosis, cystic diseases of the kidney, including polycystic kidney disease, medullary cystic disease, and medullary sponge kidney, and interstitial kidney disease, such as papillary necrosis, hydronephrosis, and drug-induced interstitial nephritis. In children with asymptomatic hematuria, hypercalciuria is the cause in 15% of cases, and 10% to 15% have IgA nephropathy. In up to 80% of children and 15% to 20% of adults with hematuria, no cause can be identified.²³

Transient hematuria has been found in a number of settings. It is present in 13% of postmenopausal women.²⁴ Episodic hematuria in a cyclic pattern during a menstrual cycle is most likely a consequence of the invasion of the urinary tract by endometrial implants.²⁵ In 1000 men between

the ages of 18 and 33 years, hematuria was present at least once in 39% and on two or more occasions in 16%. In patients with isolated asymptomatic hematuria without proteinuria or renal insufficiency, the hematuria resolves in 20% of cases; however, some of these patients will develop hypertension and proteinuria.²⁶ In older individuals, transient hematuria should raise a concern of malignancy.^{18,27,28} In some individuals, transient hematuria may be a consequence of exercise.

Glomerular hematuria, in contrast to hematuria caused by injury elsewhere in the urinary tract, is characterized by misshapen red cells that have been distorted by osmotic and chemical stress as red blood cells pass through the nephron. Hematuria with dysmorphic cells, especially cells that have membrane blebs producing the picture of acanthocyturia, is strong evidence for glomerular bleeding.²² The finding of protein (especially >2 g/day), hemoglobin, or red cell casts in the urine enhances the possibility that hematuria is of glomerular origin. Although brown or cola-colored urine is most commonly associated with glomerular hematuria, its absence does not exclude glomerular disease. Interestingly, blood clots do not occur in the urine with glomerular bleeding.

The differential pathologic diagnosis of glomerular hematuria without proteinuria, renal insufficiency, or red blood cell casts includes IgA nephropathy, thin basement membrane nephropathy, hereditary nephritis, and histologically normal glomeruli.²⁹ In a study in Europe,³⁰ 80 normotensive adults underwent kidney biopsy for the evaluation of recurrent macroscopic hematuria or persistent microscopic hematuria. Approximately 30% of these patients had IgA nephropathy, 20% had thin basement membrane nephropathy, and 30% had no discernible lesion. Hematuria disappeared in 13 of the latter patients. The remaining patients had mesangioproliferative glomerulonephritis, interstitial nephritis, or focal glomerulosclerosis. In contrast, in another study, 216 Chinese adults with isolated hematuria who underwent a kidney biopsy were much more likely to have IgA nephropathy than any other lesion.³¹

Table 31.2 provides data from an analysis of native kidney biopsy specimens from patients with hematuria performed by the University of North Carolina (UNC) Nephropathology Laboratory. Patients with SLE were excluded from the study. The patients selected for the study had a serum creatinine level of less than 1.5 mg/dL or more than 3 mg/dL. Patients with a serum creatinine level of less than 1.5 mg/dL were further divided into those with proteinuria less than 1 g/day and those with proteinuria of 1 to 3 g/day. The data showed that patients with a relatively normal serum creatinine level, hematuria, and proteinuria of less than 1 g protein/day were most likely to have thin basement membrane nephropathy, IgA nephropathy, or no identifiable renal lesion. When hematuria is accompanied by 1 to 3 g protein/day of proteinuria but no significant renal insufficiency, IgA nephropathy was the most likely cause. Patients with hematuria and a serum creatinine level higher than 3 mg/dL usually had aggressive glomerulonephritis with crescents.

Despite these overall tendencies, it is not possible to determine the cause of asymptomatic hematuria definitively without kidney biopsy, and even kidney biopsy specimen evaluation fails to reveal a cause in a minority of patients. Certain rules generally apply to the clinical prediction of

the most likely cause. Gross hematuria is usually found in IgA nephropathy or hereditary nephritis. Patients with thin basement membrane nephropathy typically do not have substantial proteinuria.

A potential benefit of kidney biopsy in patients with isolated hematuria includes the reduction of patient and physician uncertainty by confirming a specific diagnosis. Nonetheless, the role of kidney biopsy in the evaluation of asymptomatic hematuria in patients without proteinuria, hypertension, or kidney insufficiency remains unclear. In biopsy series involving patients in whom asymptomatic hematuria was accompanied by low-grade proteinuria, specific glomerular diseases, including IgA nephropathy and membranoproliferative glomerular disease, may be discovered and, when there is no proteinuria, IgA nephropathy and thin basement membrane disease or nondiagnostic minor changes remain the most common findings.^{32,33} Confirmation of a glomerular cause eliminates the need for repeated urologic studies, and a more accurate long-term prognosis can be made (e.g., thin basement membrane nephropathy is less likely to progress than is IgA nephropathy). However, isolated glomerular hematuria

without proteinuria or renal insufficiency may not warrant a kidney biopsy because the findings often will not affect management. In one study of patients with isolated hematuria, the biopsy results altered patient management in only 1 of 36 patients.³⁴

GLOMERULAR DISEASES THAT CAUSE NEPHROTIC SYNDROME AND GLOMERULONEPHRITIS

NEPHROTIC SYNDROME

The nephrotic syndrome results from proteinuria of more than 3.5 g protein/day and is characterized by edema, hyperlipidemia, hypoproteinemia, and other metabolic disorders (described in detail later). Nephrotic syndrome not only may be caused by primary (idiopathic) glomerular diseases but also may be secondary to a large number of identifiable disease states (Box 31.1). Despite the differences in these causes, the loss of substantial amounts of protein

Table 31.2 Frequency of Kidney Disease in Patients with Hematuria Undergoing Renal Biopsy (%)^a

Biopsy Findings	Hematuria		
	Urinary Protein <1 g/day, Creatinine <1.5 mg/dL	Urinary protein 1–3 g/day, Creatinine <1.5 mg/dL	Creatinine >3 mg/dL
No abnormality	30	2	0
Thin basement nephropathy	26	4	0
Immunoglobulin A nephropathy	28	24	8
Glomerulonephritis without crescents ^b	9	26	23
Glomerulonephritis with crescents ^b	2	24	44
Other kidney disease ^c	5	20	25
Total	100 (n = 43)	100 (n = 123)	100 (n = 255)

^aBased on an analysis of kidney biopsy specimens evaluated at the University of North Carolina Nephropathology Laboratory. Specimens from patients with systemic lupus erythematosus were excluded from the analysis.

^bProliferative or necrotizing glomerulonephritis other than immunoglobulin A nephropathy or lupus nephritis.

^cIncludes causes of nephrotic syndrome such as membranous nephropathy and focal segmental glomerulosclerosis.

Data from Caldas MLR, Jennette JC, Falk RJ, et al. Immunoelectron microscopic documentation of the translocation of proteins reactive with ANCA to neutrophil cell surfaces during neutrophil activation. [abstract]. Presented at the Third International Workshop on ANCA, November 29–30, 1990, Washington, DC.

Box 31.1 Classification of Disease States Associated With the Development of Nephrotic Syndrome

Idiopathic Nephrotic Syndrome Due to Primary Glomerular Disease
Nephrotic Syndrome Associated With Specific Causative Events
or in Which Glomerular Disease Arises as a Complication of Other Diseases

1. Medications and Other Chemicals

Organic, inorganic, elemental mercury^a

Organic gold

Penicillamine, bucillamine

Street heroin

Probenecid

Captopril

Nonsteroidal antiinflammatory drugs

Lithium

Interferon-α

Chlorpropamide

Rifampin

Pamidronate

Paramethadione (Paradione), trimethadione (Tridione)

Mephénytoin (Mesantoin)

Tolbutamide^b

Phenindione^b

Warfarin

Clonidine^b

Continued on following page

Box 31.1 Classification of Disease States Associated With the Development of Nephrotic Syndrome (Cont'd)

Perchlorate^b
 Bismuth^b
 Trichloroethylene^b
 Silver^b
 Insect repellent^b
 Contrast media
 Anabolic steroids

2. Allergens, Venoms, Immunizing Agents

Bee sting
Pollens
 Poison ivy and poison oak
 Antitoxins (serum sickness)
 Snake venom
 Diphtheria, pertussis, tetanus toxoid
 Vaccines

3. Infections

Bacterial: *poststreptococcal glomerulonephritis*, infective endocarditis, *shunt nephritis*, leprosy, syphilis (congenital and secondary), *Mycoplasma* infection, tuberculosis,† chronic bacterial pyelonephritis with vesicoureteral reflux

Viral: hepatitis B, hepatitis C, cytomegalovirus infection, infectious mononucleosis (Epstein-Barr virus infection), herpes zoster, vaccinia, infection with human immunodeficiency virus type 1

Protozoal: *malaria* (especially quartan malaria), toxoplasmosis

Helminthic: *schistosomiasis*, trypanosomiasis, filariasis

4. Neoplasms

Solid tumors (carcinoma and sarcoma): tumors of the *lung*, *colon*, *stomach*, *breast*, cervix, kidney, thyroid, ovary, prostate, adrenal, oropharynx, carotid body†; melanoma, pheochromocytoma, Wilms' tumor, mesothelioma, oncocytoma

Leukemia and lymphoma: *Hodgkin disease*, chronic lymphocytic leukemia, multiple myeloma (amyloidosis), Waldenström macroglobulinemia, lymphoma

Graft-versus-host disease after bone marrow transplantation

5. Multisystem Disease^c

Systemic lupus erythematosus
 Mixed connective tissue disease
 Dermatomyositis
 Rheumatoid arthritis
 Goodpasture's syndrome
IgA vasculitis (formerly *Henoch-Schönlein purpura*; see also *immunoglobulin A nephropathy*, Berger disease)
 Systemic vasculitis (including GPA, formerly Wegener)
 Takayasu arteritis
 Mixed cryoglobulinemia
 Light- and heavy-chain disease (Randall type)
 Partial lipodystrophy
 Sjögren syndrome
 Toxic epidermolysis
 Dermatitis herpetiformis
 Sarcoidosis

Ulcerative colitis
Amyloidosis (primary and secondary)

6. Hereditary—Familial and Metabolic Disease^c

Diabetes mellitus
 Hypothyroidism (myxedema)
 Graves' disease
 Amyloidosis (familial Mediterranean fever and other hereditary forms, Muckle-Wells syndrome)
 Alport syndrome
 Fabry disease
 Nail-patella syndrome
 Lipoprotein glomerulopathy
 Sickle cell disease
 α_1 -Antitrypsin deficiency
 Asphyxiating thoracic dystrophy (Jeune syndrome)
 Von Gierke disease
 Podocyte slit diaphragm mutation
 Nephrin mutation
 FAT2 mutation
 Podocin mutation
 CD2AP mutation
 Denys-Drash syndrome (*WT1* mutation)
 ACTN4 mutation
 Charcot-Marie-Tooth syndrome
 Congenital nephrotic syndrome (Finnish type)
 Cystinosis (adult)
 Galloway-Mowat syndrome
 Hurler syndrome
 Familial dysautonomia

7. Miscellaneous

Pregnancy-associated (preeclampsia, recurrent, transient)
Chronic renal allograft failure
 Accelerated or malignant nephrosclerosis
 Unilateral renal arterial hypertension
 Intestinal lymphangiectasia
 Chronic jejunoileitis^b
 Spherocytosis†
 Renal artery stenosis
 Congenital heart disease (cyanotic)^b
 Severe congestive heart failure^b
 Constrictive pericarditis^b
 Tricuspid insufficiency^b
 Massive obesity
 Vesicoureteric reflux nephropathy
 Papillary necrosis
 Gardner-Diamond syndrome
 Castleman disease
 Kartagener syndrome
 Buckley syndrome
 Kimura disease
 Silica exposure

^aDiseases and other agents in *italics* are the more commonly encountered causes of nephrotic syndrome.

^bBased on single case reports or small series in which cause-and-effect relationship cannot be established. Other factors (e.g., use of mercurial diuretics in heart failure) may have been the true inciting event.

^cSee Chapter 33 for a detailed discussion of the secondary forms of nephrotic syndrome.

ACTN4, α -Actinin-4; CD2AP, CD2-associated protein; FAT2, FAT tumor suppressor homolog 2 (*Drosophila*); WT1, Wilms' tumor 1.

in the urine results in a shared set of abnormalities that comprise nephrotic syndrome (Boxes 31.2 and 31.3 and Table 31.3). We are likely to witness an explosion of new insights into primary glomerular diseases responsible for nephrotic syndrome with the ongoing efforts of the Nephrotic Syndrome Study Network (NEPTUNE) and Cure Glomerulonephropathy (CureGN), both funded by the National Institutes of Health (NIH). NEPTUNE is a North American multicenter collaborative translational research consortium geared toward longitudinal clinical monitoring and blood, urine, and kidney tissue sample collection from 450 adults and children with MCD, FSGS, and MN.³⁵ CureGN (<https://curegn.org/>) is a longitudinal multicenter cohort observational study that is

following 2400 children and adults with MCD, FSGS, MN, and IgA nephropathy. Both NEPTUNE and CureGN provide the infrastructure for multiple ancillary studies that will advance our understanding and management of glomerular diseases.

MINIMAL CHANGE DISEASE

EPIDEMIOLOGY

MCD was first described in 1913 by Munk, who called it “lipoid nephrosis” because of the presence of lipid in the tubular epithelial cells and urine.³⁶ MCD is most common in children, accounting for 70% to 90% of nephrotic syndrome in children younger than age 10 years and 50% in adults. MCD also causes 10% to 15% of primary nephrotic syndrome in adults (Fig. 31.1).

The incidence of MCD has geographic variations; MCD is more common in Asia than in North America or Europe.³⁷ This may be a consequence of differences in kidney biopsy practices or differences in environmental or genetic influences. The disease may also affect older adult patients in whom there is a higher propensity for the clinical syndrome of MCD

Box 31.2 Alterations of Plasma Protein in the Nephrotic Syndrome

Immunoglobulins (Ig)
 Decreased IgG
 Normal or increased IgA, IgM, or IgE
 Increased α_2 and beta globulins
 Decreased α_1 globulin
 Metal-binding proteins
 Loss of metal binding proteins: iron, copper, zinc
 Loss of erythropoietin
 Depletion of transferrin
 Transcortin deficiency
 Complement deficiency
 Decreased factor B
 Decreased C3
 Decreased C1q, C2, C8, Ci
 Increased C3, C4bp
 Normal C1s, C4, and C1 inhibitor
 Coagulation components
 Decreased factors XI, XII, kallikrein inhibitor
 Decreased factors IX, XII
 Decreased antipiasmin, α_1 antitrypsin
 Plasminogen activator, endothelial prostacyclin stimulating factor
 Decreased antithrombin III
 Elevated beta-thromboglobulin
 Procoagulant

Modified from references 504 and 1434 to 1448.

Box 31.3 Procoagulant Factors in the Nephrotic Syndrome

Increased blood viscosity
 Hemoconcentration
 Increased plasma fibrinogen
 Increased intravascular fibrin formation
 Increased α_2 -macroglobulins
 Increased tissue type plasminogen activator
 Increased factors II, V, VII, VIII, X, XIII
 Decreased factors IX, XI, XII
 Decreased α -antitrypsin
 Decreased fibrinolytic activity
 Decreased plasma plasminogen
 Decreased antithrombin III
 Decreased protein S
 Thrombocytosis
 Increased platelet aggregability

Modified from references 507,1447, and 1449 to 1463.

Table 31.3 Diseases that Cause the Nephrotic Syndrome^a

Glomerular Lesion	No. of Patients	Male:Female Ratio	Caucasian:African American Ratio
Minimal change disease	522	1.1:1.0	1.9:1.0
Focal segmental glomerulosclerosis (FSGS; typical)	1103	1.4:1.0	1.0:1.0
Collapsing glomerulopathy FSGS	135	1.2:1.0	1.0:7.8
Glomerular tip lesion FSGS	94	1.0:1.0	4.7:1.0
Membranous Nephropathy	1120	1.4:1.0	1.9:1.0
C1q nephropathy	114	1.0:1.0	1.0:4.8
Fibrillary glomerulonephritis	76	1.0:1.2	14.3:1.0

^aInformation in this table is from 9605 native kidney biopsies from the UNC Nephropathology Laboratory. This laboratory evaluates kidney biopsies from a base population of approximately 10 million throughout the southeastern United States and centered in North Carolina. The expected Caucasian-to-African American ratio in this renal biopsy population is approximately 2:1. FSGS, Focal segmental glomerulosclerosis.

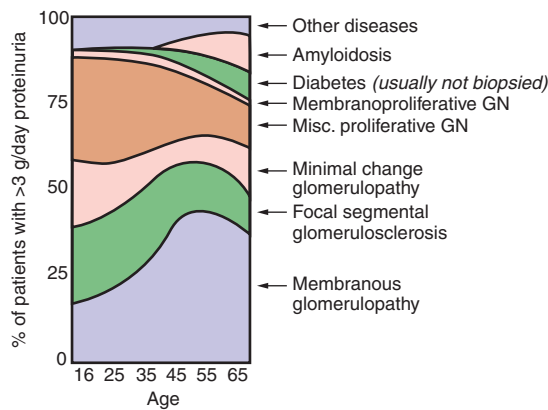


Fig. 31.1 Graphic depiction of frequencies of different forms of glomerular disease identified in kidney biopsy specimens from patients with proteinuria of more than 3 g of protein/day evaluated at the University of North Carolina Nephropathology Laboratory. Some diseases that cause proteinuria are underrepresented because they are not always evaluated by kidney biopsy. For example, in many patients, steroid-responsive proteinuria is given a presumptive diagnosis of minimal change disease, and patients do not undergo biopsy, and most patients with diabetes and proteinuria are presumed to have diabetic glomerulosclerosis and do not undergo biopsy. GN, glomerulonephritis.

and acute kidney injury (AKI; discussed later). There appears to be a male preponderance of this process in some research series, especially in children, in whom the male-to-female ratio is 2:1 to 3:1³⁸; however, data from our institution do not support this finding.

PATHOLOGY

The effacement of podocyte foot processes is accompanied by increased density of the cytoskeleton, including actin filaments, in clumps near the basement membrane surface of the podocytes. However, the extent of effacement appears to correlate more with the duration of active nephrotic syndrome than with the magnitude of the proteinuria.³⁹

Light Microscopy

In MCD, no glomerular lesions are seen by light microscopy (Fig. 31.2), or only a minimal focal segmental mesangial prominence is noted.⁴⁰ This mesangial prominence should have no more than three or four cells embedded in the matrix of a segment, and the matrix should not be expanded to the extent that capillary lumens are compromised. Capillary walls should be thin and capillary lumens patent.

The most consistent tubular lesion is increased protein and lipid resorption droplets in tubular epithelial cells. These droplets stain with periodic acid–Schiff stain. Conspicuous resorbed lipid in epithelial cells prompted the designation “lipoid nephrosis” for this disease prior to the recognition of the ultrastructural glomerular lesion. Interstitial edema is rare, even in patients with severe nephrotic syndrome and anasarca. Focal proximal tubular epithelial flattening (simplification), which is histologically identical to that seen with ischemic AKI, occurs in patients who have the syndrome of MCD with AKI.⁴¹

Focal areas of interstitial fibrosis and tubular atrophy in a specimen that otherwise looks like MCD, especially in a young person, should raise the possibility of FSGS that was not sampled in the biopsy specimen. Examination of additional

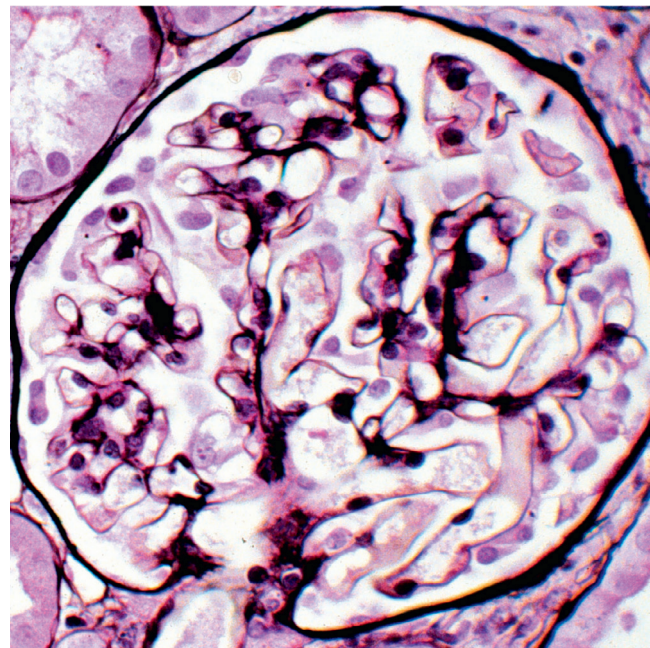


Fig. 31.2 Unremarkable light microscopy appearance of a biopsy specimen from a patient with minimal change disease. Glomerular basement membranes are thin, and there is no glomerular hypercellularity or mesangial matrix expansion. (Jones' methenamine silver stain, ×300.)

levels of section may reveal a sclerotic glomerulus. In fact, MCD and FSGS may lie along a continuous spectrum of one unifying disease with variable manifestations.⁴²

Immunofluorescence Microscopy

Glomeruli usually show no staining with antisera specific for IgG, IgA, IgM, C3, C4, or C1q. The most frequent positive finding is low-level mesangial staining for IgM, sometimes accompanied by low-level staining for C3. If the IgM staining is not accompanied by mesangial electron-dense deposits by electron microscopy, it is consistent with a diagnosis of MCD. Patients whose specimens show mesangial IgM staining by immunofluorescence (IF) microscopy (in the absence of dense deposits by electron microscopy) do not have a worse prognosis than patients whose specimens are without IgM staining.^{43,44} The presence of mesangial dense deposits identified by electron microscopy worsens the prognosis and thus justifies altering the diagnosis—for example, to IgM mesangial nephropathy.⁴⁵ Anything more than trace staining for IgG or IgA casts substantial doubt on a diagnosis of MCD. Even when no sclerotic glomerular lesions are seen by light microscopy, well-defined irregular focal segmental staining for C3 and IgM should raise the possibility of FSGS because sclerotic lesions can be enriched for C3 and IgM. Glomerular and tubular epithelial cell cytoplasmic droplets and tubular casts may stain positively for immunoglobulins and other plasma proteins when there is substantial proteinuria.

Electron Microscopy

The pathologic sine qua non of MCD is effacement of podocyte foot processes observed by electron microscopy (Figs. 31.3 and 31.4). However, this is not a specific feature because it occurs in the glomeruli of patients with severe

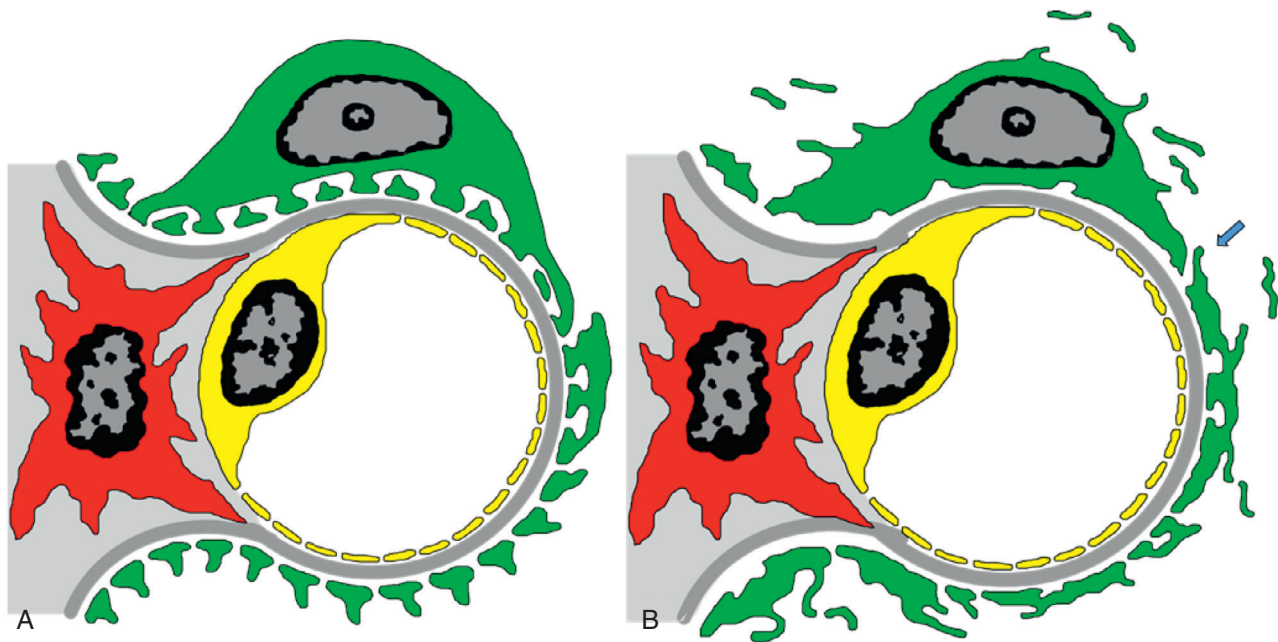


Fig. 31.3 Diagrammatic representations of the ultrastructural features of a normal glomerular capillary loop (A) and a capillary loop with features of minimal change disease (B). The latter has effacement of podocyte foot processes and microvillous projections of podocyte cytoplasm. (Courtesy JC Jennette.)

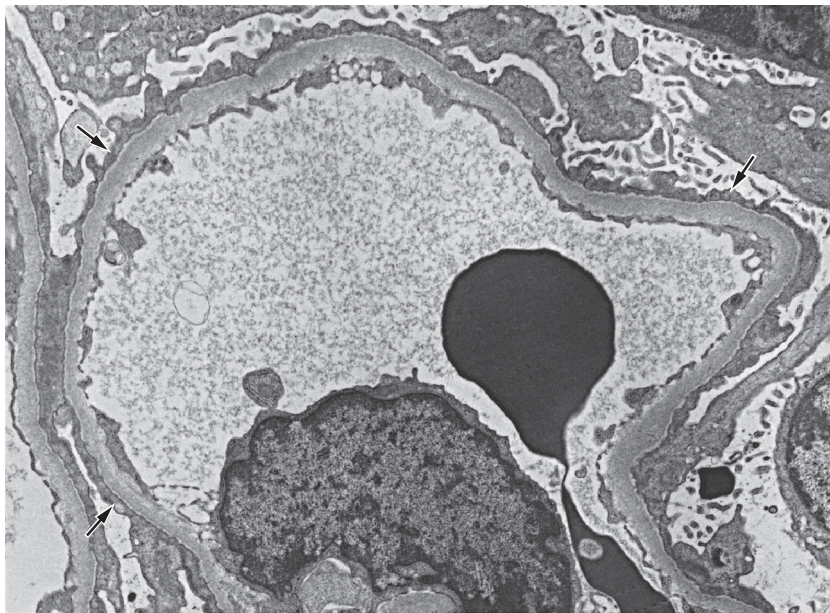


Fig. 31.4 Electron micrograph of a glomerular capillary wall from a patient with minimal change disease showing extensive podocyte foot process effacement (arrows) and microvillous transformation. ($\times 5000$.)

proteinuria of any cause. During active nephrosis, the effacement often is very extensive, with only a few scattered intact foot processes. As the patient enters remission, the extent of foot process effacement diminishes. The effacement usually is accompanied by microvillous transformation, which is the development of numerous villous projections from the podocyte surface into the urinary space. These intracytoplasmic densities should not be confused with subepithelial immune complex dense deposits as a result of resorption of increased lipids and proteins in the urine. Glomerular and proximal tubular epithelial cells have increased clear and dense cytoplasmic droplets.

All these ultrastructural glomerular changes occur in other glomerular disease when there is nephrotic-range proteinuria. Therefore, MCD is a diagnosis by exclusion that is made only when there is no evidence by light, IF, and electron microscopy for any other glomerular disease.

PATHOGENESIS

Although the pathogenesis of MCD remains unclear, this disorder is most likely a consequence of the abnormal regulation of a T cell subset^{46–50} and pathologic elaboration of a circulating permeability factor. A role for the T cell is supported by the effectiveness of corticosteroids and alkylating

drugs in the induction of remission of MCD, as well as by an association of MCD with Hodgkin disease,^{51,52} and remissions are associated with depression of cell-mediated immunity during viral infections, such as measles. Specific evidence stems from the finding that a glomerular permeability factor is produced by human T cell hybridomas obtained from a patient with MCD. When this factor was injected into rodents, proteinuria occurred, with partial fusion of glomerular epithelial cell foot processes.⁵³ Although there are no recognized abnormalities in T or B cell populations in patients with relapsing or quiescent MCD,^{54–57} lymphocytes have depressed reactivity when challenged with mitogens.^{58–66} T cells apparently produce a product, most likely a lymphokine, which increases glomerular permeability to protein. When the glomerular permeability factor is removed from the kidney, it functions normally. This is supported by the intriguing observation that transplantation of a kidney from a patient with refractory MCD results in the rapid disappearance of proteinuria.⁶⁷

A likely target of the pathogenic process is the podocyte, possibly a constituent of the slit pore membrane. Attention has focused on the role of plasma hemopexin in MCD.^{68,69} Hemopexin is present in normal plasma, but an active isoform of the protein has been suggested to cause increased glomerular permeability due to enhanced protease activity.⁶⁸ Patients with MCD in relapse demonstrate altered isoforms of plasma hemopexin with increased protease activity compared with MCD in remission, patients with other forms of nephrotic syndrome, and normal subjects.⁶⁹ It is not understood how and why the plasma hemopexin is altered in MCD or how the enhanced protease activity results in alterations in glomerular permeability.

Differential gene expression techniques have also suggested an alteration in tumor necrosis factor–related apoptosis-inducing ligand (TRAIL) in peripheral blood mononuclear cells in MCD during relapse compared with remission.⁷⁰ Many additional genes (at least 15 of the more than 20,000 examined) were upregulated during relapse, which demonstrates the complexity of events occurring in MCD. Included among these was the IgE-dependent histamine-releasing factor gene,⁷¹ and the well-known association of MCD with atopic allergic states could be the reason for this finding.

This factor may have specificity for podocytes that results in loss of the charge-selective barrier of the GBM. The loss of charge selectivity has been assessed by dextran studies.^{72,73} In these studies, there is less evidence for a defect in the size-selective barrier and more of an alteration of the basement membrane electrostatic charge. The glomerular negative charge is reduced in relapse.⁷⁴

Genetic aspects of MCD have been under intense investigation. Familial clustering of MCD has not generally been observed.⁷⁵ Heterozygous amino acid changes in nephrin and podocin are seen in about one-third of patients with typical MCD, but no amino acid changes were observed for *NEPH1* and *CD2AP* in a study involving 104 adults who presented with childhood-onset MCD.⁷⁵ Thus the genotype in MCD may be quite variable.

Polymorphisms in the genes encoding interleukins 4 and 13 (IL-4 and IL-13), activating transcription factor 6, and macrophage migration inhibitory factor have been described in MCD.^{76–78} IL-13 polymorphisms may relate to phenotype because they have been associated with relapsing forms of

MCD. IL-13 has also been suggested as a potential permeability factor in MCD. The implication of IL-13 in the pathogenesis of MCD is further suggested by the rat model of Lai and colleagues,⁷⁹ in which IL-13-transfected Wistar rats ($n = 41$) showed significant albuminuria, hypoalbuminemia, and hypercholesterolemia compared with control rats ($n = 17$). No significant histologic changes were seen in glomeruli of IL-13-transfected rats; however, electron microscopy revealed up to 80% effacement of podocyte foot process.

One observation is that podocyte-specific angiopoietin-like 4 (ANGPTL₄) overproduction in transgenic rats results in ANGPTL₄ binding to GBM with subsequent loss of GBM charge, diffuse foot process effacement, and nephrotic-range proteinuria. Furthermore, ANGPTL₄ expression has been observed in serum, glomeruli, and urine of patients with MCD,^{80,81} and further understanding of this pathway may prove to be fruitful.

CLINICAL FEATURES AND NATURAL HISTORY

The cardinal clinical feature of MCD in children is the relatively abrupt onset of proteinuria and the development of nephrotic syndrome with heavy proteinuria, hypoalbuminemia, and hyperlipidemia.⁴⁰ The edematous picture is typically what prompts the parents of these children to seek medical attention. Hematuria is distinctly unusual and, in children, hypertension is uncommon. The clinical features of MCD in adults tend to be somewhat different. In a group of 89 adults older than 60 years, hypertension, sometimes severe, as well as renal insufficiency, were more common.⁸² Because individuals older than 60 years account for almost one-quarter of adult patients with MCD, this presentation must be considered.

MCD has been associated with several other conditions, including viral infections, use of certain pharmaceutical agents, malignancy, and allergy (Box 31.4). In some patients, there is a history of a drug reaction before the onset of MCD.

Box 31.4 Common Associations With Minimal Change Disease

Infections

- Viral
- Parasitic

Pharmaceutical Agents

- Nonsteroidal antiinflammatory drugs
- Gold
- Lithium
- Interferon
- Ampicillin
- Rifampin
- Trimethadione
- Tiopronin

Tumors

- Hodgkin disease
- Lymphoma, leukemia
- Solid tumors

Allergies

- Food
- Dust
- Bee stings
- Pollen
- Poison ivy and poison oak
- Dermatitis herpetiformis

Disease and Other Associations

- Systemic lupus erythematosus
- Following allogeneic stem cell transplantation for leukemia
- Following hematopoietic cell transplantation

Modified from references 84, 101, 129 to 131, and 1464 to 1471.

The use of nonsteroidal antiinflammatory drugs (NSAIDs), and in particular fenoprofen, has been associated with and may cause MCD.⁸³ In this setting, most patients have not only proteinuria, but also pyuria and renal insufficiency as a consequence of the simultaneous development of acute tubulointerstitial nephritis. This same process has also been described with other compounds, including interferon,⁸⁴ penicillins, and rifampin. In most of these patients, discontinuation of the offending drug leads to resolution of the proteinuria, but it may take weeks to months for complete amelioration of pyuria and renal insufficiency.

MCD has been associated with lymphoid malignancy, usually Hodgkin disease or thymoma. In a retrospective study of adult patients,⁸⁵ MCD was associated with classic Hodgkin lymphoma of the nodular sclerosis type. MCD appeared before the diagnosis of lymphoma in 40% of patients.

MCD has also been associated with a variety of underlying exposures such as NSAIDs, as mentioned earlier,⁸⁶ and with syndromes such as SLE.^{87,88} It has also occurred after allogeneic stem cell transplantation for leukemia⁸⁹ and after hematopoietic cell transplantation.

There is an association of glomerular disease with simultaneous graft-versus-host disease. Nephrotic syndrome generally follows graft-versus-host disease within 5 months in approximately 60% of hematopoietic cell-transplanted patients with either MCD or MN. Compared with MN, MCD occurred earlier after hematopoietic cell transplantation, was diagnosed soon after medication change, and exhibited a better prognosis because 90% of patients attained complete remission (vs. 27% of patients with MN).

MCD is also associated with food allergy. This is an important association because in some patients, removal of the allergen resulted in the resolution of the proteinuria. Of 42 patients with idiopathic nephrotic syndrome, 16 had positive results on skin tests for food allergy. For 13 of 42, a minimally antigenic diet was prescribed that resulted in a significant reduction in proteinuria.⁹⁰ Thus, it is important to ask patients about potential allergens, especially those found in food.

A syndrome of MCD accompanied by a reversible AKI occurs at higher incidence in adults than in children.^{82,91,92} This syndrome of adult MCD with AKI was studied in 21 patients who, on presentation, had a serum creatinine level of more than 177 $\mu\text{mol/L}$ and who were compared with 50 adult patients with MCD who had a serum creatinine level of less than 133 $\mu\text{mol/L}$. Patients who had AKI were older (59 vs. 40 years), had a higher systolic blood pressure (158 vs. 138 mm Hg), and had more proteinuria (13.5 vs. 7.9 g of protein/24 hours). Importantly, kidney biopsy specimens showed evidence of focal tubular epithelial simplification compatible with ischemic AKI. Of the 18 patients with renal failure for whom follow-up data were available, all showed recovery of kidney function, but only after prolonged periods of dialytic renal replacement therapy.⁴¹

The complications of nephrotic syndrome in MCD have been well described. The development of AKI during the course of MCD, mostly in adults older than 40 years, has been well recognized, but the underlying mechanisms are debated. Explanations for this phenomenon include a marked decrease in glomerular permeability due to extensive foot process effacement, tubular obstruction from proteinaceous casts, and intrarenal hemodynamic changes. Increased endothelin-1 expression in the kidneys of patients with MCD

and AKI could indicate a hemodynamic change that underlies the pathogenesis of renal failure in these circumstances,⁹³ but the true cause of AKI in MCD remains uncertain and is probably multifactorial.

Another complication of MCD with nephrotic syndrome is the development of reduced bone mineral density, possibly due to the effects of glucocorticoids and/or vitamin D deficiency.⁹⁴ Statins may have a beneficial effect on bone mineral density, but the most recent study, in 2005, reported no beneficial effect of fluvastatin on the bone mineral density of children with MCD, although it did have some effect in lowering proteinuria.⁹⁵

LABORATORY FINDINGS

The ubiquitous laboratory feature of MCD is severe proteinuria.⁴⁰ Microscopic hematuria is seen in less than 15% of patients, with only rare episodes of macroscopic hematuria. The rapidity of the development of proteinuria in some patients is associated with evidence of volume contraction, with increased hematocrit and hemoglobin levels. The erythrocyte sedimentation rate is increased as a consequence of the hyperfibrinogenemia, as well as hypoalbuminemia. The serum albumin concentration is usually depressed, whereas the total cholesterol, low-density lipoprotein (LDL), and triglyceride levels are increased. Total serum protein concentration is usually reduced to between 4.5 and 5.5 g/dL with a serum albumin concentration generally less than 2 g/dL and, in more severe cases, less than 1 g/dL. Pseudo-hyponatremia has been observed in the setting of marked hyperlipidemia. The serum calcium level may be low, largely due to hypoproteinemia.

Several abnormalities that promote thrombosis are frequent in patients with severe nephrosis, including increased plasma viscosity, increased red cell aggregation, low plasminogen levels, and low levels of antithrombin III.⁹⁶ Kidney function is usually normal, although the serum creatinine level may be slightly increased at the time of presentation. A minority of patients (usually older adults) have substantial AKI, as discussed earlier.

The loss of albumin into the urine is largely a function of a loss of charge-selective permselectivity.^{72,73,97,98} Consequently, the fractional excretion of albumin is proportionately greater than the fractional excretion of IgG. IgG levels may be profoundly decreased, however—a condition that occurs most notably during episodes of relapse. This low level of immunoglobulin may result in susceptibility to infections. IgM levels may be elevated after a remission.⁹⁹ Mean serum IgA levels may be substantially higher in patients with MCD than in those with another kidney disease¹⁰⁰ and are also elevated in association with relapse in children.¹⁰¹ Among adult patients with MCD, over half have elevated levels of serum IgE, and two-thirds of patients have evidence of some allergic symptoms.¹⁰² Elevation of IgE levels suggests a relationship between MCD and allergy. Complement levels are typically normal in patients with MCD.

TREATMENT

The general approach to treatment of patients with MCD has been to institute corticosteroid therapy. For children, the dose of prednisone is 60 mg/m² per day. For adults, the dose of prednisone is 1 mg/kg of body weight, not to exceed 80 mg/day. In children, this form of therapy results in a

complete remission, with disappearance of proteinuria in over 90% of patients within 4 to 6 weeks of therapy. A response to prednisone therapy has occurred if the patient has had no proteinuria by dipstick analysis for at least 3 days. It should be noted that the serum albumin and serum lipid levels might not return to normal for prolonged periods of time following resolution of the proteinuria.¹⁰³

Treatment is generally continued for at least 6 weeks after complete remission of proteinuria. During those 6 weeks, the dose should be changed to alternate-day administration of prednisone or to a stepwise reduction in the daily dose of prednisone. If the dose is changed to alternate-day dosing when remission has occurred, the dosage may be decreased in children from 60 to 40 mg/m² per day.^{50,104–08}

In adult patients with MCD, a response to corticosteroid treatment may take up to 15 weeks.⁸² In a study of 89 adult patients with MCD, 75 of them were given prednisolone; remission occurred in 60% after 8 weeks, in 73% after 16 weeks, and in 77% over the course of the study. Of the 58 treated patients who showed a response (complete remission at 28 weeks), 24% never experienced relapse, 56% experienced relapse on a single occasion or infrequently, and only 21% had frequent relapses. Of the 89 patients, only 4 remained nephrotic, and 2 of these presented with AKI. Cyclophosphamide therapy was administered to 36 (2 of them as initial treatment, 11 with corticosteroids, and 23 with relapse) of the 89 patients and, in 66% of these patients, the disease was in remission at 5 years.

In a large retrospective analysis of 95 patients with primary adult-onset MCD, AKI complicated the nephrotic syndrome in 20% of patients.¹⁰⁹ The cohort was largely middle-aged, with a substantial prevalence of hypertension (45%) and microscopic hematuria (30%). Of these patients, 92 were initially treated with oral corticosteroids; two-thirds were on a daily regimen, and one-third were on an alternate-day regimen. The initial steroid dose was approximately 1 mg/kg on the daily regimen and approximately 2 mg/kg on the alternate-day regimen so that cumulative doses were similar in the two groups. There were no significant differences in demographic features between the two groups, but patients treated with alternate-day steroid regimen tended to have a lower serum albumin level at presentation than those treated with the daily regimen (1.91 ± 0.14 g/dL vs. 2.31 ± 0.1 g/dL, respectively; $P = .055$). No significant differences were seen between the daily and alternate-day treatment groups in the percentage of patients experiencing complete or partial remission (remissions in 76.8% and 73.9% of patients, respectively), or in the time to remission. It is interesting to note that the rate of relapse of nephrotic syndrome was quite elevated, at 73% of those who showed an initial response. Of the patients who were treated with at least one additional course of corticosteroids, 92% achieved a remission (complete remission in 84.4%). This nonrandomized, uncontrolled study does suggest that alternate-day and daily steroid regimens are of equivalent efficacy and safety in the treatment of MCD.

One of the most controversial issues with respect to treatment is the regimen for tapering the prednisone after the initial response. Sudden withdrawal of corticosteroids, or a rapid taper of prednisone immediately following complete remission, may prompt a relapse. Whether this is a consequence of adrenal insufficiency or depression of the

hypothalamic-pituitary-adrenal axis has been a matter of debate.^{108,110,111} At least in children, the likelihood of relapse is decreased with prolonged administration of corticosteroids over a 10- to 12-week period.^{106,112,113} Once remission has been obtained, an alternate-day schedule should begin within at least 4 weeks of the response to decrease steroid-induced side effects.

In children who have not undergone biopsy before treatment, a kidney biopsy is usually appropriate if there is failure to respond to a 4- to 6-week course of prednisone, particularly if changes have occurred in the clinical course during this period of time suggestive of another glomerular disease. Many pediatricians advocate a biopsy at the onset of the disease if there are clinical features suggesting a diagnosis other than MCD (e.g., hypertension, red blood cell casts in the urine, or hypocomplementemia), or if the nephrotic syndrome begins in the first year of life or after 6 years of age.

After the clinical response to initial treatment, as few as 25% experience a long-term remission,⁹² 25% to 30% have infrequent relapses (no more than one/year), and the remainder experience frequent relapses, steroid dependence, or steroid resistance (Box 31.5). Nephrotic patients who are steroid dependent or experience frequent relapses require additional forms of therapy. The treatment is aimed at minimizing the complications of corticosteroid therapy. In general, induction of a remission with prednisone therapy followed by the institution of cyclophosphamide treatment results in higher urine flow rates and reduced risk of hemorrhagic cystitis. When oral cyclophosphamide is given in dosages of 2 mg/kg for 8 to 12 weeks, 75% of patients remain free of proteinuria for at least 2 years.^{82,114–116} The response to cyclophosphamide may be predicted from the response to corticosteroids. Patients who have experienced an immediate relapse after the cessation of corticosteroid therapy have a greater chance of experiencing relapse immediately after the cessation of cyclophosphamide treatment. Those who have had longer remissions after corticosteroid therapy have a decreased risk of relapse after cyclophosphamide therapy.¹¹⁷ In one study of patients who were steroid dependent, the response to cyclophosphamide was improved by increasing

Box 31.5 Patterns of Response of Minimal Change Disease to Corticosteroid Treatment

- Primary responder, no relapse
- Primary responder with only one relapse in the first six months after an initial response
- Initial steroid response with two or more relapses within six months (frequent relapse)
- Initial steroid-induced remission with relapses during tapering of corticosteroid, or within 2 weeks after their withdrawal (steroid dependent)
- Steroid-induced remission, but no response to a subsequent relapse
- No response to treatment (steroid-resistant)

Modified from references 106, 107, 123, and 1472.

the duration of therapy to up to 12 weeks.¹¹⁴ In at least one other investigation, however, a 12-week course of cyclophosphamide was not proven efficacious.¹¹⁸

Cyclosporine has been emerging as a reasonable alternative to cyclophosphamide.^{119,120} Based on uncontrolled observations, complete remissions are common (>80%), and cyclosporine resistance is seen in only about 10% to 15% of patients. Side effects such as a rise in the serum creatinine level, hypertrichosis, and gingival hyperplasia are common. Relapses are very common after the cessation of cyclosporine treatment. The best method of monitoring cyclosporine levels has not been agreed on. Measurement of trough blood levels with twice-daily dosing, measurement of cyclosporine levels 1 to 2 hours after a once-daily dose (C1–C2) and abbreviated under the curve monitoring of cyclosporine have all been recommended.^{120–122} Cyclosporine may be an acceptable alternative to cyclophosphamide therapy for relapsing or steroid-dependent MCD. The optimal duration of calcineurin inhibitor (CNI)-based therapy is not known and, because relapses are common on cessation of therapy, one may consider a prolonged course of 18 to 24 months of treatment with the lowest dose that keeps a patient in clinical remission. Also, a thorough discussion with patient and continuing observation for adverse effects of prolonged CNI use is necessary.

STEROID-RESISTANT MINIMAL CHANGE DISEASE

Approximately 5% of children with MCD appear to be steroid resistant. In patients who have never undergone kidney biopsy, resistance to corticosteroid therapy is an indication for kidney biopsy. Often, evaluation of the kidney biopsy specimen will demonstrate FSGS or other forms of glomerular injury other than MCD.¹²³

If the diagnosis remains as MCD after kidney biopsy specimen evaluation, there may be several reasons for steroid resistance. Some patients, especially those for whom corticosteroid therapy is overly toxic, may skip doses or not fully adhere to the therapy regimen. For other patients, especially some adults, alternate-day therapy may not provide sufficient amounts of corticosteroid to induce clinical remission. In very edematous patients, oral corticosteroid therapy may not be well absorbed, and intravenous administration of a dose of methylprednisolone may provide a more reliable route. Available data suggest that pulse methylprednisolone may induce remission in some corticosteroid-resistant children. In one study, five of eight corticosteroid-resistant children had a remission with pulse methylprednisolone,¹²⁴ although this experience is not universal.¹²⁵

Patients with MCD may have an unrecognized lesion of focal and segmental glomerulosclerosis that requires longer courses of steroid therapy (usually >4 months) to achieve a lasting remission. A regimen of CNI (cyclosporine or tacrolimus) followed by mycophenolate mofetil (MMF) and monthly intravenous pulse cyclophosphamide therapy was demonstrated in an uncontrolled study to result in a high frequency of complete remission.¹²⁶ There are anecdotal reports of the use of combinations of sirolimus and tacrolimus to treat steroid-resistant MCD,¹²⁷ but the overall safety and efficacy of this regimen are unknown. In a small nonrandomized trial, the use of cyclosporine with steroids was associated with better outcomes in those with steroid-resistant MCD.¹²⁸

Cyclosporine can be administered at a dose of approximately 3 to 5 mg/kg per day in two divided doses. Up to 90% of patients may experience a partial or complete remission with cyclosporine.^{101,106,129–131} Unfortunately, only rarely do patients experience long-term remission once cyclosporine is discontinued.¹⁰⁷ Two trials have examined the use of cyclosporine in steroid-resistant nephrosis. A study conducted by the French Society of Pediatric Nephrology combined cyclosporine with prednisone. Prednisone was given at a dose of 30 mg/m² per day for the first month and then changed to alternate-day dosing for 5 months. Cyclosporine was administered at a dosage of 150 to 200 mg/m² per day.¹³² In this study, 48% of patients with MCD experienced complete remission, some within the first month of therapy. A minority of those who showed a response became steroid sensitive when they later experienced relapse. In a study by Ponticelli and colleagues,¹³³ 13 of 45 patients had MCD and were treated with cyclosporine. In those patients with MCD, partial or complete remission occurred within 2 months of the initiation of therapy. Unfortunately, in spite of the early positive results of this study, relapses occurred in all patients after cyclosporine was stopped.

In a summary of nine studies,¹³⁴ only 20% of children had complete remission with cyclosporine and many, if not most, relapsed with the cessation of therapy. Moreover, although cyclosporine and cyclophosphamide appear to have a similar degree of efficacy with respect to controlling nephrotic syndrome, one study found that cyclophosphamide-treated patients experience a more stable long-term remission.¹³⁵ In this study, the likelihood of a long-term remission was 63% in patients treated with cyclophosphamide but was only 25% in those treated with cyclosporine.

To counteract the usual relapse of nephrosis when cyclosporine has been used for 6 months, an alternative approach to cyclosporine treatment relies on a long-term course of this drug, using gradually lower doses to maintain the patient in remission. In one study,¹³⁶ patients who had been in complete remission for longer than 1 year while taking cyclosporine remained in remission if the cyclosporine was gradually tapered and then stopped. Serial biopsy specimens from patients treated for as long as 20 months showed no overt sign of nephrotoxicity.

An emerging treatment option in adults for steroid-resistant and relapsing MCD reported in numerous case reports and series to date is the B lymphocyte-depleting agent, rituximab. Most recently, the Rituximab in Nephrotic Syndrome of Steroid-Dependent or Frequently Relapsing MCD or FSGS (NEMO) Study Group conducted a multicenter trial in Italy in 10 children and 20 adults with recurrent MCD, mesangial proliferative glomerulonephritis, or FSGS who were treated with one or two doses of rituximab.¹³⁷ Interestingly, all patients were in remission at 1 year, 18 were treatment free and 15 never relapsed. There was also a significant decrease in the number of relapses and median prednisone dose across all disease groups over 1 year of follow-up, and rituximab was well tolerated. A more recent case series combined with a review of large previously published case series of patients with corticosteroid-resistant or corticosteroid-dependent MCD treated with rituximab revealed a very high likelihood of remission with this treatment strategy.¹³⁸ Although rituximab appears to be effective in some individuals, greater mechanistic understanding is needed.^{139–141} Mounting evidence

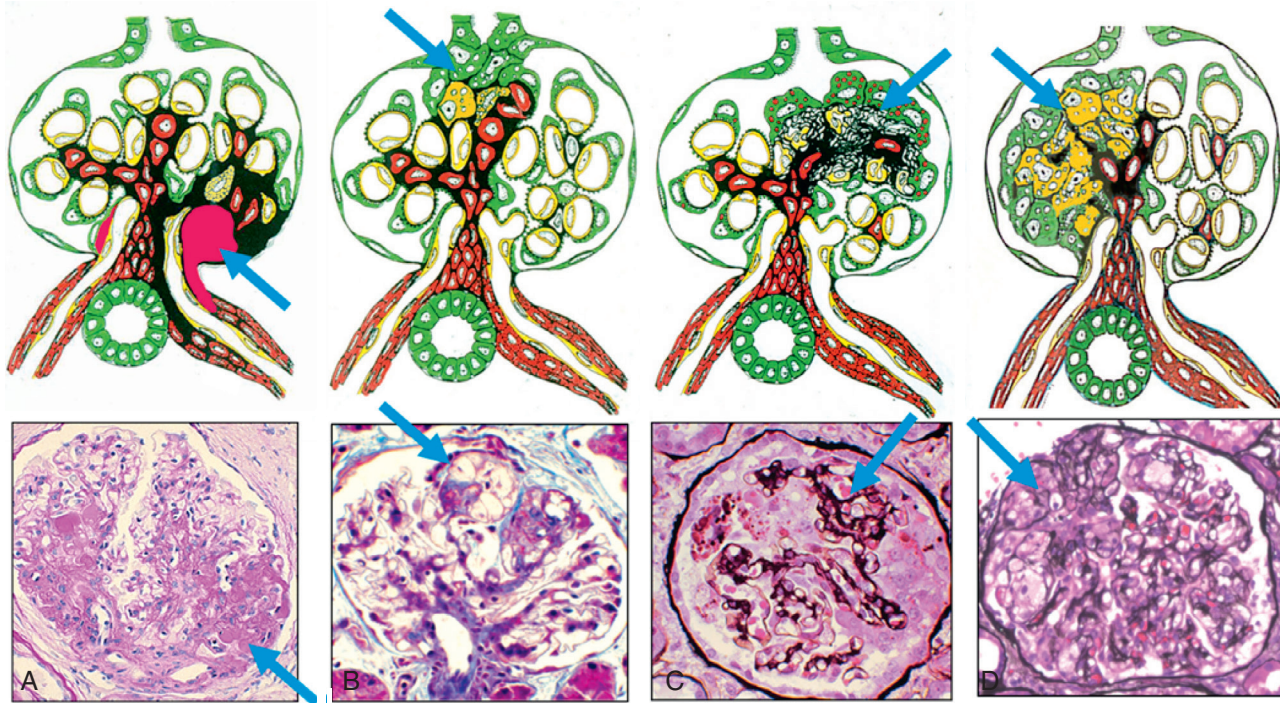


Fig. 31.5 Light micrographs and diagrams depicting histologic variants of focal segmental glomerulosclerosis (arrows). (A) Perihilar variant has a predilection for sclerosis in the perihilar regions of the glomerular (periodic acid–Schiff stain). (B) Glomerular tip lesion variant has segmental consolidation confined to the segment adjacent to the origin of the proximal tubule (Masson trichrome stain). (C) Collapsing glomerulopathy variant has segmental collapse of capillaries with hypertrophy and hyperplasia of overlying podocytes (Jones' methenamine silver stain). (D) Cellular variant with endocapillary accumulation of foam cells (Jones' methenamine silver stain).

has implied that larger, prospective, randomized controlled trials are needed to assess the utility of rituximab in MCD accurately.

FOCAL SEGMENTAL GLOMERULOSCLEROSIS

FSGS is not a single disease but rather a diagnostic term for a clinical-pathologic syndrome that has multiple causes and pathogenic mechanisms, as well as somewhat limited therapeutic options.^{142–145} The ubiquitous clinical feature of the syndrome is proteinuria, which may be nephrotic or non-nephrotic. The ubiquitous pathologic feature is focal segmental glomerular consolidation or scarring, which may have several distinctive patterns (Fig. 31.5). These patterns can be classified as collapsing FSGS, tip lesion FSGS, cellular FSGS, perihilar FSGS, and FSGS not otherwise specified (NOS).^{142,143} The collapsing variant of FSGS is a clinically aggressive variant that is much more common in African-Americans than in Caucasian populations and is characterized pathologically by the segmental collapse of capillaries accompanied by hypertrophy and hyperplasia of epithelial cells and accumulation of prominent protein resorption droplets in podocytes. The glomerular tip lesion variant of FSGS, which typically presents with marked nephrosis but often has a good outcome, is characterized by consolidation and sclerosis in the glomerular segment that is adjacent to the origin of the proximal tubule.¹⁴³ The term “cellular FSGS” is used when numerous cells, especially foam cells, are within consolidated segments. However, this term has been used in a number of ways in the literature, especially before the

publication of the Columbia classification system. For example, the term has been used to describe the collapsing variant and tip lesion variant of FSGS. Care must be taken when reading the literature to determine if this term is being used as defined by the Columbia classification system.¹⁴² The perihilar variant of FSGS is characterized pathologically by sclerosis at the hilum of the glomerulus that typically contains foci of hyalinosis.¹⁴²

As shown in Box 31.6, FSGS may appear to be a primary kidney disease or may be associated with, and possibly caused by, a variety of other conditions (see Chapter 32).

EPIDEMIOLOGY

Over the past 2 decades, the incidence of FSGS has increased, whether expressed as an absolute number of patients or as a proportion of the total incident population of patients with end-stage kidney disease (ESKD).¹⁴⁸ This trend appears to hold true, even when one accounts for a possible increase in the rate of diagnosis resulting from an increase in the frequency of kidney biopsies. Although this trend was previously reported to be most significant among African Americans, it has now been confirmed among Caucasians as well. A review of kidney biopsy findings performed between 1974 and 2003 in Olmstead County, Minnesota, in which 90% of the population are Caucasians of northern European extraction, FSGS was found to account for 17% of glomerulonephritides, second only to IgA nephropathy (22%), and was more frequent than MN (10%).¹⁴⁸ Over that period of time, the incidence of FSGS increased by 13-fold ($P < .001$), compared with a twofold increase in the incidence of all

Box 31.6 Classification of Focal Segmental Glomerulosclerosis

Primary (Idiopathic) Focal Segmental Glomerulosclerosis (FSGS)

FSGS not otherwise specified (NOS)
 Glomerular tip lesion variant of FSGS
 Collapsing variant of FSGS
 Perihilar variant of FSGS
 Cellular variant of FSGS

Secondary FSGS

With human immunodeficiency virus disease
 With intravenous drug abuse
 With other drugs (e.g., pamidronate, interferon, anabolic steroids)
 With identified genetic abnormalities (e.g., in podocin, alpha-actinin-4, TRPC6)
 With glomerulomegaly:
 Morbid obesity
 Sickle cell disease
 Cyanotic congenital heart disease
 Hypoxic pulmonary disease
 With reduced nephron numbers:
 Unilateral renal agenesis
 Oligomeganephronia
 Reflux-interstitial nephritis
 Postfocal cortical necrosis
 Postnephrectomy

glomerular diseases ($P < .001$), and a 2.5- to threefold increase in MN and IgA nephropathy, respectively.

PATHOLOGY

Light Microscopy

FSGS is characterized by focal and segmental glomerular sclerosis or consolidation.^{142,143,149} The sclerosis may begin as segmental consolidation caused by insudation of plasma proteins causing hyalinosis, by accumulation of foam cells, by swelling of epithelial cells, and by collapse of capillaries resulting in obliteration of capillary lumens. These events are accompanied by increased extracellular matrix material that accounts for the sclerosis component of the lesion.

FSGS is, by definition, a focal process when it begins. The limited number of glomeruli in a kidney biopsy specimen may not include any of the segmentally sclerotic glomeruli that are present in the kidney. In this case, focal tubulointerstitial injury or glomerular enlargement, which often accompanies FSGS, can be used as a surrogate marker. For example, FSGS should be considered in kidney biopsy specimens of patients with nephrotic syndrome when there is relatively well-circumscribed focal tubular atrophy and interstitial fibrosis with slight chronic inflammation, even when there are no light microscopic glomerular lesions, no immune deposits, and no ultrastructural changes other than foot process effacement. Segmental sclerosis that is adequate for diagnosis may be present only in the tissue examined by IF or electron microscopy.

Focal segmental glomerular scarring is nonspecific. Many injurious processes can cause focal glomerular scarring and

must be ruled out before making a diagnosis of FSGS. For example, hereditary nephritis causes progressive glomerular scarring that can mimic FSGS, as revealed by the identification of ultrastructural changes that are characteristic of hereditary nephritis. FSGS, for example caused by IgA nephropathy, lupus nephritis, or ANCA-associated glomerulonephritis, can result in focal segmental glomerular scarring that is histologically indistinguishable from that caused by FSGS. Findings by IF, electron microscopy, and serology, can reveal a glomerulonephritic basis for focal glomerular scarring.

Based on the character and glomerular distribution of lesions, five major structural variants of FSGS can be recognized that correlate, at least in part, with outcome (prognosis) and that may be have different causes and pathogenic mechanisms.^{142,143} As mentioned previously, these five pathologic variants are collapsing FSGS, tip lesion FSGS, cellular FSGS, perihilar FSGS, and FSGS NOS.^{142,143}

The characteristic feature of the collapsing variant of FSGS is focal segmental or global collapse of glomerular capillaries, with obliteration of capillary lumens. Podocytes overlying collapsed segments are usually enlarged and contain conspicuous resorption droplets. Hyperplasia of podocytes raises the possibility of crescentic glomerulonephritis. The convention among most renal pathologists is not to refer to the epithelial hyperplasia of collapsing glomerulopathy as crescent formation, although the term “pseudocrescent” has been used by some. The degree of adhesion formation relative to the extent of glomerular sclerosis is much less in collapsing glomerulopathy than in typical FSGS. This may result in contracted (collapsed) tuft basement membranes and sclerotic matrix separated from Bowman’s capsule by hypertrophied and hyperplastic epithelial cells. The collapsing glomerulopathy variant of FSGS is the major pathologic expression of human immunodeficiency virus (HIV) nephropathy^{40,147,150,151} and also occurs with intravenous drug abuse and as an idiopathic process.^{152,153} In those with a kidney transplant, this phenotype of FSGS occurs as both recurrent and de novo disease.^{154,155}

Relative to the extent of glomerular sclerosis, tubulointerstitial injury is more severe in collapsing glomerulopathy than in typical FSGS. Tubular epithelial cells have larger resorption droplets, extensive proteinaceous casts, and marked focal dilation of lumens (microcystic change). There is also more extensive interstitial infiltration by mononuclear leukocytes. IF microscopy findings are similar to those observed in typical FSGS except for the usual finding of larger protein resorption droplets in glomerular podocytes and tubular epithelial cells. Electron microscopy reveals the same structural changes as seen by light microscopy. In a specimen with the collapsing glomerulopathy variant of FSGS, an important ultrastructural assessment is for the presence or absence of endothelial tubuloreticular inclusions. Endothelial tubuloreticular inclusions are identified in over 90% of patients with HIV infection and collapsing glomerulopathy, but in fewer than 10% of patients with idiopathic collapsing glomerulopathy. Other settings in which endothelial tubuloreticular inclusions are numerous are in patients with SLE and in patients treated with interferon- α .

The glomerular tip lesion variant of FSGS was first described by Howie and colleagues and is characterized by consolidation of the glomerular segment that is adjacent to the origin of the proximal tubule and thus opposite the hilum (see Fig. 31.5B).^{156–160} The initial consolidation usually has obliteration

of capillary lumens by foam cells, swollen endothelial cells, and an increase in collagenous matrix material (sclerosis). Podocytes adjacent to the consolidated segment are enlarged and contain clear vacuoles and hyaline droplets. These altered podocytes often are contiguous to, if not attached to, adjacent parietal epithelial cells and tubular epithelial cells at the origin of the proximal tubule, which also have irregular enlargement and vacuolation. The tip lesion may project into the lumen of the proximal tubule. Some lesions are less cellular, with a predominance of matrix and collagenous adhesions to Bowman's capsule at the origin of the proximal tubule.

The cellular variant of FSGS as defined by the Columbia classification system has lesions that resemble the cellular lesion for the tip variant, but they are distributed more widely in the glomerular tuft and are not confined to the tip.¹⁴² Perihilar FSGS is characterized by the perihilar predilection of lesions and the presence of hyalinosis. When FSGS is secondary to obesity or reduced numbers of nephrons, it often has a perihilar pattern and is accompanied by glomerular enlargement. The FSGS NOS category is a nonspecific category, with lesions that do not have the distinctive features of any of the other four variants.

As discussed later, different pathologic variants of FSGS have distinctive demographic characteristics, clinical presentations, and outcomes.

Immunofluorescence Microscopy

In all the histologic variants, nonsclerotic glomeruli and segments usually show no staining for immunoglobulins or complement. As in patients with MCD, as well as individuals with no kidney dysfunction, a minority of patients with FSGS has low-level mesangial staining for IgM in nonsclerotic glomeruli. Low-level mesangial C3 staining is less frequent, and low-level IgG and IgA staining are uncommon. The presence of substantial staining of nonsclerotic glomeruli for immunoglobulins, especially if immune complex–type, electron-dense deposits are present, points toward the sclerotic phase of a focal immune complex glomerulonephritis rather than FSGS.

Sclerotic segments typically show irregular staining for C3, C1q, and IgM (Fig. 31.6). Other plasma constituents are less frequently identified in the sclerotic areas. Epithelial resorption droplets stain for plasma proteins.

Electron Microscopy

The ultrastructural features of FSGS are nonspecific. Electron microscopy plays an important role in the diagnosis of FSGS by helping identify other causes for glomerular scarring that can be mistaken for FSGS by light microscopy alone.

Foot process effacement in FSGS affects sclerotic and nonsclerotic glomeruli and usually is more focal than in MCD. Foot process effacement is less extensive in some forms of secondary FSGS than in idiopathic FSGS. Occasionally, glomerular capillaries have segmental denudation of foot processes. Podocytes adjacent to collapsed segments in collapsing FSGS are cuboidal and appear dedifferentiated. Nonsclerotic glomeruli and segments should have no immune complex–type, electron-dense deposits. One must be careful not to confuse electron-dense “insudated” lesions with immune complex deposits. These lesions correspond to the hyalinosis seen by light microscopy and result from

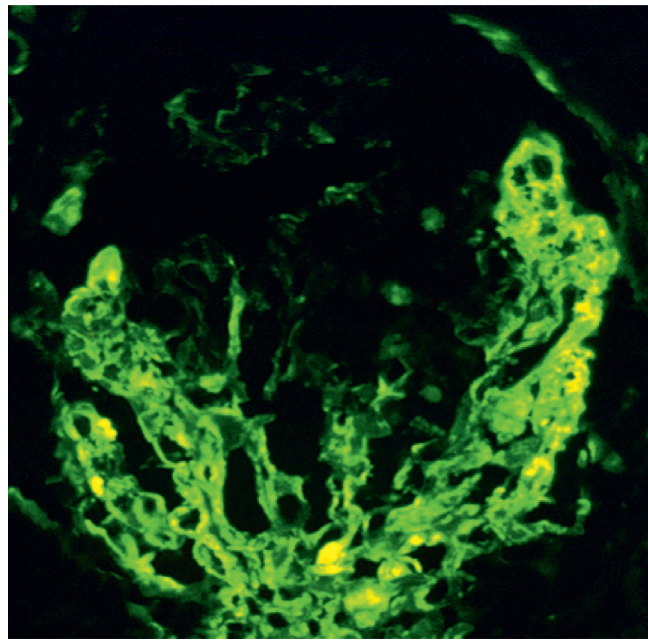


Fig. 31.6 Immunofluorescence micrograph showing irregular segmental staining for C3 corresponding to a site of segmental sclerosis in a patient with perihilar variant focal segmental glomerulosclerosis. (Fluorescein isothiocyanate anti-C3 stain, $\times 3000$.)

the accumulation of plasma proteins within sclerotic areas. When electron-dense material is present in sclerotic but not in nonsclerotic glomerular segments, it should not be considered as evidence for immune complex–mediated glomerular disease. Conversely, well-defined mesangial or capillary wall electron-dense deposits in nonsclerotic segments indicate immune complex–mediated glomerulonephritis with secondary scarring, which should be confirmed and further characterized by IF microscopy.

PATHOGENESIS

The past 30 years have witnessed an explosion of interest in the role of the podocyte in FSGS, in part due to concomitant rapidly advancing genomic technology that has revealed genetic abnormalities, as well as the establishment of well-curated cohort consortia such as NEPTUNE and CureGN. Podocytes are highly differentiated postmitotic cells whose function requires a complex cellular architecture. Tremendous interest has centered on the genetics of familial FSGS, and podocyte defects have taken center stage (reviewed by D'Agati and colleagues¹⁴⁵ and Garg and Rabelink⁹). This has led to the identification of several proteins important to the normal function of podocytes and the development of proteinuria. Mutations in several genes have been identified and linked to familial and sporadic cases of FSGS (see also Chapter 43). These include, but are not limited to, genes coding for podocin (*NPHS2*),¹⁶¹ nephrin (*NPHS1*),¹⁶² α -actinin-IV (*ACTN4*),^{163,164} transient receptor potential cation channel, subfamily C, member 6 (*TRPC6*)^{165–167} and phospholipase C ϵ_1 (*PLCE1*).¹⁶⁸ In addition to mutations in genes encoding podocyte-specific proteins, mutations in other genes are associated with syndromes of which FSGS is often a part.¹⁶⁹ These include the *COQ2* gene,¹⁷⁰ Wilms' tumor gene (*WT1*), and the gene for Lim homeobox transcription factor 1 β

Table 31.4 Gene Mutations With Causal Links to Focal Segmental Glomerulosclerosis

Gene	Protein	Mode of Inheritance
<i>NPHS1</i>	Nephrin	Autosomal recessive
<i>NPHS2</i>	Podocin	Autosomal recessive
<i>CD2AP</i>	CD2-associated protein	Autosomal dominant
<i>TRPC6</i>	Transient receptor potential cation channel 6	Autosomal dominant
<i>PTPRO</i>	Protein tyrosine phosphatase receptor type O (GLEPP1)	Autosomal recessive
<i>LAMB2</i>	Laminin- β_2	Autosomal recessive
<i>ITGB4</i>	β_4 -integrin	Autosomal recessive
<i>CD151</i>	Tetraspanin CD151	Autosomal recessive
<i>ACTN4</i>	α -actinin-4	Autosomal dominant
<i>PLCE1</i>	Phospholipase C epsilon	Autosomal recessive
<i>MYH9</i>	Nonmuscle myosin heavy chain A (NMMHC-A)	Autosomal dominant, de novo mutation
<i>INF2</i>	Inverted formin 2	Autosomal dominant
<i>MYO1E</i>	Myosin 1E	Autosomal recessive
<i>WT1</i>	Wilms' tumor suppressor protein	Autosomal dominant, de novo mutation
<i>SMARCAL1</i>	Smarca-like protein	Autosomal recessive
<i>mtDNA-A3243G</i>	tRNA ^{Leu}	maternal
<i>COQ2</i>	Coenzyme Q2 homolog, polyprenyltransferase	Autosomal recessive
<i>COQ6</i>	Coenzyme Q10 biosynthesis monooxygenase 6	Autosomal recessive
<i>SCARB2</i>	Lysosomal integral membrane protein (LIMP) type 2	Autosomal recessive
<i>APOL1</i>	Apolipoprotein L1	Autosomal recessive

(*LMX1B*), which is a transcription factor required for the expression of *CD2AP* and *NPHS2*, associated with nail-patella syndrome. The genes implicated in the pathogenesis of FSGS are summarized in Table 31.4 and have been reviewed by Rood and colleagues.¹⁷⁰ Whereas these genetic mutations were primarily identified based on cases of familial forms of FSGS, their implication in sporadic FSGS in children and adults, and their impact on treatment, outcome, and recurrence after transplantation, have been the focus of numerous investigations.

Mutations in *NPHS2*, the gene encoding podocin, are the most frequent genetic cause of steroid-resistant nephrotic syndrome and were initially described in early-onset disease (see recent review of all known mutations by Bouchireb and colleagues¹⁷¹). In a study of a large cohort of 430 patients from 404 different families with steroid-resistant nephrotic syndrome, recessive podocin mutations were present in 18.1%.¹⁷² The *R138Q* mutation (a single nucleotide substitution of G to A at position 413 in the third exon of podocin gene) was found in 57% of families with two disease-causing mutations on each of the parental gene copies. Of these podocin mutations, 70% were nonsense, frameshift, or homozygous *R138Q*. Patients with these mutations manifested symptoms at a significantly earlier age (mean onset <1.75 years) than any other patient group, with or without podocin mutations (mean onset >4.17 years). The sequence variant *R229Q* was found in 9% of families in the heterozygous mutation and 0.5% as a homozygous mutation.

The significance of the *R229Q* sequence variant as a cause of nephrotic syndrome was addressed in a study of 546 patients (from 455 families) with familial or sporadic FSGS, only 24% of whom developed nephrotic syndrome after the age of 18 years.¹⁷³ The *R229Q* allele frequency was significantly higher among European and South American patients than among control individuals (0.089 for European patients vs. 0.026; for controls, $P = .00001$; and 0.17 for South American patients vs. 0.007, $P = .000002$). Compared with individuals without

a *p.R229Q* allele, those with a *p.R229Q* allele had a significantly higher likelihood of having a single pathogenic *NPHS2* mutation, which strongly suggests a pathogenic role of *p.R229Q* in the compound heterozygous state with a *NPHS2* mutation. Patients carrying *p.R229Q* and one *NPHS2* mutation developed nephrotic syndrome significantly later than those carrying two pathogenic mutations (median, 19.0 vs. 1.1 years; $P < .01$). The frequency of *NPHS2* mutations in adults with treatment-resistant nephrotic syndrome was 11% in sporadic cases and 25% in familial cases. This study suggests that compound heterozygosity for *p.R229Q* is associated with adult-onset, steroid-resistant nephrotic syndrome. Furthermore, genetic analyses of *NPHS2* may identify individuals who are steroid-resistant before treatment initiation, as well as those who are more likely to progress to ESKD and/or have a low posttransplantation recurrence risk.¹⁷⁴

The role of *NPHS2* polymorphisms in sporadic cases of late-onset idiopathic or HIV-associated FSGS was studied in 377 biopsy-confirmed FSGS cases and 919 controls without known kidney disease.¹⁷⁵ No homozygotes or compound heterozygotes were observed for any of five missense mutations identified on gene sequencing. *R138Q* carriers were five times more frequent among FSGS cases than controls ($P = .06$), but heterozygosity for the other four missense mutations (including *R229Q*) was equally distributed among FSGS cases and controls. Genetic variation or mutation of *NPHS2* may therefore play a role in late-onset sporadic FSGS. However, given the very low frequency of *R138Q* (4–8/1000) and the lack of involvement of other mutations, the attributable risk of *NPHS2* for adult sporadic or HIV-associated FSGS is extremely small. The presence of the nonsynonymous variants of *NPHS2* (*p.R229Q* and *p.A242V*) did not significantly alter the risk of albuminuria in the Nurses' Health Study participants.¹⁷⁶

In summary, *NPHS2* gene mutations are associated with familial and childhood-onset, steroid-resistant nephrotic syndrome, but may not contribute significantly to adult-onset sporadic FSGS. Heterozygosity for *R138Q* is associated with

a fivefold increased risk of FSGS in adults. When combined with other pathogenic mutations of *NPHS2*, the R229Q variant appears to be associated with disease onset at older age.

Typically, mutations in *NPHS1*, which encodes nephrin, are associated with congenital nephrotic syndrome of the Finnish type presenting within the first 3 months of life. A study of 160 patients from 142 unrelated families who presented with nephrotic syndrome at least 3 months after birth identified *NPHS1* mutations in one familial case and in nine sporadic cases.¹⁷⁷ Kidney biopsy at presentation revealed mesangiolipofoliferative lesions in one patient, MCD in six patients, and FSGS in three patients. This study broadens the spectrum of kidney disease related to nephrin mutations and raises the possibility that mutations in *NPHS1* may contribute to sporadic nephrotic syndrome in combination with mutations in other genes associated with this syndrome.^{177,178}

Mutations in *ACTN4*, the gene encoding the actin-binding protein α -actinin-4, are a cause of familial FSGS, with an autosomal dominant pattern of inheritance, and may be associated with a distinctive ultrastructural feature of podocyte injury consisting of cytoplasmic electron-dense aggregates that help identify patients with *ACTN4* mutations.¹⁷⁹ Interestingly, mechanistic investigations have revealed that *ACTN4* protein mutants lack the ability to activate protective nuclear hormone receptors in podocytes.¹⁸⁰

African Americans have a disproportionate risk for several forms of kidney disease, including a fourfold increased risk for FSGS and an 18- to 50-fold increased risk for HIV-associated FSGS. The basis for that susceptibility is thought to be multifactorial, with a suspected genetic component. Two landmark studies, both of which used genome-wide mapping by admixture linkage disequilibrium, identified a chromosome 22 region that was associated with kidney disease in subjects of African ancestry.^{181,182} Initial efforts focused on MYH9, which encodes a nonmuscle myosin heavy-chain type IIA expressed in kidney podocytes and possibly mesangial cells, where it binds to actin to perform intracellular motor functions. Multiple MYH9 single-nucleotide polymorphisms (SNPs) and haplotypes were recessively associated with FSGS. Nine MYH9 SNPs and the same haplotype for FSGS were associated with hypertensive ESKD (odds ratio [OR] = 2.2; 95% confidence interval [CI] = 1.5–3.4; $n = 433$), but not ESKD associated with type 2 diabetes.¹⁸¹

Unlike other mutations that are directly linked to the development of FSGS, the association of MYH9 does not establish direct causality. In addition, more recent work has revealed coding variants within the apolipoprotein L-1 (APOL1) gene, which closely neighbors the MYH9 gene, that are more likely responsible for the association with kidney disease.¹⁸³ Two groups have identified sequence variants in the APOL1 gene that likely explain the genetic association of kidney disease in patients of African ancestry.^{184,185} Reexamination of the interval surrounding MYH9 has led to these newer findings. Genovese and colleagues¹⁸⁴ and Tzur and associates¹⁸⁵ both used data from the 1000 Genomes Project—which conducts the sequencing of genomic DNA derived from subjects around the world, especially the Yoruba tribe of Western Africa—and identified sequence variants in the APOL1 gene that were associated with kidney disease. These variants were more strongly associated with ESKD than the previously reported MYH9 variants.

In African Americans, Genovese and colleagues¹⁸⁴ have found that FSGS and hypertension-attributed ESKD are associated with two independent sequence variants in the APOL1 gene on chromosome 22 (FSGS, OR = 10.5; 95% CI = 6.0–18.4; hypertension-attributed ESKD, OR = 7.3, 95% CI = 5.6–9.5). The two APOL1 variants were common in African chromosomes but absent from European chromosomes, and both reside within haplotypes that harbor signatures of positive selection. The APOL1 gene product, apolipoprotein L-1, has been studied for its role in trypanosomal lysis, autophagic cell death, and lipid metabolism, as well as for its vascular effects.

APOL1 is a serum factor that lyses trypanosomes. Of interest, it was hypothesized that APOL1 variants protect patients against *Trypanosoma brucei*, the parasite that causes sleeping sickness in thousands of people in Africa. Conversely, and with great irony, these same sequences are associated with kidney disease; in vitro assays revealed that only the kidney disease-associated apolipoprotein L-1 variants lysed *T. brucei rhodesiense*, a particularly aggressive newer subspecies. Carrying two copies of the haplotype carried significantly more risk of kidney disease than carrying one copy. In some respects, therefore, kidney disease-associated APOL1 variants and protection from tsetse flies mirrors sickle cell anemia and protection from malaria.

Risk variants in MYH9 and APOL1 are in strong linkage disequilibrium, and genetic risk that was previously attributed to MYH9 may reside in APOL1. More studies are required to test whether more complex models of risk are operative. Nonetheless, this genetic association on chromosome 22 explains, at least in part, racial disparities in nondiabetic ESKD and HIV-associated kidney disease because of the high prevalence of these haplotypes in individuals of African ancestry.¹⁸⁶ (For further discussion of inherited diseases of the glomerulus, see Chapter 43.)

In addition to being the site of genetic anomalies, the podocyte may be the target of injury through several mechanisms.¹⁸⁷ These mechanisms include infections (e.g., HIV infection), exposure to certain drugs (e.g., pamidronate, interferon), metabolic disorders (e.g., diabetes), and deposition of abnormal protein (e.g., amyloid). Podocyte injury can lead to proteinuria through abnormalities of the slit diaphragm (e.g., due to genetic mutations), podocyte loss, loss of negative charges from loss of podocyte proteins (decreased podocalyxin or glomerular epithelial protein), decreased production of these proteins (decreased heparin sulfates), or destruction of the GBM (e.g., proteases), malformation of GBM, or podocyte-related endothelial cell dysfunction (e.g., decrease in vascular endothelial growth factor).¹⁸⁷ In addition, the response of the podocyte to injury may be expressed differently, depending on the nature of the insult, and can result in different clinical syndromes. Thus, it has been proposed that various clinical and histologic variants of nephrotic syndromes depend on whether the podocyte injury results in apoptosis, podocyte loss, or podocyte dedifferentiation and reentry into the cell cycle and proliferation.¹⁸⁸ Indeed, this concept has led to proposals to change the classification of nephrotic syndromes based on the understanding of the pathogenetic mechanisms of disease, in addition to a description of histologic variants.¹⁸⁹

In collapsing forms of FSGS, podocytes undergo changes in which mature podocyte markers disappear, which suggests

a dysregulated podocyte phenotype in these diseases.^{190–192} In fact, podocyte proliferation is seen in some examples of FSGS, which may be a consequence of the decrease in cyclin-dependent kinase inhibitors P27 and P57.¹⁹³ This has led to the concept that in collapsing FSGS, podocytes become dysregulated and proliferate.¹⁹⁴ However, the concept that the podocyte is a proliferating cell in collapsing FSGS has been challenged. In a mouse model of focal sclerosis, parietal epithelial and not podocytes were involved in the proliferative event.¹⁹⁵ This challenge is also supported from the findings of a study of two patients with HIV- and pamidronate-associated collapsing FSGS.¹⁹⁶

The effacement of foot processes may be a consequence of the overproduction of oxygen radicals and accumulation of lipid peroxidase.¹⁹⁷ In theory, podocyte loss could result in focal areas of GBM denudation, with diminished barrier function. Podocyte dropout may be a major factor in the development of glomerulosclerosis in general and specifically in the development of collapsing FSGS.^{198–201}

Some of the same pathogenic events that result in segmental scarring secondary to focal glomerular injury caused by a proliferative or necrotizing glomerulonephritis are probably operative in producing the sclerosis of FSGS. In this regard, the overproduction of transforming growth factor- β 1 (TGF- β 1) in glomeruli due to acute inflammatory lesions may cause glomerular sclerosis.²⁰² In experimental models of glomerular inflammation, the administration of antibodies to TGF- β or other inhibitors of TGF- β resulted in a decrease in matrix accumulation and a reduction in the severity of glomerular scarring.²⁰³ Several mechanisms are associated with the fibrosis of kidney disease. Extracellular matrix and proteoglycans such as decorin and biglycan may have a pathogenic role in fibrosing diseases through the regulation of TGF- β 1.²⁰⁴

FSGS also results from the loss of nephrons, which causes compensatory intraglomerular hypertension and enlargement in the remaining glomeruli. The compensatory capillary hypertension results in podocytes and endothelial cell injury, as well as mesangial alterations that lead to progressive focal and segmental sclerosis.^{205–211} This process, at least in experimental animals, is made worse by increased dietary protein intake and is ameliorated by protein restriction and antihypertensive therapy.

A permeability factor has been described in some patients with FSGS. In a seminal study, 33 patients with FSGS that recurred after transplantation had a higher mean value for permeability to albumin value than normal subjects.²¹² After plasmapheresis, the level of permeability factor in six patients was reduced, and proteinuria significantly decreased. The nature of the FSGS permeability factor continues to elude research efforts at identifying it. It has been hypothesized to consist of low-molecular-weight anionic proteins or proteins that alter the phosphorylation of glomerular proteins.²¹³ There is also a report of high affinity of the FSGS permeability factor to galactose, which appears to inhibit its activity *in vitro*.²¹⁴ Treatment with oral galactose has been anecdotally reported to afford remission of nephrotic syndrome.²¹⁵ Galactose treatment achieved a 50% reduction in proteinuria in three of seven patients in the FONT study,²¹⁶ but did not in another very small uncontrolled trial.²¹⁷ Another proposed permeability factor is circulating serum soluble urokinase receptor (suPAR), which has been reported to be elevated in two-thirds of patients with primary FSGS as compared to

patients with other glomerular diseases.²¹⁸ Furthermore, suPAR was shown in the same study in mice to activate podocyte β_3 integrin, resulting in foot process effacement, proteinuria, and histopathologic changes akin to human FSGS. Subsequently, serum suPAR levels were shown to be elevated in two cohorts of children and adults with FSGS and not disease controls²¹⁹; however, some more recent studies have reported an inverse correlation between the estimated glomerular filtration rate (eGFR) and suPAR, suggesting possible confounding by the eGFR.^{220–223}

Regarding the role of a circulating permeability factor in the pathogenesis of the disease, it should be noted that in a minority of patients with steroid-resistant FSGS in native kidneys, plasmapheresis may diminish proteinuria and stabilize kidney function. In most patients, however, there is no improvement in proteinuria, despite loss of the permeability factor following plasmapheresis.²²⁴

Glomerular enlargement accompanied by the development of FSGS occurs in the setting of hypoxemia—for example, in patients with sickle cell anemia, congenital pulmonary disease, or cyanotic congenital heart disease. Obesity predisposes to FSGS.^{225,226} Weight loss and the administration of an angiotensin-converting enzyme (ACE) inhibitor decreased protein excretion by 80% to 85%.^{227,228} Patients with sleep apnea may have proteinuria that is more functional in nature, but with little or no evidence of glomerular scarring or epithelial injury observed in biopsy specimens.^{229,230} The association between sleep apnea and proteinuria was examined in an analysis of 148 patients referred for polysomnography who were not diabetic and had not been treated previously for obstructive sleep apnea.²³¹ In this patient population, clinically significant proteinuria was uncommon; it was found to be associated with older age, hypertension, coronary artery disease, and arousal index by univariate analysis, but only with age and hypertension by multiple regression analysis. Body mass index and the apnea-hypopnea index were not associated with the urine protein-to-creatinine ratio. The authors concluded that nephrotic-range proteinuria should not be ascribed to sleep apnea and deserves a more thorough nephrologic evaluation.

A number of infections cause FSGS. HIV-associated FSGS is pathologically identical to idiopathic collapsing FSGS, except that endothelial tubuloreticular inclusions are present in the former but not the latter. This close association of HIV infection with collapsing FSGS, as well as experimental evidence of focal glomerular sclerosis in mice transgenic for HIV type I genes,^{146,150,232–238} raise the possibility that the HIV virus can be a causative agent of FSGS in infected patients. Whether other viral diseases, including parvovirus B19 infection, cause the idiopathic collapsing variant of FSGS remains to be elucidated.^{239,240} Parvovirus B19 has been found with greater frequency in patients with idiopathic and collapsing FSGS than in patients with other diagnoses.²³⁹ The polyomavirus SV40 may also play a role.²⁴¹

FSGS is associated with a number of malignant conditions that have been linked to lymphoproliferative disease. In one study,²⁴² an association of FSGS was found with monoclonal gammopathies of undetermined significance (MGUS) and multiple myeloma. When the lymphoproliferative disease was treated, the renal lesion improved.

Finally, FSGS has been linked to exposure to a number of medications, including pamidronate^{195,243} and interferon,²⁴⁴

and with the use of anabolic steroids.²⁴⁵ Like pamidronate and interferon therapy, the latter may also be associated with collapsing FSGS. Discontinuation of the anabolic steroids may lead to reduction of the proteinuria.

CLINICAL FEATURES AND NATURAL HISTORY

Proteinuria is the hallmark feature of all forms of primary FSGS. The degree of proteinuria varies from nonnephrotic (1–2 g of protein/day) to over 10 g of protein/day, associated with all of the morbid features of nephrotic syndrome. Hematuria occurs in over 50% of FSGS patients, and approximately one-third of patients present with some degree of renal insufficiency. Gross hematuria is more commonly seen in FSGS than in MCD.²⁴⁶ Hypertension is a presenting feature in one-third of patients. There are differences in the presentation of FSGS in adults and children.^{247–250} Children tend to have more proteinuria, whereas hypertension is more common in adults.

Differences in clinical manifestations correlate with different pathologic phenotypes of FSGS.²⁵¹ Patients with perihilar FSGS accompanied by glomerular enlargement more commonly have nonnephrotic-range proteinuria than patients without glomerular hypertrophy. In addition, there are differences in the clinical presentation of the collapsing variant of FSGS and the glomerular tip lesion variant of FSGS.

Patients with collapsing FSGS have substantially more proteinuria, a lower serum albumin level, and a higher serum creatinine level than patients with perihilar FSGS.^{152,153,252} The development of proteinuria, edema, or hypoalbuminemia may occur rapidly over the course of days to weeks, in contrast to the more indolent development of proteinuria in most patients with typical FSGS. Moreover, patients with collapsing FSGS more frequently have extrarenal manifestations of disease a few weeks prior to onset of the nephrosis, such as episodes of diarrhea, upper respiratory tract infections, or lower respiratory tract-like symptoms that are usually ascribed to viral or other infectious processes. However, the systemic symptoms of fever, malaise, and anorexia occur in fewer than 20% of patients at the time of onset of nephrosis. Pamidronate, a bisphosphonate that prevents bone disease in patients with myeloma and metastatic tumors, has been reported to be associated with collapsing FSGS.^{195,243} After discontinuation of the drug, kidney function stabilized in all patients except those with collapsing FSGS. Treatment with interferon has also been reported to be associated with the development of collapsing FSGS.²⁴⁴

The clinical presentation of a glomerular tip lesion differs from that of both perihilar FSGS and collapsing FSGS.²¹⁵ Patients with glomerular tip lesion tend to be older Caucasian males, in contrast to the younger African-American male prevalence in collapsing FSGS. The proteinuria in these patients usually is severe, and the onset is abrupt, with the sudden development of edema and hypoalbuminemia. The rapidity of onset of the disease process is similar to the clinical presentation of MCD.^{158,253,254} Patients with glomerular tip lesion may develop reversible AKI, especially at the time of initial presentation when the degree of proteinuria, edema, and hypoalbuminemia are at their peak. This also is similar to the presentation of MCD but rarely occurs with other variants of FSGS.

Several studies have addressed the clinical applicability and implications of distinguishing the variants of FSGS. An

analysis of data for 197 patients followed in the Glomerular Disease Collaborative Network between 1982 and 2001 found the FSGS NOS variant in 42% of cases, the perihilar variant in 26%, the tip lesion in 17%, the collapsing lesion in 11%, and the cellular variant in 3% of patients.¹⁴³ African Americans accounted for 91% of patients with collapsing FSGS, but only 15% of patients with the tip lesion variant. Both collapsing and tip variants were associated with significantly greater amounts of proteinuria (10 ± 5.3 g protein/day and 9.7 ± 7.0 g protein/day, respectively) than perihilar or NOS variants of FSGS (4.4 ± 3.3 g protein/day and 5.5 ± 4.6 g protein/day, respectively; $P < .001$). In this retrospective uncontrolled analysis, patients with the tip lesion variant of FSGS were significantly more likely to attain a complete remission, even after adjustment for corticosteroid exposure ($P < .001$). Collapsing FSGS had the worst 1-year (74%) and 3-year renal survival rates of all other variants, regardless of differences in the histologic severity of the injury, which suggests the possibility that the nature of the injury was inherently different.

Similarly, an analysis of data for 225 patients studied at Columbia University²⁵¹ confirmed the predilection of the tip lesion variant of FSGS for Caucasians (86.2%), although the predominance of African Americans among patients with collapsing FSGS was less pronounced (53.6%). In this large cohort, 10% of patients had the cellular variant, of whom 32% were African American. The mean proteinuria (9.5 ± 1.2 g of protein/day) in these patients was comparable to that in patients with the collapsing or tip variants (8.8 ± 1.3 g of protein/day and 7.8 ± 0.6 g of protein/day, respectively). Patients with cellular FSGS showed intermediate rates of remission (44.5%) and ESKD (27.8%) compared with patients with collapsing FSGS (remission rate, 13.2%; ESKD rate, 65.3%) and tip FSGS (remission rate, 75.8%; ESKD rate, 5.7%).

A retrospective analysis of data for a cohort of 93 adult patients in the Netherlands has confirmed the improved renal survival of patients with the tip variant compared with patients with the other variants (5-year survival of 78% for tip vs. 63% for FSGS NOS and 55% for perihilar FSGS; $P = .02$).²⁵⁵

A major prospective FSGS study has evaluated the clinical impact of these histopathologic variants among participants with steroid-resistant primary FSGS in the FSGS Clinical Trial.^{256,257} As in prior retrospective studies, the tip variant and collapsing variant had the strongest association with Caucasians (86%) and African Americans (63%, $P = .003$), respectively. This study also confirmed poor renal survival in participants with the collapsing variant and better renal survival in participants with the tip variant, although the sample size was small for these groups.

The degree of proteinuria is a predictor of long-term clinical outcome. Nonnephrotic-range proteinuria correlates with a more favorable renal survival of over 80% after 10 years of follow-up.^{258,259} In contrast, patients who have proteinuria of more than 10 g of protein/day have very poor long-term renal survival, with most patients experiencing ESKD within 3 years.^{260,261} Patients with FSGS and protein excretion that measures between nonnephrotic range and massive proteinuria have variable long-term renal outcomes. In general, these patients have a relatively poor outcome, with half experiencing ESKD by 10 years.^{249,250,262}

One of the most useful prognostic indicators for patients with FSGS is whether remission of nephrotic syndrome is

achieved.²⁴⁸ Patients who experience remission of nephrosis have a substantially greater renal survival rate than those who do not.^{248,258,259,263,264} According to Korbet and colleagues,^{249,250} fewer than 15% of patients who achieve complete or partial remission progress to ESKD within 5 years of follow-up. Up to 50% of patients who do not experience remission progress to end-stage disease within 6 years of follow-up.

As in other forms of glomerular injury, the entry serum creatinine level correlates with long-term renal survival.^{258,260,265,266} Patients with a serum creatinine level over 1.3 mg/dL have poorer renal survival than those with lower serum creatinine concentrations, irrespective of the level of proteinuria (10-year renal survival of 27% vs. 100%).²⁵⁰ Multivariate analyses have indicated that the entry serum creatinine level may be more important than proteinuria as a predictor of progression to ESKD.*

LABORATORY FINDINGS

Hypoproteinemia is common in patients with FSGS, with total serum protein reduced to varying extents. The serum albumin concentration may fall to below 2 g/dL, especially in patients with collapsing and glomerular tip variants of FSGS. As in other forms of nephrotic syndrome, levels of immunoglobulins are typically depressed, and levels of lipids are increased, especially the serum cholesterol level. Serum levels of complement components are generally in the normal range in FSGS. Circulating immune complexes have been detected in patients with FSGS,^{267,268} although their pathogenic significance has not been determined. Serologic testing for HIV infection should be obtained for patients with FSGS, especially those with the collapsing pattern.

TREATMENT

Angiotensin Inhibitors

ACE inhibitors and angiotensin II receptor blockers (ARBs) have been evaluated in the treatment of FSGS. ACE inhibitors have been shown to decrease proteinuria and the rate of progression to ESKD in diabetic and nondiabetic kidney disease.^{269–272} These results have been observed in the presence of diabetes, as well as in cases of nondiabetic kidney disease. A systematic review of studies has revealed that ACE inhibitors significantly reduce proteinuria in children with steroid-resistant nephrotic syndrome.²⁷³ In a randomized controlled trial involving normotensive children with steroid-resistant nephrotic syndrome, the addition of fosinopril to prednisone resulted in a greater reduction in proteinuria than prednisone alone.²⁷⁴

In patients with glomerulomegaly and nonnephrotic-range proteinuria, an ACE inhibitor or ARB sufficiently decreases proteinuria and potentially decreases hyperlipidemia, edema, and other manifestations of persistent loss of protein in the urine, with an excellent long-term prognosis. Regardless of what other forms of antiinflammatory or immunosuppressive therapy are used, the beneficial effects of these agents indicates that they should be added, despite the well-known side effects of hyperkalemia and reduction in GFR, especially in patients with serum creatinine levels over 3 mg/dL.

Before immunomodulatory or immunosuppressive therapy is initiated in a patient with FSGS, a careful evaluation should

be undertaken to exclude the possibility of an underlying cause, as described in the section on pathogenesis. Patients with secondary FSGS are unlikely to benefit from immunosuppressive therapy and may be at particularly high risk of complications. In addition, an assessment of the risks and benefits of immunosuppressive therapy should be undertaken for each patient. Patients with subnephrotic-range proteinuria have a generally good prognosis, and the initial therapy should be focused on blood pressure control, preferentially using maximal tolerated dosages of renin-angiotensin-aldosterone system (RAAS) blockers. Glucocorticoids or immunosuppressive therapy should be targeted to patients with idiopathic FSGS and nephrotic syndrome.

Glucocorticoids

No randomized placebo-controlled trial has been conducted to assess the role of glucocorticoids in the treatment of FSGS formally. The available data are based on case series using different treatment protocols, different definitions of remission, response, relapse, and resistance and different lengths of therapy.^{248,258,263,275} One review of studies has suggested that only 15% of patients with FSGS responded to treatment, in sharp contrast to those with MCD.²⁷⁶ More optimistic reports have been obtained by groups in Toronto and Chicago,^{248,249} which suggest that 30% to 40% of adult patients may attain some form of remission with corticosteroid treatment. A compilation of these studies by Korbet²⁵⁰ has suggested that of 177 patients who received a variety of different forms of therapy, 45% experienced complete remission, 10% experienced partial remission, and 45% showed no response.^{248,258,263,275,277}

In children, the initial treatment of FSGS is similar to that of MCD because treatment is typically initiated without histologic confirmation of the disease process. Thus, the International Study of Kidney Diseases in Children has recommended using an initial course of prednisone of 60 mg/day/m², up to 80 mg/day, for 4 weeks. This is followed with 40 mg/day/m², up to 60 mg/day, administered in divided doses for 3 consecutive days out of 7, for 4 weeks, and then tapered off for 4 more weeks. As in adult patients with MCD, a longer course of therapy at higher doses of prednisone may be necessary to induce remission. Thus, in those series and in retrospective analyses that showed an increased remission rate,[†] prednisone treatment was continued for 16 weeks to achieve remission. In adult patients, the median time for complete remission was 3 to 4 months.¹³⁶

A portion of patients showing a positive response to corticosteroid treatment will experience relapse. Guidelines for the retreatment of relapsing patients are similar to those for treatment of patients with relapsing MCD. In patients whose remission prior to relapse was prolonged (>6 months), a repeat course of corticosteroid therapy may again induce a remission. In steroid-dependent patients who develop frequent relapses, repeated rounds of high-dose corticosteroid therapy result in unacceptable cumulative toxicity. Thus, alternative strategies, such as the addition of cyclosporine, may be useful. In patients with the glomerular tip lesion variant of FSGS, a trial of corticosteroids is appropriate because many patients experience a decline in protein excretion.^{157,254,280}

Attempts at alternate-day steroid therapy have not been successful, except in older populations. The Toronto group²⁸¹

*References 258, 259, 261, 262, 265, and 266.

†References 133, 248, 258, 263, 275, 278, and 279.

has demonstrated that a 40% remission rate could be achieved in patients older than 60 years by administering up to 100 mg of prednisone on alternate days for 3 to 5 months. This therapy was well tolerated in this population, with no obvious side effects during the study period. Alternate-day prednisone most likely works in this population because of an increased susceptibility to the immunosuppressive effects of corticosteroids and altered glucocorticoid kinetics in older adults.

The benefit of corticosteroid therapy may differ among Caucasians and African Americans. In a retrospective analysis²⁸² of renal survival in predominantly African American individuals with FSGS, renal survival was found to be higher when the initial serum creatinine level was lower and blood pressure was well controlled, but treatment with steroids had no effect on renal survival.

The practice of using higher dosages of corticosteroids to induce remission with the attendant risk of toxicity has resulted in the use of alternative therapeutic approaches. Administration of very high doses of corticosteroids in children, and the continuation of daily prednisone therapy for up to 6 to 9 months in adults, is not without enormous short- and long-term side effects. In studies in which long-term, high-dose corticosteroids are administered, few analyses have been undertaken to evaluate the development of osteoporosis, short- and long-term risk of infection, cataracts, diabetes, or other long-term sequelae. Thus, the available data do not allow for a careful understanding of the risk-benefit ratio. Until the use of high-dose methylprednisolone has been studied in controlled clinical trials, this potentially useful yet dangerous approach must be viewed with caution.

Cyclophosphamide

Several studies have failed to document the effectiveness of cytotoxic drugs in the treatment of FSGS.^{263,283} In one review, only 23% of 247 children with FSGS showed a response to steroid therapy, and 70 patients were treated with cytotoxic drugs. Of these, 30% showed a response. In the final analysis, the disease was in remission in fewer than 20% of the 247 children. The use of cytotoxic drugs has been evaluated in only one series of adults.²⁶³ Although their use correlated with longer remissions and fewer relapses, no other study has corroborated these results.

The International Study of Kidney Diseases in Children carefully examined the role of cyclophosphamide in the treatment of children with FSGS.²⁸⁴ Daily oral cyclophosphamide (2.5 mg/kg) was administered in addition to prednisone (40 mg/m² every other day) for 12 months, and results were compared with those for prednisone alone. The addition of cyclophosphamide had no effect on the change in proteinuria or the likelihood of achieving complete resolution of proteinuria.²⁸⁴ Similarly, in a nonrandomized comparative study involving children with FSGS or steroid-resistant or frequently relapsing nephrotic syndrome, the addition of oral cyclophosphamide to prednisone for 3 months had no statistically significant effect on the rate of complete or partial remission or progression to ESKD.²⁸⁵ In summary, the limited, currently available data do not support the use of cyclophosphamide in patients with FSGS.

Cyclosporine

FSGS that is resistant to prednisone may be induced into remission by cyclosporine. The effectiveness of cyclosporine

in inducing remission of proteinuria in patients with FSGS has been demonstrated in two randomized controlled trials. In the study by Ponticelli and colleagues, 45 patients with steroid-resistant nephrotic syndrome were randomly assigned to receive supportive therapy or cyclosporine (5 mg/kg/day for adults, 6 mg/kg/day for children) for 6 months, with the drug then tapered off by 25% every 2 months.¹³³ Remission occurred in 13 of 22 patients receiving cyclosporine, compared with 3 of 19 patients in the control group ($P < .001$). Unfortunately, relapses occurred in 69% of patients after the withdrawal of cyclosporine.

In the North America Nephrotic Syndrome Study Group trial, 49 patients were randomly assigned to treatment with low-dose prednisone alone or in combination with oral cyclosporine for 26 weeks.²⁸⁶ At the end of 26 weeks of therapy, partial or complete remission of proteinuria occurred in 70% of patients in the cyclosporine-treated group compared with only 4% in the control group ($P < .001$); however, relapses occurred by 52 weeks in 40% of those experiencing remission. Treatment with cyclosporine was also associated with a 70% reduction in the risk that the GFR would decline by 50%.²⁸⁶

A randomized controlled trial compared the efficacy of cyclosporine with intravenous cyclophosphamide in the initial treatment of 22 children with steroid-resistant nephrotic syndrome related to MCD, FSGS, or mesangial hypercellularity.²⁸⁷ All patients also received alternate-day prednisone therapy. Treatment with cyclosporine afforded a statistically significant higher rate of partial remission of proteinuria at 12 weeks (60% of cyclosporine-treated patients vs. 17% in the cyclophosphamide group; $P < .05$). By study's end, 12 of the 14 patients who showed no response to cyclophosphamide were patients with FSGS. Of note, all 6 patients who were heterozygous for the *NPHS2* R229Q variant or the *R6QH* mutation were assigned to the cyclophosphamide group, and only one of those showed a response to treatment. These results support the use of cyclosporine over cyclophosphamide in children with steroid-resistant nephrotic syndrome.

Results from the FSGS Clinical Trial, the largest multicenter, randomized controlled trial of its kind, compared a 12-month course of cyclosporine with a combination of oral pulse dexamethasone and MMF in 138 randomized children and young adults with biopsy-proven, steroid-resistant primary FSGS.²⁵⁶ The trial was designed to determine whether treatment with MMF plus pulse steroids was superior to treatment with cyclosporine in inducing remission of proteinuria over 12 months. Partial or complete remission by the 12-month mark was achieved in only 22 MMF-treated patients and 33 cyclosporine-treated patients, with an OR of 0.59 (95% CI, 0.30–1.18) for achieving remission or better with MMF compared with cyclosporine, which was not significant. Furthermore, remission by 26 weeks after treatment cessation was not significantly different between the two groups. The investigators concluded that the lack of signal difference may have been due to insufficient sample size, and that the low rates of remission in the trial amplify the profound need for better and more targeted therapies.

Laurin and colleagues recently performed a retrospective analysis of 458 patients (173 treated with glucocorticoids alone, 90 treated with calcineurin inhibitors, with or without glucocorticoids, 12 treated with other agents, and 183 not treated with immunosuppressives) with native kidney,

biopsy-proven primary FSGS diagnosed between 1980 and 2012 in the Glomerular Disease Collaborative Network.²⁸⁹ Immunosuppressive therapy with glucocorticoids or calcineurin inhibitors was associated with better renal survival than no immunosuppression (hazard ratio [HR], 0.49; 95% CI, 0.28–0.86); however, calcineurin inhibitor use, with or without glucocorticoids, was not associated with a lower likelihood of ESKD compared with glucocorticoids alone (HR, 0.42; 95% CI, 0.15–1.18), thereby at least suggesting a role for immunosuppression in primary FSGS.²⁸⁹ Interestingly, this same group, using a similar cohort, found that patients with collapsing or NOS FSGS had similar renal survival after controlling for exposure to immunosuppressive treatment.²⁸⁸

How long should patients be treated with cyclosporine? In a study by Meyrier and colleagues,¹³⁶ when patients remained in remission for over 12 months, cyclosporine was slowly tapered and eventually removed, without subsequent relapse. Unfortunately, long-term treatment with cyclosporine was associated with increases in tubular atrophy and interstitial fibrosis, the degree of which was positively correlated with the initial serum creatinine level, the number of segmental scars on initial biopsy specimens, and a cyclosporine dose of more than 5.5 mg/kg/day. Thus there is a clear trade-off with the use of cyclosporine over the long term, given the risk of development of interstitial fibrosis and tubular atrophy.

Mycophenolate Mofetil

Data regarding the use of MMF in the treatment of FSGS has been largely anecdotal, with one small case series reporting transient improvement in proteinuria in 8 of 18 patients. However, two clinical trials have taken place within the past decade.

A randomized controlled trial compared the efficacy of an MMF-based regimen to treatment with corticosteroids, with or without cyclophosphamide, in adult patients with idiopathic MN ($n = 21$) or FSGS ($n = 33$).²⁹⁰ MMF was given at a dosage of 2 g/day for 6 months along with prednisolone, 0.5 mg/kg/day, for 2 to 3 months. Patients with FSGS in the comparison group received prednisolone, 1 mg/kg/day, for 3 to 6 months. There was no difference between the two groups with respect to the proportion of complete or partial remissions (70% vs. 69%) or the time to remission of proteinuria at any point. Remission was achieved faster in the MMF-treated FSGS patients than in the corticosteroid-only group (5.6 vs. 10.2 months, respectively), and cumulative steroid dose was lower (1.9 ± 0.3 vs. 7.3 ± 0.9 g, respectively). Although limited to a small number of patients and relatively short follow-up, this first controlled study has suggested that the addition MMF is steroid-sparing and achieves rates of remission comparable to those for corticosteroids alone in patients with FSGS. The randomized controlled FSGS Clinical Trial, discussed in the cyclosporine section, showed no significant difference in outcomes when comparing cyclosporine with oral pulse dexamethasone and MMF.²⁵⁶

Other Therapies

The use of sirolimus in the management of FSGS is poorly supported. Several reports have emerged of new-onset proteinuria in kidney transplant recipients who were switched from calcineurin inhibitor-based therapy to sirolimus.^{291–295}

In a study of 78 solid organ transplant recipients treated with sirolimus, 18 patients (23.1%) developed proteinuria in an average of 11.2 ± 2.1 months after starting sirolimus therapy.²⁹⁶ Kidney biopsy specimens obtained after the onset of proteinuria revealed various degrees of mesangial proliferation and mesangial expansion commonly seen in patients who previously had a diagnosis of chronic allograft nephropathy but showed FSGS in only 2 patients (14.3%). There was no correlation between proteinuria levels and sirolimus dose or trough blood levels. In the 6 patients in whom sirolimus was withdrawn, a complete reversal of proteinuria and edema was observed. This study certainly raises significant concern about the induction of proteinuria by sirolimus in transplant recipients.

The data pertaining to the use of sirolimus in the management of FSGS in native kidneys is rather conflicting. In a prospective open-label trial of sirolimus of 21 patients with steroid-resistant FSGS, at 6 months, 4 patients (19%) had achieved a complete remission, 8 (38%) had experienced a partial remission, and 1 patient had experienced a rapid decline in kidney function.²⁹⁷ Sirolimus therapy was associated with a substantial number of adverse events, including hyperlipidemia and anemia in 43% of patients. In patients who showed no response to sirolimus, the mean serum creatinine level increased from 1.66 mg/dL at baseline to 2.2 mg/dL at 6 months and 3.24 mg/dL at 12 months (significantly different from baseline at the $P = .028$ level). In patients who did show a response to sirolimus, the mean serum creatinine level increased from 1.76 mg/dL at baseline to 1.91 mg/dL at 12 months.

In contrast, a phase II open-label clinical trial of sirolimus had to be interrupted for safety reasons after five of six patients enrolled experienced a sharp decrease in GFR, and none achieved a complete remission.²⁹⁸ Three patients had a more than twofold increase in proteinuria during sirolimus therapy. Similar deleterious effects were reported in a cohort of 11 patients with a variety of glomerular diseases.²⁹⁹ Although inconclusive, the bulk of the data currently available suggest that sirolimus has a deleterious effect in FSGS and should be avoided.

Future approaches to the treatment of FSGS may come from the treatment of recurrent FSGS following transplantation. One study has reported the use of CTLA-4-Ig (abatacept), an inhibitor of the T cell costimulatory molecule B7-1 (CD80), in four patients with recurrent FSGS and one with primary FSGS, given that podocytes also express B7-1 in some individuals with proteinuric kidney disease, including FSGS.³⁰⁰ Surprisingly, nephrotic-range proteinuria resolved in all patients; however, much larger, more controlled studies are needed to validate this encouraging result.

Intriguing cases of recurrent FSGS after transplantation have been published, in which the proteinuria resolved after treatment with rituximab.³⁰¹ Interestingly, in one case,³⁰² the diagnosis of posttransplant lymphoproliferative disease (PTLD) was established 5 months after the transplantation, although recurrent nephrotic syndrome occurred within 2 weeks postoperatively. Proteinuria improved only after treatment of PTLD with rituximab. The authors suggested, however, that FSGS was probably unrelated to the viral infection due to the time gap between the two diagnoses. In addition, there have been a few reports of recurrent FSGS after transplantation that responded to rituximab (after plasmapheresis or

immunoabsorption) that are not related to PTLN.^{303–306} These patients may represent a specific category, and the lessons learned from their treatment may not necessarily apply to the general population of patients with FSGS.

Anecdotal reports have been emerging of treatment of primary and recurrent FSGS with rituximab, usually in combination with other immunomodulating therapies, but results have shown a mixed response to this treatment,^{307,308} with several reports of failures.³⁰⁹ In the largest case series, 5 of 8 adult patients with resistant FSGS failed to show a response to a course of rituximab, and 2 patients suffered a rapid deterioration of kidney function.³¹⁰ Only 2 patients had a clear and sustained improvement of kidney function and proteinuria. More recently, as discussed in the section on MCD, a small prospective trial in Italy of rituximab in 10 children and 20 adults with relapsing FSGS, MCD, and mesangial proliferative glomerulonephritis, (GN) showed significant benefit, which warrants further investigation with larger randomized clinical trials.¹³⁷ Interestingly, it has also been shown in the setting of recurrent FSGS that rituximab can bind to podocyte-derived sphingomyelin phosphodiesterase acid-like 3b (SMPDL-3b) to preserve its expression as a regulator of acid sphingomyelinase activity to prevent disruption of the actin cytoskeleton and apoptosis.³¹¹ It is becoming clear that the use of rituximab for the treatment of FSGS requires further evaluation in the setting of a randomized controlled trial.

Another potential future approach targets renal fibrosis, which represents a common final pathway of FSGS and other chronic kidney diseases.³¹² The orally available antifibrotic agent, pirfenidone, was evaluated in an open-label pilot study to determine its effect on the rate of decline in GFR in 18 patients with FSGS selected for having a monthly rate of decline in estimated GFR of more than 0.35 mL/min/1.73 m².³¹³ The monthly change in GFR improved from a median of -0.61 mL/min/1.73 m² during the baseline period to -0.45 mL/min/1.73 m² with pirfenidone therapy; which represented a median of 25% improvement in the rate of decline ($P < .01$). Pirfenidone had no effect on blood pressure or proteinuria and was associated with frequent dyspepsia, sedation, and photosensitive dermatitis. These enticing results provide a strong rationale for a larger placebo-controlled trial in patients with progressive chronic kidney disease.

Other forms of treatment have been used. Plasmapheresis and protein absorption strategies to remove circulating factors responsible for FSGS have led to the remission of recurrent FSGS, but do not appear to be beneficial in treating the primary disease.^{324,314}

In summary, patients with primary FSGS remain frustrating patients to treat. Clearly, FSGS is a histopathologic lesion rather than a unifying disease entity, suggesting that different subtypes of FSGS require different treatment strategies. Additional investigations into the pathogenesis of this disease entity, as well as podocyte dysfunction, are warranted to identify targets that may find success in clinical trial settings.³¹⁵ Enthusiasm for the use of high-dose, prolonged corticosteroid therapy in adults and children has prompted the use of this therapy in many FSGS patients. Only a prospective randomized trial that carefully evaluates this approach will determine its effectiveness. The first step in therapy should be geared toward excellent blood pressure control using

RAAS blockers. In patients who have nephrotic-range proteinuria, careful supportive care and consideration of a trial of oral corticosteroids in adult patients may be an acceptable approach after patients are carefully informed about the risks and potential benefits of 12 to 16 weeks of daily corticosteroid therapy. Alternatively, a trial of cyclosporine may be warranted for patients who have contraindications to corticosteroid therapy or in whom corticosteroid therapy fails to produce improvement.

C1q NEPHROPATHY

Complement component 1q (C1q) nephropathy is a relatively rare cause of proteinuria and nephrotic syndrome. Unfortunately, there has been a relative dearth of new insights into C1q nephropathy compared with other primary glomerular diseases. Since the last edition of this chapter, clinical information has remained relegated to case reports and small case series. C1q nephropathy can mimic MCD or FSGS clinically and histologically, although the clinical and pathologic presentations are quite variable.^{316,317} In a single-center retrospective analysis of kidney biopsy specimens from children and adolescents, C1q nephropathy was found in 6.6% of native kidney biopsy specimens.³¹⁸ In the largest case series, at the University of Ljubljana, Slovenia, which included both children and adults, C1q nephropathy was identified in 1.9% of native kidney biopsy specimens.³¹⁹ There appears to be a slight male predominance (56%–68%).^{319,320} Depending on the case series, there may be an association with African American or Hispanic ethnicity.³²⁰ Patients of all ages with C1q nephropathy have been described.

The diagnosis is based on the presence of mesangial immune complex deposits that show conspicuous staining for C1q in a patient with no clinical or laboratory evidence of SLE. The C1q staining usually is accompanied by staining for IgG, IgM, and C3. Electron microscopy demonstrates well-defined mesangial immune complex–type dense deposits. Light microscopic findings vary from no lesion (mimicking MCD), to focal glomerular hypercellularity, to proliferative glomerulonephritis with mesangial hypercellularity, and to focal segmental sclerosing lesions that may be indistinguishable histologically from those of FSGS. Anecdotal cases reports have described C1q nephropathy associated with a collapsing FSGS lesion.^{321,322} The findings by immunofluorescence microscopy and electron microscopy, however, readily differentiate C1q nephropathy from MCD and FSGS. The findings by IF and electron microscopy suggest an immune complex pathogenesis, but the details of the pathogenic mechanism and the cause are unknown.

Patients with C1q nephropathy generally have proteinuria, which may or may not be associated with nephrotic syndrome. Hematuria is present in at least 50% of patients and is more common among patients with a mesangial proliferative lesion on light microscopy.³¹⁹ Likewise, hypertension affects a minority of patients with no discernible lesions on light microscopy (similar to MCD), but is much more prevalent among patients with mesangial proliferation (55%).³¹⁹ Interestingly, many patients are relatively asymptomatic, and proteinuria may first be detected at the time of a physical examination in connection with sports participation or entry into the armed forces. These patients, by definition, have no clinical or serologic evidence of SLE, despite the presence

of C1q in the kidney biopsy specimen. C1q nephropathy may show spontaneous improvement.³²³

The renal outcome of patients with C1q nephropathy appears to be generally favorable,³¹⁸ especially in patients whose biopsy specimens show minimal change–like histologic features.^{319,324} Studies have suggested, however, that they may experience more frequent relapses and require additional immunosuppressants more often than patients with “pure” MCD.^{318,319,324} Patients whose biopsy specimens show FSGS-like or mesangial proliferative lesions on histologic analysis may show a less favorable response to immunosuppressive therapy, although the data are rather limited.

Interestingly, given the potential for C1q nephropathy to be a distinct autoimmune condition, a recent notable scientific advance in lupus nephritis is that the autoantigenicity of C1q itself may be associated with a conformational change in its globular head. This increased its hydrophobicity akin to a conformeropathy, as is seen in anti-GBM; that is, the self-antigen becomes an autoantigen through a *de novo* conformational change.³²⁵

MEMBRANOUS NEPHROPATHY

EPIDEMIOLOGY

Membranous glomerulopathy or nephropathy (MN) is one of the most common causes of nephrotic syndrome in adults.^{326–329} MN occurs as primary form of glomerular disease usually caused by autoantibodies specific for M-type phospholipase A2 receptor (PLA2R) or secondary to multiple nonrenal diseases, including autoimmune diseases (e.g., SLE, autoimmune thyroiditis), infection (e.g., hepatitis B, hepatitis C, malaria), drugs (e.g., penicillamine, gold), and malignancies (e.g., colon or lung cancer). Secondary MN, especially that caused by hepatitis B^{330–334} and lupus, is more frequent in children than in adults. In patients older than 60 years, MN is associated with a malignancy in 20% to 30% of patients.

MN is the cause of nephrotic syndrome in approximately 25% of adults with the syndrome.^{326,335–343} A study of patients who had urinary excretion of more than 1 g of protein/24 hours, conducted by the Medical Research Council in the United Kingdom from 1978 to 1990, determined that 20% had MN. The peak incidence of MN is in the 4th to 5th decades of life.^{326,344,345} no^{346–348} A pooled analysis of studies of patients with idiopathic MN found a 2:1 predominance of males (1190 males and 598 females).³⁴⁹ The adult-to-child ratio was 26:1 (1734 adults and 67 children); however, this low proportion of children among MN patients was biased by the exclusion of children from some of the studies included in the analysis. MN affects all races.

Although most patients with MN present with nephrotic syndrome, 10% to 20% of patients have proteinuria that remains at less than 2 g of protein /day.³⁵⁰ It is thus likely that the frequency of MN in the general population is underestimated because asymptomatic individuals with subclinical proteinuria often do not come to diagnosis or undergo kidney biopsy.

There are geographic variations in the clinical manifestations of MN. In studies in Australia and Japan, lower percentages of patients have nephrotic syndrome at entry than in Europe or North America. The geographic differences may be related to differences in the prevalence of underlying

causes of secondary MN, such as hepatitis B, malaria, and other infections.^{330,332}

The association of MN with underlying malignancy is well-recognized. In a large cohort study of 240 patients in France,³⁵¹ the incidence of cancer was significantly higher in patients with MN than in the general population (standardized incidence ratio = 9.8 [5.5–16.2] for men and 12.3 [4.5–26.9] for women). In almost half the patients, the tumor was asymptomatic and was detected only because of diagnostic procedures prompted by the diagnosis of MN. The most common malignancies were cancers of the lung and prostate. The frequency of malignancy increased with age. In a separate cohort study in Norway,³⁵² the incidence of cancer in 161 patients with MN was significantly higher compared with the age- and gender-adjusted general Norwegian population, corresponding to a standardized incidence ratio of 2.25 (95% CI = 1.44–3.35). The median time from diagnosis of MN to diagnosis of cancer was 60 months. Patients with MN who developed cancer were older (65 vs. 52 years; $P < .001$).³⁵²

Risk factors for malignancy in patients with MN include older age and a history of smoking, although the clinical presentation does not differ between patients with cancer-associated MN and those with primary MN.³⁵¹ In patients with cancer-associated MN, clinical remission of the cancer is associated with a reduction of proteinuria.³⁵¹ These studies highlight the importance of thorough cancer screening among older patients with MN, not only at the time of first diagnosis, but also during subsequent long-term follow-up.

Clinical Relevance

Cancer Screening in Membranous Nephropathy

It is important to conduct thorough cancer screening among older patients with membranous nephropathy, not only at the time of first diagnosis, but also during subsequent long-term follow-up.

PATHOLOGY

Electron Microscopy

The pathologic sine qua non of MN is the presence of subepithelial immune complex deposits or their structural consequences.³⁵³ Electron microscopy provides the most definitive diagnosis of MN, although a relatively confident diagnosis can be made based on typical light microscopic and IF microscopic findings.

Fig. 31.7 depicts the four ultrastructural stages of MN, as described by Ehrenreich and Churg.³⁴⁴ The earliest ultrastructural manifestation, stage I, is characterized by the presence of scattered or more regularly distributed small immune complex–type, electron-dense deposits in the subepithelial zone between the basement membrane and the podocyte. Podocyte process effacement and microvillous transformation occur in all stages of MN when there is substantial proteinuria. Stage II is characterized by projections of basement membrane material around the subepithelial deposits. In three dimensions, these projections surround the sides of the deposits, but when observed in cross section, they appear as spikes extending between the deposits (Fig. 31.8; see Fig. 31.7). In stage III, the new basement membrane material surrounds the deposits, and thus in cross section there is basement membrane material between the deposits and

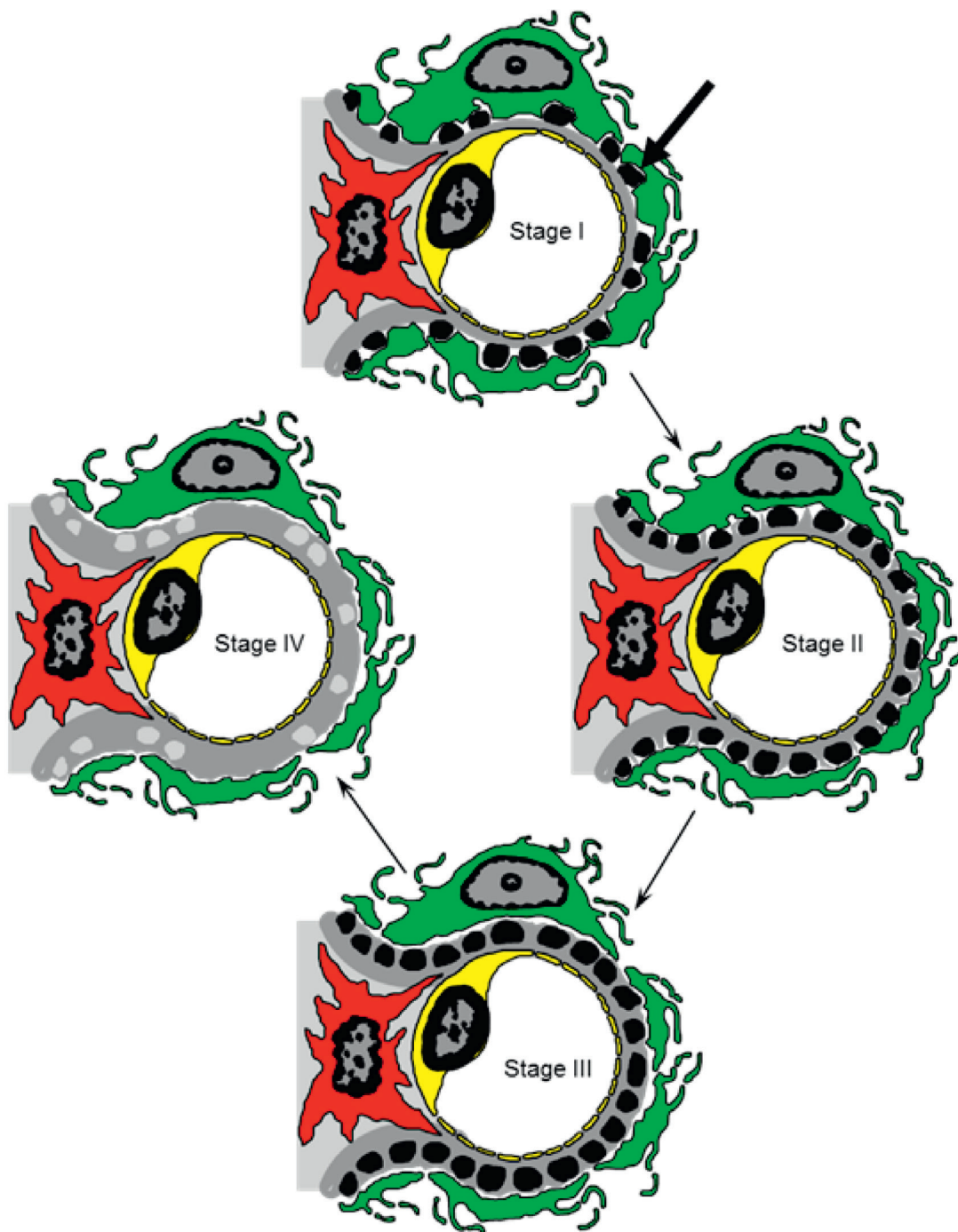


Fig. 31.7 Diagram depicting the four ultrastructural stages of membranous nephropathy. Stage I has subepithelial dense deposits (*arrow*) without adjacent basement membrane reaction. Stage II has projections of basement membrane adjacent to deposits. Stage III has deposits surrounded by basement membrane. Stage IV has thickened basement membrane with irregular lucent zones. (Courtesy J.C. Jennette.)

the epithelial cytoplasm. At this point, the deposits are, in essence, intramembranous rather than subepithelial; however, the ultrastructural appearance allows the inference that they once were subepithelial and thus indicative of MN. Stage IV is characterized by loss of the electron density of the deposits, which often results in irregular electron-lucent zones within an irregularly thickened basement membrane. Although not described by Ehrenreich and Churg, some nephropathologists recognize stage V, which is characterized by a repaired outer basement membrane zone with the only residual basement

membrane disturbance in the inner aspect of the basement membrane. At the time of kidney biopsy, most patients in the US have stage I or II disease ([Table 31.5](#)).

Mesangial dense deposits are rare in primary MN but are more frequent in secondary MN (see [Table 31.5](#)). This suggests, but does not prove, that primary MN is caused by subepithelial in situ immune complex formation with antibodies from the circulation complexing with antigens derived from the podocyte. Immune complexes formed only at this site could not go against the direction of filtration to reach

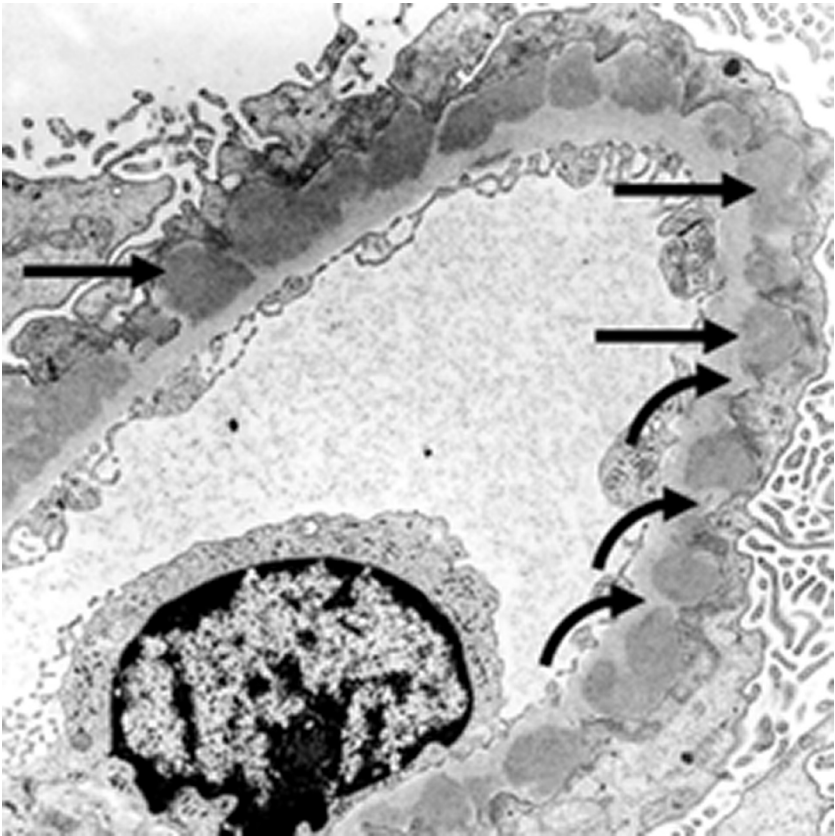


Fig. 31.8 Electron micrograph showing features of stage II membranous nephropathy with numerous subepithelial dense deposits (*straight arrows*) and adjacent projections of basement membrane material (*curved arrows*). (×100.)

Table 31.5 Pathologic Features of Nonlupus Membranous Nephropathy^a

Procedure	Feature Present (%)
Immunofluorescence Microscopy	
IgG	99 (3.5+) ^b
IgM	95 (1.2+)
IgA	84 (1.1+)
C3	97 (1.6+)
C1q	34 (1.1+)
Kappa	98 (3.1+)
Lambda	98 (2.8+)
Electron Microscopy	
Subepithelial electron dense deposits	99
Mesangial electron dense deposits	16
Subendothelial electron dense deposits	7
Endothelial tubuloreticular inclusions	3
Stage I	38
Stage II	32
Stage III	6
Stage IV	5
Stage V	1
Mixed Stage	20

^aBased on an analysis of 350 consecutive kidney biopsy specimens from patients with non-lupus membranous nephropathy evaluated at the University of North Carolina Nephropathology Laboratory.

^bValues in parentheses indicate mean intensity of positive staining on a scale of 0 to 4+.
Ig, Immunoglobulin.

the mesangium. Secondary forms of MN usually are caused by immune complexes containing antigens that are in the circulation, such as antigens derived from infections (e.g., hepatitis B), tumor antigens (e.g., colon cancer), or auto-antigens (e.g., thyroglobulin). With both the antigens and antibodies in the systemic circulation, it is likely that some immune complexes would form, which localize not only in the subepithelial zone but also in the mesangium or subendothelial zone. This is demonstrated in the secondary form of MN that occurs in patients with SLE. In over 90% of lupus MN specimens, mesangial dense deposits are identified by electron microscopy.³⁵⁴ Therefore, the presence of mesangial dense deposits should raise the index of suspicion for secondary rather than primary MN.

Immunofluorescence Microscopy

The characteristic IF microscopy finding in MN is diffuse global granular capillary wall staining for immunoglobulin and complement that often also has capillary wall staining for PLA2R in idiopathic disease (Fig. 31.9).³⁵³ IgG is the most frequent and usually the most intensely staining immunoglobulin, although less pronounced staining for IgA and IgM is common (see Table 31.5). IgG4 is the most prominent IgG subclass in the capillary wall deposits of primary MN.^{355,356} C3 staining is present over 95% of the time but typically is of relatively low intensity. C1q staining is uncommon and of low intensity in primary MN but is frequent and of high intensity in lupus MN.³⁵⁴ Although terminal complement components (i.e., components of the membrane attack complex) are not usually evaluated in routine diagnostic preparations, there is very intense staining of the capillary walls for these components. In the rare patients who have concurrent anti-GBM glomerulonephritis and MN, linear

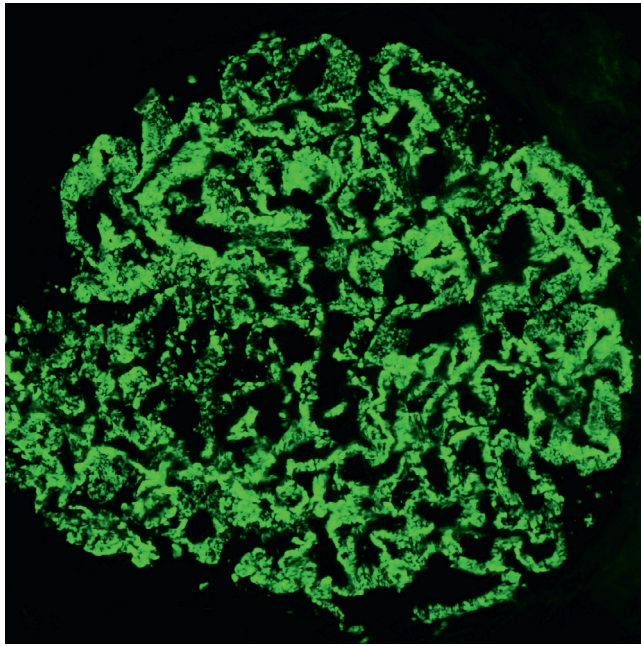


Fig. 31.9 Immunofluorescence micrograph showing global granular capillary wall staining in a glomerulus with membranous nephropathy representative of PLA2R and also seen with immunoglobulin G in a glomerulus with membranous nephropathy (Fluorescein isothiocyanate anti-PLA2R stain, $\times 300$).

staining for IgG can be discerned just below the granular staining.³⁵⁷

Tubular basement membrane staining for immunoglobulins or complement is rare in primary MN, but is common in secondary MN, especially lupus MN.³⁵⁴

Light Microscopy

The characteristic histologic abnormality by light microscopy is diffuse global capillary wall thickening in the absence of significant glomerular hypercellularity.³⁵⁸ The light microscopic features of MN, however, vary with the stage of the disease and with the degree of secondary chronic sclerosing glomerular and tubulointerstitial injury. Mild stage I lesions may not be discernible by light microscopy, especially when only a hematoxylin and eosin stain is used. Stage II, III, and IV lesions usually have readily discernible thickening of the capillary walls.

Masson trichrome stains may demonstrate the subepithelial immune complex deposits as tiny fuchsinophilic (red) grains along the outer aspect of the GBM. However, this is not a sensitive, specific, or technically reliable method for detecting glomerular immune complex deposits. Special stains that accentuate basement membrane material, such as the Jones' silver methenamine stain, may reveal the basement membrane changes that are induced by the subepithelial immune deposits. Spikes along the outer aspect of the GBM usually are seen in stage II lesions (Fig. 31.10). Stage III and IV lesions have irregularly thickened and trabeculated basement membranes, which resemble changes that occur with membranoproliferative glomerulonephritis and chronic thrombotic microangiopathy.

Overt mesangial hypercellularity is uncommon in primary MN, although it is more frequent in secondary MN.³⁵⁴ Crescent

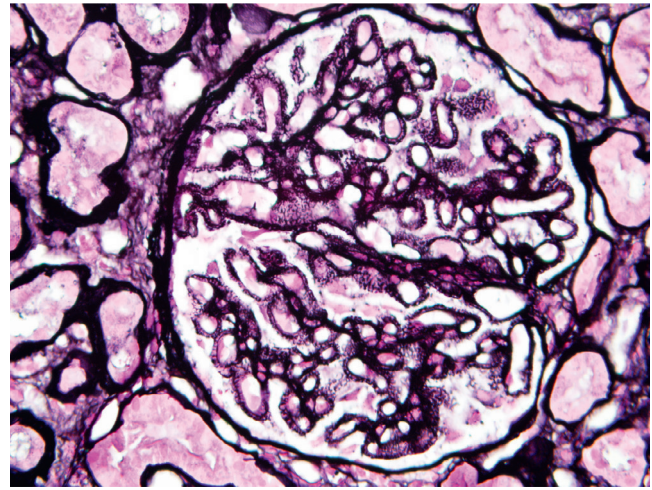


Fig. 31.10 Light micrograph of a glomerulus with features of membranous nephropathy demonstrating spikes, craters, and other irregularities along the outer aspects of the glomerular basement membrane (see Fig. 31.2). These correspond to the projections of basement membrane material between and around the immune complex deposits. (Jones' methenamine silver stain, $\times 300$.)

formation is rare unless there is concurrent anti-GBM disease or ANCA disease.^{359–365}

With disease progression, chronic sclerosing glomerular and tubulointerstitial lesions develop. Glomeruli become segmentally and globally sclerotic, and develop adhesions to Bowman's capsule. Worsening tubular atrophy, interstitial fibrosis, and interstitial infiltration by mononuclear leukocytes parallel the progressive loss of kidney function.³⁵⁸

PATHOGENESIS

MN is caused by immune complex localization in the subepithelial zone of glomerular capillaries. The nephritogenic antigens can be endogenous to the glomerulus itself (e.g., podocyte autoantigens) or can be exogenous (e.g., hepatitis B antigens). In the latter case, the antigen may be deposited in the subepithelial zone as part of preformed circulating immune complexes or could be produced in or planted in the subepithelial zone as free antigen to which antibodies bind to form immune complexes in situ. In rat Heymann nephritis, an animal model that closely resembles human primary MN, there is convincing evidence that the subepithelial immune deposits form in situ as a result of the binding of antibodies to glycoproteins produced by podocytes, followed by the accumulation of masses of the immune complexes in the subepithelial zone.^{366–368}

The long-standing search for antigens targeted in a substantial proportion of patients with MN has witnessed significant breakthroughs over the past decade. The podocyte neutral endopeptidase was the first to be identified as the endogenous target of autoantibodies in a neonate with nephrotic syndrome. This antibody crossed the placenta and was induced in the mother, who lacked the neutral endopeptidase epitope because of a mutational deletion. Sensitization to the nascent antigen was induced during a previous pregnancy.^{369–371} These findings provided the first direct support for the paradigm of in situ immune complex formation in the pathogenesis of human primary MN and constitute an example of alloimmunization leading to the generation

of immune complex–mediated glomerulopathy.^{371,372} It is now clear that most cases of primary MN are due to antibodies directed against the phospholipase A2 receptor (PLA₂R1) and the thrombospondin type 1 domain containing 7A (THSD7A; see next section).

The current understanding of these mechanisms, whereby immune complex deposition leads to proteinuria, is largely based on data emerging from studies of passive Heymann nephritis.^{368,373} In this model, immune complex formation in the subepithelial zone initiates activation of the complement pathway, leading to the formation of the C5b-C9 membrane attack complex. This results in complement-mediated injury to the epithelial cells.^{374–376} The proposed sequence of events includes complement activation and sublytic complement C5b-9 attack on podocytes, resulting in upregulated expression of genes for the production of oxidants, proteases, prostanoids, growth factors, connective tissue growth factor, TGF, and TGF receptors, leading to the overproduction of extracellular matrix.^{370,371,377} C5b-9 also causes alterations of the cytoskeleton that lead to abnormal distribution of slit diaphragm proteins and detachment of viable podocytes. These events result in disruption of the functional integrity of the GBM and the protein filtration barrier of podocytes.

The characteristic findings of a predominance of IgG4 with less IgG3 and no IgG1 in subepithelial deposits,^{355,356} and the paucity of C1q and C4 in these deposits,³⁷⁸ argues against a predominant role for the classic or lectin pathways of complement activation in MN and, instead, points to a role of the alternative pathway.³⁷⁹ The fact that the alternative pathway is spontaneously active in turn points to the likely importance of the complement regulatory proteins. Podocytes primarily rely on membrane complement receptor 1 (CR1; Crry in rodents) and decay-accelerating factor and have the capability to make their own factor H. The importance of complement-mediated injury (at least in passive Heymann nephritis) comes from evidence that nephritogenic serum contains antibodies to membrane complement-regulatory proteins (Crry).^{380,381} In a model of active Heymann nephritis, immunization with fraction 1A (Fx1A) lacking Crry leads to the formation of anti-Fx1A antibodies and subepithelial immune complex deposits, but no complement activation or the development of proteinuria.³⁸² Conversely, the overexpression of Crry or treatment with exogenous Crry has a salutatory effect on immune complex–mediated glomerulonephritis.^{383,384} Subsequent injury to the epithelial cell membrane and to the GBM has been hypothesized to be mediated, at least in part, by the production of reactive oxygen species and lipid peroxidation of cell membrane proteins and type IV collagen.³⁸⁵

Proteinuria may also be mediated by mechanisms independent of the formation of the C5b-9 membrane attack complex, as is suggested by the generation of proteinuria in passive Heymann nephritis in PVG rats that are deficient in complement factor 6 (PVG/C6 rats). These rats are incapable of generating the membrane attack complex. In this study, PVG/C6 and normal PVG rats developed similar levels of proteinuria after the injection of Fx1a antisera. Isolated glomeruli showed similar deposition of rat Ig and C3 staining in both groups of rats, but C9 deposition was not detected in the glomeruli of C6-deficient rats, which indicates that the C5b-9 membrane attack complex had not formed.³⁸⁶

Furthermore, the alteration in the glomerular extracellular matrix seen in MN may be caused, at least in part, by a decrease in fibrinolytic activity, due to the stabilization of active plasminogen activator inhibitor I in conjunction with vitronectin in the subepithelial deposits.³⁸⁷

Complement activation also results in tubular epithelial cell injury and mediates progressive interstitial disease in MN.^{388–390} Proteinuria itself may lead to tubulointerstitial damage through activation of the alternative complement pathway. Strong staining for properdin, a soluble complement regulator also known as “complement factor P,” on the luminal surface of the tubules was observed in kidney biopsy specimens from patients with primary MN, but not from healthy kidney donors.³⁹¹ After spontaneous hydrolysis of C3, properdin binds to C3b and enhances complement activation by stabilizing C3 convertase. Target-bound properdin may serve as a focal point for amplification of C3 activation. Properdin was shown in vitro to bind proximal tubular epithelial cells. Exposure of proximal tubular epithelial cells with normal human serum as a source of complement, but not to properdin-depleted serum, resulted in complement activation, with the deposition of C3 and generation of C5b-9. This led to the hypothesis that in proteinuric kidney disease, filtered properdin may bind to proximal tubular epithelial cells and act as a focal point for alternative pathway activation.

The human leukocyte antigen (HLA) class II antigen DR3 has been linked with MN,^{392–394} and its presence is associated with a relative risk of 12.³⁹² In a Japanese population, there is an increased frequency of HLA-DR2^{395,396} and DQW1³⁹⁷ in patients with MN. It is possible that a haplotype containing HLA-DR3 and specific HLA class I antigens may be common in these patients as well.³⁹² For example, HLA-B18 and HLA-DR3 haplotypes may confer an even greater risk of the development of MN.³⁹⁸ Also associated with an increased susceptibility to MN are polymorphisms of the tumor necrosis factor (TNF)- α gene.^{399,400} C4-null alleles are also more frequently found in patients with MN, especially in Caucasian populations.⁴⁰¹ A genome-wide association study (GWAS) of Caucasian patients with primary MN from the United Kingdom ($n = 335$), France ($n = 75$), and The Netherlands ($n = 146$) has identified two significant genomic loci—namely, chromosome 2q24 containing the gene encoding the culprit autoantigen PLA2R and chromosome 6p21 containing the gene encoding HLA-DQA1.⁴⁰² Despite the relative risk associated with some of these genetic markers, there are relatively few examples of familial MN.^{403–408}

Other studies have also suggested a possible role for APOL1 risk alleles in increasing risk or accelerating progression of PLA₂R-associated MN in association with collapsing nephropathy.⁴⁰⁹

PLA₂R, THSD7A, AND MEMBRANOUS NEPHROPATHY

Primary membranous nephropathy (PMN) is an autoimmune disease characterized by circulatory immunoglobulin G (IgG) autoantibodies against two major podocyte autoantigens. Muscle (M)-type phospholipase A2 receptor (PLA₂R1) has been detected in 70% of cases, and thrombospondin type 1 domain containing 7A (THSD7A) in 5% to 10% of cases. Autoantibodies to other podocyte intracytoplasmic antigens (e.g., aldose reductase, superoxide dismutase, alpha-enolase) have been detected in primary and secondary membranous

nephropathy. The pathogenesis and relationship of these autoantibodies to disease activity remain to be defined.^{410,411}

The team of Beck and colleagues has identified M-type PLA₂R as a target autoantigen common to about 70% of patients with primary MN. In contrast, none of the sera from normal controls, patients with MN secondary to SLE or hepatitis B, patients with proteinuric conditions other than MN, or other autoimmune disorders ($n = 7$) reacted with this antigen. PLA₂R expression in podocytes was confirmed by IF microscopy and in cultured immortalized human podocytes, indicating that the target antigen is intrinsic to glomeruli rather than deposited from sera of patients with PMN.⁴¹²

An international team has investigated a cohort of 154 patients with idiopathic membranous nephropathy and found, using mass spectrometry, that 15 patients (10%) harbor autoantibodies to the glomerular protein THSD7A and were anti-PLA₂R-negative.⁴¹³ A subsequent study has demonstrated that human anti-THSD7A autoantibodies administered to mice bind to murine THSD7A on murine podocytes, resulting in proteinuria and a glomerular lesion akin to MN, suggesting a causative role. Similar to PLA₂R autoantibody (PLA₂RAb), autoantibodies against THSD7A recognize conformation-dependent epitopes.^{413,414} Patients with primary membranous nephropathy may have an autoantibody targeting PLA₂R or THSD7A, but rarely against both autoantigens.

PLA₂R is one of the four members of the mannose receptor family.⁴¹⁵ PLA₂R is a 185-kDa transmembrane glycoprotein consisting of an extracellular N-terminal, cysteine-rich region (CysR), a fibronectin-like type II domain, and eight carbohydrate recognition domains (c-type lectin domain [CTLD1-8]), followed by a transmembrane and C-terminal intracellular tail.⁴¹⁶ The function of this human protein in podocyte remains undefined.⁴¹⁷ IgG4 is the predominant autoantibody against PLA₂R, although IgG1 and IgG3 have been reported, especially during the early phase of the disease process. This autoantibody may be present in the circulation and also colocalizes with PLA₂R on the glomerular capillary wall in PMN. The genesis of this autoantibody is currently not known, but molecular mimicry between microbial or environmental antigen and PLA₂R is a possibility.⁴¹⁸ Although the pathogenic role of this autoantibody remains to be elucidated, multiple studies have shown a correlation between the antibody titer and disease activity, suggesting its role as a useful biomarker.

When first described, Beck and colleagues⁴¹² performed Western blot analysis after nonreducing SDS-PAGE for the detection of PLA₂R antibody (PLA₂RAb) with recombinant PLA₂R antigen. Although this method is very sensitive and specific, it is impractical for clinical use. In 2014, the U.S. Food and Drug Administration (FDA) approved recombinant cell indirect IF and enzyme-linked immunosorbent assay (ELISA). IF uses transfected cells as a substrate, whereas ELISA is based on purified recombinant human PLA₂R receptors from transfected cells. Based on a metaanalysis, positive serum PLA₂RAb in PMN has a summary sensitivity of 78% (95% CI, 66%–87%) and specificity of 99% (95% CI, 96%–100%). Heterogeneity in diagnostic accuracy could be related to multiple factors, including race (lower in Japanese cohort), sample size, different ELISA methodology and, most importantly, the stage and activity of the disease process.⁴¹⁹

PLA₂R is expressed on the cell body and foot processes of human podocytes and is absent in rodents. THSD7A is expressed at the basal surface of podocytes in humans and rodents.⁴²⁰ After binding to PLA₂RAb, the immune complexes are shed into the lamina rara externa of the GBM. On routine immunohistochemistry (IHC) of kidney biopsy samples in healthy controls, there is a weak expression of glomerular PLA₂R.⁴²¹

IHC may be performed after antigen retrieval of paraffin-embedded renal biopsy samples using polyclonal anti-PLA₂RAb. Patients with PMN may have enhanced expression of glomerular PLA₂R. Qin and colleagues have reported the presence of serum PLA₂RAb in 70% of cases of PMN, of which 98% also had positive glomerular PLA₂R antigen staining by IHC. In contrast, only 70% of patients who were seronegative for PLA₂RAb had podocyte expression of PLA₂R.⁴²² This may be attributed to the rapid clearance of serum antibodies or immunologic remission with glomerular structural changes. Moreover, due to the high affinity of anti-PLA₂R antibodies to the cysteine-rich (CysR) epitope, patients who are low producers of antibody may remain seronegative until they have fully saturated PLA₂R binding sites on podocytes.^{417,422} Patients may also have circulatory PLA₂RAb with negative glomerular staining, suggesting that either the antibody is not pathogenic or there are inaccessible epitopes during the antigen retrieval process of kidney biopsy samples.⁴²³

In a number of studies, including the initial landmark study by Beck and colleagues, circulatory PLA₂RAb and/or glomerular staining of PLA₂R were negative in secondary membranous nephropathy.^{412,424} Moreover, PLA₂RAb was absent in a European cohort of SLE with membranous nephropathy and active proliferative lupus nephritis.⁴²⁵ It begs the question whether positive PLA₂R serology and glomerular staining obviate the need for screening of secondary causes, including malignancy. Subsequent studies have shown positive PLA₂R glomerular staining in patients with many systemic disorders associated with membranous nephropathy, including sarcoidosis, chronic viral B hepatitis, hepatitis C, Sjögren syndrome, and neoplasms.^{426,427} PLA₂R-positive glomerular staining and its colocalization with hepatitis B surface antigen (HBsAg) along capillary loops have been reported in hepatitis B virus (HBV)-associated MN.^{428,429} Whether these secondary disorders induce an immunologic response against PLA₂R or only a coincidental association is uncertain, but with the current body of evidence, it may be premature to obviate the need for screening of secondary causes. In contrast to PLA₂R-positive membranous nephropathy, THSD7A-associated membranous nephropathy has a female predominance and is associated with malignancy. Hoxha and colleagues⁴³⁸ have described a case of THSD7A-MN with the concomitant presence of gallbladder carcinoma.⁴³⁰ On IHC, both the tumor and lymph node metastases were positive for THSD7A. Also, messenger RNA for THSD7A was detected in the tumor but not in normal tissue. Initiation of chemotherapy led to decreased THSD7A antibody, with improvement in proteinuria. In a screening of 1000 patients with membranous nephropathy, 25 had positive THSD7A antibodies and 7 of these had a malignant tumor. It has been hypothesized that autoantibodies generated against tumor-expressed THSD7A cross-react with podocyte THSD7A *in situ*.⁴³⁰

Although most patients with PMN present with nephrotic range proteinuria, one-third of patients present with subnephrotic range proteinuria and usually have a favorable prognosis.⁴³¹ In a small retrospective study of biopsy-proven PMN with subnephrotic range proteinuria, the authors reported that after multivariate analysis, patients with positive PLA₂Rab at entry were more likely to progress to nephrotic syndrome than PLA₂Rab-negative patients (HR, 3.66; 95% CI, 1.39–9.64). Moreover, PLA₂R-positive patients were more likely to receive immunosuppression and had a more significant rise in serum creatinine levels than PLA₂Rab-negative patients. Future prospective controlled studies with longer follow-up will be required to confirm these associations. If proven, patients with subnephrotic range proteinuria with positive PLA₂R would require close observation and possible early initiation of immunosuppression for a favorable outcome.⁴³²

Based on many retrospective and observational studies, a relationship between circulatory PLA₂Rab, glomerular staining of PLA₂R1, and disease activity has been proposed. In a large study of 117 Caucasian patients with PMN and nephrotic-range proteinuria, anti-PLA₂Rab was positive in about 74% of cases when measured by IF and ELISA, with a 94% agreement between the two tests. IgG4 anti-PLA₂Rab titer significantly correlated with baseline proteinuria, and spontaneous remission occurred less frequently in patients with high titers (38% vs. 4% in the lowest and highest tertiles, respectively).⁴³³ However, not all antibody titers were measured at the time of diagnosis and may have been delayed by 6 months after biopsy. In another retrospective analysis, after Cox multivariate analysis, anti-PLA₂Rab titers were inversely and independently associated with the likelihood of spontaneous remission (HR, 0.37; CI, 0.17–0.84; *P* = .017).⁴³⁴ In contrast to the previous study, no significant association was found between anti-PLA₂Rab titers at diagnosis with baseline proteinuria, kidney function, or progression of chronic kidney disease. In primary MN, serum PLA₂Rab may be detected before the development of proteinuria, and resolution of proteinuria may lag behind immunologic remission by months.^{435,436} Therefore, in appropriate clinical scenarios of PLA₂Rab-positive PMN, monitoring of the PLA₂Rab level may assist clinicians in anticipating a spontaneous remission and avoid “premature” initiation of immunosuppression. PLA₂Rab values are usually high in patients with nephrotic-range proteinuria and decrease substantially with either spontaneous or treatment-induced remission. Also, antibody titers may increase again with disease relapse.⁴³⁷

In their observational study, Hoxha and colleagues reported that patients with a low antibody titer reach remission faster than patients with a high titer, although the cumulative rate of patients with remission of proteinuria was similar at the end of the study period at 24 months.⁴³⁸

The PLA₂Rab recognizes only the nonreduced form of PLA₂R autoantigen, suggesting the need for disulfide bonds for epitope conformation. Kao and colleagues studied sera from patients with PLA₂Rab-positive PMN with truncated PLA₂R extracellular domains and identified the three-domain protein complex CysR, fibronectin-like type II (FnII-), and C-type lectin-like domain 1 (CTLD1) as the immunodominant humoral epitope.⁴³⁹ Taking this a step further, Fresquet and colleagues have identified the CysR domain as the major humoral epitope.⁴⁴⁰

Epitope spreading is the diversification of epitope specificity from the initial dominant epitope-specific immune response to subdominant epitopes on that protein.^{441,442} This phenomenon has been described in many autoimmune disorders, including human anti-GBM disease, rheumatoid arthritis, and Heyman nephritis. Seitz-Polski and colleagues⁴⁴³ have described similar phenomena among three domains of the PLA₂R1 molecule in PMN. Using a series of PLA₂R1 deletion mutants of the extracellular domains, three reactive epitopes have been identified, with CysR being the primary dominant epitope and CTLD1 and CTLD7 being additional distinct epitopes. Patients with anti-PLA₂R1 reactivity against CysR had a favorable outcome, with less severe disease, younger age of onset, and a higher chance of going into spontaneous remission compared with those with antibody reactivity against other domains. The authors hypothesized that anti-PLA₂R reactivity is restricted to the CysR domain in the initial stage and, after additional immune challenges, this reactivity spreads to the C-terminal region (CTLD11 or CTLD7), resulting in progressive disease that may be less responsive to therapy. Further studies are required to confirm this finding; if proven, it could guide physicians to personalize therapy based on anti-PLA₂R reactivity.

In an observational study of PMN treated with cyclophosphamide or MMF, baseline anti-PLA₂Rab titers did not predict initial response, but those at the end of therapy predicted long-term outcome. After a median follow-up of 5 years, 58% patients with negative antibody titers at the end of therapy maintained remission, and none of the patients with persistent seropositivity went into remission. Moreover, cyclophosphamide-based therapy was more effective in reducing antibody titers than MMF.⁴⁴⁴ However, this observation has not been confirmed in other studies. Radice and associates could not detect statistical differences in the probability of negativization of PLA₂Rab during follow-up based on different initial immunosuppressive therapies.⁴⁴⁵ In a longitudinal observational study, the response to treatment with rituximab therapy was similar, irrespective of the presence or absence of anti-PLA₂Rab.⁴⁴⁶ Among PLA₂Rab-seropositive patients, the probability of complete or partial remission of proteinuria progressively decreased in parallel with progressive increasing PLA₂Rab titers.^{446,447} The probability plateaus at a level more than 500 to 600 relative units (RU)/mL (1 RU/mL roughly equates to 2 pg/mL). Additionally, a 50% decrease in antibody titer preceded an equivalent decrease in proteinuria by 10 months.⁴⁴⁶

A combination of clinical parameters and PLA₂R serology and glomerular staining may become very important for making appropriate and timely clinical decisions. Future randomized controlled studies with long-term follow-up are required to understand the relationship among PLA₂Rab titers, glomerular staining, and clinical parameters of membranous nephropathy. With current evidence, it may not be appropriate to develop a treatment algorithm purely based on PLA₂Rab titer. The process and relationship are complicated and need to be studied further and refined to put in an algorithm.

CLINICAL FEATURES AND NATURAL HISTORY

Patients with MN usually have nephrotic syndrome with hypoalbuminemia, hyperlipidemia, peripheral edema, and lipiduria. This presentation occurs in 70% to 80% of

patients.^{346,448} The onset of nephrotic syndrome is usually not associated with any prodromal disease process or antecedent infections. Hypertension may be present at the outset of disease in 13% to 55% of patients.³⁴⁸ Most patients present with normal or slightly decreased kidney function at presentation.

If progressive renal insufficiency develops, it is usually relatively indolent. An abrupt change to more acute renal insufficiency should prompt investigation of a superimposed condition, such as a crescentic glomerulonephritis.⁴⁴⁹ One-third of such patients have anti-GBM antibodies, and some have ANCA.

Other causes of sudden deterioration of kidney function include acute bilateral renal vein thrombosis and hypovolemia in the setting of massive nephrosis. The incidence of renal vein thrombosis in MN varies from 4% to 52%.⁴⁵⁰ The diagnosis of renal vein thrombosis may be clinically apparent based on the sudden development of macroscopic hematuria, flank pain, and reduction in kidney function, but a more insidious development is also common. Although ultrasonography with Doppler studies may demonstrate the renal thrombus,⁴⁵¹ venography with contrast remains the gold standard. Spiral computed tomography⁴⁵² or magnetic resonance imaging with contrast have also been used.⁴⁵³

Drug-induced kidney injury is another reason for the sudden deterioration in kidney function in a patient with MN. The use of nonsteroidal antiinflammatory drugs (NSAIDs), diuretics, and antimicrobials has been linked to the occurrence of acute interstitial nephritis or acute tubular necrosis.^{454,455}

An estimate of renal survival in patients with MN can be obtained from a pooled analysis of outcomes in clinical studies.³⁴⁹ In this analysis of 1189 pooled patients,^{326,338,347,456–468} the probability of renal survival was 86% in 5 years, 65% in 10 years, and 59% years at 15 years. Although 35% of patients may progress to ESKD by 10 years, 25% may experience a complete spontaneous remission of proteinuria within 5 years.⁴⁶⁹

In a retrospective study of 328 patients with MN who were not treated with immunosuppressive agents, spontaneous remission of proteinuria occurred in 32% of patients: partial remission (proteinuria ≤ 3.5 g/day) occurred at a mean of 14.7 $\pm$ 11.4 months, and the mean time to complete remission was 38.0 $\pm$ 25.2 months.⁴³¹ Importantly, severe proteinuria at onset of disease does not preclude the possibility of spontaneous remission because it occurred in 26% of patients with a baseline proteinuria of 8 to 12 g/day and in 21% of patients with a baseline proteinuria of more than 12 g of protein/day. Multivariate analysis has revealed that the best predictor of spontaneous remission was a decrease in proteinuria of more than 50% in the first year of follow-up (HR, 12.6; 95% CI, 5.2–30.5; $P < .0001$). Other predictors were the baseline serum creatinine level, baseline proteinuria, and the use of angiotensin II inhibitors.⁴³¹

Persistent proteinuria is more predictive of renal insufficiency than proteinuria at a single time point. Thus, persistent proteinuria of 8 g or more/day for 6 months or more was associated with a 66% probability of progression to chronic renal failure. Patients with at least 6 g of protein/day for 9 months or longer had a 55% probability of developing chronic renal insufficiency. Persistent proteinuria of 4 g or more/day for longer than 18 months was associated with an even greater risk of chronic renal insufficiency.⁴⁷⁰ Patients with

overtly declining kidney function are at higher risk for progressive renal deterioration.⁴⁶⁹

In addition to renal insufficiency and proteinuria, other factors may be associated with an increased risk of progressive renal failure. Male gender, advanced age (>50 years), poorly controlled hypertension, and reduced GFR at presentation have been reported as risk factors for progressive decline in kidney function.^{345,463,467,469–474} In addition to the clinical prognostic features, the presence of advanced MN on kidney biopsy specimens (stage III or IV), tubular atrophy, and interstitial fibrosis can also be associated with increased risk. In fact, chronic interstitial fibrosis and tubular atrophy have been shown to be independent predictors of progressive renal failure in primary MN.^{462,475–477} The presence of crescents on kidney biopsy specimens may also portend a poor long-term prognosis. The stage of glomerular lesions detected by electron microscopy has also been suggested as a risk factor for poor prognosis in some^{478–480} but not all studies.^{335,467,475,481} Similarly, FSGS superimposed on MN may have a worse long-term renal prognosis than MN without sclerosis.^{482,483} However, the importance of these demographic and histologic risk factors was not substantiated in a retrospective analysis of a large cohort of patients from the University of Toronto that examined the rate of progression (slope).⁴⁸⁴ Of the histologic variables, only a greater degree of complement deposition appeared to be associated with a more rapid decline in GFR.⁴⁸⁵

In a prospective study, urinary excretion of β_2 -microglobulin level of more than 0.5 μ g/min and a urinary IgG level of more than 250 mg/24 hr, assessed in a timed urine sample, were found to predict the progressive loss of GFR in a prospective cohort of 57 patients with primary MN and normal kidney function.⁴⁸⁶ In a multivariate analysis, urine β_2 -microglobulin excretion was the strongest independent predictor of the development of renal insufficiency, with a sensitivity and specificity of 88% and 91%, respectively. Unfortunately, the measurement of urine β_2 -microglobulin is cumbersome because it is unstable in urine and requires alkalization of the urine prior to collection. More recently, there have been several studies from different groups demonstrating an association between high anti-PLA₂R autoantibodies and worse long-term outcomes in that there appears to be growing evidence for an inverse correlation between the level of the autoantibodies and disease.^{444,487–489}

In summary, one of the strongest indicators of progressive disease appears to be the persistence of at least moderate proteinuria.⁴⁷³ Impaired kidney function, severe proteinuria at presentation, the presence of substantial interstitial infiltrates on biopsy specimen, superimposed crescentic glomerulonephritis, and segmental sclerosis also portend a poorer outcome.

LABORATORY FINDINGS

Proteinuria is the hallmark of MN. More than 80% of MN patients excrete more than 3 g of protein/day. In some patients, the amount of urinary protein may exceed 20 g/day. A Medical Research Council study has reported that 30% of patients with MN excreted more than 10 g of protein per day at the time of presentation.³⁵⁰ Microscopic hematuria is present in 30% to 50% of patients at the time of presentation.^{346,490,491} Macroscopic hematuria, on the other hand, is distinctly uncommon and occurs in less than 4% of adult

patients,^{492,493} although it may be common in children.⁴⁹⁴ Most patients have either normal or only slightly decreased kidney function. Impaired kidney function is found in fewer than 10% of patients at the time of presentation.^{490,495}

In patients with severe nephrosis, hypoalbuminemia is common, as is the loss of other serum proteins, including IgG. Serum lipoprotein levels are characteristically elevated, as they are in other forms of nephrotic syndrome. Elevated levels of low-density and very low-density lipoproteins are common in MN. In one study, elevated levels of lipoprotein(a) normalized in patients whose disease was in remission.⁴⁹⁶

Levels of complement components C3 and C4 are typically normal in patients with MN. The complex of terminal complement components known as C5b-9 is found in the urine of some patients with active MN. There is increased excretion of this complex in patients with active immune complex formation. The excretion may decrease during disease inactivity.^{374–376,497–502}

To exclude common causes of secondary MN, one should carry out serologic tests for nephritogenic infections such as hepatitis B, hepatitis C, and syphilis, as well as tests for immunologic disorders such as lupus, mixed connective tissue disease, and cryoglobulinemia. MN has been associated with graft-versus-host disease following allogeneic stem cell transplantation, and this should be considered as well.⁵⁰³

Although hypercoagulability appears to be present in patients with nephrosis in general, this tendency may be enhanced in patients with MN.^{504–506} The exact mechanisms leading to thrombophilia in this group of patients are poorly understood. Patients with MN have hyperfibrinogenemia with increased levels of circulating procoagulants and decreased levels of anticoagulant factors, such as antithrombin III.⁵⁰⁷ The thrombotic tendency may be increased by the erythrocytosis that occurs in some patients, as well as by the effect of lipoprotein(a) to retard thrombolysis. Other possible contributors to the thrombophilic state include volume depletion, diuretic and/or steroid use, venous stasis, immobilization, and immune complex activation of the clotting cascade and anti- α -enolase antibodies.^{508–510} Renal vein thrombosis has been reported more frequently in patients with MN than in those with nephrotic syndrome of other causes.^{506,511–514} The prevalence of renal vein thrombosis in patients with MN ranges from approximately 5% to 63%, depending on the mode of diagnosis used and whether or not systematic screening is performed. The prevalence of all forms of deep vein thrombosis in patients with MN ranges from 9% to 44%. The combined burden of deep vein and renal vein thrombosis has been estimated to be as high as 45%.⁵¹⁰ Renal vein thrombosis is often silent, with pulmonary embolism being the first presenting sign. The risk of venous thromboembolic events appears to be higher when the serum albumin concentration is less than 2.5 g/dL; such events occur in as many as 40% of these patients.^{510,515}

It is the concern for the morbidity and, at times, mortality associated with pulmonary embolism that has led to the use of prophylactic anticoagulation for patients with severe nephrotic syndrome and MN. A decision analysis has suggested that the risk of life-threatening complications of pulmonary embolism outweigh the risks associated with anticoagulant therapy.⁵¹⁶ However, this analysis may be based on an overestimate of the true incidence of thromboses among patients with MN. No direct controlled data are available to support

or refute such a contention. More recently, using data from an inception cohort of 898 patients with primary MN, as well as literature review-based risk estimates of hemorrhage, a clinical tool to estimate the likelihood of benefit of anticoagulation, based primarily on bleeding risk and serum albumin concentration, was proposed as a way to begin to personalize prophylactic anticoagulation (www.gntools.com).⁵¹⁷ A retrospective analysis of a treatment regimen to prevent venous thromboembolism in 143 patients with nephrotic syndrome (low-molecular-weight heparin or low-dose heparin for patients with a serum albumin level <2.0 g/dL and aspirin 75, mg daily, for patients with serum albumin level of 2.0–3.0 g/dL) has appeared to be effective, with very few complications.⁵¹⁸ Unfortunately, there is no direct controlled support for the routine use of prophylactic anticoagulation in patients with primary MN; however, the case could be made for the judicious use of warfarin in patients with severe nephrotic syndrome who have a profoundly decreased serum albumin level (probably <2 mg/dL) if no contraindications are present. Randomized controlled trials are warranted.

TREATMENT

Corticosteroids

Despite numerous studies, the optimal treatment of MN remains incompletely defined. The difficulty in treating MN is a consequence of the chronic nature of the disease, the tendency for spontaneous remission and relapse, the variability of clinical severity, and the only partial efficacy of existing treatment protocols. The role of corticosteroids and alkylating agents in the treatment of this disease has been debated for decades. Common therapeutic approaches for new-onset disease include the following: (1) conservative therapy with RAAS blockade; (2) corticosteroid therapy (usually prednisone or methylprednisolone); and (3) administration of alkylating agents, such as chlorambucil or cyclophosphamide, with or without concurrent corticosteroid treatment.

Numerous studies using corticosteroid treatment have demonstrated different outcomes.*

In a pooled analysis of these studies, corticosteroid therapy was found to have no beneficial effect on renal survival.³⁴⁹ Three large prospective randomized trials have examined the efficacy of oral corticosteroid therapy in adult patients with MN, with different results.^{458,459,521} Findings of the US Collaborative Study³³⁷ have suggested that 8 weeks treatment with 100 to 150 mg of prednisone, given on alternate days, resulted in a transient decrease in urinary protein excretion to less than 2 g compared with placebo. Prednisone was discontinued after 3 months unless proteinuria recurred after a partial or complete remission. Relapses were treated by reinstitution of high-dose prednisone for 1 month, followed by a taper. The results of this study have suggested that patients treated with prednisone are less likely to experience a doubling of their entry serum creatinine level and were more likely to experience a transient decrease in proteinuria to less than 2 g of protein/day, and that even a partial remission of proteinuria was associated with well-preserved, long-term kidney function. This seminal study was criticized because the control group fared substantially worse than untreated patients in several other studies.

*References 326, 337, 338, 347, 456–463, 466–469, 481, 519, and 520.

A British Medical Research Council study⁴⁵⁸ used a similar regimen except that prednisolone was discontinued after 8 weeks without tapering and without treatment of the relapse of proteinuria. Patients with lower creatinine clearance (≤ 30 mL/min) were included in the study. Three to 9 months after study entry, patients showed no improvement in kidney function, and the urine protein excretion and albumin level improved only transiently.

A third prospective randomized study of corticosteroid reported by Cattran and colleagues⁴⁵⁹ included patients with relatively low levels of proteinuria (≤ 0.3 g of protein/day). In this study, alternate-day prednisone (45 mg/m² of body surface area) afforded no benefit with regard to proteinuria or renal function.

In a metaanalysis³⁴⁹ of the U.S. Collaborative study³³⁷ and the studies by Cameron,⁴⁵⁸ Cattran,⁴⁵⁹ and Kobayashi and colleagues⁴⁶⁰ comparing glucocorticoid with supportive therapy, corticosteroid therapy was associated with a trend toward achievement of complete remission at 24 to 36 months, but this result did not reach statistical significance. A pooled analysis of randomized trials and prospective studies again demonstrated a lack of benefit of corticosteroid therapy in inducing a remission of nephrotic syndrome or preserving kidney function.³⁴⁹

An alternative to oral glucocorticoid therapy has been treatment with pulse methylprednisolone, largely in patients with deteriorating kidney function. Treatment of patients with renal insufficiency using pulse methylprednisolone at 1 g/day for 5 days followed by oral prednisone was associated with an improvement in kidney function for 6 months and a reduction in proteinuria.⁵²² The long-term outcomes for over 50% of these patients were discouraging; one-third experienced renal failure and 13% developed myocardial infarction with kidney dysfunction. A similar study⁵²³ combined pulse methylprednisolone with azathioprine or cyclophosphamide. Although there may have been some improvement in proteinuria and kidney function in a minority of patients, substantial side effects were experienced by almost the entire study population. The evidence to date does not support the use of oral corticosteroids alone for the treatment of primary MN.

Cyclophosphamide or Chlorambucil

Cytotoxic drugs, including cyclophosphamide and chlorambucil, have been used for the treatment of primary MN in conjunction with intravenous and/or oral corticosteroids. In a number of studies, Ponticelli and colleagues have demonstrated that chlorambucil has a beneficial effect in the treatment of MN.^{462,479,481,524} In these studies, patients with primary MN were treated initially with intravenous pulse methylprednisolone at 1 g/day for the first 3 days of each month, with daily oral glucocorticoid therapy (methylprednisolone, 0.4 mg/kg/day, or prednisone, 0.5 mg/kg/day), given on an alternating monthly schedule with chlorambucil at a dosage of 0.2 mg/kg/day. Both groups received a low-salt diet, diuretics, and antihypertensive agents, as needed. In patients randomly assigned to the treatment group, nephrotic syndrome lasted for a significantly shorter duration, and a complete or partial remission of proteinuria occurred in 83% of MN patients compared with 38% of control patients.⁵²⁴ The slope of the mean reciprocal plasma creatinine level remained stable in the treatment group, but declined in

the untreated patients beginning at 12 months. At 10-year follow-up, the probability of having a functioning kidney was 92% in the treated patients and 60% in the control patients. In only 10% of patients was therapy discontinued because of side effects. Compared with treatment with glucocorticoids alone, treatment with a combination of chlorambucil and methylprednisolone was associated with an earlier remission of nephrotic syndrome and a greater stability of complete or partial remission of proteinuria.⁴⁸¹ Interestingly, the overall decline in kidney function was no different in the two treatment groups. Unfortunately, although a difference in favor of the chlorambucil-treated patients persisted for the first 3 years of follow-up, it was no longer statistically significantly different by 4 years (62% without nephrotic syndrome in the group receiving combination therapy vs. 42% in the steroid-only group; $P = .102$). In a study comparing cyclophosphamide with chlorambucil, cyclophosphamide was found to be at least as effective as chlorambucil when used in a similar dosing protocol and appeared to have somewhat fewer side effects.⁵²⁵

A prospective, open-label, randomized study in India⁵²⁶ involving 93 patients followed for a median of 11 years (range, 10.5–12 years) compared supportive therapy (dietary sodium restriction, diuretics, and antihypertensive agents) with a 6-month course of alternate months of steroid and cyclophosphamide treatment, similar to the Ponticelli protocol.^{524,527} Unfortunately, angiotensin II blockade was withheld in all patients for at least 1 year. Study endpoints were the doubling of the serum creatinine level, development of ESKD, or patient death. Of the 47 patients who received the immunosuppressive protocol, 34 experienced remission compared with 16 of 46 in the control group ($P < .0001$). The 10-year dialysis-free survival was 89% in the immunosuppression group and 65% in the supportive treatment group ($P = .016$), and the likelihood of survival without death, dialysis, or doubling of the serum creatinine level was 79% and 44% ($P = .0006$), respectively. A significant divergence between the two groups in terms of proteinuria became apparent within the first year, and the estimated GFR was significantly lower in the control group than in the cyclophosphamide-treated group from 4 years onward. This study confirms, in a different patient population, the short- and long-term benefits associated with treatment with cyclophosphamide and corticosteroids, according to the Ponticelli protocol.^{524,527}

A more recent randomized controlled trial of 108 patients with biopsy-documented primary MN in the United Kingdom compared prednisolone in combination with chlorambucil to either cyclosporine or supportive therapy alone. The primary endpoint was a further 20% decline in excretory kidney function from baseline readings, which was significantly lower in the prednisolone and chlorambucil group (HR, 0.44; 95% CI, 0.24–0.78; $P = .0042$); however, this group also had a higher frequency of serious adverse events.⁵²⁸

Despite these reported benefits, there has been a lack of corroboration in other trials regarding the salutary effects of alkylating agents combined with prednisone or other agents^{461,465,468,529}; however, two metaanalyses have suggested that the use of cytotoxic agents improves the chance of a complete remission of proteinuria by four- to fivefold, but had no long-term protective effect on renal survival.^{349,530} A further metaanalysis of 36 clinical trials investigating the effects of immunosuppression on adults with idiopathic MN

has concluded that corticosteroids combined with alkylating agents significantly reduce all-cause mortality or ESKD (8 randomized controlled trials [RCTs], 448 participants; risk ratio, 0.44; 95% CI, 0.26–0.75; $P = .002$) and increased partial or complete remission (7 RCTs, 422 participants; risk ratio 1.46; 95% CI, 1.13–1.89; $P = .004$), but led to more adverse events (4 RCTs, 303 participants; risk ratio, 4.20; 95% CI, 1.15–15.32; $P = .03$).⁵³¹

Calcineurin Inhibitors

Despite the most recent trial showing the superiority of prednisolone in combination with chlorambucil over cyclosporine,⁵²⁸ there has been interest in the use of cyclosporine, which has resulted in improvement in proteinuria and stability of kidney function in many patients.^{532–534} In an RCT comparing 26 weeks of treatment with cyclosporine plus low-dose prednisone to treatment with placebo plus prednisone, 75% of the cyclosporine group but only 22% of the control group ($P < .001$) experienced a partial or complete remission of proteinuria by 26 weeks.⁵³⁵ Relapse occurred in about 40% of patients achieving remission in both treatment groups. The fraction of patients achieving sustained remission remained significantly different between the groups until the end of the study (cyclosporine treatment, 39%; placebo, 13%; $P = .007$). Kidney function was unchanged and equal in the two groups over the test medication period.⁵³⁵ This study was criticized for the rapid discontinuation of cyclosporine over 4 weeks at the end of the 26-week treatment period.

In a prospective study, treatment with cyclosporine alone (2–3 mg/kg/day) was compared with treatment with a combination of cyclosporine and oral prednisolone in 51 patients.⁵³⁶ Prednisolone was started at 0.6 mg/kg/day and then gradually tapered to 10 to 15 mg/day at 6 months and continued to 12 months. Patients who experienced complete or partial remission then received long-term treatment with lower doses of cyclosporine (1–1.5 mg/kg/day) plus prednisolone (0.1 mg/kg/day) or cyclosporine alone. This study did not have a randomized design because patients with contraindications to corticosteroid use were assigned to the cyclosporine-only group. During the follow-up phase of the study, relapses were more common in patients treated with cyclosporine alone than in patients receiving cyclosporine plus oral prednisolone (47% vs. 15%, respectively; $P < .05$). However, the results suggested that the risk of relapse may be determined by the levels of cyclosporine because patients in both groups who experienced relapse had lower cyclosporine trough levels than those who did not experience relapse (72 ± 48 ng/mL vs. 194 ± 80 ng/mL, respectively; $P < .03$).

A prospective RCT was undertaken to evaluate monotherapy with tacrolimus versus supportive therapy alone in 48 patients with MN who had preserved kidney function and had persistent nephrotic syndrome for longer than 9 months, despite treatment with an ACE inhibitor or ARB.⁵³⁷ Treatment with tacrolimus consisted of 0.05 mg/kg/day divided into two daily doses and adjusted to achieve a whole-blood, 12-hour trough level between 3 and 5 ng/mL. The target trough level was increased to between 5 and 8 ng/mL if a remission was not achieved after the first 2 months of treatment. Tacrolimus was continued for a total of 12 months, followed by a 6-month taper. The probability of

remission in the treatment group was 58%, 82%, and 94% after 6, 12, and 18 months, but was only 10%, 24%, and 35%, respectively, in the control group. The decrease in proteinuria was significantly greater in the treatment group. Notably, 6 patients in the control group and only 1 in the treatment group reached the secondary endpoint of a 50% increase in serum creatinine level. Unfortunately, as in the previously published study of cyclosporine, almost half of the patients who had achieved remission experienced a recurrence of nephrotic syndrome by the 18th month after tacrolimus withdrawal.

An interesting pilot study from Spain⁵³⁸ has studied combination therapy with corticosteroids, MMF, and tacrolimus in patients with persistent nephrotic syndrome after 6 months of treatment with full-dose RAAS blockers and a creatinine clearance of more than 60 mL/min/1.73 m². The initial dose of prednisone was 0.5 mg/kg/day for the first month, and the drug was then tapered to 7.5 mg/day at month 6. The starting dose of tacrolimus was 0.05 mg/kg/day to achieve target whole-blood trough levels of 7 to 9 ng/mL. If the level of proteinuria was less than 1 g of protein/day at the end of 3 months of this therapy, the tacrolimus dosage was reduced to maintain blood levels between 5 and 7 ng/mL and continued for a period of 9 more months. If, however, the level of proteinuria was greater than 1 g of protein/day, the dose of tacrolimus was reduced, and MMF was added at a dose of 0.5 g twice daily and adjusted to achieve target whole-blood trough levels of 2 to 4 mg/L. Triple therapy was then maintained for 9 additional months, after which immunosuppressants were tapered off over 3 months in all patients. Of the 21 adult patients enrolled, 11 had proteinuria of less than 1 g of protein/day at the end of 3 months and then received maintenance dosages of prednisolone plus tacrolimus. MMF was added after the third month in 9 patients and was associated with complete or partial remission in 5. Unfortunately, the relapse rate was very high in all groups of patients, whether treated with double or triple therapy. Clearly, additional controlled trials are required to determine the optimal duration of treatment and the benefit of adding MMF in patients who show only partial response to treatment with calcineurin-inhibitors alone.

Adrenocorticotrophic Hormone

The use of synthetic adrenocorticotrophic hormone (ACTH) has been assessed in patients with nephrotic syndrome, including those with MN.⁵³⁹ In one RCT,⁵⁴⁰ 32 patients were treated with either corticosteroids and chlorambucil or cyclophosphamide administered according to the Ponticelli protocol, or with ACTH (tetracosactide) administered intramuscularly twice weekly for 1 year. Of patients in the ACTH arm, 87% experienced complete or partial remission with a dramatic reduction in proteinuria and mean serum cholesterol and remission at almost 3 years.⁵⁴⁰ Few patients developed symptoms and signs of excess glucocorticoids.

Given lack of availability of the synthetic, long-acting ACTH analogue formulation in the United States, a retrospective case series has examined the role of a natural ACTH gel formulation for nephrotic syndrome.⁵⁴¹ In this study, 11 patients with primary MN out of 21 patients with nephrotic syndrome were treated with ACTH gel and 9 of 11 MN patients (82%) achieved complete (3 of 11, 27%) or partial (6 of 11, 55%) remission. A subsequent prospective,

open-label study evaluated the efficacy of ACTH gel (80 units subcutaneously twice weekly for 6 months) in patients with resistant glomerular disease, defined as failure to achieve sustained remission of proteinuria off immunosuppressive therapy with at least two treatment regimens.⁵⁴² In this study, 5 patients with primary MN out of 15 patients were treated; 2 patients achieved partial remission and 3 patients achieved immunologic remission. The mechanism responsible for the beneficial effect of ACTH is unknown. Studies have pointed to the expression of the melanocortin 1 receptor on podocytes, which suggests a possible direct effect of ACTH on these cells.⁵⁴³

Mycophenolate Mofetil

Interest has arisen in the use of mycophenolate mofetil (MMF) for the management of MN in a few small studies yielding disparate results. In an open-label study in the Netherlands, 32 patients with primary MN treated with MMF were compared with a historical matched control group treated with oral cyclophosphamide for 12 months.⁵⁴⁴ Both groups received intermittent methylprednisolone and alternate-day prednisone. Although, on average, the degree of proteinuria decreased similarly in the MMF-treated group and in the cyclophosphamide-treated group, and although the cumulative incidence of remission of proteinuria at 12 months was comparable in the two groups, the percentage of patients who showed no response to therapy was statistically significantly higher in the MMF-treated group, as was the proportion of patients who experienced a relapse.

In a 1-year RCT of 36 patients, treatment with MMF (target dose, 2 g/day) for 12 months was compared with conservative care alone.⁵⁴⁵ The change in the mean urine protein-to-creatinine ratio from baseline to month 12 was measured; the ratio decreased by 1834 mg/g in the control group and increased by 213 mg/g in the MMF group ($P = .3$). Complete or partial remission at month 12 was observed in 37% of patients in the MMF group and 41% in the control group.

In a separate prospective, randomized, controlled, open-label study involving 20 patients in Hong Kong and Shanghai,⁵⁴⁶ treatment with MMF and prednisolone given for 6 months was compared with treatment that followed a modified Ponticelli regimen.⁵²⁵ Over a total follow-up of 15 months, proteinuria decreased to a similar extent in both treatment groups. The rates of complete and partial remissions were 27.3% and 36.4%, respectively, in the MMF group, and 33.3% and 33.3% in the control group. These results are in marked distinction to those of the previously described study comparing MMF with conservative therapy.⁵⁴⁵ These two studies differ in the makeup of the patient populations (Caucasians vs. Asians) and by the concomitant use of prednisolone in the Hong Kong and Shanghai study. Overall, the results of studies of MMF have been disappointing, with a notable exception of those involving Chinese patients.

Rituximab

Given the high likelihood of antibody-mediated injury in patients with MN, there has been an explosion of interest in the use of rituximab, a humanized anti-CD20 monoclonal antibody. In an initial report of eight patients, treatment with rituximab (4 weekly doses of 375 mg/m² body surface area) was associated with prompt and sustained reduction in proteinuria.^{547,548} There have been additional positive

open-label studies,^{447,549–552} but the effects on long-term renal outcome are unproven. The available uncontrolled data have suggested that rituximab, 375 mg/m² once weekly for 4 weeks or at 1 g on days 1 and 15, achieves a 15% to 20% rate of complete remission and 40% to 45% rate of partial remission.⁵⁵³

A cohort study has reported on the effect of rituximab in 13 patients with MN deemed to be calcineurin inhibitor-dependent (defined as the occurrence of at least four calcineurin inhibitor-responsive relapses of nephrotic proteinuria while the patient was being weaned off these drugs).⁵⁵⁴ After rituximab therapy (375 mg/m² weekly for 4 weeks, with each dose preceded by 125 mg of methylprednisolone), proteinuria decreased significantly, and calcineurin inhibitors and other immunosuppressant drugs could be withdrawn in all patients.

Although the aggregate of these uncontrolled case series suggests a beneficial effect of rituximab in the management of MN, whether, when, and how (and how long) to use rituximab in treating MN remains to be determined. A large pharmaceutical company-funded, randomized controlled study comparing rituximab with calcineurin inhibitor therapy across 19 medical centers in the United States is now underway (www.clinicaltrials.gov, NCT01180036)⁵⁵⁵; however, results may be met with skepticism given prior unfavorable results seen with calcineurin inhibitors as compared with corticosteroids in combination with chlorambucil.⁵²⁸

Other Therapies

Other forms of therapy have been tried in primary MN, with varying results. These include the use of azathioprine,^{519,520} which demonstrated no positive effect either alone or in combination with prednisone. The use of pooled intravenous immunoglobulin has been evaluated only in a small case series⁵⁵⁶ and a retrospective study.⁵⁵⁷

Based on the greater appreciation of the role of complement activation, and especially that of complement-regulatory proteins in the pathogenesis of MN, a great deal of interest exists in targeting this pathway for therapy. Several compounds are under development. To date, human trials have been conducted only for eculizumab, a monoclonal antibody directed against the fifth component of complement (C5). In a randomized trial of patients with de novo MN, treatment with eculizumab was not associated with a statistically significant improvement in proteinuria or preservation of kidney function. These disappointing results were likely due to insufficient dosing because consistent inhibition of complement was achieved only in a minority of patients.⁵⁵⁸ Nevertheless, this general approach is thought to hold a great deal of promise based on early animal studies.

In the absence of a full understanding of the pathogenesis of MN, and thus an effective targeted therapy, the current approach to the treatment of MN must rely on risk stratification. The indolent disease process that results in spontaneous remissions in 25% of patients, coupled with the known adverse consequences of long-term treatment with oral glucocorticoids, alkylating agents, and calcineurin inhibitors, should prompt a careful analysis of the risk-benefit ratio in the treatment of any given patient. All patients should receive excellent supportive care, including the use of RAAS blockers^{269,559–562} and lipid-lowering agents. Most patients should be observed for the development of adverse prognostic factors or the

occurrence of spontaneous remissions. Adult patients with good prognostic features should be managed conservatively without the use of immunomodulatory or suppressive agents.

Patients at moderate risk (persistent proteinuria between 4 and 6 g of protein/day, despite RAAS blockade and normal kidney function) or at high risk of progression (persistent proteinuria of more than 8 g of protein/day, with or without renal insufficiency) should be considered for immunosuppressive therapy with a combination of glucocorticoids and cyclophosphamide (or chlorambucil) in alternating monthly pulses (Ponticelli protocol). This decision must be individualized to each patient with consideration of the patient's comorbidities and assessment of the risk associated with each type of therapy. The data currently available do not suggest that MMF alone is effective. Whether ACTH and/or rituximab are viable effective alternatives awaits further confirmation. Some of the difficulties associated with determining drug efficacy in clinical trials focused on MN, not to mention the other prior glomerular diseases, are related to variable endpoint definitions. Furthermore, surrogate endpoints for a treatment effect on ESKD, such as complete and/or partial remission of proteinuria used early in the disease process, could make clinical trial design more effective. A recent study has found that available data suggest that complete remission of proteinuria could potentially be used as a surrogate endpoint in primary MN in future clinical trial settings.⁵⁶³

Individuals who have advanced chronic renal failure are best managed by supportive care while awaiting dialysis and kidney transplantation. Acute renal insufficiency in this population should prompt evaluation for interstitial nephritis, crescentic nephritis, and renal vein thrombosis.

MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS AND C3 GLOMERULOPATHY

Primary membranoproliferative glomerulonephritis (MPGN) is a collection of morphologically similar but pathogenetically distinct disorders. Overall, MPGN is identified in approximately 10% of kidney biopsy specimens.^{564,565} MPGN has traditionally been classified based on electron microscopic findings into three types; MPGN type I is the most common form and is characterized by mesangial and subendothelial deposits. MPGN type II (dense deposit disease, DDD) has electron-dense deposits along the GBM. MPGN type III has both subepithelial and subendothelial deposits.^{566,569,570} Recent advances in the understanding of the different pathogenetic mechanisms underlying the MPGN subtypes have resulted in a new classification system.^{542,566–568} In the new classification, MPGN is classified based on glomerular immune deposit staining on IF, both immunoglobulin and complement (immune-complex mediated MPGN), and those with predominant C3 staining (complement-mediated MPGN, C3 glomerulopathy; Fig. 31.11). Immune complex glomerular deposits may activate the classical complement pathway, resulting in deposition of C1q, C4, and C3.^{566,570} The category MPGN type I, and probably type III, based on immunoreactant staining, involves cases of immune complex-mediated GN and cases of C3G. C3 glomerulopathy is characterized by variable light microscopic findings (with or without MPGN pattern) and defined by dominant C3 staining of glomerular deposits; the intensity of C3 staining has to be twofold greater than other immunoreactants. C3G is further subdivided based on electron microscopy into DDD and C3 glomerulonephritis.⁵⁷¹

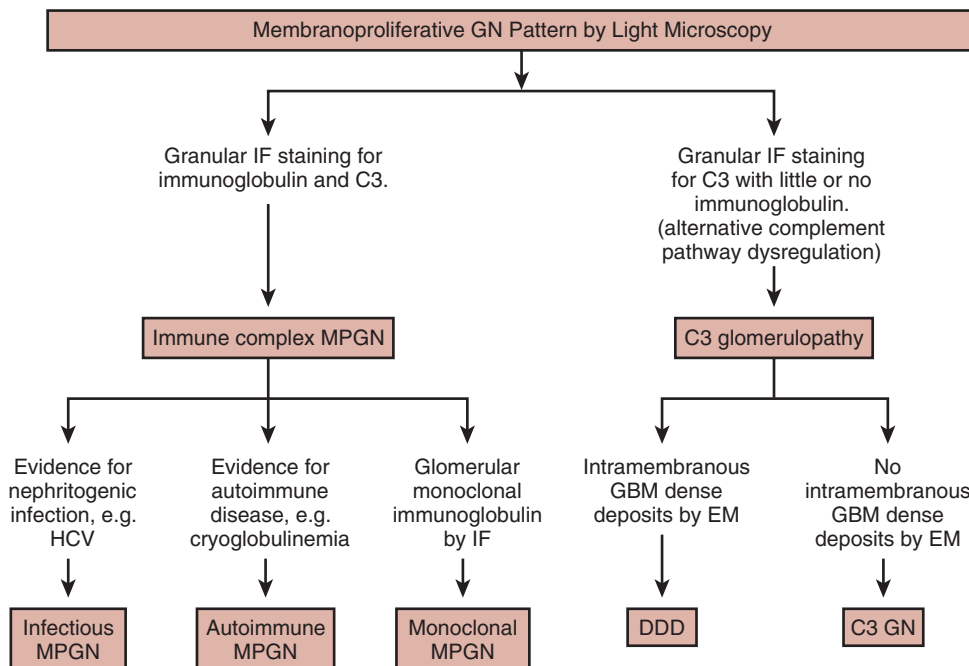


Fig. 31.11 Algorithm for the histopathologic classification of membranoproliferative glomerulonephritis (MPGN) by immunofluorescence microscopy findings based on the presence of both immunoglobulin and C3 versus the presence of C3 with little or no immunoglobulin. This modified classification system also includes a category of diseases termed “C3 glomerulopathy,” that includes dense deposit disease (DDD, formerly type II MPGN) and C3 glomerulonephritis, which comprises C3 glomerulopathy that does not have an MPGN pattern.

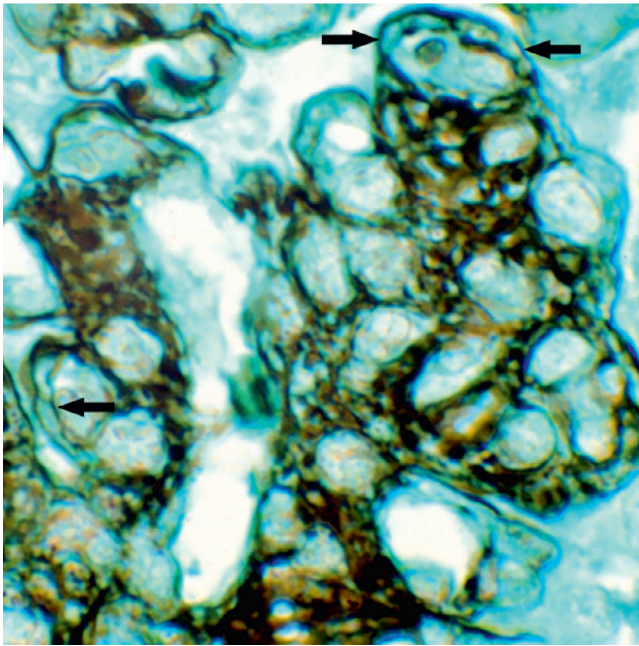


Fig. 31.12 Light micrograph of a glomerular segment from a patient with type I membranoproliferative glomerulonephritis demonstrating doubling (arrows) and more complex replication of glomerular basement membranes. (Jones' methenamine silver stain, $\times 300$.)

A host of disorders may present as immune complex-mediated MPGN, including infections, malignancy, autoimmune disorders, and cryoglobulinemia. Rarely, immunotactoid-fibrillary GN and proliferative GN with monoclonal IgG deposits may present with a membranoproliferative pattern.⁵⁷⁰ C3G results from a genetic or acquired dysregulation of the alternate complement pathway. C3 nephritic factor, anti-FH antibody, and monoclonal immunoglobulin are some of the acquired elements.^{570–572}

MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS TYPE I

Pathology

Light Microscopy. The typical histologic features of type I MPGN are diffuse global capillary wall thickening, an increased mesangial matrix, and mesangial and endocapillary hypercellularity.^{573,574} Infiltrating mononuclear leukocytes and neutrophils contribute to the glomerular hypercellularity. The consolidation of glomerular segments that results from these changes often causes an accentuation of the segmentation referred to as “hypersegmentation or lobulation.” As a consequence, an earlier name for this phenotype of glomerular injury was “lobular glomerulonephritis.” Markedly expanded mesangial regions may develop a nodular appearance with a central zone of sclerosis that may resemble that of diabetic glomerulosclerosis or monoclonal immunoglobulin deposition disease. However, the integration of light, IF, and electron microscopy findings differentiates type I MPGN from other diseases that can mimic it by light microscopy.

A distinctive but not completely specific feature of type I MPGN is doubling or more complex replication of GBMs that can be seen with stains that highlight basement membranes, such as Jones silver methenamine stain or periodic acid–Schiff stain (Fig. 31.12). This change is caused by the

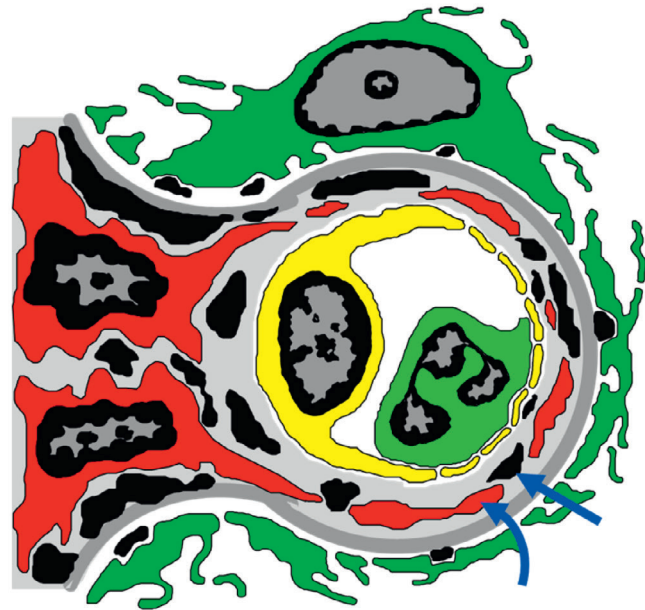


Fig. 31.13 Diagram depicting the ultrastructural features of type I membranoproliferative glomerulonephritis. Note the subendothelial dense deposits (straight arrow), subendothelial mesangial cytoplasm interposition (curved arrow), and production of new basement material (asterisk). (Courtesy JC Jennette.)

production of basement membrane material between and around projections of mesangial cytoplasm that extend into an expanded subendothelial zone, probably in response to the presence of subendothelial immune complex deposits (Fig. 31.13). The presence of so-called hyaline thrombi in capillary lumens should raise the possibility of cryoglobulinemia or lupus as the cause for the MPGN. Hyaline thrombi are not true thrombi but are aggregates of immune complexes filling capillary lumens. A minority of patients with type I MPGN have crescents, but these rarely involve more than 50% of glomeruli.^{575,576} As with other types of glomerulonephritis, substantial crescent formation correlates with a more rapid progression of disease.⁵⁷⁴

As noted earlier, C3 glomerulopathy is divided into dense deposit disease (formerly type II MPGN) and C3 glomerulonephritis (see Fig. 31.11). C3 glomerulonephritis in turn is divided into the C3 glomerulopathy variant of type I MPGN, as well as other patterns of glomerulonephritis that do not fulfill the pathologic criteria for MPGN. Non-MPGN C3 glomerulonephritis often has focal or diffuse proliferative glomerulonephritis, with varying degrees of endocapillary and mesangial hypercellularity (Fig. 31.14).⁵⁶⁸ Crescents may be present.

Immunofluorescence Microscopy. The characteristic pattern of IF staining is peripheral granular to band-like staining for complement, especially C3 (Fig. 31.15). The C3 staining is accompanied by substantial immunoglobulin staining in patients with immune complex MPGN. This positive IF corresponds to the prominent subendothelial immune deposits seen by electron microscopy. The staining pattern is less granular and less symmetric than that seen in MN. Mesangial granular staining may be conspicuous or inconspicuous. The hypersegmentation or lobulation seen by light microscopy often can be discerned by IF microscopy. A minority of patients

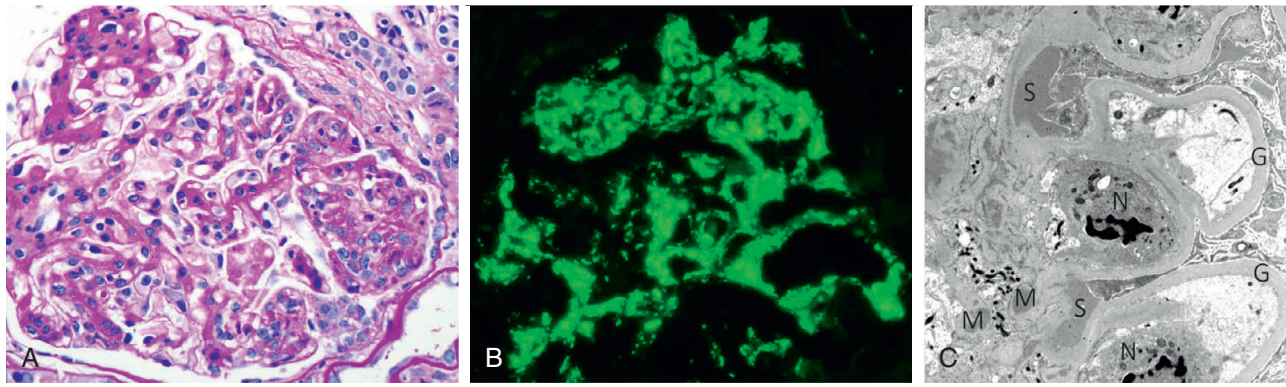


Fig. 31.14 C3 glomerulonephritis variant of C3 glomerulopathy with light microscopy (A) showing segmental mesangial and endocapillary hypercellularity (periodic acid–Schiff stain). (B) Immunofluorescence (IF) microscopy showing granular capillary wall staining for C3 (there was no staining for immunoglobulin). (C) Electron microscopy demonstrating subepithelial electron dense deposits (S), mesangial dense deposits (M), influx of capillary neutrophils (N), and no intramembranous dense deposits in glomerular basement membranes (G).

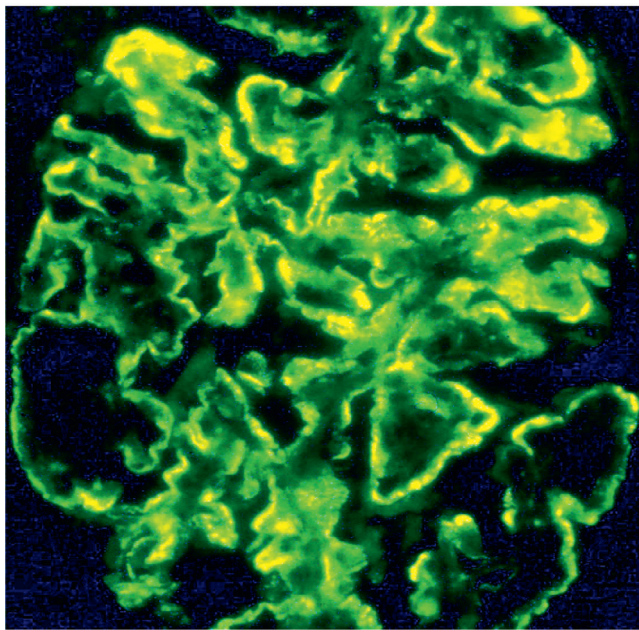


Fig. 31.15 Immunofluorescence micrograph of a glomerulus with features of type I membranoproliferative glomerulonephritis showing global bandlike capillary wall staining for C3, as well as irregular mesangial staining. (Fluorescein isothiocyanate anti-C3 stain, $\times 300$.)

with type I immune complex MPGN have staining along tubular basement membranes, in extraglomerular vessels, or both.

Most specimens with immune complex MPGN have more intense staining for C3 than for any immunoglobulin, but some specimens have more intense staining for IgG or IgM. Rare specimens have a predominance of IgA and can be considered an MPGN expression of IgA nephropathy. Intracapillary globular structures that stain intensely for immunoglobulin and complement correspond to the hyaline thrombi seen by light microscopy and raise the possibility of MPGN caused by lupus or cryoglobulinemia.

The C3 glomerulopathy variant of MPGN has little or no staining for immunoglobulin. However, Hou and associates⁵⁶⁷ have noted that some specimens that seem appropriate for the C3 glomerulopathy category have minor staining for

immunoglobulin rather than no staining. They have proposed a definition for C3 glomerulopathy that requires C3-dominant staining that is at least two orders of magnitude (on a scale of 0–3 or 0–4) more intense than any other immune reactant. In line with the evidence that C3 glomerulopathy is mediated by dysregulated activation of the alternative complement pathway, there is little or no staining for C4 or C1q in the C3 glomerulopathy variant of MPGN or in other histologic expressions of C3 glomerulonephritis.

Electron Microscopy. The ultrastructural hallmark of type I MPGN is mesangial interposition into an expanded subendothelial zone that contains electron-dense immune deposits (Fig. 31.16; see Fig. 31.13). This distinct pattern of mesangial and capillary involvement has prompted a synonym for type I MPGN, called “mesangiocapillary glomerulonephritis.” New basement membrane material is formed around the subendothelial deposits and around the projections of mesangial cytoplasm, which is the basis for the basement membrane replication seen by light microscopy (see Fig. 31.12). Scattered mesangial dense deposits are usually found in association with mesangial hypercellularity and mesangial matrix expansion. Variable numbers of subepithelial electron-dense deposits occur. When they are numerous enough to resemble MN, some nephropathologists apply the diagnosis mixed membranous and proliferative glomerulonephritis or type III MPGN, as proposed by Burkholder and colleagues.⁵⁶⁹ The term “type III MPGN” also has been applied to a very rare pattern of glomerular injury that resembles type I MPGN by light microscopy and IF microscopy, but is characterized ultrastructurally by irregularly thickened GBMs, with numerous intramembranous deposits of variable density.^{577,578} Type III MPGN can be classified further as either immune complex disease or C3 glomerulopathy based on the presence or absence of staining for immunoglobulin. In the early application of the new classification system, type III MPGN as proposed by Burkholder and associates⁵⁶⁹ tends to have an immune complex phenotype, whereas the other variant described by Strife and colleagues tends to have a C3 glomerulopathy phenotype.^{577,578}

The hyaline thrombi seen by light microscopy appear as intraluminal spherical densities. When these structures, or any of the other electron-dense deposits, have a microtubular

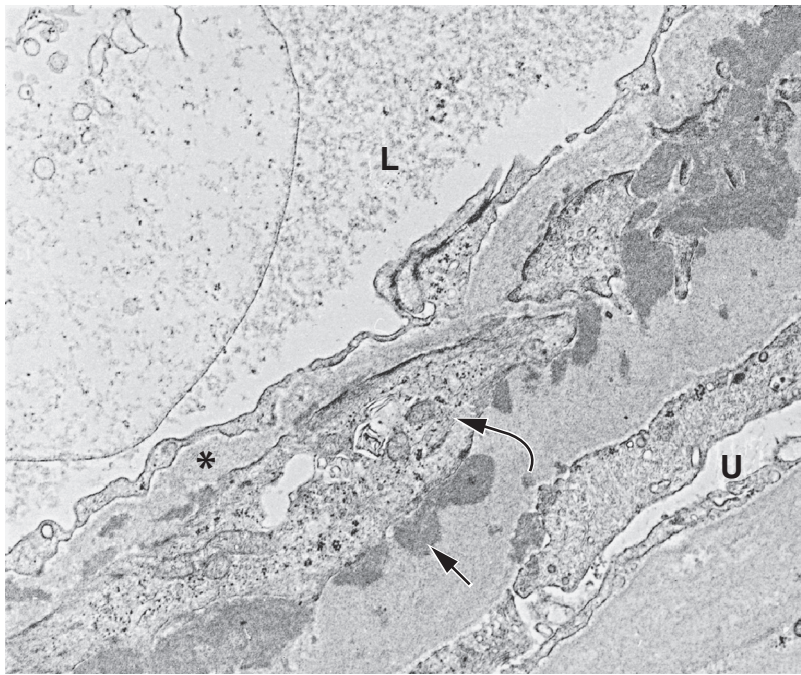


Fig. 31.16 Electron micrograph of a capillary wall from a glomerulus with features of type I membranoproliferative glomerulonephritis. The capillary lumen (L) is in the upper left and the urinary space (U) is in the lower right. In the subendothelial zone are dense deposits (*straight arrow*), extensions of mesangial cytoplasm (*curved arrow*), and new basement membrane material (*asterisk*) (see Fig. 31.12). (×10,000.)

substructure, the possibility of cryoglobulinemic glomerulonephritis or immunotactoid glomerulopathy should be considered.

Pathogenesis

As noted earlier, two major pathogenic pathways can cause the pathology pattern of glomerular injury called MPGN—immune complex localization and dysregulation of alternative pathway complement activation. A rare cause for this pattern is glomerular deposition of monoclonal immunoglobulin.

In the minority of patients with immune complex MPGN in whom the antigen has been identified, the sources have included infections, autoimmune diseases, neoplasms, and hereditary diseases (Box 31.7). The pathological finding of intense immune complex deposition with hypercellularity suggests that the inflammation caused by the immune complexes has resulted in both proliferation of mesangial and endothelial cells and the recruitment of inflammatory cells, including neutrophils and monocytes. These leukocytes are attracted to the glomerulus by activation of multiple mediator systems, including the complement system, cytokines, and chemokines.

Immune complex type I MPGN may be secondary to recognizable causes, such as cryoglobulinemia, hepatitis C, hepatitis B, osteomyelitis, subacute bacterial endocarditis, infected ventriculoatrial shunt, malignancies,^{579–581} autoimmune disease (SLE or autoimmune thyroiditis⁵⁸²), light chain nephropathy,⁵⁸³ and celiac sprue.⁵⁸⁴ Serologic and clinical evidence of these processes should be sought. The precise percentage of patients with MPGN due to hepatitis C may vary according to geographic area and cultural factors. The observation that upper respiratory tract infections precede the onset of MPGN in as many as 50% of patients⁵⁸⁵ raises the possibility that infectious agents contribute to the pathogenesis of many cases of type I MPGN.

Box 31.7 Secondary Causes of Membranoproliferative Glomerulonephritis

Associated With Infection

Hepatitis B and C
Visceral abscesses
Infective endocarditis
Shunt nephritis
Quartan malaria
Schistosoma nephropathy
Mycoplasma infection

Associated With Rheumatologic Disease

Systemic lupus erythematosus
Scleroderma
Sjögren syndrome
Sarcoidosis
Mixed essential cryoglobulinemia with or without hepatitis C infection
Anti-smooth muscle syndrome

Associated With Malignancy

Carcinoma
Lymphoma
Leukemia

Associated With an Inherited Disorder

α_1 -Antitrypsin deficiency
Complement deficiency (C2 or C3), with or without partial lipodystrophy

Modified from references 1106 and 1473 to 1481.

Table 31.6 Selected Serologic Findings in Patients With Primary Glomerular Disease

Disease	C4	C3	ASO, ADNase B	Cryo Ig	aGBM	ANCA
Minimal change disease	N	N	—	—	—	—
Focal glomerulosclerosis	N	N	—	—	—	—
Membranous nephropathy	N	N	—	—	—	—
Membranoproliferative GN ^a						
Type I	N or ↓↓	↓↓	+	++	—	—
Type II	N	↓↓↓	+	—	—	—
Fibrillary GN	N	N	—	—	—	—
IgA nephropathy	N	N	—	—	—	—
Acute poststreptococcal GN	N or ↓	↓↓	+++	++	—	—
Crescentic GN						
Anti-GBM	N	N	—	—	+++	±
Immune complex	N or ↓	N or ↓↓	—	N/++	—	±
ANCA-SVV	N	N	—	—	±	+++

^aFormer MPGN classification schema.

ANCA, Antineutrophil cytoplasmic autoantibody; aGBM, anti-glomerular basement membrane antibody; ASO, antistreptolysin O; ADNase B, anti-deoxyribonuclease B; cryo Ig, cryoglobulins; GBM, glomerular basement membrane; GN, glomerulonephritis; IgA, immunoglobulin A; N, normal levels; SVV, small vessel vasculitis.

Modified from Jennette JC, Nickleleit V. Anti-glomerular basement membrane glomerulonephritis and Goodpasture's syndrome. In: Jennette JC, Olson JL, Schwartz MM, eds. *Heptinstall's Pathology of the Kidney*. 6th ed. Philadelphia: Lippincott Williams & Wilkins; 2006.

When immune complex type I MPGN is secondary to other disease processes such as malignancy or a rheumatic condition, the laboratory results associated with the systemic disease (e.g., SLE) are positive (e.g., antibodies to double-stranded DNA⁵⁸⁶; see Chapters 32 and 33). MPGN type I is also associated with underlying complement deficiency, notably of C2 and C3,⁵⁸⁷ as well as deficiency in α_1 -antitrypsin.⁵⁸⁶

Even before recognition of the distinct C3 glomerulopathy variant of type I MPGN, alterations in circulating complement levels had been observed in MPGN patients (Table 31.6). The C3 level is persistently depressed in approximately 75% of MPGN patients.^{564,573,588–590} This is in contrast to poststreptococcal glomerulonephritis, in which depressed C3 levels typically return to normal within 2 months.^{591–593} The persistent depression of C3 levels and the presence of nephritic syndrome should suggest type I MPGN. In some patients, activation of the alternative pathway is suggested by the observation that C3 levels are depressed, whereas levels of the classical pathway activators C1q and C4 are normal. Patients with this complement profile are likely to have a C3 glomerulopathy variant of MPGN. However, when MPGN is caused by cryoglobulinemia, which is an immune complex MPGN, there may be more depression of C4 than C3.⁵⁹³

Epidemiology

Most patients with MPGN are children between the ages of 8 and 16 years⁵⁸⁵ accounting for 90% of cases of MPGN type I. The proportions of males and females are nearly equal.^{564,565,573,577,594–600}

Clinical Features

Type I MPGN, characterized by deposits of electron-dense material in the subendothelial zones of glomeruli, is a very heterogeneous disorder. Some combination of proteinuria (often nephrotic range), hematuria, hypertension, and renal failure are usually present.

Treatment

The prognosis of type I MPGN has been reviewed and described in several reports.^{564,573,601,602} Actuarial renal survival 10 years after renal biopsy has been calculated with life-table analysis to be 60% to 65% in type I MCGN, without significant differences between treated and untreated patients.⁵⁶⁴ Nonnephrotic patients have an improved 10-year renal survival of 85%.⁵⁷³ A minority of patients may have a spontaneous remission.⁵⁶⁴ The features suggestive of poor prognosis in MPGN type I include hypertension,^{600,603} impaired GFR,^{600,603–605} and the appearance of cellular crescents in biopsy specimens.^{573,605,606}

The treatment of type I MPGN is based on the underlying cause of the disease process. Thus, the therapy for MPGN associated with cryoglobulinemia and hepatitis C should be aimed at treating the hepatitis C virus infection, whereas the treatment of MPGN associated with lupus or scleroderma should be based on the principles of care for those rheumatologic conditions. Most recommendations for the treatment of type I MPGN have been derived from studies involving children.^{607–613} West has touted the benefits of continuous prednisone therapy for improved renal survival.⁶⁰⁹ Whether the benefit of low-dose prednisone therapy is seen only in children or whether similar effects can be achieved in adults has never been investigated in a prospective randomized trial. However, low-dose, alternate-day prednisone therapy may improve renal function.^{611,612}

In addition to glucocorticoids, a number of other immunosuppressive and anticoagulant therapies have been used in the treatment of type I MPGN. Initial reports have indicated that treatment with aspirin and dipyridamole has a positive effect on renal survival.⁵⁶⁵ This approach was widely accepted; however, statistical design flaws led to a reanalysis of the data, which revealed no difference in the treatment and control groups with respect to long-term outcome.⁵⁸⁸ A subsequent study using aspirin with dipyridamole demonstrated a slight

decrease in urine protein excretion by 3 years but no effect on renal function.⁶¹⁴ Treatment with dipyridamole, aspirin, and warfarin, with and without cyclophosphamide, was examined in both controlled and uncontrolled studies.^{461,565,613–618}

A regimen of warfarin, dipyridamole and cyclophosphamide⁶¹³ was suggested to improve long-term renal survival based on a retrospective analysis; however, a controlled trial in Canada has demonstrated no benefit of this approach.²⁸⁶

The use of MMF and corticosteroids has been suggested based on uncontrolled and anecdotal observations.⁶¹⁹ In patients with a defined underlying disease (e.g., neoplasia or hepatitis B or C), treatment should be directed at the underlying condition.^{620,621} Type I MPGN is significantly ameliorated by the use of cyclosporine in the very rare condition known as “Buckley syndrome.”^{104,105,622}

MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS TYPE III

Type III MPGN occurs in a very small number of children and young adults. Regardless of the pathologic distinctions of MPGN type III of Burkholder and colleagues⁵⁹⁶ and Strife and colleagues,⁵⁷⁸ few distinguishing clinical characteristics are noted in these patients. These patients may have clinical features of disease quite similar to those of type I MPGN, and the long-term clinical course is similar as well. Patients with MPGN described by Strife and associates⁵⁷⁸ have low C3 levels in the absence of C3 nephritic factor. Patients with nonnephrotic proteinuria do better than patients who have nephrotic syndrome.

C3 GLOMERULOPATHIES (DENSE DEPOSIT DISEASE AND C3 GLOMERULONEPHRITIS)

Epidemiology

DDD (formerly MPGN II) accounts for about 25% of MPGN in children but is much less common in adults. The large majority of patients are children between the ages of 8 and 16 years⁵⁸⁵ and account for about 70% of cases. It is estimated to affect 2 to 3 persons per million. Males and females were reported to be similarly affected in some studies,⁶²³ whereas others reported a female predominance.^{573,624,625} A retrospective of 32 cases identified through the nephropathology laboratory at Columbia University between 1977 and 2007 reported that 43% of patients were children, of whom 65% were between the ages of 5 and 10 years, and 22% of patients were adults older than 60 years. The female-to-male ratio was 1.9, and 85% of patients were Caucasian.⁶²⁶ It is unclear whether the unexpectedly high representation of older adults in this cohort reflects a change in the demographics of the disease or a change in biopsy practices or selection or referral bias in this or the older studies.

C3 glomerulonephritis includes all patients with C3 glomerulopathy who do not have intramembranous dense deposits that warrant a diagnosis of DDD; this includes the C3 glomerulopathy variant of MPGN, as well as other patterns of proliferative glomerulonephritis. There is a relative paucity of epidemiologic data given that this is a rather new diagnostic category; however, one study of 19 patients with C3 glomerulonephritis revealed a median age at onset of 29.9 years (range, 7–70 years).⁶²⁷ A larger and more recent study of 88 patients with C3 glomerulopathy in the United Kingdom, including 59 patients with C3 glomerulonephritis and 21

with DDD, has revealed that patients with C3 glomerulonephritis are significantly older (median age, 26 years vs. 12 years; $P = .002$).⁶²⁸

Pathology

The term *dense deposit disease* emphasizes the pathognomonic feature of discontinuous electron-dense bands in the GBM^{568,574} (Figs. 31.17 and 31.18). These are accompanied by spherical to irregular mesangial dense deposits, some of which resemble the humps seen in postinfectious glomerulonephritis. Dense

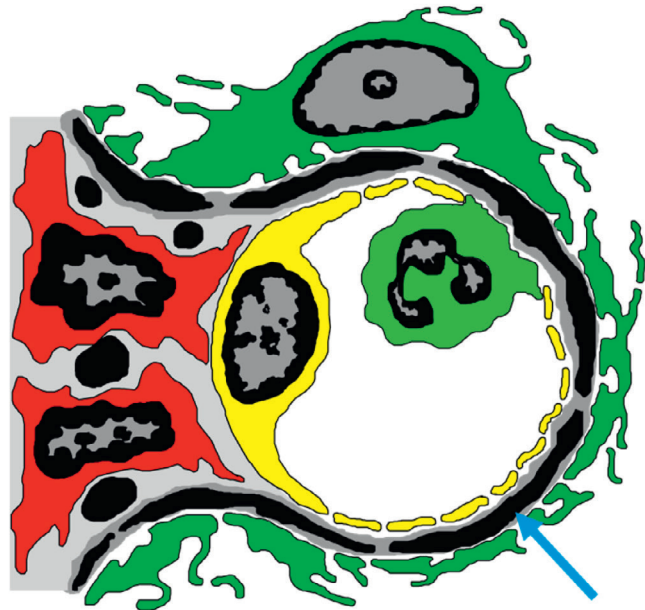


Fig. 31.17 Diagram depicting a glomerular capillary loop with features of dense deposit disease, with bandlike intramembranous dense deposits (arrow) and spherical mesangial dense deposits. (Courtesy JC Jennette.)

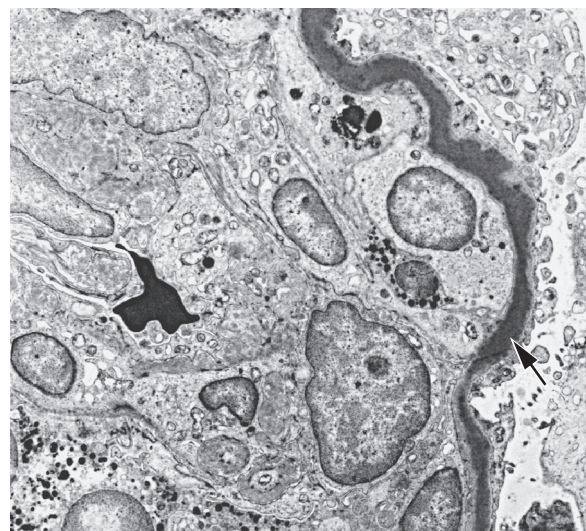


Fig. 31.18 Electron micrograph of a glomerular capillary from a patient with dense deposit disease showing a bandlike intramembranous dense deposit that has essentially replaced the normal glomerular basement membrane. Also note the endocapillary hypercellularity. (×5000.)

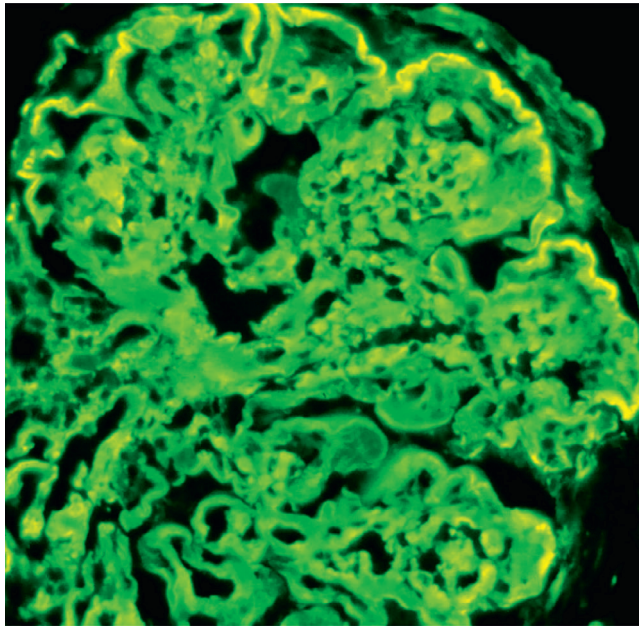


Fig. 31.19 Immunofluorescence micrograph of a glomerulus with dense deposit disease demonstrating discontinuous band-like capillary wall staining and coarsely granular mesangial staining for C3. (Fluorescein isothiocyanate anti-C3 stain, $\times 500$.)

deposits also may be identified in Bowman's capsule and tubular basement membranes.

Immunofluorescence microscopy demonstrates intense capillary wall linear to bandlike staining for C3 (**Fig. 31.19**), with little or no staining for immunoglobulin.^{567,568,629,630} The capillary wall staining may have a fine double contour, with outlining of the outer and inner aspects of the dense deposits. The mesangial deposits usually appear as scattered spherules or rings, with the latter resulting from staining of the outer surface but not the interior of the spherical deposits. There typically is intense staining for C3, with little or no staining for C4, C1q, or immunoglobulins.

The light microscopic appearance of DDD is much more variable than that of type I MPGN and often does not have a membranoproliferative appearance.⁵⁶⁸ Thus the term "DDD" is preferable to the term "type II MPGN."^{630,631} In a review of a large number of renal biopsy specimens obtained in North America, Europe, and Japan from patients with DDD, the light microscopy findings could be classified into five distinct patterns: (1) membranoproliferative pattern; (2) mesangioproliferative pattern; (3) crescentic pattern; (4) acute proliferative and exudative pattern; and (5) unclassified DDD. Of these, the mesangioproliferative lesion characterized by focal segmental and mesangial hypercellularity accounted for about 50% of the cases reviewed, with 28% presenting with a membranoproliferative pattern (type I) and 20% with a crescentic lesion. Although the patients' ages ranged from 3 to 67 years, nearly 75% of the patients were younger than 20 years of age, and all patients with crescentic DDD or acute proliferative DDD were between the ages of 3 and 18 years.⁶³¹ Therefore, the histologic appearance of DDD can mimic many other categories of glomerulonephritis, and the findings by IF and especially electron microscopy are required for accurate diagnosis.

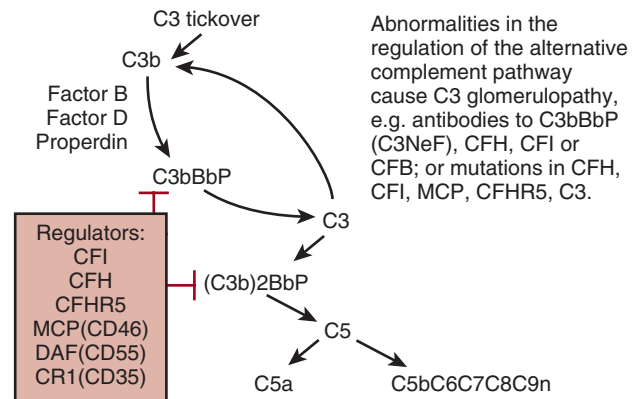


Fig. 31.20 Regulators of the alternative complement pathway and abnormalities that cause C3 glomerulopathy. *CF*, Complement factor.

As described earlier in this chapter, the pathology of the non-DDD category of C3 glomerulopathy (i.e., C3 glomerulonephritis) is extremely variable by light microscopy, but falls in the spectrum of type I MPGN, to focal or diffuse proliferative to mesangial proliferative glomerulonephritis (see **Fig. 31.14**).⁵⁶⁸ As one of the variants of C3 glomerulopathy, there is C3 dominant staining by IF microscopy, with little or no staining for immunoglobulin, C4, or C1q. By definition, electron microscopy of C3 glomerulonephritis does not reveal the typical intramembranous dense deposits that are diagnostic for DDD. However, this is a somewhat subjective division, and borderline specimens are difficult to classify as DDD versus C3 glomerulonephritis. From a practical perspective, this may not be important because both are variants of C3 glomerulopathy and have similar pathogenesis and clinical features.

Pathogenesis

As C3 glomerulopathies, DDD and C3 glomerulonephritis share a pathogenic mechanism that involves abnormal regulation of the alternative complement pathway (reviewed by Bomback and Appel).⁶³²

A porcine model of DDD has suggested that there is massive deposition of C3 and the terminal C5b-9 complement complex—the membrane attack complex. In the circulation, there is extensive complement activation, with very low C3 levels and high levels of circulating terminal complement components. No immunoglobulin deposits were detected in kidney tissue. In this animal model of DDD, the pathogenetic mechanism does not appear to involve immune complexes, but rather uses some other mechanism for the activation of complement and the trapping of activating complement components in the GBM.⁶³³

The hypocomplementemia in DDD and other forms of C3 glomerulopathy reflects dysregulated activation of the alternative complement pathway (**Fig. 31.20**). Under normal circumstances, the alternative complement pathway maintains low-level basal activity, termed *tickover*, which maintains C3 convertase. Any disruption in C3 convertase activity can thereby alter tickover and result in overactivation. Three distinct mechanisms result in uncontrolled activation of C3 convertase: (1) the development of an autoantibody, the C3 nephritic factor (C3NeF); (2) the absence of circulating regulators (e.g., factor H); and (3) the presence of a

circulating inhibitor of factor H.⁶³³ The most common of these is the presence of the autoantibody C3NeF. C3NeF is an antibody that protects C3 convertase (C3bBb) from dissociation by factor H and thus prolongs its half-life by 10-fold.⁶³⁴ It does so in one of two ways, by binding to C3bBb or to IgG-C3b-C3bBb of the assembled convertase. The stabilization of this complex results in perpetual C3 breakdown. It is tempting to incriminate this factor as central to the pathogenesis of MPGN, especially DDD, because most affected patients harbor C3NeF in their circulation. However, C3NeF does not always correlate with disease activity and, more importantly, progressive kidney damage still occurs in patients who have normal levels of complement.^{635–637} Interestingly, autoantibodies to other complement components such as factor B, factor H, factor I, or component of C3 convertase (C3b and/or CFB) can also be seen in some individuals.^{632,638,639} More recently, reclassification of MPGN was strongly proposed by Sethi and colleagues,^{640,641} who found higher level alternative and terminal complement pathway components, as identified by mass spectrometry, directly within the glomeruli of patients with DDD as compared with those from patients with immune complex-mediated forms of MPGN, as well as healthy individuals.

Normal protective, or regulatory, mechanisms control C3bBb levels and complement deposition, of which factor H is one of the most important. Factor H is a soluble glycoprotein that regulates complement in the fluid phase and on cell surfaces by binding to C3b.⁶⁴² Some mutations in factor H result in C3 glomerulopathy that does not have an MPGN pattern.^{643,644}

The genetics of DDD are complex. Only a few families have been identified with more than one affected member, although families exist with one patient with DDD and other members affected by other autoimmune diseases. The most robust genetic association with DDD is a deficiency of factor H, associated with a mutation of the complement factor H (*CFH*) gene.^{645,646} Another study has identified a family with DDD in which a C3 convertase-resistant C3 gene mutation was identified in a mother and her identical twin sons, thereby resulting in alternative complement pathway activation.⁶⁴⁷

C3 glomerulonephritis is a result of overactivity of the alternative complement pathway due to autoantibodies to and/or mutations in critical regulators of the pathway.⁵⁶⁸ The underlying genetic defect in C3 glomerulonephritis has been identified in some hereditary forms of the disease such as CFHR5 nephropathy, and further insight is likely to occur rapidly given intense interest in this area.^{648–650}

Clinical Features

Patients with DDD may have hematuria, proteinuria, or both. They may have a nephrotic or acute nephritic syndrome. At least one-third of patients have all the components of the nephrotic syndrome on presentation. Microhematuria is present in the overwhelming majority of patients, whereas gross hematuria occurs only in about 15%.⁶²⁶ Finally, 25% of patients have acute nephritic syndrome associated with red cells and red cell casts in the urine, hypertension, and renal insufficiency.^{564,573,588,589} Hypertension is typically mild, but may be severe in some cases. Kidney dysfunction occurs in at least 50% of cases and is more common among adults than in children.⁶²⁶ When present at the outset of disease, kidney dysfunction portends a poor prognosis.

Hypocomplementemia of the C3 factor is present in 80% to 90% of patients with DDD. In a retrospective review from Columbia University, depressed C3 levels occurred in 100% of children, but only in 41% of adults ($P = .001$).⁶²⁶ Depressed C4 levels were very uncommon in both age groups. C3 hypocomplementemia is prolonged in patients with DDD⁴⁶⁷ and is associated with decrements in terminal complement components C5b-9. C3NeF is present in more than 80% of patients.

Respiratory tract infections precede cases of MPGN in 50% of patients, especially in children.⁶²⁶ On rare occasions, the onset can be triggered by a streptococcal infection, so that the disease mimics acute poststreptococcal glomerulonephritis except for the persistence of C3 hypocomplementemia beyond 8 weeks from onset.⁶⁵¹ Comorbid conditions may be seen in adult patients with DDD, including plasma cell dyscrasias, which were noted in 4 of 18 patients (22%) in the Columbia University review.⁶²⁶

Patients with DDD may have deposits in the retina, along Bruch's membrane, that are similar in structure and composition to the deposits in GBM. These whitish yellow drusen develop at an early age. Initially, drusen have little impact on visual acuity, but visual loss can occur in about 10% of patients.⁶⁵² A careful retinal examination is therefore indicated in all patients with proven DDD. There is no correlation between the severity of kidney and ocular involvement; however, patients with other forms of MPGN do not typically have drusen.⁶⁵³

DDD may be associated with the syndrome of acquired partial lipodystrophy.⁶³⁷ About 80% of patients with this syndrome have low C3 levels and C3NeF. About 20% of patients develop MPGN, although the lipodystrophy and glomerular disease may occur several years apart.⁶⁵⁴ The link between acquired partial lipodystrophy and MPGN stems from the production by adipocytes of C3, factor B, and factor D (adeptine) whose function is the cleavage of factor B. In the presence of C3NeF, the alternative pathway of complement activation is dysregulated, leading to the destruction of adipocytes.⁶⁵⁴

Patients with C3 glomerulonephritis, as also seen with DDD, can present with hypertension, azotemia, hematuria, and proteinuria, with or without nephrotic syndrome. Interestingly, although C3 levels are usually low and C4 levels are normal, the large study of 59 patients with C3 glomerulonephritis in the United Kingdom found that C3 was more likely to be normal when compared with patients with DDD (52% vs. 11%; $P = .003$); therefore, a normal C3 level alone does not exclude C3 glomerulonephritis.^{628,641} As with DDD, some patients may have C3NeF.^{627,632,655}

Treatment

The prognosis for DDD is worse than that for type I MPGN. The less favorable prognosis is in accord with a higher frequency of crescentic glomerulonephritis and chronic tubulointerstitial nephritis at the time of biopsy in DDD.^{603,656,657} In DDD, clinical remissions are rare,^{573,594} occurring in fewer than 5% of children. Patients generally reach ESKD in 8 to 12 years from the onset of disease. In DDD, the prognosis is worse in adults than in children.⁶²⁶

There is currently no widely agreed on treatment of DDD, and the available information that remains is based on small case series. Inhibition of angiotensin II may be helpful, but

it has not been formally tested.⁶²⁶ The use of corticosteroid therapy is probably not effective.⁶⁴⁵ Immunosuppressive therapy with agents such as MMF and rituximab aimed at reducing C3NeF in DDD has been suggested.⁶⁵⁸ It is not likely that immunosuppression, with or without plasma exchange, will be beneficial, except perhaps if an inhibitory autoantibody to complement factor H is present. There are reports of effective treatment of patients with defined deficiency of complement factor H with infusion of fresh-frozen plasma every 14 days to provide functionally intact factor H.^{655,659–662} In a proof of concept study in mice genetically deficient in complement factor H, treatment with purified human complement factor H resulted in rapid normalization of plasma C3 levels and resolution of the GBM C3 deposition.⁶⁶³ Based on the current understanding of the pathogenesis of DDD, inhibition of C5 activation and formation of the terminal component of complement activation (C5b-9) with the monoclonal antibody eculizumab would be theoretically beneficial. However, evidence for this approach had been lacking until a proof of concept efficacy and safety study in three patients with DDD and three patients with C3 glomerulonephritis treated in an open-label fashion with eculizumab.⁶⁶¹ This resulted in reduced serum creatinine levels in two subjects, reduction in proteinuria in one subject, and histopathologic improvement in one subject. It also appeared that elevated levels of the serum membrane attack complex might predict treatment response.⁵⁴²

Given that C3 glomerulonephritis represents a new diagnostic category, treatment protocols outside of supportive care are limited but will likely involve targeted therapies that correct complement dysregulation now that pathogenetic mechanisms are becoming increasingly understood. As one example, a proof of concept efficacy and safety study in three patients with DDD and three patients with C3 glomerulonephritis treated in an open-label fashion with eculizumab resulted in reduced serum creatinine levels in two subjects, reduction in proteinuria in one subject, and histopathologic improvement in one subject. It also appeared that elevated levels of the serum membrane attack complex might predict treatment response.⁶⁶⁴

Data are also limited with regards to prognosis; however, the study by Servais and colleagues would suggest that prognosis is poor, but possibly better than that of DDD. Of the 19 patients with C3 glomerulonephritis, 3 progressed to ESKD at the time of the study, and 6 had a creatinine clearance less than 60 mL/min.⁶⁶⁵ It was also found in a cohort of 134 patients with C3 glomerulopathy in France, in which roughly 25% of patients with C3 glomerulonephritis progressed to ESKD over a follow-up period of 10 years.⁶⁶⁵ Sethi and colleagues reported on 12 patients with C3 glomerulonephritis, with a mean follow-up of 26.4 months, that kidney function remained stable in every patient, except for 1 who progressed to dialysis soon after presentation.⁶⁴¹ With regard to kidney transplantation, one group reported their experience with 21 patients with ESKD due to C3 glomerulonephritis who underwent kidney transplantation. In this study, 14 patients (68%) developed recurrent C3 glomerulonephritis in the allograft, which was typically manifested by hematuria and proteinuria, and median time to recurrence was 28 months. Furthermore, 50% of patients with recurrent C3 glomerulonephritis experienced graft failure, with a median time to graft failure of 77 months.⁶⁶⁶ Recurrences in those with kidney

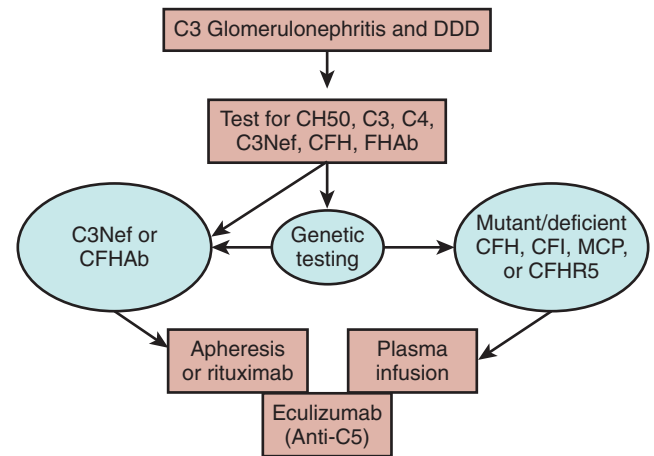


Fig. 31.21 A framework for the diagnosis and management of C3 glomerulopathies, C3 glomerulonephritis, and dense deposit disease (DDD). Ab, Antibody; CF, complement factor; C3NeF, C3 nephritic factor.

transplants for DDD are common, 80% or higher,^{594,667–669} especially in the presence of C3NeF or CFH mutations.⁶⁷⁰ Prophylactic plasma infusions or simultaneous liver transplantation can be beneficial in the latter cases.^{659,660} Levels of C3 in the serum do not seem to predict recurrences.⁶⁷⁰ Fig. 31.21 depicts a current framework for the diagnosis and management of C3 glomerulopathy, which is likely to be refined even further as additional pathogenetic insight occurs.

ACUTE POSTSTREPTOCOCCAL GLOMERULONEPHRITIS

EPIDEMIOLOGY

Acute poststreptococcal glomerulonephritis (PSGN) is a disease that affects primarily children, with a peak incidence in those between the ages of 2 and 6 years. Children younger than 2 years and adults older than 40 years account for only about 15% of patients with acute PSGN. Interestingly, evidence has pointed to acute PSGN as the cause of Wolfgang Amadeus Mozart's early death at the age of 35.⁶⁷¹ Subclinical microscopic hematuria may be four times more common than overt acute PSGN, as documented in studies of family members of affected patients.^{672–674} Only rarely do PSGN and rheumatic fever occur concomitantly.⁶⁷⁵ Males are more likely than females to have overt nephritis. Acute PSGN may occur as part of an epidemic or sporadic disease. During epidemic infections of streptococci of proven nephrogenicity, the clinical attack rate appears to be about 12%,^{676–678} but has been reported at 33%⁶⁷⁹ or even as high as 38% in certain affected families.⁶⁷⁴

Differences in incidence rates among different families argue for the existence of host genetic susceptibility factors affecting the propensity for overt nephritis.⁶⁸⁰ An association was found between PSGN and HLA-DRW4,⁶⁸⁰ HLA-DPA*02:022, and DPB1*05:01⁶⁸¹ and, more recently, DRB1*03:011.⁶⁸²

The rate of acute PSGN after sporadic infections with group A streptococci of potentially nephritogenic types is quite variable,^{683,684} which again points to an effect of ill-defined host factors. A minority of streptococcal infections lead to nephritic syndrome, which argues for the presence of certain nephritogenic characteristics of the offending agent. In the 1950s, Rammelkamp and colleagues^{683,684}

identified certain strains of streptococci within Lancefield group A, in particular type XII, that are capable of leading to an acute glomerulonephritis. Other nephritogenic serotypes include M types 1, 2, 3, 4, 18, 25, 31, 49, 52, 55, 56, 57, 59, 60 and 61. There are differences among these serotypes in their propensity to be associated with nephritis, depending on the site of infection. Certain strains, such as types 2, 49, 55, 57, and 60, are usually associated with nephritis after pyoderma,^{685,686} whereas M type 49 can lead to nephritis after pharyngitis or pyoderma.⁶⁸⁶ In addition to occurring after infection with group A beta-hemolytic streptococci, acute PSGN has also been described after infection with group C streptococci and possibly group G streptococci.^{687,688}

Acute PSGN has been on the decline in developed countries, but it continues to occur in developing communities.^{689,690} Epidemic PSGN is frequently associated with skin infections, whereas pharyngitides are associated with sporadic PSGN in developed countries. Overt glomerulonephritis is found in about 10% of children at risk, but when one includes subclinical disease, as evidenced by microscopic hematuria, about 25% of children at risk are found to be affected.^{691,692} In some developing countries, acute PSGN remains the most common form of acute nephritic syndrome among children. The incidence rate appears to follow a cyclic pattern, with outbreaks occurring about every 10 years.⁶⁹³ A review of 11 published population-based studies has estimated the median incidence of PSGN in children to be 24 cases per 100,000 person-years, based on studies that examined populations in less developed countries or included substantial minority populations in more developed countries; the incidence in adults was conservatively estimated to be 2 cases per 100,000 person-years in less developed countries and 0.3 cases per 100,000 person-years in more developed countries.⁶⁹⁴ The authors of this review have estimated that over 470,000 cases of acute PSGN occur annually, leading to approximately 5000 deaths (1% of total cases), 97% of which are in less developed countries.

Interestingly, the epidemiology of PSGN in Florida appears to have changed in the last 2 decades. Compared with the 1960s and 1970s, pharyngitis has replaced impetigo as the predominant underlying infection, a shift has occurred in racial distribution (now predominantly Caucasians are affected) and in seasonal variation, and the severity of disease has decreased.⁶⁹⁵ These changes are thought to reflect a change in the causal agent of impetigo.⁶⁹⁵

PATHOLOGY

Light Microscopy

The pathologic appearance of acute PSGN varies during the course of the disease. The acute histologic change is the influx of neutrophils, which results in diffuse global hypercellularity (Fig. 31.22).⁶⁹⁶⁻⁷⁰⁰ Endocapillary proliferation of mesangial cells and endothelial cells also contributes to the hypercellularity. The hypercellularity often is very marked and results in enlarged consolidated glomeruli. The description “acute diffuse proliferative glomerulonephritis” often is used as a pathologic designation for this stage of acute PSGN. A minority of patients has a crescent formation, which usually affects only a small proportion of glomeruli.⁷⁰¹ Extensive crescent formation is rare.^{702,703} Special stains that have differential reactions with immune deposits may demonstrate

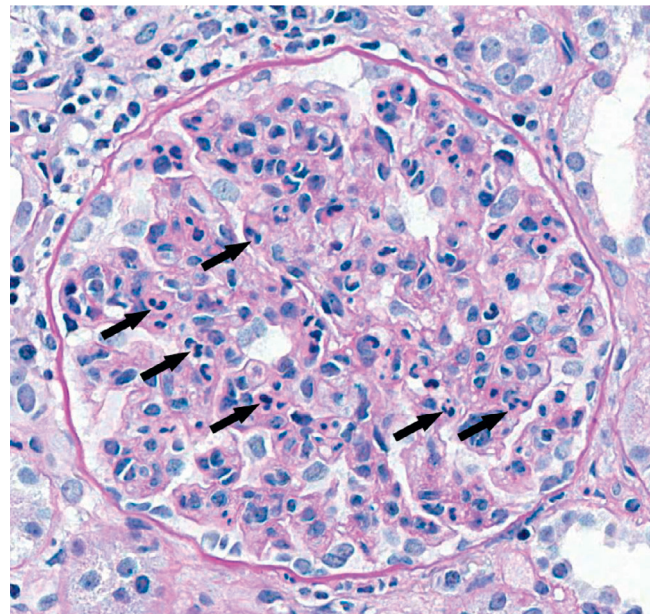


Fig. 31.22 Light micrograph of a glomerulus with acute post-streptococcal glomerulonephritis demonstrating marked influx of neutrophils (arrows). (Periodic acid–Schiff stain, $\times 300$.)

subepithelial deposits. For example, the subepithelial deposits may stain red (fuchsinophilic) with Masson trichrome stain.

Interstitial edema and interstitial infiltration of predominantly mononuclear leukocytes usually are present and occasionally are pronounced, especially with unusually severe disease associated with crescents. Focal tubular epithelial cell simplification (flattening) also may accompany severe disease. Arteries and arterioles typically have no acute changes, although preexisting sclerotic changes may be present in older patients.

During the resolving phase of self-limited PSGN, which usually begins within several weeks of onset, the infiltrating neutrophils disappear, and endothelial hypercellularity resolves, leaving behind only mesangial hypercellularity.^{696,704} This mesangioproliferative stage often is present in patients with PSGN who have had resolution of nephritis but have persistent isolated proteinuria, and it may persist for several months in patients who have complete clinical resolution. There may be focal segmental glomerular scarring as sequelae of particularly injurious inflammation, but this is seldom extensive, except in the rare patient with crescentic PSGN. Ultimately, the pathologic changes of acute PSGN can resolve completely.^{704,705}

Immunofluorescence Microscopy

IF microscopy demonstrates glomerular immune complex deposits in PSGN.^{697,699,700,706} The pattern and composition of deposits change during the course of PSGN. During the acute diffuse proliferative phase of the disease, there is diffuse global coarsely granular capillary wall and mesangial staining, which usually is very intense for C3, and of varying degrees for IgG from intense to absent (Fig. 31.23). Staining for IgM and IgA is less frequent and is usually less intense. In self-limited disease, biopsy should be performed later in the disease course because it is more likely that the staining will be predominantly or exclusively for C3, with little or

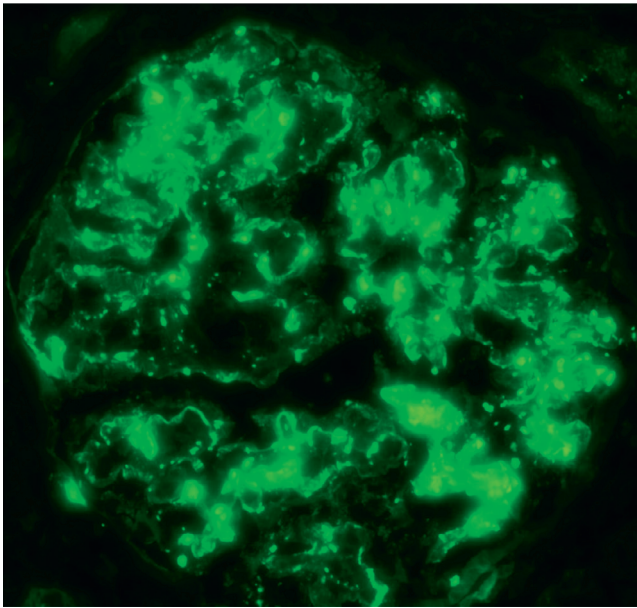


Fig. 31.23 Immunofluorescence micrograph of a glomerular segment from a patient with acute poststreptococcal glomerulonephritis showing coarsely granular capillary wall staining for C3. Compare this to the finely granular capillary wall staining of membranous nephropathy in Fig. 31.9. (Fluorescein isothiocyanate anti-C3 stain, $\times 300$.)

no immunoglobulin staining. Because most patients with uncomplicated new-onset acute PSGN do not undergo kidney biopsy, most biopsy specimens are obtained later in the course, when there is diagnostic uncertainty due to equivocal serologic confirmation or unusually aggressive or persistent clinical features. At this time, IF microscopy staining is usually predominantly for C3. This may reflect termination of nephritogenic immune complex localization in the kidney, with masking of residual complexes by complement. An alternative explanation for the presence of C3 in the absence of immunoglobulin is activation of complement or blockade of complement regulatory mechanisms by factors released by the infection. This could initiate a pathogenic mechanism similar to that in C3 glomerulopathy. The continued presence of intense staining for IgG 1 month or more into the course of what otherwise looks like pathologically typical PSGN is cause for concern that the process will not be self-limited.

Several patterns of immune staining have been described but are of limited prognostic value.^{697,700,707} The garland pattern is characterized by numerous large, closely apposed granular deposits along the capillary walls. Patients with this pattern usually have nephrotic-range proteinuria as a component of their disease. The starry sky pattern has more scattered granular staining, which corresponds somewhat to less severe disease. The mesangial pattern, especially when it is predominantly C3 staining, corresponds to the resolving phase, with a mesangioproliferative light microscopic appearance.

Electron Microscopy

The hallmark ultrastructural feature of PSGN is the subepithelial humplike dense deposits (Figs. 31.24 and 31.25).^{699,704–706,708} However, small subendothelial and mesangial dense deposits can usually be identified with careful observation and theoretically may be more important in the pathogenesis of the

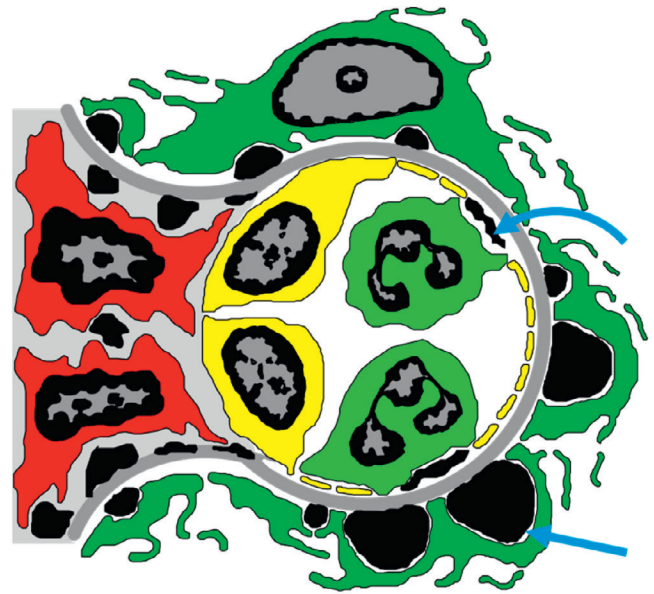


Fig. 31.24 Diagram of the ultrastructural features of acute poststreptococcal glomerulonephritis. Note the subepithelial humplike dense deposits (straight arrow), subendothelial deposits (curved arrow), and mesangial deposits. There is endocapillary hypercellularity caused by neutrophil infiltration and endothelial and mesangial proliferation. (Courtesy JC Jennette.)

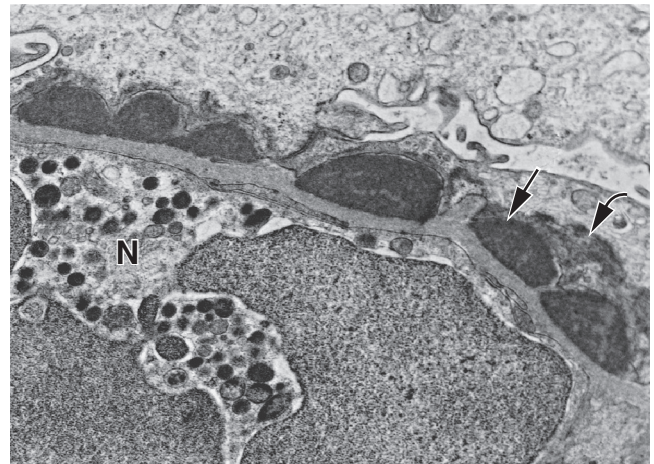


Fig. 31.25 Electron micrograph of a portion of a glomerular capillary from a patient with acute poststreptococcal glomerulonephritis showing subepithelial dense deposits (straight arrow), condensation of cytoskeleton in adjacent podocyte cytoplasm (small curved arrow), and a neutrophil (N) margined against the basement membrane, with no intervening endothelial cytoplasm. ($\times 5000$.)

disease, especially the neutrophilic influx and endocapillary proliferative response, than are the subepithelial humps. The subepithelial humps are covered by effaced epithelial foot processes, which usually contain condensed cytoskeletal filaments (including actin) that form a corona around the immune deposits (Fig. 31.25). Similar humps are observed in C3 glomerulopathy. During the acute phase, capillary lumens often contain margined neutrophils, some of which are in direct contact with GBMs (see Fig. 31.25). Lesser numbers of monocytes and macrophages contribute to the leukocyte influx. Mesangial regions are expanded by increased numbers

of mesangial cells and leukocytes, as well as increased matrix material and varying amounts of electron-dense material.

During the resolution phase, usually 6 to 8 weeks into the course, the subepithelial humps disappear, leaving behind only mesangial and sometimes a few scattered subendothelial and intramembranous dense deposits. The subepithelial deposits first become electron-lucent and then disappear completely. The humps in peripheral capillary loops disappear before the humps in the subepithelial zone adjacent to the perimesangial basement membrane.

PATHOGENESIS

Acute PSGN is the prototype disease of an acute glomerulonephritis associated with an infectious cause. The first description of this link dates back to the early 19th century after scarlet fever epidemics in Florence and Vienna. Richard Bright first described the association in 1836, reporting that scarlet fever was sometimes followed by hematuria and kidney disease.⁷⁰⁹ In 1907, Schick described an asymptomatic interval of 12 days to 7 weeks between the onset of streptococcal infection and the onset of nephritis.⁷¹⁰ In the early 1950s, Rammelkamp and colleagues further defined the association of PSGN with specific serotypes of streptococci.^{584,711}

Despite the early recognition of an association between streptococcal infection and acute glomerulonephritis, the pathogenic mechanism of disease remains incompletely understood. Conceptually, either acute PSGN could be secondary to a direct pathogenic effect of a streptococcal protein (e.g., a protein that activates or blocks regulation of the alternative complement pathway), or the streptococcal product could induce an immune complex-mediated injury. This could occur by a number of different mechanisms: (1) by introducing an antigen into the glomerulus (planted antigen); (2) by the deposition of circulating immune complexes; (3) by alteration of a normal renal antigen that causes it to become a self-antigen; or (4) by induction of an autoimmune response to a self-antigen by way of antigenic mimicry. It is conceivable that more than one streptococcal antigen may be involved in the pathogenesis of acute PSGN, and more than one pathogenic mechanism may be at play simultaneously.

Several streptococcal proteins have been implicated in the pathogenesis of acute PSGN.⁷¹² M protein molecules protruding from the surface of group A streptococci contain epitopes that cross-react with glomerular antigens. Shared sequences of M protein types 5, 6, and 19 have been shown to elicit antibodies that react with several myocardial and skeletal muscle proteins.⁷¹³ Conversely, monoclonal antibodies raised against human kidney cortex have been shown to cross-react with types 6 and 12 M proteins, which provides evidence that certain M proteins may share antigenic determinants between all glomeruli.⁷¹⁴ The glomerular cross reactivity of the amino-terminal region of type 1 M protein was further localized to a tetrapeptide sequence at position 23-26.⁷¹⁵ Antibodies raised against the amino-terminus of type 1 M protein was shown to cross-react with the cytoskeletal protein of glomerular mesangial cells—namely, the filament protein vimentin.⁷¹³ Two antigens have been found within the glomerular deposits in kidney biopsy specimens from patients with PSGN and have been reported to induce an antibody response characteristic for nephritogenic streptococcal infections—streptococcal proteinase exotoxin B (SPEB, zymogen)⁷¹⁶ and the glycolytic enzyme glyceraldehyde 3-phosphate dehydrogenase (GAPDH),

which shares exact homology with the nephritis-associated plasmin receptor (NaPlr) that binds plasminogen.⁷¹⁷ In a study that tested antigen deposition in 17 kidney biopsy specimens and circulating antibodies in sera from 53 patients, responses to SPEB were more consistently found than deposits and antibody responses to GAPDH.⁷¹⁸

Currently, the spectrum of infectious agents associated with postinfectious or periinfectious glomerulonephritis includes many more bacterial pathogens than streptococci. Other agents include staphylococci, gram-negative rods, and intracellular bacteria.^{719,720} Likewise, the population at risk for periinfectious glomerulonephritis has changed to include alcoholic individuals, intravenous drug users, and patients with ventricular atrial shunts and on immunosuppressive therapy.⁷²⁰ However, PSGN remains one of the most extensively studied and documented infection-associated glomerulonephritides.

CLINICAL FEATURES AND NATURAL HISTORY

Classically, the syndrome of acute PSGN presents abruptly with hematuria, proteinuria, hypertension, and azotemia. This syndrome can show a wide spectrum of severity, from asymptomatic disease to oliguric AKI.⁷²¹ A latent period occurs from the onset of pharyngitis to the onset of nephritis. In postpharyngitic cases, the latent period averages 10 days, with a range of 7 to 21 days. The latent period may be longer after a skin infection (from 14–21 days), although this period is harder to define after impetigo.⁷²² The latency period can exceed 3 weeks.⁷²³ Short latency periods less than 1 week are suggestive of a so-called synpharyngitic syndrome, typically corresponding to exacerbation of an underlying IgA nephropathy.

The hematuria is microscopic in more than two-thirds of cases but may be macroscopic on occasion. Patients commonly report gross hematuria and transient oliguria. Anuria is infrequent, however and, if persistent, may indicate the development of crescentic glomerulonephritis.

Mild to moderate hypertension occurs in more than 75% of patients. It is most evident at the onset of nephritis and typically subsides promptly after diuresis.⁶⁷⁵ Antihypertensive treatment is necessary in only about 50% of patients. Signs and symptoms of congestive heart failure may occur and may dominate the clinical picture. These include jugular venous distention, the presence of an S3 gallop, dyspnea, and signs of pulmonary congestion.^{723–726} Frank heart failure may be a complication in as many of 40% of older patients with PSGN.

Edema is the presenting symptom in two-thirds of patients and is present in as many as 90% of cases.⁶⁷² The presence of edema is based on primary renal sodium and fluid retention. The edema typically appears in the face and upper extremities. Ascites and anasarca may occur in children.

Encephalopathy presenting as confusion, headache, somnolence, or even convulsion is not common and may affect children more frequently than adults. Encephalopathy is not always attributable to severe hypertension but may be the result of central nervous system vasculitis instead.^{723,725–727}

The clinical manifestations of acute PSGN typically resolve in 1 to 2 weeks as the edema and hypertension disappear after diuresis, and the patient remains typically remains asymptomatic. Both the hematuria and proteinuria may persist for several months but are usually resolved within 1 year. However, proteinuria may persist in patients who initially had nephrotic syndrome.⁶⁷² The long-term persistence of

proteinuria, and especially albuminuria, may be an indication of the persistence of proliferative glomerulonephritis.⁶⁷⁸

The differential diagnosis of acute PSGN includes the following: (1) IgA nephropathy⁷²⁸ and IgA vasculitis (formerly Henoch-Schönlein purpura), especially when the acute nephritic syndrome is associated with gross or rusty hematuria; (2) MPGN and C3 glomerulopathy; and (3) acute crescentic glomerulonephritis—rapidly progressive glomerulonephritis: immune complex-mediated, anti-GBM-mediated, or pauci-immune). The occurrence of acute nephritis in the setting of persistent fever should raise the suspicion of a periinfectious glomerulonephritis, especially with persistence of an infection such as an occult abscess or infective endocarditis.

Although rheumatic fever and PSGN rarely occur together, their concurrence has been described.⁷²⁹

LABORATORY FINDINGS

Hematuria, microscopic or gross, is nearly always present in acute PSGN. There are, however, rare cases of documented acute PSGN with no associated hematuria.^{674,730} Microscopic examination of urine typically reveals the presence of dysmorphic red blood cells⁷³¹ or red blood cell casts. Other findings on microscopy are leukocytes, renal tubule epithelial cells, and hyaline and granular casts.⁶⁷⁵ When the hematuria is macroscopic, the urine typically has a rusty or tea color.

Proteinuria is nearly always present but is typically in the subnephrotic range. In 50% of patients, it may be less than 500 mg of protein/day.^{732,733} Nephrotic-range proteinuria may occur in as many as 20% of patients and is more frequent in adults than in children.⁶⁷² The excreted proteins may include large amounts of fibrin degradation products and fibrinopeptides.^{730,734}

A pronounced decline in urine GFR is common in older patients with acute PSGN, affecting nearly 60% of patients 55 years of age and older.⁷²⁵ This profound decrease in GFR is uncommon in patients from childhood to middle age. Indeed, because of the accompanying fluid retention and increase in circulatory volumes, a mild decrease in GFR may not be accompanied by an increase in the serum creatinine concentration above laboratory limits of normal. Renal plasma flow, tubular reabsorptive capacity, and concentrating ability are typically not affected. On the other hand, urinary sodium excretion and calcium excretion are greatly reduced.⁷³⁵

A transient hyporeninemic hypoaldosteronism may lead to mild to moderate hyperkalemia. This may be exacerbated by a concomitant decrease in GFR and reduced distal delivery of solute. This type 4 renal tubular acidosis may resolve with the resolution of nephritis in the event of diuresis but may be persistent beyond that point in some patients.⁷³⁶ The suppressed plasma renin activity may be a consequence of the volume expansion present in those patients.⁷³⁷

Cultures of throat or skin samples frequently reveal group A streptococci.^{675,738} The sensitivity and specificity of these tests are likely affected by the method of obtaining a culture specimen and the test used.⁷³⁹ Such cultures may be less satisfactory than serologic studies to evaluate for the presence of recent streptococcal infection in patients suspected of having PSGN.⁷⁴⁰ The antibodies most commonly studied for the detection of a recent streptococcal infection are antistreptolysin O, antistreptokinase, antihyaluronidase, antideoxyribonuclease B, and anti-nicotinamide adenine dinucleotidase.⁷⁴¹ Of these, the most commonly used is the

antistreptolysin O test. An elevated antistreptolysin O titer above 200 units may be found in 90% of patients with pharyngeal infection.⁶⁷⁵ In the diagnosis of an acute PSGN, however, a rise in titer is more specific than the absolute level of the titer. The latter is likely affected by the geographic and socioeconomic prevalence of pharyngeal infections with group A streptococci. Increased antistreptolysin O titers are present in about two-thirds of patients with upper respiratory tract infection, but in only about one-third of patients following streptococcal impetigo.⁶⁷² Serial antistreptolysin O titer determinations with a twofold or greater rise in titer are highly indicative of a recent infection.^{672,675}

The streptozyme test combines several antistreptococcal antibody assays and may be a useful screening test.⁷⁴⁰ Because certain strains of type 12 group A streptococci do not produce streptolysin S or O, and in patients in whom impetigo-associated PSGN is suspected, testing for antideoxyribonuclease B and antihyaluronidase is a useful procedure.⁶⁸⁵ Antibodies to other streptococcal cell wall glycoproteins may also increase, including those for endostreptosin.^{675,742–745} On occasion, autoantibodies to collagen and laminin may be detected.^{675,746} Cultures of throat or skin specimens may yield positive results in as few as 25% of patients.

The serial measurement of complement component levels is important in the diagnosis of PSGN. Early in the acute phase, the levels of hemolytic complement activity (CH-50 and C3) are reduced. These levels return to normal, usually within 8 weeks.^{675,723,747–752} The reduction in serum C3 levels is especially marked in patients with C3NeF, which is capable of cleaving native C3.^{591–593} The finding of low properdin and C3 levels, and concomitant normal to modestly reduced levels of C1q, C2, and C4,^{747,748,753} all point to the importance of the activation of the alternate pathway of the complement cascade.⁷⁴⁷ Immunohistochemical analysis of mannose-binding protein and mannose-binding protein-associated serine proteinase 1 suggests that the lectin pathway of complement activation is engaged in about one-third of patients.⁷⁵⁴ There is also some evidence for activation via the classic pathway.⁷⁵⁵ Another complement level abnormality is a mild depression of C5 levels, whereas levels of C6 and C7 are usually normal.^{591,675,753} The plasma level of soluble terminal complement components (C5b-9) rises acutely and then falls to normal.⁷⁴⁸ Because complement levels typically return to normal within 8 weeks, the persistent depression of C3 levels may be indicative of another diagnosis, such as MPGN, endocarditis, occult sepsis, SLE, atheromatous emboli, or congenital chronic complement deficiency.⁷⁴⁷

Circulating cryoglobulins^{756,757} as well as circulating immune complexes,^{758–761} may be detected in some patients with PSGN. The pathophysiologic importance of these circulating immune complexes for the development of acute nephritis is unclear.^{760–762}

Abnormalities in blood coagulation systems may be detected in acute PSGN; thus, thrombocytopenia may be seen.⁷⁶³ Elevated levels of fibrinogen, factor VIII, plasmin activity, and circulating high-molecular-weight fibrinogen complexes may be seen and correlate with disease activity and an unfavorable prognosis.^{764–768}

Although complement studies have suggested that the alternative pathways are primarily involved in acute PSGN, there is some evidence also for activation via the classic pathway.⁷⁵⁵

TREATMENT

Treatment of acute PSGN is largely supportive care. Children almost invariably recover from the initial episode.^{769–771} Of concern to clinicians are patients who have AKI at presentation. An initial episode of AKI is not necessarily associated with a bad prognosis.⁷²¹ In a study of 20 adult patients with diffuse proliferative glomerulonephritis, 11 had AKI and 9 had normal or mild renal insufficiency. There were no differences between these groups in clinical, immunologic, or histological features. After 18 months of follow-up, outcomes were similar in the two groups. Thus, there is little evidence to suggest the need for any form of immunosuppressive therapy. Because of the profound salt and water retention observed in these patients—and, in some, pulmonary congestion—it is important to use loop diuretics such as furosemide to avoid volume expansion and hypertension. When volume expansion does occur, antihypertensive agents are frequently useful to ameliorate the hypertension. Interestingly, plasma renin levels are reduced; however, captopril has been shown to lower blood pressure and improve GFR in patients with PSGN.⁷⁷²

Some patients with substantial volume expansion and marked pulmonary congestion show no response to diuretic therapy. In those individuals, dialytic support is appropriate, either hemodialysis or continuous venovenous hemofiltration in adults or peritoneal dialysis in children. Some patients develop substantial hyperkalemia. In those patients, treatment with exchange resins or dialysis may be useful. Importantly, so-called potassium-sparing agents, including triamterene, spironolactone, and amiloride, should not be used in this disease state. Usually, patients undergo a spontaneous diuresis within 7 to 10 days after the onset of their illness and no longer require supportive care.^{721,773} There is no evidence to date that early treatment of streptococcal disease, either pharyngitic or cellulitic, will alter the risk of PSGN. It has long been speculated that treatment with penicillin can control the spread of outbreaks of epidemic PSGN. In studies from Aboriginal communities in Australia, the use of benzathine penicillin prevented new cases of PSGN, especially in children with skin sores and household contact with affected cases.⁷⁷⁴

The long-term prognosis of patients with PSGN is not as benign as previously considered. Widespread crescentic glomerulonephritis results in an increased number of obsolescent glomeruli associated with tubulointerstitial disease that results in the progressive reduction of the renal mass over time.⁷⁷⁵ A proportion of patients with streptococcal glomerulonephritis develop hypertension, proteinuria and renal insufficiency from 10 to 40 years after the illness.^{775–777} Nonetheless, it is most common that the long-term disease process is marked only by mild hypertension.

In some patients, there is evidence to suggest that the original diagnosis of PSGN may have been in error. This is especially true for individuals in whom a kidney biopsy was never performed. For example, a patient who has an upper respiratory tract infection and then develops a glomerulonephritis may be considered to have PSGN when, in fact, the patient has another proliferative form of glomerulonephritis. In these patients, lack of resolution of the kidney disease should prompt a kidney biopsy to elucidate the underlying cause of the glomerular injury.

IMMUNOGLOBULIN A NEPHROPATHY

EPIDEMIOLOGY

IgA nephropathy remains one of, if not the, most common forms of glomerulonephritis, especially in developed countries with a low prevalence of infectious diseases.⁷⁷⁸ Initially described in the late 1960s by Berger and Hinglais,^{779,780} the disorder is characterized by deposition predominantly of IgA (and, to a lesser extent, of other immunoglobulins) in the mesangium, with mesangial proliferation and with clinical features that span the spectrum from asymptomatic hematuria to rapidly progressive glomerulonephritis. Although it was previously considered a benign disease, it is now clear that up to 40% of patients may progress to ESKD. Moreover, it has become recognized that in addition to a primary form of the disorder, IgA nephropathy is also associated with and may be secondary to a variety of disease processes (Box 31.8).

IgA nephropathy occurs in individuals of all ages, but it is still most common in the 2nd and 3rd decades of life and is much more common in males than females (Table 31.7). IgA nephropathy is uncommon in children younger than 10 years of age. In fact, 80% of patients are between the ages of 16 and 35 years at the time of kidney biopsy.^{781–786} The male to female ratio has been described as anywhere from 2:1 to 6:1.^{781–786}

The distribution of IgA nephropathy varies in different geographic regions throughout the world.⁷⁸⁷ It is the most common form of primary glomerular disease in Asia,

Box 31.8 Classification of Immunoglobulin A Nephropathy

Primary immunoglobulin A (IgA) nephropathy
 Secondary IgA nephropathy: Associated disorders
 IgA vasculitis (formerly Henoch-Schönlein purpura)
 Human immunodeficiency virus infection
 Toxoplasmosis
 Seronegative spondyloarthropathy
 Celiac disease
 Dermatitis herpetiformis
 Crohn disease
 Liver disease
 Alcoholic cirrhosis
 Ankylosing spondylitis
 Reiter syndrome
 Neoplasia
 Mycosis fungoides
 Lung CA
 Mucin-secreting CA
 Cyclic neutropenia
 Immunothrombocytopenia
 Gluten-sensitive enteropathy
 Scleritis
 Sicca syndrome
 Mastitis
 Pulmonary hemosiderosis
 Berger disease
 Leprosy
 Familial IgA nephropathy

CA, Cytoplasmic antibody.

Modified from references 826, 850, 862, 979, 980, and 1482 to 1509.

Table 31.7 Diseases That Cause Glomerulonephritis^a

Glomerular Lesion	N	Male:Female Ratio	Caucasian:African American Ratio
IgA nephropathy	693	2.0:1.0	14.0:1.0
MPGN I	248	1.2:1.0	3.3:1.0
Anti-GBM	82	1.1:1.0	7.9:1.0
ANCA-GN	257	1.0:1.0	6.7:1.0
Fibrillary glomerulonephritis	76	1.0:1.2	14.3:1.0

^aBased on the analysis of 9605 native kidney biopsy specimens evaluated at the University of North Carolina Nephropathology Laboratory. This laboratory evaluates kidney biopsy specimens from a base population of approximately 10 million throughout the southeastern United States and centered in North Carolina. The expected Caucasian:African American ratio in this kidney biopsy population is approximately 3:2.

ANCA-GN, Antineutrophil cytoplasmic antibody- glomerulonephritis; Anti-GBM, anti- glomerular basement membrane; MPGN I, membranoproliferative glomerulonephritis type 1.

accounting for up to 30% to 40% of all biopsies performed for the diagnosis of glomerular disease, and it accounts for 20% of all biopsies in Europe and 10% of all biopsies in North America.⁷⁸⁷ This wide variation in incidence is partly attributable to the differing indications for kidney biopsy in Asia compared with those in North America. In Asia, urinalyses are performed routinely in school-age children. Those with asymptomatic hematuria typically undergo biopsy, which may lead to an increased number of diagnoses of IgA nephropathy.

Genetic factors may also be important in the geographic differences. A Japanese study of biopsy specimens obtained from kidney donors immediately before transplantation showed that 16% of donors had covert mesangial deposition of IgA.⁷⁸⁸ IgA nephropathy has been reported to be rare in African-Americans,^{789,790} although population-based incidence rates of newly diagnosed IgA nephropathy have been found to be similar in African American and Caucasian populations.⁷⁹¹ IgA nephropathy is quite common in Native Americans of the Zuni and Navajo tribes.⁷⁹² The prevalence of IgA in the general population has been estimated to be between 25 and 50 cases/100,000^{787,793} although, notably, almost 5% of all patients undergoing kidney biopsy have at least some IgA deposits in their glomeruli.⁷⁹⁴ Population studies in Germany and in France calculated an incidence of 2 cases/10,000,^{795–798} but autopsy studies performed in Singapore⁷⁹⁹ have suggested that 2.0% to 4.8% of the population had IgA deposition in their glomeruli.

GENETICS

IgA nephropathy is a histologically based diagnosis. It is unlikely to be related to a single genetic locus, but is probably due to the interactions of multiple susceptibility and progression genes in combination with environmental factors.⁸⁰⁰ A number of studies have suggested that there are genes that render an individual susceptible to IgA nephropathy and genes that portend a more rapid progression of IgA nephropathy. Polymorphisms in a number of genes, including those coding for ACE, angiotensin, angiotensin II receptor, T cell receptor, IL-1 and IL-6, IL receptor antagonist, TGF, mannose-binding lectin, uteroglobin, nitric oxide synthase, and TNF, as well as major histocompatibility loci, have been evaluated as possibly affecting both the susceptibility to and progression of disease.^{778,801–807} A number of studies have examined the

role of ACE gene in IgA nephropathy, with or without progressive disease. The D allele of the ACE gene may be associated with susceptibility to IgA nephropathy in Asians but not in Caucasians.⁸⁰⁸ The Polymorphism Research to Distinguish Genetic Factors Contributing to Progression of IgA Nephropathy (PREDICT-IgA nephropathy) study, which investigated associations between the progression of IgA nephropathy and 100 atherosclerotic disease-related gene polymorphisms using a retrospective candidate gene approach, has found significant associations between polymorphisms in the glycoprotein Ia and intracellular adhesion molecule-1 genes and progression of disease.⁸⁰⁹ In this study, the association between the ACE I/D polymorphism and progression of disease was not found to be significant after adjustment for multiple comparisons.

Familial IgA nephropathy has been reported in multiple ethnic groups worldwide, including Africa and Central America. Some studies have suggested that 4% to 14% of patients with IgA nephropathy may have a family history of kidney disease,^{795,810,811} and systematic screening of asymptomatic first-degree relatives has detected hematuria in more than 25% of them.⁸¹² Disease findings in most pedigrees are consistent with autosomal dominant transmission with incomplete penetrance.⁸¹⁰ However, in some families, IgA nephropathy may aggregate with other glomerular diseases.^{810,813}

Linkage studies have suggested an association of IgA nephropathy with genes at several loci.⁸¹⁰ Based on a genome-wide linkage study of 30 kindreds with IgA nephropathy, a locus on chromosome 6q22-23 was identified, with a logarithm of the odds ratio (LOD) score of 5.6.⁸¹⁴ This was named IGAN1 (a LOD score ≥ 3 signifies odds $\geq 1000:1$ in favor of linkage).⁸¹⁰

Genomewide association studies have successfully identified candidate genes and single-nucleotide polymorphisms (SNPs) that may correlate with susceptibility to or protection against IgA nephropathy. A genomewide association study (GWAS) was first performed using DNA in the UK Glomerulonephritis DNA Bank from individuals with IgA nephropathy⁸¹⁵ and then later in Chinese cohorts.^{816,817} These studies led to the identification of seven susceptibility loci—three loci within the MHC region on chromosome 6p21, DEFA locus on chromosome 8p23, TNFSF13 locus on chromosome 17p23,

HORMAD2 locus on chromosome 22q12 and CFH/CFHR locus on chromosome 1q32—which implicate defects in innate immunity, adaptive immunity, and the alternative complement pathway.^{818,819} A larger GWAS of 20,612 individuals comprising discovery and follow-up cohorts of European and East Asian descent has revealed six new associations, four in ITGAM-ITGAX, VAV3, and CARD9 and two new signals at the HLA-DQB1 and DEFA loci. The biologic nature of the risk alleles has strongly suggested a role for host–intestinal pathogen interactions, and almost all risk alleles were associated with inflammatory bowel disease risk and/or intestinal epithelial barrier maintenance and response to mucosal pathogens.⁸²⁰ Notably, the major histocompatibility complex has been implicated in every GWAS to date.

The prevailing hypothesis regarding the pathogenesis of IgA nephropathy focuses on defects in protein glycosylation, particularly in B cells secreting IgA1. Studies measuring serum levels of galactose-deficient IgA1 in patients with IgA nephropathy, their relatives, and unrelated controls have found a higher level of aberrantly glycosylated IgA1 in patients with familial or sporadic IgA nephropathy and in their at-risk relatives than in unrelated individuals or control subjects.^{821,822} This finding suggests that abnormal IgA1 glycosylation is an inherited rather than an acquired trait. Polymorphisms of the genes for the enzymes responsible for glycosylation of IgA1 may thus be associated with increased susceptibility to IgA nephropathy. Such genes include the core 1 β -galactosyltransferase gene (*C1GALT1*),^{823,824} and the molecular chaperone COSMC gene (*C1GALT1C1*), although the findings are inconsistent across studies. Interestingly, a recent GWAS for serum levels of galactose-deficient IgA1 in 2633 subjects of European and East Asian descent has identified two novel loci in *C1GALT1* and *C1GALT1C1* that encode molecular partners required for the enzymatic O-glycosylation of IgA1.⁸²⁵

PATHOLOGY

Immunofluorescence Microscopy

IgA nephropathy can be definitively diagnosed only by the immunohistologic demonstration of glomerular immune deposits that stain dominantly or codominantly for IgA compared with IgG and IgM (Fig. 31.26).^{826–829} The staining is usually exclusively or predominantly mesangial, although a minority of specimens, especially from patients with severe disease, will have substantial capillary wall staining. By definition, 100% of IgA nephropathy specimens stain for IgA. On a scale of 0 to 4+, the mean intensity of IgA staining is approximately 3+.⁸²⁸ IgM staining is observed in 84% of specimens with a mean intensity (when present) of only approximately 1+. IgG staining is observed in 62% of specimens, also with a mean intensity (when present) of approximately 1+. Early studies of IgA nephropathy described more frequent and more intense IgG staining than is seen today, but this probably was caused by the use of less specific antibodies that cross-react between IgA and IgG. Almost all IgA nephropathy specimens have substantial staining for C3. In contrast, staining for C1q is rare and weak when present. If there is intense staining in a specimen that shows substantial IgA and IgG, the possibility of lupus nephritis rather than IgA nephropathy should be considered.⁸²⁸ An additional relatively distinctive feature of IgA nephropathy is that unlike

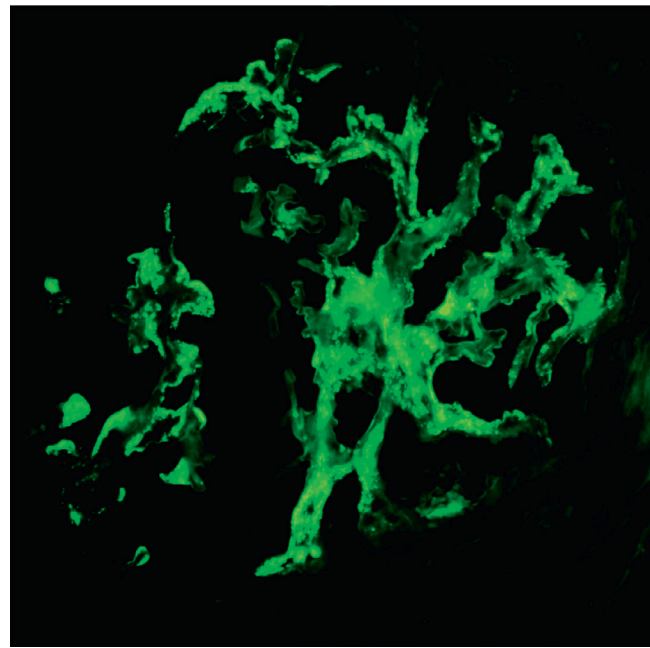


Fig. 31.26 Immunofluorescence micrograph of a glomerulus with features of immunoglobulin A (IgA) nephropathy showing intense mesangial staining for IgA. (Fluorescein isothiocyanate anti-IgA stain, $\times 300$.)

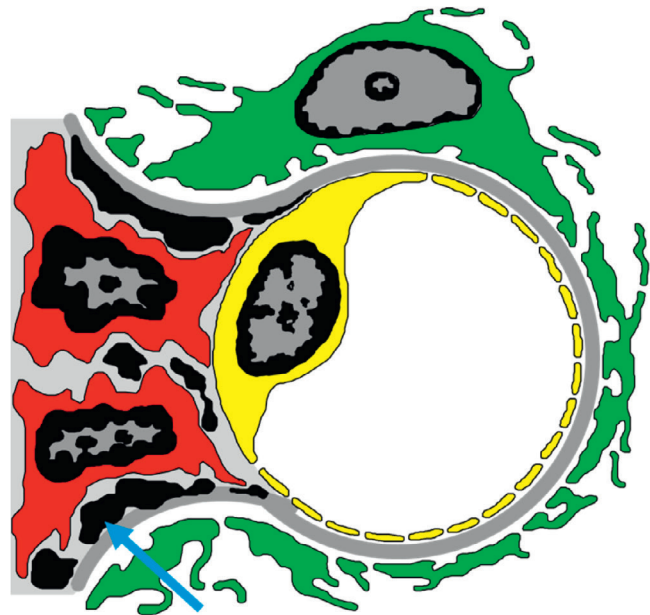


Fig. 31.27 Diagram depicting the ultrastructural features of immunoglobulin A nephropathy. Note the mesangial dense deposits (blue arrow) and mesangial hypercellularity. (Courtesy JC Jennette.)

any other glomerular immune complex disease, the immune deposits usually have more intense staining for gamma (λ) light chains than kappa (κ) light chains.^{826,828}

Electron Microscopy

The ubiquitous ultrastructural finding is mesangial electron-dense deposits that correspond to the immune deposits seen by immunohistologic analysis (Figs. 31.27 and 31.28).⁸²⁷ The mesangial deposits often are immediately beneath the

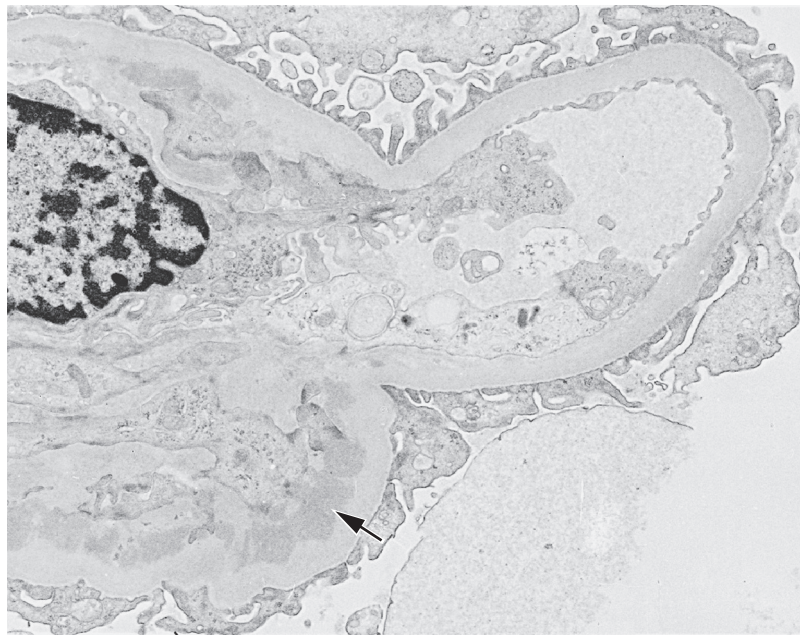


Fig. 31.28 Electron micrograph of a capillary and adjacent mesangium from a patient with immunoglobulin A nephropathy showing mesangial dense deposits immediately beneath the perimesangial basement membrane (arrow). ($\times 7000$.)

perimesangial basement membrane. They are accompanied by varying degrees of mesangial matrix expansion and hypercellularity. Most specimens do not have capillary wall deposits, but a minority, especially from patients with more severe disease, show scattered subendothelial dense deposits or subepithelial dense deposits or both. The extent of endocapillary proliferation and leukocyte infiltration parallel the pattern of injury observed by light microscopy. Epithelial foot process effacement is observed in patients with substantial proteinuria.

Light Microscopy

IgA nephropathy can cause any of the light microscopic phenotypes of proliferative glomerulonephritis (Fig. 31.29) or may cause no discernible histologic changes.^{827–834} As depicted in Fig. 31.30, this spectrum of glomerular inflammatory responses is shared by a variety of glomerulonephritides that have different causes but that induce similar or identical light microscopic alterations in glomeruli. Fig. 31.30 also depicts the most frequent clinical manifestations at the time of biopsy of the different histologic phenotypes of glomerulonephritis, all of which can be caused by IgA nephropathy. At the time of biopsy, IgA nephropathy usually manifests as a focal or diffuse mesangioproliferative or proliferative glomerulonephritis, although specimens from a few patients will have no lesion by light microscopy, those from a few will show aggressive disease with crescents, and occasional specimens will already demonstrate chronic sclerosing disease.

Different criteria for performing a kidney biopsy result in different frequencies of the various phenotypes of IgA nephropathy among distinct populations of patients. Of 668 consecutive native kidney IgA nephropathy specimens diagnosed in the UNC Nephropathology Laboratory, 4% showed no lesion by light microscopy, 13% had exclusively mesangioproliferative glomerulonephritis, 37% had focal proliferative glomerulonephritis (25% of these had <50%

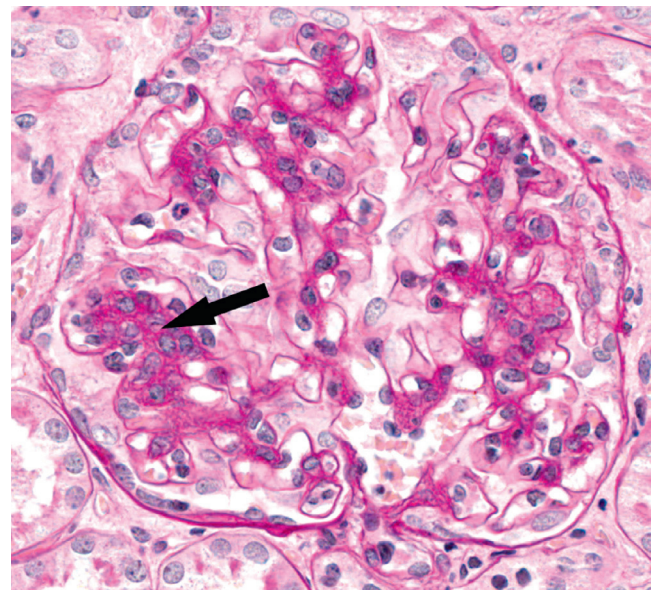


Fig. 31.29 Light micrograph of a glomerulus with immunoglobulin A (IgA) nephropathy showing segmental mesangial matrix expansion and hypercellularity (arrow). (Periodic acid–Schiff stain, $\times 300$.)

crescents), 28% had diffuse proliferative glomerulonephritis (45% of these had <50% crescents), 4% had crescentic glomerulonephritis ($\geq 50\%$ crescents), 6% had focal sclerosing glomerulonephritis without residual proliferative activity, 6% had diffuse chronic sclerosing glomerulonephritis, and 2% had lesions that did not fall into any of these categories.

The mildest light microscopic expression of IgA nephropathy, other than no discernible lesion, is focal or diffuse mesangial hypercellularity without more complex endocapillary hypercellularity, such as endothelial proliferation or influx of leukocytes. This is analogous to International

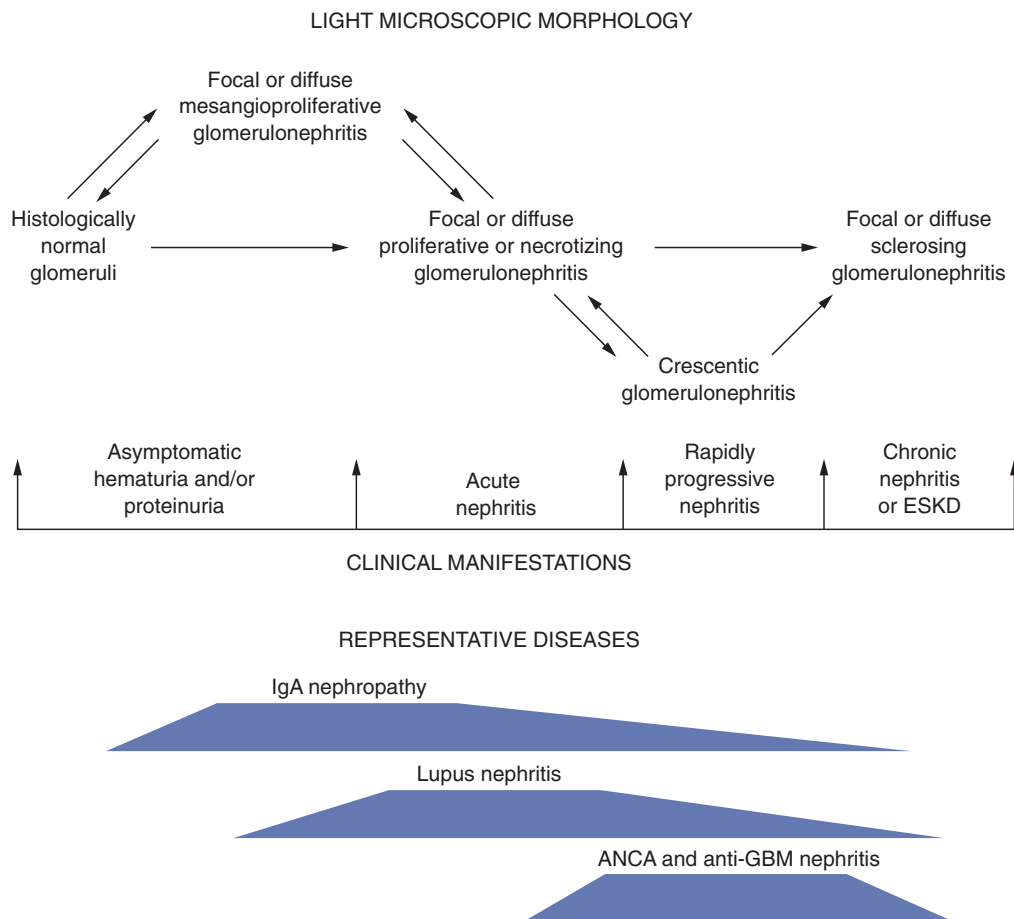


Fig. 31.30 Diagram depicting the continuum of histological changes that can be caused by glomerular inflammation (*top*), the usual clinical syndromes caused by each expression of glomerular injury (*middle*), and the portion of the continuum that most often corresponds to several specific categories of glomerular disease (*bottom*). ANCA, Antineutrophil cytoplasmic autoantibody; ESKD, end-stage kidney disease; GBM, glomerular basement membrane; IgA, immunoglobulin A.

Society of Nephrology/Renal Pathology Service class II lupus nephritis.⁸³⁵ More severe inflammatory injury causes focal (involving <50% of glomeruli) or diffuse proliferative glomerulonephritis as the pathologic expression of IgA nephropathy, which is pathologically analogous to classes III and IV lupus nephritis. The lesions are characterized not only by mesangial hypercellularity, but also some degree of endothelial proliferation or leukocyte infiltration that distorts or obliterates some capillary lumens. Extensive necrosis is rare in IgA nephropathy, although slight focal segmental necrosis with karyorrhexis can occur in severely inflamed glomeruli. With time, destructive glomerular inflammatory lesions progress to sclerotic lesions that may form adhesions to Bowman's capsule. Occasional patients with IgA nephropathy will have focal glomerular sclerosis by light microscopy that is indistinguishable from FSGS until the IF microscopic findings are taken into consideration. Because of the episodic nature of IgA nephropathy, many patients have combinations of focal sclerotic lesions and focal active proliferative lesions. Patients with the most severe IgA nephropathy have crescent formation because of extensive disruption of capillaries.⁸³⁴ Advanced chronic disease is characterized by extensive glomerular sclerosis associated with marked tubular atrophy, interstitial fibrosis, and interstitial infiltration by mononuclear leukocytes.

Whether or not histologic features detected on kidney biopsy can be used to predict the progression of IgA nephropathy has been studied for many years. Nephropathologic findings have previously provided limited prognostic value over and above that of simple clinical parameters such as blood pressure, serum creatinine level, and degree of proteinuria. The Oxford classification of IgA nephropathy study is a seminal investigation that assessed the value of specific pathologic features in predicting the risk of progression of kidney disease in IgA nephropathy.^{836,837} The study population was a multiethnic cohort of patients with IgA nephropathy that included children ($n = 59$) and adults ($n = 206$) who were followed for a mean of 69 months. By an iterative process, pathologic variables selected based on reproducibility, least susceptibility to sampling error, ease of scoring, and independent association with outcome were then correlated through multivariate analysis with three clinical outcomes—the rate of kidney function decline, survival from a 50% decline in kidney function or ESKD, and proteinuria during follow-up.⁸³⁶

Four parameters emerged as independently predictive of clinical outcomes: mesangial hypercellularity, endocapillary hypercellularity, segmental glomerulosclerosis, and tubular atrophy–interstitial fibrosis. Of these predictors, mesangial

hypercellularity was significantly associated with ESKD or a 50% reduction in GFR, segmental sclerosis was associated with the rate of decline in kidney function, and tubular atrophy/interstitial fibrosis was statistically associated with both the rate of decline and ESKD or a 50% decline in kidney function. Endocapillary hypercellularity was not significantly predictive of the rate of decline of kidney function or survival from ESKD or a 50% reduction in kidney function. However, patients with endocapillary (or extracapillary) hypercellularity were more likely to receive immunosuppressive therapy, and the relationship between this pathologic variable and the rate of decline in kidney function may have been influenced by the use of immunosuppression. This latter finding indirectly suggests that patients with this type of lesion are responsive to immunosuppressive therapy.

These conclusions led to a proposal to incorporate the following scoring of the four identified parameters (known as the “Oxford-MEST score”) into the pathology report for IgA nephropathy⁸³⁶:

- Mesangial hypercellularity—score $\leq 0.5 = 0$ or score $>0.5 = 1$
- Endocapillary hypercellularity—absent = 0 or present = 1
- Segmental glomerulosclerosis: absent = 0 or present = 1
- Tubular atrophy–interstitial fibrosis—percentage of cortical area $\leq 25\% = 0$, 26% to 50% = 1, or $>50\% = 2$

This scoring system was shown to be valid for both children and adults,⁸³⁸ but true proof of its utility required validation in separate cohorts, especially in comparison with clinical features such as serum creatinine level and proteinuria.⁸³⁹ Since the development of the Oxford classification of IgA nephropathy, a metaanalysis of 16 subsequent retrospective cohort studies between January 2009 and December 2012, assessing the utility of the scoring system, has found that M, S, T, and crescent (C) lesions (not E lesions) were strongly associated with progression to kidney failure, defined as a doubled serum creatinine level, 50% decline in eGFR, or ESKD.⁸⁴⁰ A large validation study in Europe examined 1147 patients who had the whole spectrum of IgAN.⁸⁴¹ Over a median follow-up of 4.7 years, in patients with an eGFR less than 30 mL/min/1.73 m², M and T lesions independently predicted a poor survival and, in patients with proteinuria less than 0.5 g/day, both M and E lesions were associated with a rise in proteinuria to 1 or 2 g/day or more. The combination of M, S, and T lesions with clinical parameters significantly increased the ability to predict progression in the absence of substantial immunosuppressive therapy. An update from the IgA Nephropathy Classification Working Group in 2016 also found that the crescents are predictive of outcome, thereby changing the MEST score to the MEST-C score (C for crescents).⁸⁴² Additional prospective studies using the Oxford classification are warranted, and the CureGN cohort is poised to address the utility of the MEST-C score prospectively.

PATHOGENESIS

A great deal of progress has been made in our understanding of the pathogenesis of IgA nephropathy.^{778,843,844} The characteristic pathologic finding by IF microscopy of granular deposits of IgA and C3 in the glomerular mesangium, as well as the dermal capillaries in IgA vasculitis (formerly known as

Henoch-Schönlein purpura), suggests that this disease is the result of the deposition of circulating immune complexes leading to the activation of the complement cascade via the alternate pathway. The deposited IgA is predominantly polymeric IgA1.^{845–848} The fact that polymeric IgA1 is usually derived mainly from the mucosal immune system, as well as the association of clinical flare-ups in some cases of IgA nephropathy with syndromes that affect the respiratory tract or gastrointestinal tract, has led to the suggestion that IgA nephropathy is a consequence of defective mucosal immunity.⁸⁴⁹ This concept was supported by the finding in some patients with IgA nephropathy of antibodies to dietary antigens or various infectious agents, both viral and bacterial,^{850–863} and the clinical observation that hematuria increases acutely in some patients at the time of upper respiratory tract or gastrointestinal infections. However, it has now been determined that the elevation in polymeric IgA1 antibody synthesis does not occur in the mucosa, and polymeric IgA levels are increased after systemic immunization with tetanus toxoid.^{846,864,865} In addition, an increase in IgA-secreting B cells was documented in both peripheral blood⁸⁶⁶ and bone marrow⁸⁶⁷ of patients.

Serum levels of IgA do not correlate with disease activity or mesangial deposits; therefore, it is unlikely that the pathogenesis of IgA nephropathy is related to a quantitative increase in serum levels of polymeric IgA1. Rather, it relates to an anomaly in the IgA molecule itself—namely, in its glycosylation, as discussed earlier.⁸⁶⁸ This is best exemplified by patients with IgA-secreting multiple myeloma, among whom only those patients with aberrant IgA glycosylation develop glomerulonephritis.

In humans, the heavy chain of IgA1, but not that of IgA2, contains an 18-amino acid hinge region that is rich in proline, serine, and threonine residues. O-linked monosaccharides or oligosaccharides consisting of N-acetylgalactosamine can be posttranslationally added to these amino acid residues. This N-acetylgalactosamine is usually substituted with a terminal galactose.⁸⁶⁹ Lectin-binding studies and carbohydrate composition analysis have demonstrated that the IgA1 in patients with IgA nephropathy contains less terminal galactose than that of healthy control subjects.^{846,870} The addition of a galactose residue to the glycosyl side chain is blocked by premature sialylation of the N-acetylgalactosamine residues on the hinge region of IgA1. The precise cause of this has not been fully elucidated. Three mechanisms have been postulated—excessive activity of α -2,6-sialyltransferase, decreased activity of β -1,3-galactosyltransferase, and decreased stability of β -1,3-galactosyltransferase due to decreased activity of its chaperone (Cosmc).^{871,872} Whether these abnormalities are acquired or genetically determined remains unclear.^{824,873,874}

IgA glycosylation may also be influenced by acquired abnormalities such as the polarity of the T cell cytokine milieu.⁸⁷⁵

Galactose-deficient IgA1 (Gd-IgA1) results in IgA1 hinge region exposure of N-acetylgalactosamine and neoepitope formation. This neoepitope is the target of IgG autoantibodies, as demonstrated in studies of immortalized B cells from patients with IgA nephropathy,^{872,876} and IgG autoantibodies specific for Gd-IgA1 are found in the circulation of such patients.⁸⁷⁷ These autoantibodies were found to be of highly restricted heterogeneity directed at the unique epitopes present on the abnormally glycosylated IgA1.⁸⁷⁶ The autoantibodies and target Gd-IgA1 form circulating immune

complexes^{878,879} that escape removal by the reticuloendothelial system and deposit in glomeruli via an interaction with a mesangial IgA receptor, possibly the transferrin receptor.^{880–883} Once deposited in glomerular mesangium, these immune complexes provoke mesangial proliferation.⁸⁸⁴ The deposition of IgA in glomeruli may also occur independently of immune complex formation⁸⁸⁵ because abnormally glycosylated IgA1 also leads to an increased binding in the kidney.^{868,879,886} However, aberrantly glycosylated IgA in isolation does not cause mesangial proliferation in tissue culture.⁸⁷⁶ Mesangial Gd-IgA1 molecules induce a variety of inflammatory mediators, including cytokines, chemokines, and growth factors, as well as complement activation through the mannose-binding lectin pathway.^{887–889}

Of particular interest is the clinical relevance of Gd-IgA1 and anti-Gd-IgA1 autoantibodies. One study has demonstrated that serum levels of IgG- and IgA-based antiglycan autoantibodies from 97 patients with IgA nephropathy, compared with 30 patients with non-IgA nephropathy disease and 30 healthy controls, correlate with disease progression and poor prognosis.⁸⁹⁰ A larger subsequent study followed 275 patients with IgA nephropathy for a median of 47 months (range, 12–96 months) and assessed renal survival. Not only was the level of Gd-IgA1 elevated compared with healthy controls, but higher levels were independently associated with a greater risk of renal failure (HR per standard deviation of a natural log-transformed Gd-IgA1, 1.44; 95% CI, 1.11–1.88; $P = .006$).⁸⁹¹

Another component of the pathogenesis of IgA nephropathy pertains to direct cytokine- or chemokine-induced podocyte injury, which is reflected by increased podocyturia. Podocyte damage may be the result of local complement activation and elevated levels of platelet-derived growth factor or TNF- α levels.^{892–894}

It has been hypothesized that a subsequent autoantibody response may be triggered by an environmental cross-reacting antigen and lead to in situ immune complex formation by the interaction of circulating autoantibodies to the “planted” autoantigen. The formation of circulating immune complexes with abnormally glycosylated IgA and circulating IgA receptor molecules could also be involved.⁸⁹⁵ Because IgA1 is normally cleared from the circulation by the liver via the asialoglycoprotein receptor,^{896–898} it is also thought that the defective galactosylation of the hinge region in IgA1 may lead to decreased clearance of IgA1 molecules in patients with IgA nephropathy.^{886,899}

The existence, nature, and role of other autoantibodies in IgA nephropathy has also been under investigation. A number of autoantibodies to various putative autoantigens have been described in IgA nephropathy.⁹⁰⁰ Such autoantigens include a mesangial cell membrane antigen,⁹⁰¹ endothelial cells (human umbilical vein endothelial cells),^{851,900} single-stranded DNA,⁸⁵¹ and cardiolipin.^{851,902} Most of these autoantibodies were found in subsets of patients that rarely exceeded 3% of patients with IgA nephropathy and may sometimes have been the result of high circulating levels of IgA in these patients.⁹⁰² A more recent integrative “anti-biomics” approach found 117 specific autoantibodies, many of which were directed to proteins in the kidney.⁹⁰³ The presence of IgG ANCA has been described to occur in a minority of patients with IgA nephropathy.⁹⁰⁴ In addition, IgA ANCA have been rarely associated with a systemic

vasculitis, IgA vasculitis.^{905–907} In the setting of IgA ANCA, the autoantigen seems to be different from the major ANCA autoantigens—namely, myeloperoxidase (MPO) and proteinase 3 (PR3). Circulating IgA-fibronectin complexes have also been described in the circulation of patients with IgA nephropathy and HIV infection.⁹⁰⁸ These complexes may not be true immune complexes, however, and may be directly related to increased IgA levels in patients with IgA nephropathy.⁹⁰⁹ A special form of IgA-dominant immune complex glomerulonephritis that resembles postinfectious glomerulonephritis occurs secondary to staphylococcal infection, especially but not exclusively in patients with diabetes.⁹¹⁰

Given the lack of efficacy of immunosuppressive treatments focused on autoantibody production, such as plasma exchange and B cell-depleting agents, interest has greatly expanded with regard to the alternative and lectin complement pathways in IgA nephropathy.⁹¹¹ As one example of many, deletion polymorphisms of complement factor H-related protein 1 (FHR-1), an endogenous antagonist of factor H, and FHR-3 are strongly associated with protection against IgA nephropathy, and plasma samples from a large cohort of patients with IgA nephropathy were found to have elevated levels of FHR-1, suggesting a role in disease progression.⁹¹² An open-label study investigating the efficacy of an oral C5a receptor antagonist is now underway in patients with IgA nephropathy on stable RAAS blockade (www.clinicaltrials.gov, NCT02384317).

CLINICAL FEATURES AND NATURAL HISTORY

Approximately 40% to 50% of these patients have macroscopic hematuria at the time of their initial presentation. The episodes tend to occur in close temporal relationship to upper respiratory infection, including tonsillitis and pharyngitis. This synchronous association of pharyngitis and macroscopic hematuria has been given the name *synpharyngitic nephritis*. Much less commonly, episodes of macroscopic hematuria follow infections that involve the urinary tract or gastroenteritis. Macroscopic hematuria may be entirely asymptomatic, but more often is associated with dysuria that may prompt the treating physician to consider bacterial cystitis. Systemic symptoms are frequently found, including nonspecific symptoms such as malaise, fatigue, myalgia, and fever. Some patients have abdominal or flank pain.^{913,914} In a minority of patients (<5%), malignant hypertension may be an associated presenting feature.⁹¹⁵ In the most severe cases (fewer than 10%), acute glomerulonephritis results in acute renal insufficiency and failure.^{916,917} Recovery typically occurs with the resolution of symptoms, even in patients who have been temporarily dialysis-dependent.⁹¹⁷

Macroscopic hematuria due to IgA nephropathy occurs more often in children than in young adults. When it occurs in older individuals, it should raise the possibility of the more common causes of urinary tract bleeding, such as stones or malignancy.

A presentation with asymptomatic microscopic hematuria, with or without proteinuria, occurs in 30% to 40% of patients.⁹¹⁸ Patients with IgA nephropathy come for the evaluation of asymptomatic hematuria, with or without the presence of proteinuria. In addition to glomerulonephritis, these patients may commonly have hypertension. In fact, in Caucasian patients with hypertension and hematuria, IgA nephropathy is the most common form of hematuria.⁹¹⁹ Intermittent macroscopic hematuria occurs in 25% of these

patients. Microscopic hematuria and proteinuria persist between episodes of macroscopic hematuria.

Patients with nephrotic syndrome at presentation may have widespread proliferative glomerulonephritis or coexisting IgA nephropathy and minimal-change glomerulopathies.⁹²⁰ Finally, some patients with IgA nephropathy have reached ESKD at the time of their first presentation. These individuals typically have had asymptomatic microscopic hematuria and proteinuria that remained undetected.⁹¹⁶

In addition to idiopathic IgA nephropathy, there is secondary IgA nephropathy that is the glomerular expression of a systemic disease (see Box 31.8). For example, patients with IgA vasculitis have abdominal pain, arthritis, a vasculitic rash, and a glomerulonephritis that is indistinguishable from that of primary IgA nephropathy. This condition is discussed more fully in Chapter 32.

Although IgA nephropathy was earlier thought to carry a relatively benign prognosis, it is estimated that measured from the time of diagnosis, 1% to 2% of all patients with IgA nephropathy will develop ESKD each year. In a review encompassing 1900 patients derived from 11 separate series, long-term renal survival was estimated to be 78% to 87% at 1 decade after presentation.⁹²¹ Similarly, European studies have suggested that renal insufficiency may occur in 20% to 30% of patients within 2 decades of the original presentation.⁷⁸⁵ In a study of the natural history of IgA nephropathy and “isolated” microscopic hematuria in 135 Chinese children,³¹ spontaneous clinical remission occurred in 12%, whereas 88% had persisting hematuria. Almost 30% developed new-onset proteinuria, and hypertension developed in 32%. Eventually, 20% developed renal insufficiency of varying severity. A poor outcome was associated with persistent hematuria, microalbuminuria, and tubulointerstitial changes on the kidney biopsy specimen. This study clearly demonstrates that careful follow-up is required for all patients given the diagnosis of IgA nephropathy.

Overall, about 25% of patients develop ESKD within 10 to 25 years from diagnosis, depending on the initial severity of disease. Patients with episodes of gross (macroscopic) hematuria generally have a more favorable prognosis than those with persisting microhematuria; however, after an episode of microhematuria associated with AKI, some patients ($\approx 25\%$) may not recover normal kidney function.⁹²² The proliferative forms of IgA nephropathy seem to be associated with better outcomes in children than in adults.⁹²³ It is unclear whether gender affects the prognosis of IgA nephropathy,⁹²⁴ with some studies suggesting that the prognosis may be worse for males. Older age at disease onset may also connote a poor prognosis.^{921,925–929}

Several studies have assessed features that predict a poor prognosis. Sustained hypertension, persistent proteinuria (especially proteinuria of >1 g protein/24 hours), impaired kidney function, and nephrotic syndrome are markers of a poor prognosis.^{811,829,925} Controversy persists with respect to the issue of recurring bouts of macroscopic hematuria.⁹³⁰ It is possible that macroscopic hematuria is an overt manifestation of disease and therefore identifies patients earlier in the course of their disease. Alternatively, macroscopic hematuria may represent an episodic process that results in self-limited inflammation, in contrast to persistent hematuria, which represents ongoing, low-grade inflammation. In general, persistent microscopic hematuria is associated with a poor

prognosis.⁹³¹ In fact, remission of hematuria improves kidney survival and has a favorable effect of outcomes associated with IgA nephropathy.⁹³² It is important to note that AKI associated with macroscopic hematuria does not affect the long-term prognosis. The fact that AKI does occur during gross episodes of hematuria has been confirmed.^{933–935} In these patients, the AKI is most likely associated with acute tubular damage and not with true crescentic disease. After the episodes of gross hematuria, kidney function typically returns to baseline, and the long-term prognosis is good.

The degree of proteinuria is more than likely an additional marker of glomerular disease. Whether this is a consequence of the relationship between proteinuria and the tubular dysfunction found in many forms of glomerular disease or is specific to IgA nephropathy is not clear. In a study by Chen and colleagues,⁹³⁶ mice that had been made proteinuric by various methods had enhanced deposition of administered IgA immune complexes. This suggests that these complexes might be more easily deposited in proteinuric states. More importantly, the amount of protein excretion 1 year after diagnosis was highly predictive of the development of ESKD within 7 years of subsequent follow-up. Individuals with less than 500 mg/dL/day had no renal failure within 7 years, whereas those with over 3 g of protein excretion had approximately a 60% chance of ESKD.⁹³⁷

Many formulas have been advanced to predict the progression of IgA nephropathy in individual patients that yield different results for the same patient. The Toronto formula, based on average mean arterial pressure and proteinuria during the first 2 years of observation, is the best-validated in Caucasian U.S. and European subjects,⁹³⁸ but a large fraction of the variation in progression remains unexplained by these two factors. Risk stratification for progression can be aided by algorithms using a small set of variables—age, gender, family history of chronic kidney disease, reduced estimated GFR at diagnosis, proteinuria, serum albumin and total serum protein levels, hematuria, systolic or diastolic blood pressure, and histologic variables.^{939–942} An absolute renal risk score to predict dialysis or death, developed by analysis of a prospective cohort of 332 patients with biopsy-proven IgA nephropathy followed for an average of 13 years, allowed for significant risk stratification ($P < .0001$) by counting the number of risk factors present at diagnosis—hypertension, proteinuria 1 g or more/day, and severe pathologic lesions.⁹⁴³ One of the most recent risk prediction scores based on a large cohort of Chinese patients with IgA nephropathy, using four baseline variables (lower eGFR, serum albumin, hemoglobin, and higher systolic blood pressure) had a significant independent effect on the risk of ESKD.⁹⁴⁴ Treatment has also favorably influenced the long-term trends of progression in IgA nephropathy,⁹⁴⁵ and a postdiagnosis decline in the level of protein excretion to less than 1.0 g/day is a very reliable surrogate measure of a more favorable long-term prognosis.⁹⁴⁶

A large number of factors other than simple clinical assessment have been examined to predict outcomes. Some have been independently correlated with outcomes, whereas others have failed to demonstrate any added value in prognostication or therapeutic decision making.⁹⁴⁷ Some of the more recently described factors include autophagy in podocytes,⁹⁴⁸ CD19⁺CD5⁺ B cells in kidney biopsy specimens and in blood,⁹⁴⁹ C5b-9 glomerular deposition,⁹⁵⁰ extensive C4d deposition in

the mesangium,⁹⁵¹ tubular $\alpha_3\beta_1$ -integrin expression,⁹⁵⁰ granule membrane protein of 17 kDa (GMP-17)-positive T cells in renal tubules,⁹⁵² glomerular density and size,⁹⁵³ urinary epidermal growth factor-to-monocyte chemotactic peptide 1 ratios,^{954,955} urinary growth arrest and DNA damage-45 γ (GADD45 γ) expression,⁹⁵⁶ analysis of the urinary proteome (kininogen, trypsin inhibitor chain 4, transthyretin),⁹⁵⁷ and the fractional urinary excretion of IgG (in combination with assessment of nephron loss) in crescentic IgA nephropathy.⁹⁵⁸ Likewise, hematuria associated with podocyturia may be associated with a poorer prognosis.⁹⁵⁹ The clinical utility and applicability of these assays in prognostication and treatment decision making remains to be established, and individual patient-level risk prediction still remains limited until risk scoring systems are validated in multiple diverse cohorts.

In addition to these variables, obesity,⁹⁶⁰ elevated nocturnal blood pressure,⁹⁶¹ increased uric acid levels,⁹⁶¹ and elevated levels of C4-binding protein⁹⁶² have been associated with a poorer prognosis. Moderate alcohol consumption is associated with an improved prognosis in IgA nephropathy.⁹⁶³ A mildly elevated serum bilirubin level (>0.6 mg/dL) was associated with an improved long-term outcome in Korean patients with IgA nephropathy,⁹⁶⁴ a finding that has not been confirmed in a non-Asian population. Prolonged, high-level exposure to organic solvents may also confer a worse prognosis to patients with IgA nephropathy.⁹⁶⁵

Women with IgA nephropathy tolerate pregnancy well.⁹⁶⁶ Only women with uncontrolled hypertension, a GFR of less than 70 mL/min, or severe arteriolar or interstitial damage on kidney biopsy are at risk for kidney dysfunction.^{967,968} Women with creatinine levels higher than 1.4 mg/dL have a greater propensity for hypertension and a progressive increase in creatinine level during the course of pregnancy, and pregnancy-related loss of maternal kidney function occurs in 43% of these patients.⁹⁶⁹ The infant survival rate was 93% in this study; preterm delivery occurred in almost two-thirds and growth retardation in one-third of infants.⁹⁶⁹ A more recent study of 223 women (136 and 87 in the pregnancy and nonpregnancy groups) with biopsy-proven IgA nephropathy in Italy has shown no difference in long-term outcomes of IgA nephropathy between the two groups over a minimum follow-up period of 5 years.⁹⁷⁰

LABORATORY FINDINGS

To date, there are no specific serologic or laboratory tests diagnostic of IgA nephropathy or IgA vasculitis. The identification of abnormally galactosylated IgA1 has led to the development of a potential diagnostic test based on the detection of increased lectin binding in patients with IgA nephropathy.⁹⁷¹

Although the serum IgA levels are elevated in up to 50% of patients, the presence of elevated IgA in the circulation is not specific for IgA nephropathy. The detection of IgA-fibronectin complexes was initially thought to be a marker in patients with IgA nephropathy but has not proven to be a useful clinical test.^{972,973} As noted earlier, polymeric IgA also appears to be found in some patients with IgA nephropathy.^{845,974–978} The polymeric IgA itself is of the IgA1 subclass. IgA may also be contained in circulating immune complexes that are not complement binding. Similar immune complexes have been described in IgA vasculitis.^{979–996} The level of circulating immune complexes wax and wane and may sometimes correlate with episodes of macroscopic hematuria. In one

interesting study, the level of circulating immune complexes was increased after patients drank cow's milk. This phenomenon occurred in 10% to 15% of patients and possibly suggests sensitivity to bovine serum albumin. Unfortunately, none of these findings is pathognomonic of IgA nephropathy.

Antibodies to the GBM,⁹⁹⁷ the mesangium,^{998,999} glomerular endothelial cells,^{850,903} neutrophil cytoplasmic constituents,^{852,853} IgA rheumatoid factor,^{1000,1001} and a number of infectious agents, bovine serum proteins, and soy proteins^{855–862,1002} have been found in patients with IgA nephropathy. Until studies demonstrate that certain patients have sensitivity to a particular pathogen or food allergen, it is difficult to know whether to perform antibody testing to identify certain foods that should be eliminated from the patient's diet. None of these antibody tests has been standardized in large patient populations. Therefore their applicability to all patients with IgA nephropathy is not known. Levels of complements, such as C3 and C4, are typically normal and, in some patients, even elevated,¹⁰⁰³ as are complement components C1q and C2–C9.^{783,979,980,1003,1004} The fact that these complement levels are normal may belie the fact that either the alternate or classical pathway of complement may be activated. In this regard, C3 fragments are increased in 50% to 75% of patients,^{1005,1006} and C4-binding protein concentrations are also increased.¹⁰⁰⁴ It has also been suggested that both a decreased C3 level and an elevated IgA/C3 ratio may have diagnostic utility for IgA nephropathy³¹ and may be associated with a higher risk of progression.^{1007,1008} Interestingly, a large study in Korea of 343 patients with biopsy-proven IgA nephropathy also demonstrated an independently predictive correlation between decreased C3 and mesangial C3 deposition and renal outcomes.¹⁰⁰⁹

A typical finding is microscopic hematuria on urinalysis that may persist, even at very low levels of macroscopic hematuria. The finding of dysmorphic erythrocytes in the urine is typical.¹⁰¹⁰ Proteinuria is found in many patients with IgA nephropathy although, in most of them, protein excretion is less than 1 g/day. Mesangial and endocapillary hypercellularity, segmental glomerulosclerosis, and extracapillary proliferation are strongly associated with proteinuria.⁸³⁶

Although older studies suggested that the detection of dermal capillary IgA deposits in the skin may be of diagnostic utility in IgA nephropathy,¹⁰¹¹ this test has not gained widespread acceptance, largely because of the substantial variation in sensitivity and specificity of skin biopsy findings in IgA in patients with nephropathy.¹⁰¹²

TREATMENT

In part because of the outcome variability of patients with IgA nephropathy, the best approach to therapy remains incompletely established.^{1013–1015} Treatment is indicated for patients with urinary protein excretion of more than 0.5 g of protein/day.¹⁰¹⁶ Three major approaches have emerged and are supported by substantial direct evidence: (1) RAAS blockade; (2) oral and/or intravenous glucocorticoids; and (3) combined immunosuppressive (cytotoxic) therapy. The latter is usually reserved for patients with documented progressive disease.¹⁰¹⁷ Combinations of these approaches have been under intense evaluation.

Angiotensin II Inhibition

In retrospective studies, angiotensin II inhibition has been associated with a slower rate of loss of kidney function and

a higher frequency of remission of proteinuria compared with either no therapy¹⁰¹⁸ or the use of beta blockers.¹⁰¹⁹ Several RCTs of angiotensin II inhibition in patients with IgA nephropathy have been undertaken.^{1020–1025}

A metaanalysis of 11 studies (totaling 585 subjects) has revealed that the use of angiotensin II inhibition is associated with a reduction in proteinuria and preservation of GFR.¹⁰²⁶ The antiproteinuric effects of the ACE inhibitor appear to be more profound in patients with the ACE gene DD genotype.¹⁰²⁷ Observational studies of patients with IgA nephropathy have suggested that an elevated fractional excretion of IgG is a powerful predictor of the renoprotective response to angiotensin II inhibition.¹⁰²⁸ Higher doses of angiotensin II inhibitors may afford additional renoprotective effects. In an RCT involving 207 patients, a high-dose ARB (losartan, 200 mg/day) was compared with an ARB given at the usual dose (losartan, 100 mg/day), as well as with a usual-dose ACE inhibitor and a low-dose ACE inhibitor (equivalent to enalapril, 20 and 10 mg/day, respectively).¹⁰²⁹ High-dose ARB therapy was most efficacious in reducing proteinuria and slowing the rate of loss of decline of estimated GFR. The current first line of treatment consists of escalating doses of an ARB to achieve a target urinary protein excretion of less than 1 g/day, along with dietary sodium restriction, for patients of any age with IgA nephropathy and proteinuria of more than 500 mg of protein excretion/day.

Glucocorticoids

Although prednisone was initially considered to be without effect,⁹²¹ some cohort studies have suggested that corticosteroids may afford some benefit.^{1030,1031} For example, a RCT has demonstrated that a 6-month course of intravenous plus oral glucocorticoids may be useful in patients with IgA nephropathy who have well-preserved kidney function (serum creatinine level <1.5 mg/dL and proteinuria of 1–3.5 g of protein/day).¹⁰³² After a 5-year follow-up, the risk of a doubling in plasma creatinine concentration was significantly lower in the corticosteroid-treated patients, who also showed a significant decrease in mean urinary protein excretion after 1 year that persisted throughout the follow-up.¹⁰³³ This beneficial effect was maintained after 10 years of follow-up, as reflected by a rate of renal survival (failure to double the serum creatinine level) of 97% in the treated group compared with 53% in the placebo group (log rank test, $P = .0003$).¹⁰³² On the other hand, no benefit of corticosteroids over placebo could be demonstrated in the multicenter RCT conducted by the Southwest Pediatric Nephrology Study Group,¹⁰³⁴ although this negative result was mitigated by a statistically significant lower degree of proteinuria at baseline among placebo-treated patients. A metaanalysis of seven RCTs involving 366 patients suggested that glucocorticoid therapy was effective in reducing proteinuria and preventing loss of kidney function.¹⁰³⁵ A subsequent metaanalysis of 15 clinical trials that measured ESKD, doubling of the serum creatinine level, and urinary protein excretion as outcomes has found that corticosteroid therapy is associated with a decrease of proteinuria and with a statistically significant reduction in ESKD risk.¹⁰³⁶

Another circumstance in which prednisone has demonstrated a substantial beneficial effect is in the treatment of patients with IgA nephropathy and concurrent MCD. These patients had nephrotic-range proteinuria and diffuse foot

process effacement. They responded to prednisone in a manner very similar to that of patients with MCD.^{129,920,1037,1038}

Low doses of prednisone (20–30 mg/day, tapered to 5–10 mg/day over 2 years) may also be effective in lowering proteinuria in patients with mild inflammatory glomerular lesions.¹⁰³⁹ Conversely, a poor response to glucocorticoids can be predicted in patients with extensive glomerular obsolescence, tuft adhesions, severe interstitial fibrosis, low serum albumin level, low estimated GFR, and marked proteinuria.¹⁰⁴⁰ A high number of fibroblast-specific protein 1 (FSP1)-positive cells in the interstitium (>33 FSP1+ cells/high-power field) is highly predictive of a poor response to steroids.¹⁰⁴¹

The Therapeutic Evaluation of Steroids in IgA nephropathy (TESTING) study, assessing, as the name implies, the effects of corticosteroids in patients with IgA nephropathy and persistent proteinuria, was recently completed to assess the risk of disease progression.¹⁰⁴² This multicenter, double-blind, RCT randomized 262 participants to oral methylprednisolone (0.6–0.8 mg/kg/day; maximum, 48 mg/day) or placebo for 2 months, weaned over 4 to 6 months with a primary composite outcome (ESKD, death due to kidney failure, or 40% decrease in eGFR). However, recruitment was discontinued because of excess serious adverse events, primarily in the steroid group, including serious infections and two deaths.

Of considerable recent import, the NEFIGAN trial was a phase 2b double-blind, randomized, placebo-controlled clinical trial that randomized 150 patients with native kidney biopsy-proven IgA nephropathy and persistent proteinuria despite maximal RAAS blockade to one of three arms—placebo, targeted-release formulation (TRF) budesonide, 8 mg orally daily or TRF budesonide, 16 mg daily. Primary outcome was defined as the mean change from baseline for the 9 months of treatment. Patients in the TRF budesonide arms, regardless of dose, achieved a 24.4% decrease from baseline in mean urine protein-to-creatinine ratio (UPCR; change in UPCR vs. placebo, 0.74; 95% CI, 0.59–0.94; $P = .0066$), an effect that was sustained throughout follow-up, and adverse events were similar across all groups (www.clinicaltrials.gov, NCT01728035). These results would suggest that nonsystemic corticosteroid use that more specifically targets intestinal mucosal immunity locally may prove to be an additional strategy in the treatment toolkit for IgA nephropathy.¹⁰⁴³

In summary, glucocorticoid therapy is a reasonable option for the treatment of patients with adverse prognostic features with well-preserved kidney function (GFR >60 mL/min/1.73 m²) who remain proteinuric despite a 3- to 6-month trial of angiotensin II inhibitors or patients with features of MCD and nephrotic syndrome.¹⁰⁴⁴

Combinations of Angiotensin II Inhibition and Glucocorticoid Therapy

Two RCTs in patient with IgA nephropathy have compared the combined use of glucocorticoids and angiotensin II inhibition to angiotensin II inhibition alone, but not with glucocorticoids alone.^{1045,1046} In the larger of the two studies¹⁰⁴⁶ (97 subjects with IgA nephropathy, urinary protein excretion >1.0 g/day, eGFR >50 mL/min/1.73 m²), 27% of the subjects receiving ramipril alone developed a doubling of the baseline serum creatinine level or ESKD, whereas only 4% of subjects in the ramipril plus steroid group developed these endpoints ($P = .003$) after a follow-up of 8 years. These studies demonstrate an

added benefit of glucocorticoids over angiotensin II inhibitors alone. Whether such combined therapy should be instituted as initial therapy or only after a trial of angiotensin II inhibition alone remains to be investigated.^{1047–1049}

In a retrospective analysis of the VALIGA cohort, using propensity score matching, 184 patients with IgA nephropathy treated with corticosteroids (CS) and RAAS blockade were compared with a similar number of patients on only RAAS blockade.¹⁰⁴⁷ During a median follow up of almost 5 years, the CS-treated group had a decline in renal function of -1.0 ± 7.3 mL/min/1.73 m² per year compared with 3.2 ± 8.3 mL/min/1.73 m² per year in the control group ($P = .004$). Also, among patients with more than 3 g/day of proteinuria, 64% in the CS group achieved proteinuria less than 1 g/day compared with only 4% in the RAAS blockade only group ($P < .001$). Interestingly, CS treatment in patients with an eGFR of less than 50 mL/min/1.73 m² resulted in a slower rate of renal function decline. Unfortunately, the study was limited by its retrospective nature and lack of important information, including, dosing of CS, compliance, adverse effects, and duration and timing of the CS regimen.

More aggressive treatment may be appropriate in patients with severe crescentic or progressive IgA nephropathy.^{1050–1052} In an RCT, patients with a serum creatinine concentration more than 1.5 mg/dL and a GFR declining at a rate of more than 15%/year either received no immunosuppression or were treated with oral prednisolone (initially, 40 mg/day) and cyclophosphamide (1.5 mg/kg/day) for 3 months, followed by 2 years of treatment with azathioprine (1.5 mg/kg/day).¹⁰⁵³ Over a follow-up of 2 to 6 years, 5-year renal survival was 72% in treated patients versus only 6% in untreated patients.¹⁰⁵³ This approach of prednisone coupled with oral azathioprine for 2 years in patients with proteinuria of more than 2.5 g of protein/day was also observed in a retrospective survey.^{1054,1055}

The use of pulse methylprednisolone, oral prednisone, and/or cyclophosphamide to treat patients who have rapidly progressive glomerulonephritis with widespread crescentic transformation has been reported.^{1056–1058} It is reasonable to treat crescentic disease in IgA nephropathy in a manner similar to that for other forms of crescentic glomerulonephritis (e.g., ANCA glomerulonephritis). Of concern, however, was the finding in 12 patients of the persistence of crescents on repeat biopsy, despite the early and aggressive treatment with pulse methylprednisolone and oral prednisone, and a short-term reversal of the acute crescentic glomerulonephritis.¹⁰⁵⁸ This study suggests that there was only a diminution in the rate of progression to ESKD.

Combinations of Angiotensin II Inhibition, Glucocorticoid Therapy, and Immunosuppression

This approach to therapy was recently the subject of a large multicenter RCT in Germany, the supportive versus immunosuppressive therapy for the treatment of progressive IgAN (STOP-IgAN) study.^{1092,1093} This multicenter, open-label, randomized, controlled clinical trial enrolled 309 patients with biopsy-proven IgA nephropathy who were considered at high risk of progression (proteinuria >0.75 g/day plus hypertension, eGFR <90 mL/min/1.73 m², or both). All patients began with a 6-month run-in phase during, which they received supportive care with RAAS blockade. Over 6 months, 94 patients (30%) achieved and maintained

proteinuria less than 0.75 g/day and were not included in the study, per protocol. Of the remaining patients, 162 consented and were randomized to continuing supportive therapy ($n = 80$) or immunosuppression group ($n = 82$). Patients received immunosuppression, consisting of glucocorticoids for patients with an eGFR more than 60 mL/min/1.73 m² or glucocorticoids in combination with oral cyclophosphamide, followed by azathioprine maintenance for patients with an eGFR between 30 and 59 mL/min/1.73 m². Patients with proteinuria more than 3.5 g/day, eGFR less than 30 mL/min/1.73 m², crescentic IgA, and secondary and rapidly progressive diseases were excluded. The two primary endpoints assessed at 36 months, in hierarchic order, were full clinical remission, defined by a UPCR less than 0.2 and a stable eGFR, with less than a 5-mL/min/1.73 m² decline and decrease in the eGFR of at least 15 mL/min/1.73 m² from baseline to the end of 36 months. At the final visit, 4 of 80 patients (5%) in the supportive group and 14 of 82 patients (20%) in the immunosuppression group had achieved full clinical remission. The higher rate of clinical remission was attributable solely to the reduction in proteinuria because there was no significant differences in renal function. The total number of adverse events was not significantly different between the groups, but the rate of infections were significantly higher in the immunosuppression group.

In a posthoc analysis, the same authors concluded that although corticosteroid monotherapy may reduce proteinuria, neither immunosuppressive regimen prevented GFR loss. Some of the factors noted in this study include the short-term follow-up (which may have been too early to show any beneficial effect on the kidney function), lack of histologic (MEST) scores, and modest degree of proteinuria.

Other Treatment Modalities

Aliskiren, a direct inhibitor of renin, has been investigated as an antiproteinuric agent in IgA nephropathy. An open-label pilot study in 25 consecutive patients with IgA nephropathy in Hong Kong treated for 12 months with aliskiren resulted in a 26.3% mean reduction in the UPCR and a significant reduction in plasma renin activity (95% CI, 20.1–43.6; $P = .001$ vs. baseline); however, there were mild allergic reactions in 2 patients (8%) and transient hyperkalemia in 6 patients (24%).¹⁰⁵⁹ A subsequent randomized crossover study using aliskiren or placebo in 22 patients in China with biopsy-proven IgA nephropathy and persistent proteinuria, despite ACE inhibition or angiotensin receptor blockade, demonstrated significant reductions in proteinuria at 4 weeks (1.76 ± 0.95 – 1.03 ± 0.69 g protein/g creatinine; $P < .0001$) and from 4 to 16 weeks, but there was a modest but statistically significant reduction in the eGFR (57.2 ± 29.1 – 54.8 ± 29.3 mL/min/1.73 m²; $P = .013$).¹⁰⁶⁰

In a 3-year prospective controlled trial of cyclophosphamide, dipyridamole, and low-dose warfarin, it was reasonably clear that this treatment¹⁰⁶¹ has very little long-term benefit in patients with IgA nephropathy. Five years after the end of a small controlled trial (total, 48 patients), there was no significant difference in the rate of ESKD between patients previously treated with cyclophosphamide, dipyridamole, and warfarin (22%) and the control group (33%). Of note, it has since been proposed that the apparent lack of benefit of cyclophosphamide therapy may have been clouded by the potential risk of a recently recognized mechanism of AKI in

study subjects, termed “warfarin-related nephropathy.”¹⁰⁶² This observation warrants additional clinical trials using combination therapy without this potential confounder.

Whether MMF is useful in the treatment of IgA nephropathy is currently unknown. Three randomized trials of MMF have shown conflicting results.^{1063–1066} The studies, based in China and Hong Kong, reported a beneficial effect of MMF on proteinuria and hyperlipidemia,^{1064,1067} but no effect on kidney function in the short term.¹⁰⁶³ Long-term follow-up of this cohort suggested better preservation of kidney function in the MMF-treated group.¹⁰⁶⁸ On the other hand, the two placebo-controlled studies of MMF in Caucasian populations of 32 and 34 patients failed to demonstrate a benefit of MMF on proteinuria or the preservation of kidney function.^{1065,1066} It is noteworthy that in one study,¹⁰⁶⁵ patients had relatively advanced renal insufficiency (mean serum creatinine level, 2.4 mg/dL). Collectively, these underpowered studies have failed to establish a role for MMF in the treatment of IgA nephropathy and raise the question as to whether certain ethnic groups (e.g., Asians) may be more responsive to this form of therapy.

With regard to azathioprine use, an RCT in Italy of 207 patients with biopsy-proven IgA nephropathy compared steroids alone or in combination with azathioprine for 6 months and found no difference in renal survival, defined as time to a 50% increase in the plasma creatinine level from baseline over a median follow-up of 4.9 years. In this study, the 5-year cumulative renal survival was not significantly different (88 vs. 89%; $P = .83$).¹⁰⁶⁹ A separate randomization list with a longer treatment course of 1 year in patients with impaired kidney function (plasma creatinine >2.0 mg/dL; proteinuria ≥ 1 g/day) involving 253 patients with biopsy-proven IgA nephropathy from the same group of investigators compared steroids alone or in combination with azathioprine and, similarly, found no difference in renal survival at the 6-year mark.¹⁰⁷⁰

There has been much discussion in the literature about the use of tonsillectomy in IgA nephropathy. Results of retrospective trials have been inconsistent.^{1071–1074} A single-center retrospective cohort study of 200 patients with biopsy-proven IgA nephropathy, 70 (35%) of whom received tonsillectomy, revealed by multiple regression modeling that tonsillectomy prevented GFR decline during the follow-up period (regression coefficient, 2.0; $P = .01$). This effect was also observed in non-steroid-treated patients.¹⁰⁷⁴ Based on a retrospective multivariate analysis¹⁰⁷² of a large cohort of 329 patients from Japan, treatment with tonsillectomy and pulse glucocorticoid therapy (methylprednisolone, 0.5 g/day for 3 days for three courses, followed by oral prednisolone at an initial dose of 0.6 mg/kg on alternate days, with a decrease of 0.1 mg/kg every 2 months) was associated with clinical remission. Similarly, in a multivariate analysis¹⁰⁷⁵ focusing on the subgroup of 70 patients from the same cohort, with a baseline serum creatinine concentration of more than 1.5 mg/dL, treatment with the combination of tonsillectomy and pulse glucocorticoids was associated with improved long-term renal survival. Another retrospective analysis,¹⁰⁷¹ however, showed no benefit of tonsillectomy on the clinical course of IgA nephropathy. Interestingly, a retrospective analysis of 365 patients with biopsy-proven IgA nephropathy in Japan has revealed that tonsillectomy delays disease progression to ESKD (OR, 0.09; 95% CI, 0.01–0.75; $P = .026$).¹⁰⁷⁶

In a controlled nonrandomized trial, tonsillectomy plus pulse glucocorticoids was associated with a higher rate of remission of proteinuria and hematuria (but not kidney function) than pulse glucocorticoids alone.¹⁰⁷⁷ A multicenter RCT of patients with biopsy-proven IgA nephropathy randomly allocated patients to receive tonsillectomy with steroid pulses ($n = 33$) versus steroid pulses alone ($n = 39$).¹⁰⁷⁸ Although urinary protein excretion was significantly lower in the tonsillectomy group at the 12-month mark ($P < .05$), clinical remission (defined as the disappearance of hematuria, proteinuria, or both) was not. In contrast, in a large cohort of subjects from the European validation study of the Oxford classification of IgAN (VALIGA), no significant differences were found in respect to proteinuria or decline in renal function in patients with or without tonsillectomy.¹⁰⁷⁹ Long-term follow-up results will possibly provide further clarity.

Omega-3 Fatty Acids. Despite a great deal of interest in the past decade, the value of treatment with omega-3 long-chain polyunsaturated fatty acids (eicosapentaenoic and docosahexaenoic acids, found in fish oils) in IgA nephropathy remains unproven. This is based on the premise that omega-3 polyunsaturated fatty acids, eicosapentaenoic and docosahexaenoic acids, after biologic transformation into trienoic eicosanoids, may modulate the vascular reactivity and inflammatory mediators. By inhibiting cytokines, leukotriene Br, and possibly platelet-activating factor formation, fish oil may play a role in suppressing cell-mediated inflammation and limit glomerular injury.^{1080,1081} In a study by the Mayo Clinic,¹⁰⁸¹ 106 patients were randomly assigned to 12 g of omega-3 fatty acids or olive oil for 2 years. Only 6% of patients treated with fish oil experienced a doubling of their plasma creatinine concentration, compared with 33% of those treated with olive oil. In the fish oil-treated patients, only 14% excreted over 3.5 g of protein/day, in contrast to 65% of those treated with olive oil. The enthusiasm for this approach, however, was tempered by subsequent studies that showed no benefit of fish oil therapy.^{1082,1083}

A metaanalysis of published trials on omega-3 fatty acids encompassing 17 trials and 626 subjects with a variety of kidney diseases, including five trials in IgA nephropathy,¹⁰⁸⁴ has revealed no beneficial effects on proteinuria or slowing in the rate of GFR decline. In an RCT involving 30 patients, the addition of omega-3 fatty acids to angiotensin II inhibition was more effective than angiotensin II inhibition alone in decreasing proteinuria and erythrocyturia over 6 months.¹⁰⁸⁵ A subsequent metaanalysis of five RCTs (totaling 233 patients) has found a significant reduction in proteinuria with omega-3 fatty acid use, but there was no significant benefit in preservation of kidney function.¹⁰⁸⁶ Unfortunately, no benefit was found in a metaanalysis published the same year.¹⁰⁸⁷ Therefore, if omega-3 fatty acids should be used at all in the treatment of IgA nephropathy, they should be used adjunctively in combination with angiotensin II inhibition and not as monotherapy.

In summary, until better long-term follow up studies are available, patients with IgA nephropathy with preserved renal function, with proteinuria more than 1 g/day, should be treated with supportive care, including RAAS blockade.^{1088–1091} Should proteinuria persist despite angiotensin II inhibition, the addition of glucocorticoids (oral or intravenous plus oral) should be considered for patients with well-preserved

kidney function (GFR ≥ 60 mL/min/1.73 m²). In patients with progressive renal insufficiency, the use of prednisone and cyclophosphamide followed by azathioprine should be considered.¹⁰⁵³ In patients at high risk of progression (e.g., those fitting STOP-IgAN inclusion criteria), the use of corticosteroids or other immunosuppression requires a multifaceted approach.¹⁰⁹² This challenge is best solved by assessing the risks of disease progression, MEST score, patient's clinical profile, and risks of adverse effects of immunosuppression. Also, an open discussion with patients about the study trials, risks, and benefits must be undertaken to reach a conclusion. It must be noted that the STOP-IgAN study did not address the role of immunosuppression in patients with crescentic, rapidly progressive glomerulonephritis or with nephrotic-range proteinuria because they were excluded from the study.^{1093–1096}

It remains plausible that high-dose corticosteroids and/or cyclophosphamide could also be considered for patients with widespread crescentic glomerulonephritis, whereas patients with AKI associated with tubular necrosis and little glomerular damage should be treated conservatively, because these individuals have an excellent long-term response. Although there is no conclusive evidence of efficacy, the relatively benign side effect profile of omega-3 fatty acid therapy permits its use in patients who have an unfavorable prognosis. Patients with nephrotic syndrome and MCD may benefit from oral glucocorticoids.

FIBRILLARY GLOMERULONEPHRITIS AND IMMUNOTACTOID GLOMERULOPATHY

NOMENCLATURE

Fibrillary glomerulonephritis and immunotactoid glomerulopathy are glomerular diseases that are characterized by patterned deposits seen by electron microscopy (Figs. 31.31 and 31.32).^{1097–1104} Most renal pathologists prefer to distinguish fibrillary glomerulonephritis from immunotactoid

glomerulopathy based on the presence of fibrils of approximately 20 nm in diameter in fibrillary glomerulonephritis and larger, 30- to 40-nm-diameter microtubular structures in immunotactoid glomerulopathy.^{1097,1099–1102} (see Figs. 31.31 and 31.32). A minority of pathologists, however, have advocated grouping glomerular diseases with fibrillary deposits or microtubular deposits under the term “immunotactoid glomerulopathy.”^{1101,1104}

FIBRILLARY GLOMERULONEPHRITIS PATHOLOGY

Electron Microscopy

The diagnosis of fibrillary glomerulonephritis (FGN) requires the identification by electron microscopy of irregular accumulations of randomly arranged, nonbranching fibrils of approximately 20-nm diameter in glomerular mesangium, capillary walls, or both.^{1097–1103,1105} (see Fig. 31.31A). In capillary walls, the fibrillary deposits can be subepithelial, subendothelial, or intramembranous. The fibrillary deposits often contain blotchy electron-dense material, but only rarely have associated well-defined, electron-dense deposits. The fibrils are distinctly larger than the actin filaments in adjacent cells, which is a useful observation that helps distinguish the fibrils of FGN from those of amyloidosis, which are only slightly larger than actin. The fibrils of FGN are not as large as the microtubular deposits of immunotactoid glomerulopathy or cryoglobulinemia, and they do not have the “fingerprint” configuration occasionally observed in lupus nephritis dense deposits. Most patients with FGN have substantial proteinuria, and therefore there usually is extensive effacement of visceral epithelial foot processes.

Light Microscopy

In FGN, extensive localization of fibrils in capillary walls causes capillary wall thickening. Mesangial localization results in an increased mesangial matrix and usually stimulates mesangial hypercellularity. Varying distributions of the

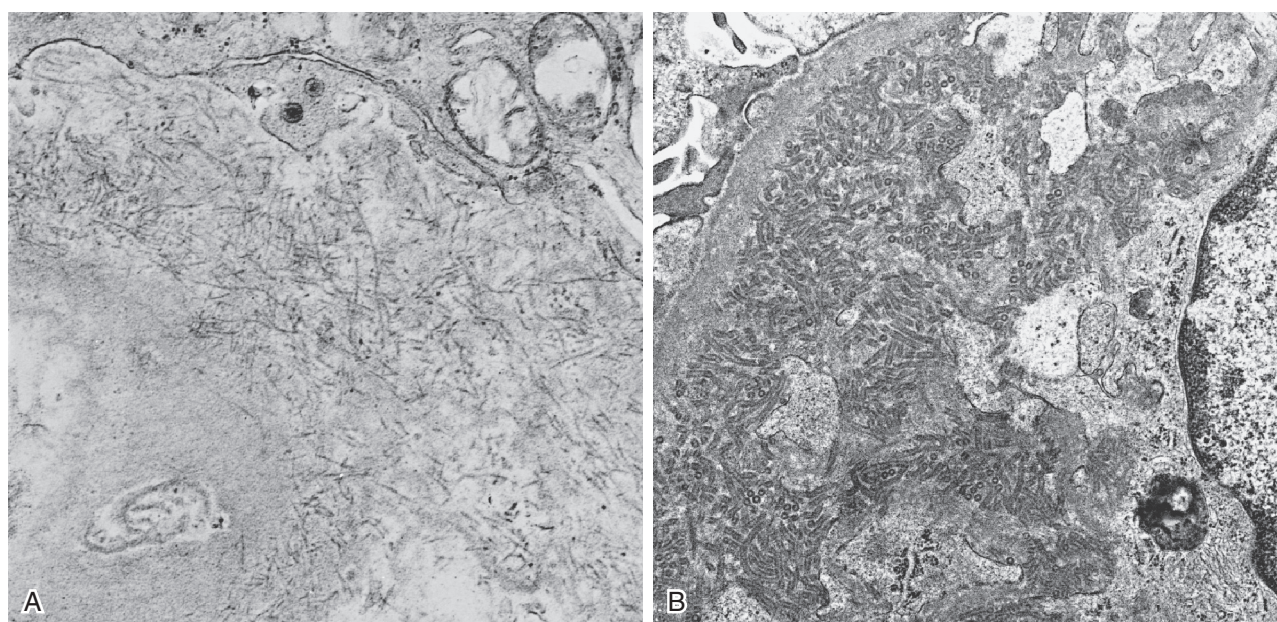


Fig. 31.31 Electron micrographs showing the glomerular deposits of fibrillary glomerulonephritis (A) and immunotactoid glomerulopathy (B). Note the random orientation of the former and the microtubular appearance and greater organization of the latter. ($\times 20,000$.)

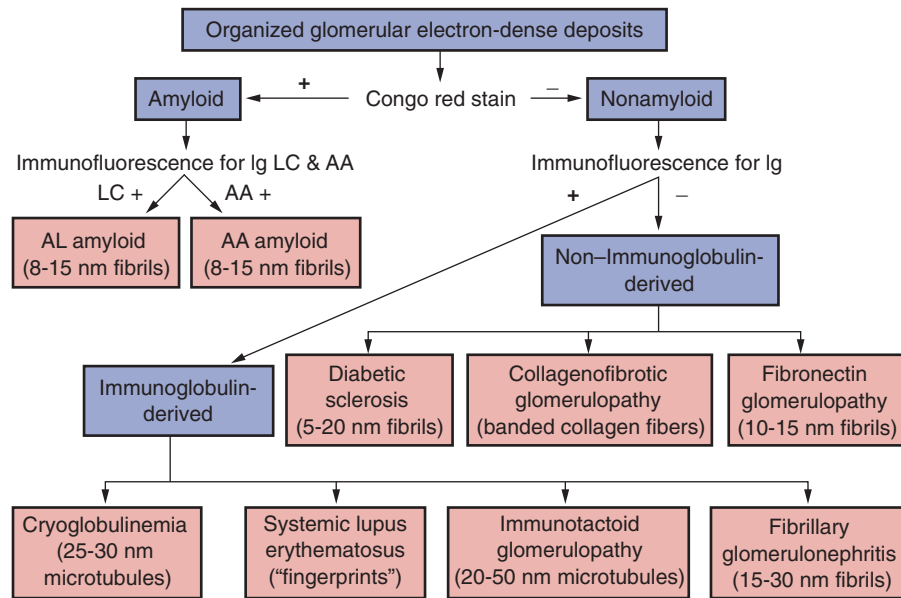


Fig. 31.32 Algorithm for the pathologic categorization of glomerular diseases with patterned or organized deposits. The first division is into amyloid versus nonamyloid disease and the second is into diseases caused by immunoglobulin (Ig) molecule deposition and those that are not. By the approach illustrated, fibrillary glomerulonephritis is distinguished from immunotactoid glomerulopathy based on the ultrastructural characteristics of the deposits.

fibrillary deposits cause the light microscopic appearance of fibrillary glomerulonephritis to be extremely variable.^{1097,1103} Therefore FGN can mimic the light microscopic appearance of MPGN, proliferative glomerulonephritis, or MN. Crescents occur in the most aggressive phenotypes. Of 74 sequential fibrillary glomerulonephritis specimens evaluated at UNC, 28% had crescents with an average involvement of 29% of glomeruli (range, 5%–80%). The fibrillary deposits typically have a moth-eaten appearance when stained with a Jones silver methenamine stain. They do not stain with Congo red, which distinguishes them from amyloid deposits.

Immunofluorescence Microscopy

The deposits of FGN almost always stain more intensely for IgG than for IgM or IgA, and many specimens have little or no staining for IgM and IgA.^{1097–1103} IgG4 is the dominant subclass. Only rare specimens have staining for only one light-chain type. C3 staining usually is intense. The IF staining pattern of fibrillary glomerulonephritis is relatively distinctive (Fig. 31.33). It is not granular or linear, but rather has an irregular bandlike appearance in capillary walls and an irregular shaggy appearance in the mesangium.

IMMUNOTACTOID GLOMERULOPATHY PATHOLOGY

Electron Microscopy

The tubular substructure of the deposits of immunotactoid glomerulopathy is readily discerned at 5,000 to 10,000 magnification (see Fig. 31.31B). At this magnification, the deposits of FGN have no tubular structure. The microtubules of immunotactoid glomerulopathy also have a greater tendency to align in parallel arrays, whereas the fibrils of FGN always are randomly distributed.⁸²⁷ The ultrastructural deposits of immunotactoid glomerulonephritis resemble those seen in cryoglobulinemic glomerulonephritis; thus the latter must be ruled out before making a diagnosis of immunotactoid

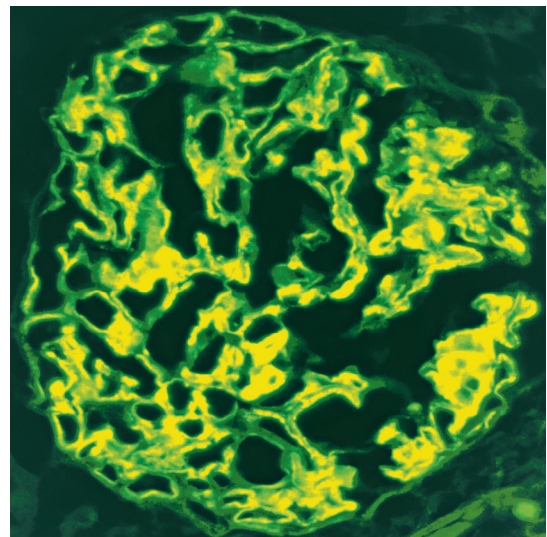


Fig. 31.33 Immunofluorescence micrograph of a glomerulus with fibrillary glomerulonephritis showing mesangial and bandlike capillary wall staining for immunoglobulin G (IgG). (Fluorescein isothiocyanate anti-IgG stain, $\times 300$.)

glomerulopathy. However, cryoglobulinemic microtubules typically are shorter and less well designed than immunotactoid microtubules.

Light Microscopy

Immunotactoid glomerulopathy has a varied light microscopic appearance. Combined capillary wall thickening and mesangial expansion is most common, which often gives a membranoproliferative appearance. Immunotactoid deposits may be massive, resulting in nodular mesangial expansion in some specimens.

Immunofluorescence Microscopy

The deposits of immunotactoid glomerulopathy usually are IgG dominant, with staining for both κ and λ light chains; however, the immunoglobulin in the deposits of immunotactoid glomerulopathy is more often monoclonal than in fibrillary glomerulonephritis.¹⁰⁹⁹ Monoclonality warrants a clinical workup for a B cell dyscrasia.

PATHOGENESIS

The cause and pathogenesis of FGN and immunotactoid glomerulopathy are not known. Fibrillary glomerulonephritis and immunotactoid glomerulonephritis have been associated with lymphoproliferative disease (e.g., chronic lymphocytic leukemia, B cell lymphomas).^{1099,1105,1106} Immunotactoid glomerulonephritis is more frequently associated with a monoclonal gammopathy.¹¹⁰⁷ On rare occasions, FGN can also be associated with a monoclonal gammopathy.^{1108,1109} The possible oligoclonal character of the deposits of FGN may facilitate self-association and fibrillar organization in a fashion analogous to that of the monoclonal light chains of immunoglobulin light-chain (AL) amyloidosis.¹¹⁰² The resemblance of immunotactoid deposits to those of cryoglobulinemia, which often contain a monoclonal component, also raises the possibility that the presence of some type of uniformity of the immunoglobulin in the deposits may be causing the patterned organization in immunotactoid glomerulopathy. Rarely, FGN may be associated with concomitant hepatitis C virus infection¹¹¹⁰ or an unusual IgM glomerular deposit disease.¹¹¹¹

In a recent study, FGN was postulated as an autoimmune disease caused by autoantibodies targeting putative autoantigen DNAJB9.¹¹¹² Molecular chaperones are proteins that facilitate proper protein folding and the assembly needed for biologic functions. DNAJs are cochaperones that regulate activity of the heat shock protein (HSP), which is the major chaperone in humans. DNAJB9 (also known as “Mdg-1” or “ERdj4”) is a member of the DNAJ protein located in the endoplasmic reticulum (ER). It functions as a cochaperone for Bip, which is the ER member of HSP. It is induced by many external stimuli, including heat, and also by p53 in response to DNA damage via the Ras/Raf/ERK pathway.

In two recent studies using laser capture microdissection, glomeruli were isolated from paraffin-embedded biopsy samples.^{1113,1114} The proteins present in the samples were analyzed using liquid chromatography–assisted tandem mass spectrometry. DNAJB9 was a highly sampled protein found only in cases of FGN, along with IgG1 and proteins of the classical complement pathway. This was specific for FGN because DNAJB9 was not detected in other glomerular diseases, including immunotactoid. Also, IF staining using DNAJB9 antibody showed specific staining in the GBM and/or mesangial regions. Dual staining was evident for the colocalization of DNAJB9 and IgG in glomerular and extra-glomerular deposits. Further studies are required to understand the role of DNAJB9 and anti-DNAJB9 antibodies in the pathogenesis, of FGN.

EPIDEMIOLOGY AND CLINICAL FEATURES

An analysis of 9085 consecutive native kidney biopsy specimens evaluated in the UNC Nephropathology Laboratory has revealed a frequency of 0.8% for fibrillary glomerulonephritis and 0.1% for immunotactoid glomerulonephritis,

compared with 14.5% for MN, 7.5% for IgA nephropathy, 2.6% for type I MPGN, 1.5% for amyloidosis, and 0.8% for anti-GBM glomerulonephritis. Thus, fibrillary glomerulonephritis is about as common as anti-GBM glomerulonephritis and much more frequent than immunotactoid glomerulopathy.

Patients with fibrillary glomerulonephritis present with a mixture of nephrotic and nephritic syndrome features.^{1100,1102,1105} Patients may have microscopic or macroscopic hematuria, renal insufficiency (including rapidly progressive glomerulonephritis in a few patients), hypertension, and proteinuria, which may be in the nephrotic range. In a series of 28 patients with fibrillary glomerulonephritis seen at UNC, the mean age was 49 years (range, 21–75 years), the ratio of males to females was 1:1.8, and the ratio of Caucasians to African Americans was 8.3:1.¹¹⁰² After 24 months of follow-up, renal survival was only 48%.¹¹⁰² In a subsequent series of 66 patients with fibrillary glomerulonephritis seen at the Mayo Clinic in Rochester, MN, between 1993 and 2010, the mean age at diagnosis was 53 years, and the male-to-female ratio was 1:1.2. At presentation, 100% of patients had proteinuria, 52% had hematuria, 71% were hypertensive, and 66% had renal insufficiency. Underlying malignancy (23%), dysproteinemia (17%), or autoimmune disease (15%) was common. Of 61 patients with available data followed for an average of 52.3 months, 13% had complete or partial remission, 43% had persistent kidney dysfunction, and 44% progressed to ESKD. Not surprisingly, older age, higher creatinine level, proteinuria at biopsy, and a higher percentage of global glomerulosclerosis were independent predictors of ESKD by multivariate analysis.¹¹¹⁵ Overall, proteinuria is common at the time of presentation, as are hematuria, renal insufficiency, and hypertension. In patients in whom these disorders are diagnosed, malignancy, Crohn disease, SLE, and cryoglobulinemia must be ruled out. Such patients have progressive renal failure in less than 5 years, although long-term patient survival is more than 80% at 5 years.^{1107,1116}

In a group of six patients with immunotactoid glomerulopathy, the mean age was 62 years.¹¹⁰⁰ At presentation, the clinical features in these patients looked very much like those in patients with fibrillary glomerulonephritis; they included proteinuria, hematuria, and renal insufficiency. In the largest series to date, 16 patients with immunotactoid glomerulopathy were identified from the pathology archives at the Mayo Clinic. Proteinuria was present in 100% of patients; 80% had microhematuria, 69% had nephrotic syndrome, and 50% had renal insufficiency. Interestingly, 38% of patients had a hematologic malignancy. Over an average of 48 months of follow-up of 12 patients, 50% remitted, 33% had persistent kidney dysfunction, and 17% progressed to ESKD.¹¹¹⁷

Importantly, patients with immunotactoid glomerular disease are more likely to have an associated hematopoietic process and poor long-term survival.¹¹⁰⁰ In a review study of 67 patients presenting with fibrillary glomerulopathy ($n = 61$) or immunotactoid glomerulopathy ($n = 6$), all patients had proteinuria and 50% had nephrotic syndrome, whereas hematuria occurred in approximately two-thirds of patients and hypertension in about 75% of patients.¹¹¹⁸ Renal insufficiency was discovered in 50% of the patient population. There were no statistically significant differences in clinical presentation between patients with fibrillary glomerulonephritis and those with immunotactoid

glomerulonephritis. In regard to causative factors, patients with immunotactoid glomerulonephritis were statistically more likely to have an underlying lymphoproliferative disease, a monoclonal spike on serum protein electrophoresis, and hypocomplementemia.¹¹¹⁸

Fibrillary glomerulonephritis with associated pulmonary hemorrhage has been reported anecdotally.¹¹¹⁹ One patient with immunotactoid glomerulopathy also had extrarenal deposits in both the liver and bone.¹¹²⁰

TREATMENT

At this time, there is no convincingly effective form of treatment for patients with fibrillary glomerulonephritis or immunotactoid glomerulopathy.¹¹⁰⁵ The dismal prognosis in patients with either of these diseases has prompted physicians to search for some immunosuppressive form of treatment. Fully 40% to 50% of patients with these diseases develop ESKD within 6 years of presentation.^{1097,1098,1100,1102} Efforts at treatment with glucocorticoids or alkylating agents such as cyclophosphamide have typically shown no response or, at best, some amelioration of proteinuria.¹¹²¹ In our experience, prednisone therapy alone has had no benefit. One small case series (three patients) reported significant improvement in proteinuria in response to rituximab, either alone or in combination with corticosteroids, or tacrolimus.¹¹²² In fibrillary glomerulonephritis and other forms of glomerulonephritis associated with chronic lymphocytic leukemia or other forms of lymphocytic lymphoma, there has been a report of improvement in a minority of patients treated with chlorambucil. Thus, it is possible that the treatment of the underlying malignancy, if present, may improve the glomerulonephritis.¹¹⁰⁶ Of note, a recent case series and review of previously published large case series have revealed a role for rituximab in patients with fibrillary glomerulonephritis if administered early in the disease process with a preserved eGFR.¹¹²³

The recurrence rate of fibrillary glomerulonephritis after kidney transplantation is unclear. One report has described recurrent disease in three of four patients who had received five transplants.¹¹²⁴ In a larger case series, recurrent disease occurred in none of five patients with fibrillary glomerulonephritis, but in five of seven patients with monoclonal gammopathy and fibrillary deposits.¹¹²⁵

RAPIDLY PROGRESSIVE GLOMERULONEPHRITIS AND CRESCENTIC GLOMERULONEPHRITIS

NOMENCLATURE AND CATEGORIZATION

The term “rapidly progressive glomerulonephritis” (RPGN) refers to a clinical syndrome characterized by a rapid loss of kidney function, often accompanied by oliguria or anuria and features of glomerulonephritis, including dysmorphic erythrocyturia, erythrocyte cylindruria, and glomerular proteinuria.¹¹²⁶ Aggressive glomerulonephritis that causes RPGN usually has extensive crescent formation.¹¹²⁷ For this reason, the clinical term “rapidly progressive glomerulonephritis” is sometimes used interchangeably with the pathologic term “crescentic glomerulonephritis.” Crescentic glomerulonephritis is the most aggressive structural phenotype in the continuum of injury that results from glomerular

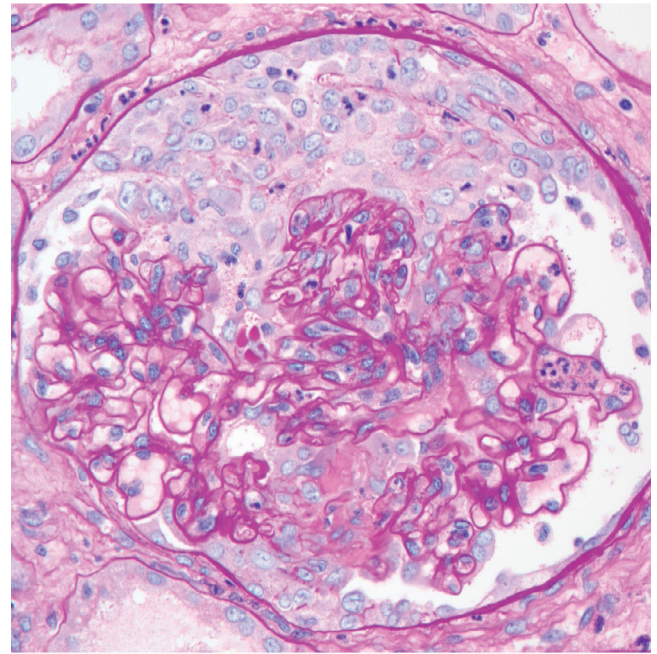


Fig. 31.34 Light micrograph showing a cellular crescent in Bowman's space. The underlying glomerular tuft is delineated by the glomerular basement membranes. (Periodic acid-Schiff stain, $\times 500$.)

inflammation (see Fig. 31.30). This pathologic feature can be seen on light, IF, and electron microscopy.^{1127–1129} It is the result of focal rupture of glomerular capillary walls that allows inflammatory mediators and leukocytes to enter Bowman's space, where they induce epithelial cell proliferation and macrophage influx and maturation, which together produce cellular crescents (Fig. 31.34).^{1130–1132}

Kidney diseases other than crescentic glomerulonephritis can cause the signs and symptoms of RPGN. Two examples are acute thrombotic microangiopathy and atheroembolic kidney disease. Although acute tubular necrosis and acute tubulointerstitial nephritis may cause rapid loss of kidney function and oliguria, these processes typically do not cause dysmorphic erythrocyturia, erythrocyte cylindruria, or substantial proteinuria.

A small minority of all patients with glomerulonephritis develops RPGN, except patients with anti-GBM disease and ANCA disease who have a high frequency of crescents. The incidence of rapidly progressive glomerulonephritis has been estimated to be as low as 7 cases/million population per year.^{675,1133} The three major immunopathologic categories of crescentic glomerulonephritis have different frequencies in different age groups (Table 31.8).^{1126–1128,1134} In a patient who has RPGN clinically and in whom crescentic glomerulonephritis is identified by light microscopy in a kidney biopsy specimen, the precise diagnostic categorization of the disease requires the integration of clinical, serologic, immunohistologic, and electron microscopic data (Fig. 31.35).

Immune complex crescentic glomerulonephritis is caused by immune complex localization within glomeruli. It is the most common cause of RPGN in children (see Table 31.8).¹¹²⁷ The major clinical differential diagnosis in children is hemolytic uremic syndrome, which also can cause rapid loss of kidney function, hypertension, hematuria, and proteinuria.

Table 31.8 Relative Frequency of Immunopathologic Categories of Crescentic Glomerulonephritis In Different Age Groups (%)^a

Immunopathologic Category	Age in Years			
	All Ages (n = 632)	1–20 (n = 73)	21–60 (n = 303)	>60 (n = 256)
Antiglomerular basement membrane CGN	15	12	15	15
Immune complex CGN	24	45	35	6
Pauci-immune CGN ^b	60	42	48	79
Other	1	0	30	0

^aCGN is defined as the presence of crescents in >50% of glomeruli. Frequency is determined with respect to age in patients whose kidney biopsy specimens were evaluated at the University of North Carolina Nephropathology Laboratory. Note the very high frequency of pauci-immune disease (usually antineutrophil cytoplasmic antibody [ANCA]-associated) in older adults.

^bApproximately 90% associated with ANCA.

CGN, Crescentic glomerulonephritis.

Data from Jennette JC, Nickleleit V: Anti-glomerular basement membrane glomerulonephritis and Goodpasture's syndrome. In Jennette JC, Olson JL, Silva FG, D'Agati V, eds. *Heptinstall's pathology of the kidney*. 7th ed. Wolters Kluwer: Philadelphia; 2015:657–684.

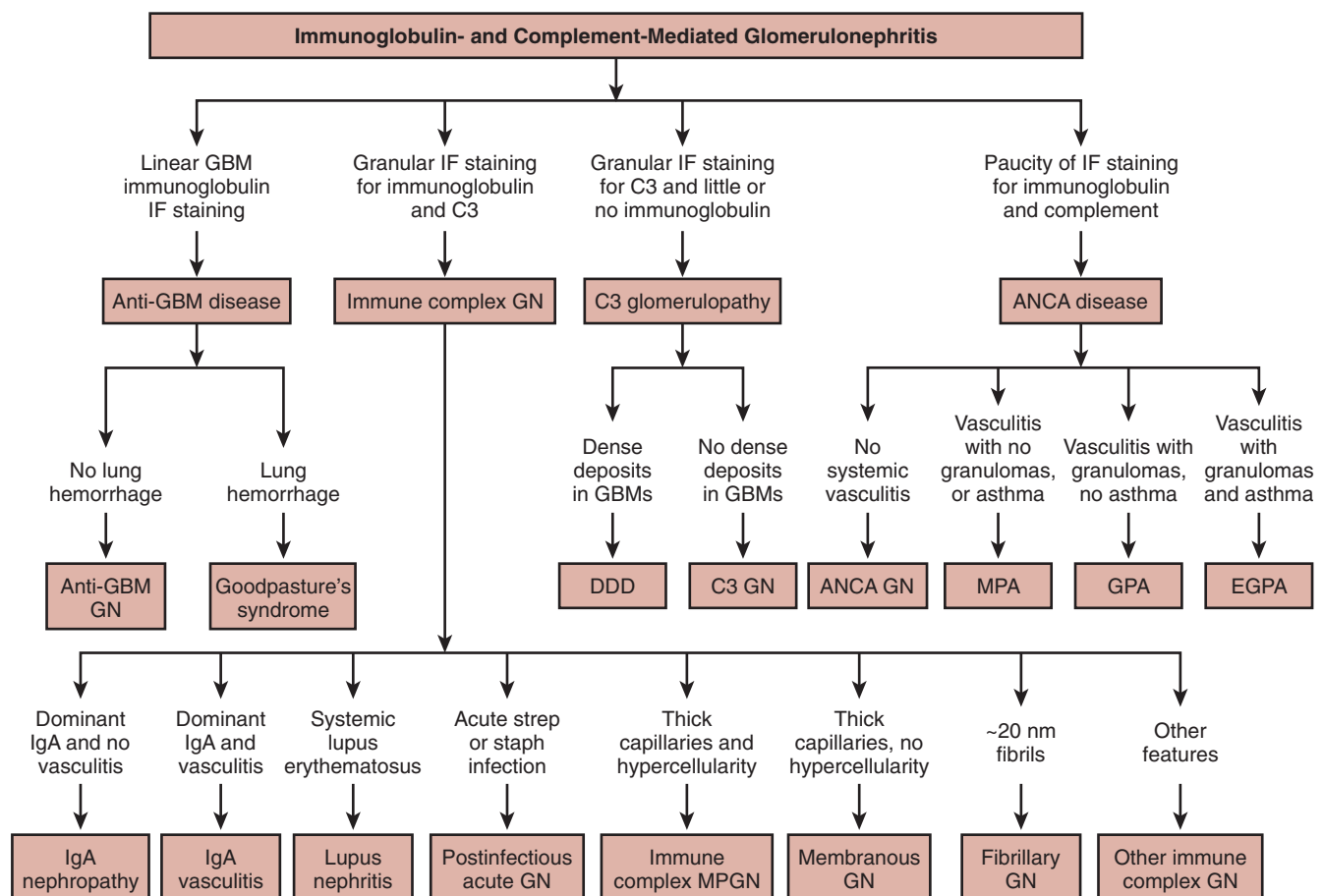


Fig. 31.35 Algorithm for the diagnostic classification of glomerulonephritis that is known or suspected of being mediated by antibodies and complement. Note that the integration of light microscopy, immunofluorescence (IF) microscopy, electron microscopy, laboratory data, and clinical manifestations is required to diagnose glomerulonephritis (GN) precisely. ANCA, Antineutrophil cytoplasmic autoantibody; DDD, dense deposit disease; EGPA, eosinophilic granulomatosis with polyangiitis; GBM, glomerular basement membrane; GPA, granulomatosis with polyangiitis; MPA, microscopic polyangiitis.

Table 31.9 Frequency of Immunopathologic Categories of Glomerulonephritis in Kidney Biopsy Specimens Evaluated by Immunofluorescence Microscopy^a

Immunohistology	All Proliferative Glomerulonephritis (n = 1093)	Any Crescents (n = 540)	>50% Crescents (n = 195)	Arteritis in Biopsy (n = 37)
Pauci-immune (<2+ Ig)	45% (496/1093)	51% (227/540)	61% (118/195) ^b	84% (31/37)
Immune complex (≥2+ Ig)	52% (570/1093)	44% (238/540)	29% (56/195)	14% (5/37) ^c
Anti-GBM	3% (27/1093)	5% (25/540) ^d	11% (21/195)	3% (1/37) ^e

^aBased on the analysis of over 3000 consecutive nontransplant renal biopsy specimens evaluated at the University of North Carolina Nephropathology Laboratory.

^bOf 77 patients, 70 (91%) tested positive for antineutrophil cytoplasmic antibody (ANCA), (44 for perinuclear ANCA [P-ANCA] and 26 cytoplasmic ANCA [C-ANCA]).

^cFour patients had lupus and one had poststreptococcal glomerulonephritis.

^dThree of 19 patients (16%) tested positive for ANCA (2 for P-ANCA and 1 for C-ANCA).

^eThis patient also tested positive for P-ANCA (myeloperoxidase ANCA).

GBM, Glomerular basement membrane; IF, immunofluorescence.

Modified from Jennette JC, Nickleleit V: Anti-glomerular basement membrane glomerulonephritis and Goodpasture's syndrome. In Jennette JC, Olson JL, Silva FG, D'Agati V, eds. *Heptinstall's pathology of the kidney*. 7th ed. Wolters Kluwer: Philadelphia; 2015:657–684.

The presence of microangiopathic hemolytic anemia and thrombocytopenia are indicators that the rapid loss of kidney function is more likely caused by hemolytic uremic syndrome than crescentic glomerulonephritis. Pauci-immune crescentic glomerulonephritis, which shows little or no evidence of the localization of immune complex or anti-GBM antibodies in glomeruli, is usually associated with the presence of ANCAs and is the most common cause for RPGN and crescentic glomerulonephritis in adults, especially older adults (Table 31.9; see Table 31.8).^{1126,1134–1136} In most patients, pauci-immune crescentic glomerulonephritis is a component of a systemic small-vessel vasculitis, such as GPA or MPA; however, some patients have renal-limited (primary) disease.^{1127,1137} Anti-GBM disease is the least frequent cause of crescentic glomerulonephritis (see Tables 31.8 and 31.9).^{1126,1127,1134,1135}

IMMUNE COMPLEX-MEDIATED AND C3 GLOMERULOPATHY CRESCENTIC GLOMERULONEPHRITIS

EPIDEMIOLOGY

Most patients with immune complex-mediated crescentic glomerulonephritis have clinical or pathologic evidence of a specific category of primary glomerulonephritis, such as IgA nephropathy, postinfectious glomerulonephritis, or MPGN, or they have glomerulonephritis that is a component of a systemic immune complex disease, such as SLE, cryoglobulinemia, or IgA vasculitis. A minority of patients with immune complex crescentic glomerulonephritis, however, do not have patterns of immune complex localization that readily fit into these specific categories of immune complex glomerulonephritis.¹¹³⁸

Immune complex crescentic glomerulonephritis accounts for most crescentic glomerulonephritides in children, but for only a minority of crescentic glomerulonephritis in older adults (see Table 31.8). The higher frequency in children and young adults reflects a similar trend in other types of immune complex glomerulonephritides, such as IgA nephropathy, PSGN, MPGN, DDD, and lupus nephritis.

PATHOLOGY

Light Microscopy

The light microscopic appearance of crescentic immune complex glomerulonephritis depends on the underlying category of glomerulonephritis. For example, in their most aggressive expressions, MPGN, acute postinfectious glomerulonephritis, or proliferative glomerulonephritis, including IgA nephropathy, can all have crescent formation.*

This underlying phenotype of immune complex glomerulonephritis is recognized best in the intact glomeruli or glomerular segments. Immune complex-mediated glomerulonephritis and C3 glomerulopathy usually have varying combinations of capillary wall thickening and endocapillary hypercellularity in the intact glomeruli. This is in contrast to anti-GBM glomerulonephritis and ANCA glomerulonephritis, which tend to have surprisingly few alterations in intact glomeruli and segments, in spite of the severe necrotizing injury in involved glomeruli and segments. In glomerular segments adjacent to crescents in immune complex glomerulonephritis, there usually is some degree of necrosis with karyorrhexis; however, the necrosis is rarely as extensive as that typically seen with anti-GBM or ANCA glomerulonephritis. In addition, there is less destruction of Bowman's capsule associated with crescents in immune complex glomerulonephritis, as well as less pronounced periglomerular tubulointerstitial inflammation. Crescents in immune complex glomerulonephritis have a higher proportion of epithelial cells to macrophages than crescents in anti-GBM or ANCA glomerulonephritis, which may be related to the less severe disruption of Bowman's capsule and thus less opportunity for macrophages to migrate in from the interstitium.¹¹³⁸

Immunofluorescence Microscopy

IF microscopy, as well as electron microscopy, provides the evidence that crescentic glomerulonephritis is immune complex-mediated or complement-mediated versus anti-GBM

*References 359–363, 697, 703, 831, 1058, 1133, and 1139.

antibody-mediated or ANCA-mediated. The pattern and composition of immunoglobulin and complement staining depend on the underlying category of immune complex glomerulonephritis or C3 glomerulopathy that has induced crescent formation.^{360,702,1140} For example, crescentic glomerulonephritis with predominantly mesangial IgA-dominant deposits is indicative of crescentic IgA nephropathy, C3-dominant deposits with peripheral bandlike configurations suggest crescentic MPGN, coarsely granular capillary wall deposits raise the possibility of crescentic postinfectious glomerulonephritis, and finely granular IgG-dominant capillary wall deposits suggest crescentic MN. The latter may be a result of concurrent anti-GBM disease, which also causes linear GBM staining beneath the granular staining, or concurrent ANCA disease, which can be documented serologically. About 25% of all patients with crescentic immune complex glomerulonephritis are ANCA-positive, whereas fewer than 5% of patients with noncrescentic immune complex glomerulonephritis are ANCA positive. This suggests that the presence of ANCAs in patients with immune complex glomerulonephritis may predispose to a disease that is more aggressive.

Electron Microscopy

As with the findings by IF microscopy, the findings by electron microscopy in patients with crescentic immune complex glomerulonephritis depend on the type of immune complex disease that has induced crescent formation. The hallmark ultrastructural finding is immune complex–type, electron-dense deposits. These can be mesangial, subendothelial, intramembranous, subepithelial, or any combination of these. The pattern and distribution of deposits may indicate a particular phenotype of primary crescentic immune complex glomerulonephritis, such as postinfectious, membranous, membranoproliferative, or DDD.^{360,702,1140} Ultrastructural findings also may suggest that the disease is secondary to some unrecognized systemic process. For example, endothelial tubuloreticular inclusions suggest lupus nephritis, and microtubular configurations in immune deposits suggest cryoglobulinemia.

In all types of crescentic glomerulonephritis, breaks in GBMs usually can be identified if sought carefully, especially in glomerular segments adjacent to crescents. Dense fibrin tactoids occur in thrombosed capillaries, in sites of fibrinoid necrosis, and in the interstices between the cells in crescents. In general, the extent of fibrin tactoid formation in areas of fibrinoid necrosis is less conspicuous in crescentic immune complex glomerulonephritis than in crescentic anti-GBM or ANCA glomerulonephritis.

PATHOGENESIS

Crescentic glomerulonephritis is the result of a final common pathway of glomerular injury that results in crescent formation. Multiple causes and pathogenic mechanisms can lead to the final common pathway, including many types of immune complex disease. The general dogma is that immune complex localization in glomerular capillary walls and mesangium, by deposition, in situ formation, or both, activates multiple inflammatory mediator systems.^{211,1126,1127} This includes humoral mediator systems, such as the coagulation system, kinin system, and complement system, as well as inflammatory cells, such as neutrophils, monocytes and macrophages, lymphocytes, platelets, endothelial cells, and

mesangial cells. The activated cells also release soluble mediators, such as cytokines and chemokines. If the resultant inflammation is contained internally to the GBM, a proliferative or membranoproliferative phenotype of injury ensues, with only endocapillary hypercellularity. However, if the inflammation breaks through capillary walls into Bowman's space, extracapillary hypercellularity (crescent formation) results.

Complement activation has often been considered a major mediator of injury in immune complex glomerulonephritis; however, experimental data also indicate the importance of Fc receptors in immune complex-mediated injury.^{1141,1142} For example, mice deficient for the FcγR1 and FcγRIII receptors have a markedly reduced tendency to develop immune complex glomerulonephritis.^{1143,1144}

TREATMENT

The therapy for crescentic immune complex glomerulonephritis is influenced by the nature of the underlying category of immune complex glomerulonephritis. For example, acute PSGN with 50% crescents might not prompt the same therapy as IgA nephropathy with 50% crescents. However, there have been an inadequate number of controlled prospective studies to guide therapy for most forms of crescentic immune complex glomerulonephritis. Some nephrologists extrapolate from the lupus nephritis experience and choose to treat patients with crescentic immune complex disease with immunosuppressive drugs that they would not use if the glomerular lesions appeared less aggressive. For the minority of patients who have idiopathic immune complex crescentic glomerulonephritis, the most common treatment is immunosuppressive therapy with pulse methylprednisolone, followed by prednisone at a dosage of 1 mg/kg daily tapered over the second to third month to an alternate-day regimen until completely discontinued.^{675,1145–1147} In patients with a rapid decline in kidney function, cytotoxic agents, with or without plasma exchange, in addition to corticosteroids, may be considered. As with anti-GBM and ANCA disease, immunotherapy should be initiated as early as possible during the course of crescentic immune complex glomerulonephritis to reduce the likelihood of reaching the irreversible stage of advanced scarring. There is evidence, however, that crescentic glomerulonephritis with an underlying immune complex proliferative glomerulonephritis is less responsive to aggressive immunosuppressive therapy than is anti-GBM or ANCA crescentic glomerulonephritis.^{1058,1138}

ANTI-GLOMERULAR BASEMENT MEMBRANE GLOMERULONEPHRITIS

EPIDEMIOLOGY

Anti-GBM disease accounts for about 10% to 20% of crescentic glomerulonephritides.⁶⁷⁵ This disease is characterized by circulating antibodies to the GBM (anti-GBM) and deposition of IgG or, rarely, IgA along the GBM (see also Chapter 32).^{675,1138,1148–1160} Anti-GBM antibodies may be eluted from kidney tissue samples from patients with anti-GBM disease, which allows verification that the antibodies are specific to the GBM.^{675,1154,1158} The antibodies eluted from kidney tissue bind to the same epitope of type IV collagen as the circulating anti-GBM antibodies from the same patient.¹¹⁶¹

Anti-GBM disease occurs as a renal-limited disease (anti-GBM glomerulonephritis) and as a pulmonary-renal vasculitic

syndrome (Goodpasture syndrome).^{675,1138,1148–1160,1162} The incidence of anti-GBM disease has two peaks with respect to age. The first peak is in the 2nd and 3rd decades of life, and anti-GBM disease in this age group shows a higher frequency of pulmonary hemorrhage (Goodpasture syndrome). The second peak is in the 6th and 7th decades, and this later onset disease is more common in women, who more often have renal-limited disease. Interestingly, anti-GBM autoantibodies were detected in multiple serum samples from the Department of Defense Serum Repository before diagnosis in a case-control study involving 30 patients with anti-GBM disease and 30 healthy controls (50% vs. 0%; $P < .001$),¹¹⁶³ which suggests the development of the autoimmune response prior to onset of disease.

Genetic susceptibility to anti-GBM disease is associated with HLA-DR2 specificity.¹¹⁶⁴ More detailed analysis of the association with HLA-DR2 has revealed a link with the DRB1 alleles, DRB1*1501 and DQB*0602.^{1165–1169} Further refinement of this association has shown that polymorphic residues in the second peptide-binding region of the HLA class II antigen segregated with disease, supporting the hypothesis that the HLA association in anti-GBM disease reflects the ability of certain class II molecules to bind and present anti-GBM peptides to helper T (T_H) cells.¹¹⁶⁵

This concept is further supported by mouse models of anti-GBM disease in which crescentic glomerulonephritis and lung hemorrhage are restricted to only certain major histocompatibility complex (MHC) haplotypes, despite the ability of mice of all haplotypes to produce antibodies to the α_3 -NC1 (“noncollagenous”) domain of type IV collagen.¹¹⁷⁰ Analyses of gene expression in the kidneys of mouse strains susceptible to anti-GBM antibody-induced nephritis, compared with those of control strains, have revealed that 20% of the underexpressed genes in these mice belonged to the kallikrein gene family, which encodes serine esterases implicated in the regulation of inflammation, apoptosis, redox balance, and fibrosis.¹¹⁷¹ Antagonizing the kallikrein pathway by blocking the bradykinin receptors B1 and B2 augmented disease, whereas bradykinin administration reduced the severity of anti-GBM, antibody-induced nephritis in a susceptible mouse strain. Nephritis-sensitive mouse strains had kallikrein haplotypes that were distinct from those of control strains, including several regulatory polymorphisms. These results suggest that kallikreins are protective disease-associated genes in anti-GBM antibody-induced nephritis.¹¹⁷¹ Whether these findings pertain to susceptibility to or severity of anti-GBM disease in humans is unknown. It should also be noted that another genetic association study of 48 Chinese patients with anti-GBM disease compared with 225 matched healthy controls revealed a genetic association of an FC γ RIIB polymorphism (I232T) with disease susceptibility.¹¹⁷² This same polymorphism has been identified in patients with SLE and is thought to alter this inhibitory receptor responsible for the maintenance of B cell tolerance and activation thresholds.¹¹⁷³

PATHOLOGY

Immunofluorescence Microscopy

The pathologic finding of linear staining of the GBMs for immunoglobulin is indicative of anti-GBM glomerulonephritis (Fig. 31.36).^{1155,1158,1159,1174–1177} The immunoglobulin is predominantly IgG; however, rare patients with IgA-dominant,

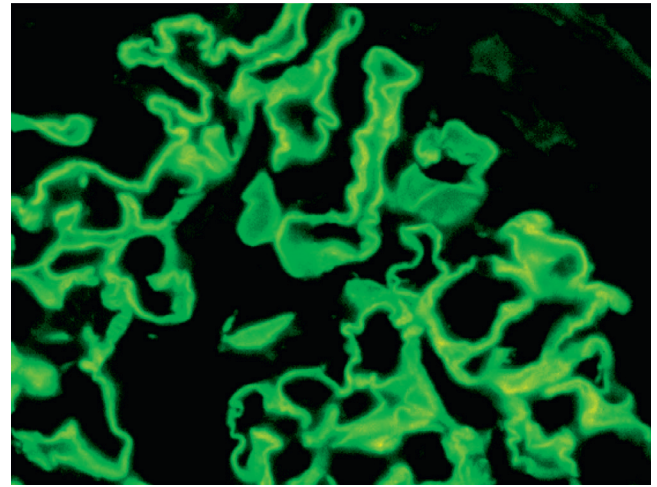


Fig. 31.36 Immunofluorescence micrograph of a portion of a glomerulus with antiglomerular basement membrane (anti-GBM) glomerulonephritis showing linear staining of GBMs for immunoglobulin G (IgG). (Fluorescein isothiocyanate anti-IgG stain, $\times 500$.)

anti-GBM glomerulonephritis have also been reported.^{1156,1178} Linear staining for both κ and λ light chains typically accompanies the staining for γ heavy chains. Linear staining for γ heavy chains alone indicates γ heavy-chain deposition disease. Most specimens with anti-GBM glomerulonephritis have discontinuous linear to granular capillary wall staining for C3, but a minority show little or no C3 staining. Linear staining for IgG may also occur along distal tubular basement membranes.¹¹⁵⁹

The linear IgG staining of GBMs frequently seen in patients with diabetic glomerulosclerosis and the less intense linear staining seen in older patients with hypertensive vascular disease, must not be confused with that in anti-GBM disease. Clinical data and light microscopic findings should help make this distinction. Serologic confirmation should always be obtained to substantiate the diagnosis of anti-GBM disease.

Serologic testing for ANCA should be ordered simultaneously because one-quarter to one-third of patients with anti-GBM disease are also ANCA-positive. This may modify the prognosis and likelihood of systemic small-vessel vasculitis.^{1179,1180}

Light Microscopy

At the time of biopsy, 97% of patients with anti-GBM disease have some degree of crescent formation, and 85% have crescents in 50% or more of glomeruli (Table 31.10).^{1127,1174} On average, 77% of glomeruli have crescents. Glomeruli with crescents typically have fibrinoid necrosis in adjacent glomerular segments. Nonnecrotic segments may look entirely normal by light microscopy or may have slight infiltration by neutrophils or mononuclear leukocytes. This differs from crescentic immune complex glomerulonephritis and C3 glomerulopathy, which typically have capillary wall thickening and endocapillary hypercellularity in the intact glomeruli. Special stains that outline basement membranes, such as Jones' silver methenamine or periodic acid–Schiff stain, often demonstrate focal breaks in GBMs in areas of necrosis and also show focal breaks in Bowman's capsule. The most severely injured glomeruli have global glomerular necrosis,

Table 31.10 Frequency of Crescent Formation in Various Glomerular Diseases^a

Disease	Patients With Crescents (%)	Patients With ≥50% Crescents	Average No. of Glomeruli With Crescents (%)
Anti-GBM glomerulonephritis	97	85	77
ANCA-associated glomerulonephritis	90	50	49
Immune complex-mediated glomerulonephritis			
Lupus glomerulonephritis (classes III and IV)	56	13	27
IgA vasculitis (formerly Henoch-Schönlein purpura glomerulonephritis) ^b	61	10	27
IgA nephropathy ^b	32	4	21
Acute postinfectious glomerulonephritis ^b	33	3	19
Fibrillary glomerulonephritis	23	5	26
Type I membranoproliferative glomerulonephritis	24	5	25
Membranous lupus glomerulonephritis (class V)	12	1	17
Membranous glomerulonephritis (nonlupus)	3	0	15

^aBased on analysis of over 6000 native kidney biopsy specimens evaluated at the University of North Carolina Nephropathology Laboratory. In general, diseases in which crescents are most often seen also have the largest percentage of glomeruli involved by crescents when they are present.

^bBecause more severe cases of immunoglobulin A nephropathy and postinfectious glomerulonephritis are more often evaluated by kidney biopsy, the extent of crescent involvement is higher in the patients included in this table than in the general group of patients with these diseases.

ANCA, Antineutrophil cytoplasmic antibody; GBM, glomerular basement membrane; GN, glomerulonephritis; IgA, immunoglobulin A. Modified from Jennette JC: Rapidly progressive and crescentic glomerulonephritis. *Kidney Int.* 2003;63:1164–1177.

circumferential cellular crescents, and extensive disruption of Bowman's capsule.

The acute necrotizing glomerular lesions and the cellular crescents evolve into glomerular sclerosis and fibrotic crescents, respectively.¹¹⁷⁴ If the kidney biopsy specimen is obtained several weeks into the course of anti-GBM disease, the only lesions may be these chronic sclerotic lesions. There may be a mixture of acute and chronic lesions; however, the glomerular lesions of anti-GBM glomerulonephritis tend to be more in synchrony than those of ANCA glomerulonephritis, which more often show admixtures of acute and chronic injury.

Tubulointerstitial changes are commensurate with the degree of glomerular injury. Glomeruli with extensive necrosis and disruption of Bowman's capsule typically have intense periglomerular inflammation, including occasional multinucleated giant cells. There also is focal tubular epithelial acute simplification or atrophy, focal interstitial edema and fibrosis, and focal interstitial infiltration of predominantly mononuclear leukocytes. There are no specific changes in arteries or arterioles. If necrotizing inflammation is observed in arteries or arterioles, the possibility of concurrent anti-GBM and ANCA disease should be considered.

Electron Microscopy

The findings by electron microscopy reflect those seen by light microscopy.^{1174,1181} In acute disease, there is focal glomerular necrosis with disruption of capillary walls. Bowman's capsule also may have focal gaps. Leukocytes, including neutrophils and monocytes, often are present at sites of necrosis, but are uncommon in intact glomerular segments. Fibrin tactoids, which are electron-dense curvilinear accumulations of polymerized fibrin, accumulate at the sites of coagulation system activation, including sites of capillary thrombosis, fibrinoid necrosis, and fibrin formation in Bowman's space (Fig. 31.37). Cellular crescents contain cells with ultrastructural

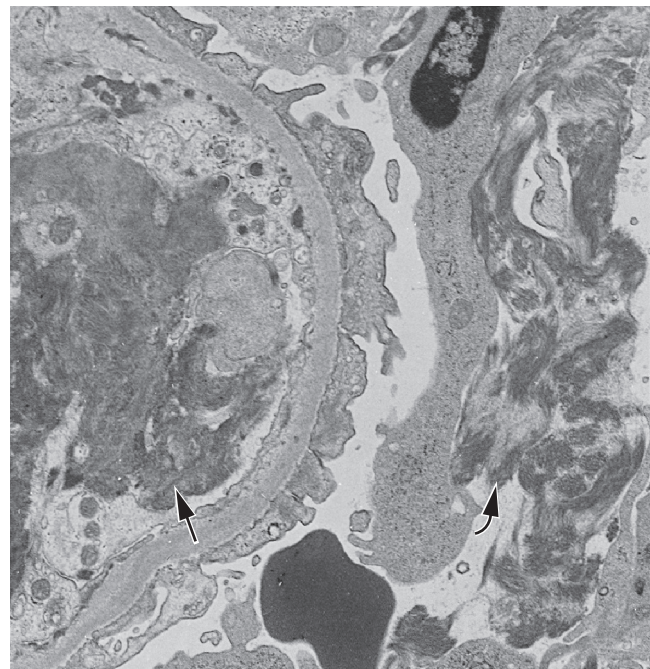


Fig. 31.37 Electron micrograph of a portion of a glomerular capillary wall and adjacent urinary space from a patient with antiglomerular basement membrane (anti-GBM) glomerulonephritis. Note the fibrin tactoids within a capillary thrombus (*straight arrow*) and in Bowman's space (*curved arrow*) between the cells of a crescent. Also note the absence of immune complex-type, electron-dense deposits in the capillary wall. (×6000.)

features of macrophages and epithelial cells. An important negative observation is the absence of immune complex–type, electron-dense deposits. These occur only in specimens from patients with anti-GBM disease patients who have concurrent immune complex disease. Glomerular segments that do not have necrosis may appear remarkably normal, with only focal effacement of visceral epithelial foot processes. There may be slight lucent expansion of the lamina rara interna, but this is an inconstant and nonspecific feature. In chronic lesions, amorphous and banded collagen deposition distorts or replaces the normal architecture.

PATHOGENESIS

The landmark studies opening the way to an understanding of the pathogenesis of anti-GBM disease were those of Lerner and colleagues.¹¹⁵⁴ In these studies, antibodies eluted from kidneys of patients with Goodpasture syndrome and injected into monkeys led to the induction of glomerulonephritis, proteinuria, renal failure, and pulmonary hemorrhage, along with intense staining of the GBM for human IgG.

The antigen to which anti-GBM antibodies react was initially found to be in the collagenase-resistant part of type IV collagen, the so-called noncollagenous domain, or NC1 domain.^{1182–1184} The antigenic epitopes found in the NC1 domain are in a cryptic form, as evidenced by the fact that little reactivity is found against the native hexameric structure of the NC1 domain. However, when the hexameric NC1 domain is denatured and dissociates into dimers and monomers, the reactivity of antibodies increases 15-fold.¹¹⁸⁴ About 90% of anti–type IV collagen antibodies are directed against the α_3 chain of type IV collagen.^{1146,1185} The Goodpasture epitopes in the native autoantigen are sequestered within the NC1 hexamers of the $\alpha_3\alpha_4\alpha_5$ (IV) collagen network and are a feature of the quaternary structure of two distinct subsets of $\alpha_3\alpha_4\alpha_5$ (IV) NC1 hexamers. Goodpasture antibodies breach only the quaternary structure of hexamers containing only monomer subunits, whereas hexamers composed of both dimer and monomer subunits (D-hexamers) are resistant to autoantibodies under native conditions.^{1186,1187} The epitopes of D-hexamers are structurally sequestered by dimer reinforcement of the quaternary complex.¹¹⁸⁷ Extensive work over the past several decades that has focused on elucidating autoantibody-specific epitopes along the quaternary structure of the $\alpha_3\alpha_4\alpha_5$ (IV) NC1 hexamer has reinforced the paradigm that this disease process is an autoimmune conformationopathy.¹¹⁸⁸ It is presumed that environmental factors, such as exposure to hydrocarbons,¹¹⁸⁹ tobacco smoke,¹¹⁹⁰ and endogenous oxidants¹¹⁹¹ can also expose the cryptic Goodpasture epitopes. In patients with anti-GBM disease who do not have antibodies to the classic epitope on the α_3 chain, antibodies to entactin have been detected.¹¹⁹² A small percentage of patients with anti-GBM disease may also have limited reactivity with the NC1 domains of the α_1 or α_4 chains of type IV collagen. These additional reactivities seem to be more frequent in patients with anti-GBM–mediated glomerulonephritis alone.¹¹⁹³

Most patients with anti-GBM disease express antibodies to two major conformational epitopes (E_A and E_B) located within the carboxy-terminal noncollagenous (NC1) domain of the α_3 chain of type IV collagen.^{1194–1197} The immunodominant target epitope, E_A , is encompassed by α_3 -NC1 residues 17 to 31. A homologous region at α_3 NC1 residues 127 to

141 encompasses the E_B epitope, recognized by the autoantibodies of only a small number of patients.¹¹⁹⁸ In a large cohort of Chinese patients,¹¹⁹⁹ the levels of antibody against E_A and E_B were strongly correlated with each other. Antibody levels against α_3 , E_A , and E_B correlated with serum creatinine level and with death or ESKD at 1 year, but not with gender, age, presence of ANCA, or hemoptysis. Interestingly, a more recent study has found that autoantibodies against E_A and E_B are crucial for kidney dysfunction; multivariate Cox regression analysis revealed that autoantibody reactivity to E_B was an independent risk factor for renal failure (HR, 6.91; $P = .02$).¹²⁰⁰ The stimuli and mechanism(s) leading to the formation of autoantibodies remain unclear, as is the mechanism whereby the normally hidden target epitopes become accessible to circulating autoantibodies.

About one-third of patients with anti-GBM–Goodpasture syndrome also have circulating ANCA, with most being to MPO (MPO-ANCA).^{1180,1193,1196,1201,1202} In a study of a large cohort of Chinese patients with anti-GBM disease, with or without ANCA, no differences in reactivity to the E_A , E_B , and S2 epitopes (a recombinant construct expressing the nine amino acid residues critical for the anti-GBM epitope)¹²⁰³ were detected between patients with anti-GBM plus ANCA compared with anti-GBM alone.¹²⁰⁴ The mechanism whereby some patients develop both anti-GBM and ANCA is unknown. It has been speculated that in such patients, ANCA may appear first and cause damage to the GBM, thus exposing the normally hidden target epitopes of anti-GBM antibodies. The coexistence of ANCA in patients with anti-GBM antibodies is associated with small-vessel vasculitis in organs in addition to the lung and kidney. In experimental models, the presence of antibodies to MPO aggravate experimental anti-GBM disease.^{1180,1205,1163}

Some patients with X-linked Alport syndrome (XLAS) develop anti-GBM antibodies and glomerulonephritis post-transplantation. The main structural component of mature GBM is type IV collagen, expressed as a heterotrimer composed of three α chains, α_3 , α_4 , and α_5 ,^{1206,1207} with a central collagenous domain and noncollagenous (NC) domain at the N- and C-terminal ends. Self-reactive B cells are negatively regulated at different stages of B cell development. Deletion, anergy (functional inactivation), and receptor editing are some of the mechanisms for B cell tolerance.¹²⁰⁸ Patients with XLAS lack the network of $\alpha_3\alpha_4\alpha_5$ in the GBM; therefore, B cells specific for epitopes in $\alpha_3\alpha_4\alpha_5$ -NC1 cannot undergo tolerance. Unlike the autoantibodies seen in anti-GBM disease that are directed to the NC1 domain of the α_3 chain of type IV collagen, the anti-GBM alloantibodies that cause post-transplantation nephritis in some patients with XLAS are directed against conformational epitopes in the NC1 domain of α_5 (IV) collagen only, which is normally expressed in the allograft but absent in the recipient. Allograft-eluted alloantibodies mainly targeted two epitopes accessible in the $\alpha_3\alpha_4\alpha_5$ -NC1 hexamers of human GBM, unlike the sequestered α_3 -NC1 epitopes of anti-GBM autoantibodies.¹²⁰⁹

A number of animal models of anti-GBM disease have been developed over the years, based on the immunization of animals with heterologous or homologous GBM.¹²¹⁰ Alternatively, anti-GBM antibody-induced injury can be produced passively by the intravenous injection of heterologous anti-GBM antibodies.¹²¹⁰ This leads to two phases of injury. The first, or so-called heterologous phase, occurs in

the first 24 hours and is mediated by direct deposition of the heterologous antibodies on the GBM, with subsequent recruitment of neutrophils. This is usually followed by an autologous phase, depending on the host's immune response to the heterologous immunoglobulin bound to the GBM.¹²¹⁰

The rat model of anti-GBM disease induced by the injection of heterologous anti-GBM antibodies has permitted the study of the roles of various inflammatory mediators in the development of anti-GBM disease.^{1211–1214} Thus in Wistar-Kyoto (WKY) rats injected with a rabbit antiserum to rat GBM, impairing leukocyte recruitment and monocyte-macrophage glomerular infiltrate by blocking the chemokine C-X-C motif ligand 16 (CXCL16) with a polyclonal anti-CXCL16 antiserum in the acute inflammatory or progressive phase of established glomerulonephritis, has significantly attenuated glomerular injury and improved proteinuria.¹²¹⁵ Similarly, the depletion of CD8⁺ cells has prevented the initiation and progression of anti-GBM crescentic glomerulonephritis. In the same animal model, treatment with an antibody to perforin resulted in a significant reduction in the amount of proteinuria, frequency of glomerular crescents, and number of glomerular monocytes and macrophages, although the number of glomerular CD8⁺ cells was not changed.¹²¹⁶ These results suggest that CD8⁺ cells play a role in glomerular injury as effector cells, in part through a perforin-granzyme-mediated pathway.

The more recent development of analogous murine models of anti-GBM disease opens the way for more specific evaluations of the inflammatory processes with the use of strains of mice with specific gene knockouts.¹¹⁷⁰ For example, the role of protease-activated receptor 2 (PAR-2) in kidney inflammation has been studied using PAR-2-deficient (PAR-2^{-/-}) mice.¹²¹⁷ PAR-2 is a cellular receptor expressed predominantly on epithelial, mesangial, and endothelial cells in the kidney and on macrophages. PAR-2 is activated by serine proteases and coagulation factors VIIa and Xa. In the kidney, PAR-2 induces endothelium-dependent and endothelium-independent vasodilation of afferent renal arteries and renal mesangial cells proliferation in vitro. Glomerulonephritis was induced in mice by the intravenous injection of sheep antimouse GBM globulin. In this model, PAR-2-deficient mice had reduced crescent formation, proteinuria, and serum creatinine level compared with wild type mice, but this was not associated with a difference in the glomerular accumulation of CD4⁺ T cells or macrophages or with the number of proliferating cells in glomeruli. These results demonstrate a proinflammatory role for PAR-2 in crescentic glomerulonephritis that is independent of effects on glomerular leukocyte recruitment and mesangial cell proliferation.

The theory of autoantigen complementarity states that the initiator of an autoimmune response is not necessarily the autoantigen or its mimic, but is instead an exogenous or endogenous peptide that is antisense or complementary to the autoantigen. This theory has been applied to a WKY rat model of anti-GBM disease. Rats immunized with a complementary peptide corresponding to the critical immunodominant epitope of the α_3 chain of type IV collagen developed crescentic glomerulonephritis within 8 weeks, and sera from patients with anti-GBM disease were found to have antibodies to the complementary peptide of the α_3 chain of type IV collagen, suggesting a role for autoantigen complementarity in anti-GBM disease. This was also previously implicated in ANCA vasculitis.^{1218,1219}

Although anti-GBM disease is considered a prototypical antibody-mediated glomerulonephritis, several lines of evidence have indicated an important role for T cells in the initiation or pathogenesis of this disease. A role for T cells in the autoimmune response is suggested by the increased susceptibility to the disease associated with the presence of HLA class II antigens DRB1*1501 and DQB*0602.^{1165–1169} Further evidence of the involvement of T cell activation in the development of the autoimmune response to the NC1 domain of the α_3 chain of type IV collagen comes from studies of T cell proliferation in response to other monomeric components of the GBM¹²²⁰ and synthetic oligopeptides.¹²²¹ The transfer of CD4⁺ T cells specific to a recombinant GBM antigen into syngeneic rats resulted in a crescentic glomerulonephritis, without linear anti-GBM IgG deposition.¹²²² Furthermore a single nephritogenic T cell epitope of type IV collagen α_3 -NC1 was demonstrated to induce glomerulonephritis in WKY rats.¹²²³ More recently, CD4⁺ T cell clones generated from HLA-DRB*1501 transgenic mice immunized with a peptide corresponding to amino acids 3136–3146 of the NC1 domain of α_3 (IV) were capable of transferring disease into HLA-DRB*1501 transgenic mice.¹²²⁴ Interestingly, cross-reactive peptides from human infection-related microbes could be identified that also induced severe proteinuria and moderate to severe glomerulonephritis in immunized rats.¹²²⁵ One peptide derived from *Clostridium botulinum* also induced pulmonary hemorrhage.¹²²⁵

On immunization of mice with α_3 -NC1 domains of type IV collagen, the development of glomerulonephritis and lung hemorrhage depends on certain MHC haplotypes and the ability of mice to mount a T_H1 response.¹¹⁷⁰ The role of T cells in this model was further documented by the fact that the passive transfer of lymphocytes or antibodies from nephritogenic strains to syngeneic recipients led to the development of nephritis, whereas the passive transfer of antibodies to T cell receptor-deficient mice failed to do so.¹¹⁷⁰

CD4⁺CD25⁺ regulatory T cells may play an important role in regulating the immune response in anti-GBM disease. Thus, the transfer of regulatory T cells into mice that were previously immunized with rabbit IgG, and before an injection of anti-GBM rabbit serum, significantly attenuated the development of proteinuria and dramatically decreased glomerular damage. On histologic analysis, there was reduced infiltration of CD4⁺ T cells, CD8⁺ T cells, and macrophages, but the deposition of immune complexes was not prevented.¹²²⁶ In humans, the action of regulatory T cells may explain, in part, the uncommon occurrence of disease relapses and the eventual disappearance of anti-GBM antibodies in patients, even without the use of immunosuppressant medications.¹²²⁷ Thus analyses of peripheral blood mononuclear cells from patients with Goodpasture syndrome have revealed the emergence of GBM-specific CD25⁺ regulatory T cells in the convalescent period, whereas they were undetected at the time of presentation.¹²²⁸

The role of complement in the pathogenesis of anti-GBM disease is evidenced by the deposition of C3 along the GBM. The role of complement activation has been examined largely in studies of passive injection of heterologous antibodies to GBM. Investigations using this model have suggested that the terminal components of the complement system are not involved in the pathogenesis of disease.¹²²⁹ Results of further studies in rabbits that are congenitally deficient in the sixth

component of complement have also suggested that the terminal components of complement do not play a major part in the pathogenesis of the disease, except in leukocyte-depleted animals.^{1230,1231} The role of complement cascade activation in a murine model of heterologous anti-GBM previously led to conflicting results in regard to the role of complement activation in this model.¹²³² More recent studies involving the same model, using mice completely deficient of complement components C3 or C4, have revealed a greater protective effect of C3 deficiency, more than C4 deficiency. Both protective effects could be overcome if the dose of nephritogenic antibodies was increased.¹²³³

To evaluate the role of complement activation and of Fcγ receptors further, an attenuated mouse model of anti-GBM was developed using a subnephritogenic dose of rabbit antimouse GBM antibody, followed 1 week later with an injection of mouse monoclonal antibody against rabbit IgG, which resulted in albuminuria.¹²³⁴ In this model, albuminuria was absent in Fcγ chain-deficient mice and reduced in C3-deficient mice, which indicates a role for both Fcγ receptors and complement. C1q- and C4-deficient mice did develop proteinuria, which is suggestive of involvement of the alternative complement pathway.¹²³⁴ The role of Fcγ receptors is also evidenced by the occurrence of severe lung hemorrhage in mice deficient in the inhibitory Fcγ2b receptor that were treated with bovine type IV collagen.¹²³⁵

Conclusions about the pathogenesis of human anti-GBM disease from animal models must be tempered because animal models may not accurately replicate human disease.

CLINICAL FEATURES AND NATURAL HISTORY

The onset of renal anti-GBM disease is typically characterized by an abrupt, acute glomerulonephritis, with severe oliguria or anuria. There is a high risk of progression to ESKD if appropriate therapy is not instituted immediately. Prompt treatment with plasmapheresis, corticosteroids, and cyclophosphamide results in patient survival of approximately 85% and renal survival of approximately 60%.^{1145,1236–1240}

Rarely, the disorder has a more insidious onset, in which patients remain essentially asymptomatic until the development of uremic symptoms and fluid retention.^{675,1162,1177,1241} The onset of disease may be associated with arthralgias, fever, myalgias, and abdominal pain; however, neurologic disturbances and gastrointestinal complaints are rare.

Goodpasture syndrome is characterized by the presence of pulmonary hemorrhage concurrent with glomerulonephritis. The usual pulmonary manifestation is severe pulmonary hemorrhage, which may be life threatening; however, patients may have milder disease, which can be focal. The absence of hemoptysis does not rule out diffuse alveolar hemorrhage. For patients with early or focal disease, a high level of suspicion is necessary to establish the diagnosis, especially in the presence of unexplained anemia. The diagnosis may be aided by measurements showing an increased diffusing capacity of carbon monoxide and by findings on computed tomography of the chest. Ultimately, the diagnostic evaluation of alveolar hemorrhage usually includes bronchoscopic examination and bronchoalveolar lavage.¹²⁴² This approach also allows exclusion of airway sources of bleeding and possible associated infections. In patients with anti-GBM disease, the occurrence of pulmonary hemorrhage is far more common in smokers than nonsmokers¹²⁴³ and may be associated with

environmental exposures to hydrocarbons^{1243–1246} or upper respiratory tract infections.¹²⁴⁷ Occupational exposure to petroleum-based mineral oils is essentially a risk factor for the development of anti-GBM antibodies.¹²⁴⁸ The association of pulmonary hemorrhage with environmental exposures and infection raises the theoretic possibility that they expose the cryptic antigen in the alveolar basement membrane, thereby allowing its recognition by circulating anti-GBM antibodies.

LABORATORY FINDINGS

Kidney involvement by anti-GBM disease typically causes an acute nephritic syndrome with hematuria that includes dysmorphic erythrocytes and red blood cells casts. Although nephrotic-range proteinuria may occur, full nephrotic syndrome is rarely seen.^{1159,1172,1174,1177,1241}

The diagnostic laboratory finding in anti-GBM disease is the detection of circulating antibodies to GBM, specifically to the α₃ chain of type IV collagen. These antibodies are detected in approximately 95% of patients by immunoassays using various forms of purified or recombinant substrates.¹²⁴⁹ The anti-GBM antibodies are most often of the IgG1 subclass, but may also be of the IgG4 subclass, with the latter being more often seen in females.¹²⁵⁰

TREATMENT

The standard treatment for anti-GBM disease is intensive plasmapheresis combined with corticosteroids and cyclophosphamide.^{1145,1237,1251–1254} Plasmapheresis consists of removal of 2 to 4 L of plasma and its replacement with a 5% albumin solution, continued on a daily basis until circulating antibody levels become undetectable. In patients with pulmonary hemorrhage, clotting factors should be replaced by administering fresh-frozen plasma at the end of each treatment. Prednisone should be administered starting at a dose of 1 mg/kg of body weight for at least the first month and then tapered to alternate-day therapy during the second and third months of treatment. Cyclophosphamide is administered orally (2 mg/kg/day, adjusted with consideration for the degree of impairment of kidney function and white blood cell count) for 8 to 12 weeks. The role of high-dose intravenous methylprednisolone pulses remains unproven in the treatment of anti-GBM disease.^{1255–1259} Nonetheless, the urgent nature of the clinical process prompts some nephrologists to administer methylprednisolone (7 mg/kg daily for 3 consecutive days) as part of induction therapy in this and other forms of crescentic glomerulonephritis.

When the regimen of aggressive plasmapheresis with corticosteroids and cyclophosphamide is used, patient survival is approximately 85%, with 40% progression to ESKD.^{1145,1236–1241} These results are better than those achieved before the introduction of plasmapheresis, when patient survival was less than 50% with an almost 90% rate of ESKD. In a study at the Hammersmith Hospital in the United Kingdom, Gaskin and Pusey have demonstrated that aggressive plasmapheresis, even in patients with severe renal insufficiency, may have an ameliorative effect and provide improved long-term patient and renal survival.¹²⁶⁰ In that cohort, among patients who had a creatinine concentration of 500 μmol/L or more (>5.7 mg/dL) at presentation but did not require immediate dialysis, patient and renal survival were 83% and 82% at 1 year and 62% and 69% at last follow-up, respectively. The renal prognosis of patients who presented with dialysis-dependent

renal failure was poor—92% of patients had ESKD at 1 year. All patients who required immediate dialysis and whose kidney biopsy specimens had crescents involving 100% of glomeruli remained dialysis dependent.¹²⁶¹

The major prognostic marker for the progression to ESKD is the serum creatinine level at the time of initiation of treatment. Patients with a serum creatinine concentration above 7 mg/dL are unlikely to recover sufficient kidney function to discontinue renal replacement therapy.¹¹⁵⁷ At issue is whether and for how long aggressive immunosuppression should persist in dialysis-dependent patients. Aggressive immunosuppression and plasmapheresis are warranted in patients with pulmonary hemorrhage. Aggressive immunosuppression should be withheld in patients with disease limited to the kidney, whose kidney biopsy specimens show widespread glomerular and interstitial scarring and who have a serum creatinine concentration of more than 7 mg/dL at presentation. In such patients, the risks of therapy outweigh the potential benefits. In patients who have an elevated serum creatinine level, yet whose biopsy specimens show active crescentic glomerulonephritis, aggressive treatment should continue for at least 4 weeks. If there is no restoration of kidney function by 4 to 8 weeks, and in the absence of pulmonary bleeding, immunosuppression should be discontinued.

Patients who have both circulating anti-GBM antibodies and ANCA may have a better chance of recovery of kidney function than patients with anti-GBM antibodies alone. In these patients, immunosuppressive therapy should not be withheld, even with serum creatinine levels above 7 mg/dL, because the concomitant presence of ANCA was associated with a more favorable renal outcome in some studies,^{1259,1262} although not in all.¹²⁶³ In a retrospective analysis comparing patients with anti-GBM autoantibodies, MPO-ANCA, and both, so-called double-positive patients and those with anti-GBM autoantibodies had significantly higher serum creatinine levels at presentation (10.3 ± 5.6 and 9.6 ± 8.1 mg/dL, respectively) than patients with MPO-ANCA alone (5.0 ± 2.9 mg/dL). One-year renal survival was better in patients with MPO-ANCA alone (63%) than in the double-positive group (10.0%; $P = .01$) and the anti-GBM group (15.4%; $P = .17$).¹²⁶³

Once remission of anti-GBM disease has been achieved with immunosuppressive therapy, recurrent disease occurs only rarely.^{1264–1267} Similarly, the recurrence of anti-GBM disease after kidney transplantation is also rare, especially when transplantation is delayed until after the disappearance or substantial diminution of anti-GBM antibodies in the circulation.¹²⁶⁸

PAUCI-IMMUNE CRESCENTIC GLOMERULONEPHRITIS

EPIDEMIOLOGY

In pauci-immune crescentic glomerulonephritis, the characteristic feature of the glomerular lesion is focal necrotizing and crescentic glomerulonephritis, with little or no glomerular staining for immunoglobulins by IF microscopy.^{1127,1149,1174,1252,1254} Pauci-immune crescentic glomerulonephritis usually is a component of a systemic small-vessel vasculitis; however, some patients have renal-limited (primary), pauci-immune crescentic glomerulonephritis.^{1127,1137,1179,1269} ANCA-associated vasculitis is also discussed in Chapter 32. Pauci-immune crescentic glomerulonephritis, including that accompanying

small-vessel vasculitis, is the most common category of RPGN in adults, especially older adults (see Table 31.8). The disease has a predilection for Caucasians compared with African Americans (see Table 31.7). There are no gender differences (see Table 31.7).

PATHOLOGY

Light Microscopy

The light microscopic appearance of ANCA-associated pauci-immune crescentic glomerulonephritis is indistinguishable from that of anti-GBM crescentic glomerulonephritis.^{360,701,1128,1138,1269–1272} Renal-limited (primary) pauci-immune crescentic glomerulonephritis also is indistinguishable from pauci-immune crescentic glomerulonephritis that occurs as a component of a systemic small-vessel vasculitis, such as GPA (formerly Wegener granulomatosis), microscopic polyangiitis (MPA), or EGPA, formerly Churg-Strauss syndrome). As illustrated in Fig. 31.30, ANCA glomerulonephritis and anti-GBM glomerulonephritis most often manifest as crescentic glomerulonephritis.

At the time of biopsy, approximately 90% of kidney biopsy specimens with ANCA-associated pauci-immune glomerulonephritis have some degree of crescent formation, and approximately half of the specimens have crescents involving 50% or more of glomeruli (see Tables 31.9 and 31.10). Over 90% of specimens have focal segmental to global fibrinoid necrosis (Fig. 31.38). As with anti-GBM disease, the intact glomerular segments often have no light microscopic abnormalities. The most severely injured glomeruli not only have extensive necrosis of glomerular tufts but also have extensive lysis of Bowman's capsule, with resultant periglomerular inflammation. The periglomerular inflammation contains varying mixtures of neutrophils, eosinophils, lymphocytes, monocytes, and macrophages, including occasional multinucleated giant cells. This periglomerular inflammation area

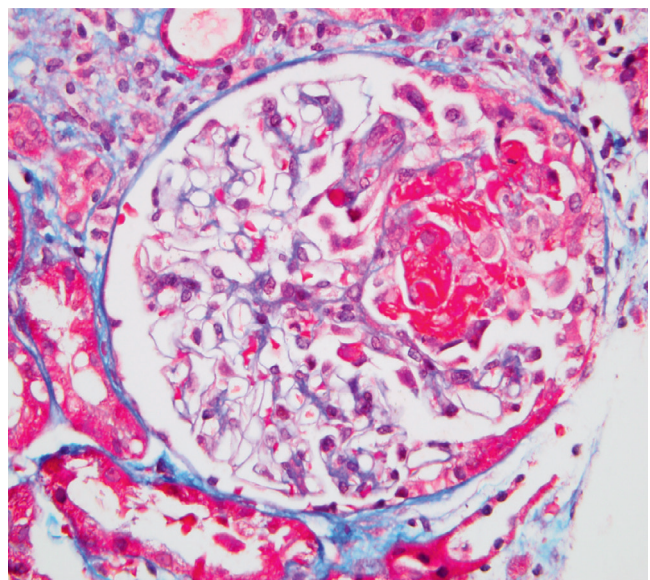


Fig. 31.38 Light micrograph showing segmental fibrinoid necrosis (segmental bright red staining) in a glomerulus from a patient with antineutrophil cytoplasmic autoantibody-associated pauci-immune crescentic glomerulonephritis. (Masson trichrome stain, $\times 300$.)

may have a granulomatous appearance, especially when the glomerulus that was the nidus of inflammation has been destroyed or is not in the plane of section. This granulomatous appearance is a result of the periglomerular reaction to extensive glomerular necrosis and is not specific for a particular category of necrotizing glomerulonephritis.

This pattern of injury can be seen with anti-GBM glomerulonephritis, renal-limited pauci-immune crescentic glomerulonephritis, and crescentic glomerulonephritis secondary to MPA, GPA and EGPA. Necrotizing granulomatous inflammation that is not centered on a glomerulus, but rather is in the interstitium or centered on an artery, raises the possibility of GPA or EGPA. The presence of arteritis in a biopsy specimen that has pauci-immune crescentic glomerulonephritis indicates that the glomerulonephritis is a component of a more widespread vasculitis, such as MPA, GPA, or EGPA.

The acute necrotizing glomerular lesions evolve into sclerotic lesions. During completely quiescent phases, a kidney biopsy specimen may have only focal sclerotic lesions that may mimic FSGS. ANCA-associated glomerulonephritis is also often characterized by many recurrent bouts of exacerbation. Therefore, combinations of active acute necrotizing glomerular lesions and chronic sclerotic lesions often occur in the same kidney biopsy specimen.

Immunofluorescence Microscopy

By definition, the distinguishing pathologic difference between pauci-immune crescentic glomerulonephritis and anti-GBM and immune complex crescentic glomerulonephritis is the absence or paucity of glomerular staining for immunoglobulins. How pauci-immune is pauci-immune crescentic glomerulonephritis? One basis for categorizing the disorder as pauci-immune crescentic glomerulonephritis is to determine whether the patient is likely to be ANCA-positive, which increases the likelihood of certain systemic small-vessel vasculitides.^{365,1269,1273,1274} The likelihood of positivity for ANCA is inversely proportional to the intensity of glomerular immunoglobulin staining by IF microscopy in a specimen with crescentic glomerulonephritis.¹²⁷² The likelihood of positive results on an ANCA serologic assay is approximately 90% if there is no staining for immunoglobulin, approximately 80% if there is trace to 1+ staining (on a scale of 0–4+), approximately 50% if there is 2+ staining, approximately 30% if there is 3+ staining, and less than 10% if there is 4+ staining. Thus, even patients with definite evidence for immune complex-mediated glomerulonephritis have a higher than expected frequency of ANCA, but the highest frequency is in patients with little or no evidence for immune complex- or anti-GBM-mediated disease.

The presence of ANCA at a higher than expected frequency in immune complex disease is intriguing and raises the possibility that ANCA contributes to the pathogenesis of not only pauci-immune crescentic glomerulonephritis, but also the most severe examples of immune complex disease.³⁶⁵ Considering this issue from a different perspective, approximately 25% of patients with idiopathic immune complex crescentic glomerulonephritis (i.e., immune complex glomerulonephritis that does not fit well into one of the categories of primary or secondary immune complex disease) are ANCA-positive, compared with less than 5% of patients who have idiopathic immune complex glomerulonephritis with no crescents.³⁶⁵

Glomerular capillary wall or mesangial staining usually accompanies immunoglobulin staining and is present in occasional specimens that do not have immunoglobulin staining. There is irregular staining for fibrin at sites of intraglomerular fibrinoid necrosis and capillary thrombosis and in the interstices of crescents. Foci of glomerular necrosis and sclerosis also may have irregular staining for C3 and IgM.

Electron Microscopy

The findings by electron microscopy are indistinguishable from those described earlier for anti-GBM glomerulonephritis.¹¹⁸¹ Specimens with pure pauci-immune crescentic glomerulonephritis have no or only a few immune complex-type, electron-dense deposits. Foci of glomerular necrosis have leukocyte influx, breaks in GBMs, and fibrin tactoids in capillary thrombi and sites of fibrinoid necrosis.

PATHOGENESIS

The pathogenesis of pauci-immune crescentic glomerulonephritis is currently not fully understood, but there is strong evidence that ANCA IgG is a major pathogenic factor.^{1275–1277} In the absence or paucity of immune complex deposition in glomeruli or other vessels, classic mechanisms of immune complex-mediated damage are not implicated in the pathogenesis of pauci-immune crescentic glomerulonephritis. On the other hand, the substantial accumulation of polymorphonuclear leukocytes at sites of vascular necrosis has led to examination of the role of neutrophil activation in this disease. A large body of in vitro data has implicated a pathogenic role for ANCA based on the demonstration that these autoantibodies activate normal human polymorphonuclear leukocytes.^{1272,1275,1278–1280}

For anti-MPO autoantibodies, anti-PR3 autoantibodies, or autoantibodies to other neutrophil antigens contained in the azurophilic granules to interact with their corresponding antigens, either the antibodies must penetrate the cell or those antigens must translocate to the cell surface. Indeed, small amounts of cytokine (e.g., TNF- α , IL-1) at concentrations too low to cause full neutrophil activation are capable of inducing such a translocation of ANCA antigens to the cell surface.¹²⁸¹ This translocation of ANCA antigens to the cell surface has been demonstrated in vivo on the neutrophils of patients with GPA and in patients with sepsis.^{1282–1284} Patients with ANCA disease aberrantly express *PR3* and *MPO* genes, and this expression correlates with disease activity.¹²⁸⁵ Despite the fact that these genes exist on different chromosomes, their expression appears coordinately upregulated during disease activity and downregulated during remission. Epigenetic changes occur as a result of increased unmethylated DNA at the *MPO* and *PR3* loci, and as a result of loss of recruitment of histone methylase PRC2 (polycomb repressive complex 2) by RUNX3 (Runt-related transcription factor 3) in both *MPO* and *PR3* genes with depressed gene transcription. In addition, JMJD3 (Jumonji domain-containing protein 3) appears to be expressed in these patients, which further diminishes histone H3K27me3 methylation status.¹²⁸⁶ In further support of epigenetic modifications contributing to autoantigen gene expression and disease state in ANCA vasculitis, measurement of gene-specific DNA methylation of the *MPO* and protein-coding proteinase 3 (*PRTN3*) genes in leukocytes of patients with ANCA vasculitis over their

disease course has revealed that patients with active disease have hypomethylation of *MPO* and *PRTN3* and increased autoantigen expression, whereas DNA methylation during times of remission increased.¹²⁸⁷ Furthermore, patients with increased DNA methylation at the *PRTN3* promoter had a significantly greater probability of a relapse-free period ($P < .001$), regardless of ANCA serotype, suggesting the potential use of these types of measurements as bioindicators of disease activity.¹²⁸⁷

Regardless of whether the antigen is expressed on the surface of the cell as a consequence of cytokine stimulation or gene expression, in the presence of circulating ANCAs, the interaction of the autoantibody with its externalized antigen results in full activation of the neutrophil, which leads to the respiratory burst and degranulation of primary and secondary granule constituents.^{1288,1289} The current hypothesis stipulates that ANCAs induce a premature degranulation and activation of neutrophils at the time of their margination and diapedesis, which leads to the release of lytic enzymes and toxic oxygen metabolites at the site of the vessel wall, thereby producing a necrotizing inflammatory injury. This view has been supported by in vitro studies demonstrating that neutrophils activated by ANCAs lead to the damage and destruction of human umbilical vein endothelial cells in culture.^{1290–1292}

Not only does neutrophil degranulation cause direct damage of the endothelium, but ANCA antigens released from neutrophils and monocytes enter endothelial cells and cause cell damage. PR3 can enter the endothelial cells by a receptor-mediated process^{1293–1295} and result in the production of IL-8¹²⁹⁶ and chemoattractant protein-1. PR3 also induces an apoptotic event from both proteolytic and nonproteolytic mechanisms.^{1297,1298} Interestingly, PR3-mediated apoptosis appears to be in part related to cleavage of the cell cycle inhibitor p21^{CIP1/WAF1} and nuclear factor-kappa B (NF- κ B).^{1299,1300} Similarly, MPO enters endothelial cells by an energy-dependent process¹³⁰¹ and transcytoses intact endothelium to localize within the extracellular matrix. There, in the presence of the substrates H_2O_2 and NO_2^- , MPO catalyzes the nitration of tyrosine residues on extracellular matrix proteins,¹³⁰² which results in the fragmentation of extracellular matrix protein.¹³⁰³ It also appears that endothelial cells inhibit superoxide generation by ANCA-activated neutrophils and that serine proteases may play a more important role than reactive oxygen species as mediators of endothelial injury during ANCA-associated vasculitis.¹³⁰⁴

Neutrophil activation by ANCA is likely mediated by both the antigen-binding portion of the autoantibodies ($F(ab')_2$) and by the engagement of their Fc fraction to Fc gamma receptors on the surface of neutrophils.^{1137,1291,1305,1306} Human neutrophils constitutively express the IgG receptors Fc γ RIIa and Fc γ RIIIb.¹³⁰⁷ ANCAs have been shown to engage both types of receptors.^{1291,1308} Engagement of the Fc receptors results in a number of neutrophil activation events, including respiratory burst, degranulation, phagocytosis, cytokine production, and upregulation of adhesion molecules.¹³⁰⁹ In particular, Fc γ RIIa engagement by ANCAs appears to increase neutrophil actin polymerization in neutrophils, which leads to distortion in their shape and possibly decreases their ability to pass through capillaries (the primary site of injury in ANCA vasculitis).¹³¹⁰ Furthermore, polymorphisms of the Fc γ RIIIb receptors^{1311,1312} (but not

of Fc γ RII^{1313,1314}) appear to influence the severity of ANCA vasculitis.

In addition to the Fc receptor-mediated mechanism, substantial data support a role for the $F(ab')_2$ portion of the antibody molecule in leukocyte activation. ANCA $F(ab')_2$ portions induce oxygen radical production¹³⁰⁶ and the transcription of cytokine genes in normal human neutrophils and monocytes. Microarray gene chip analysis has shown that ANCA IgG and ANCA- $F(ab')_2$ stimulate the transcription of a distinct subset of genes, some unique to whole IgG, some unique to $F(ab')_2$ fragments, and some common to both.¹³¹⁵ It is most likely that $F(ab')_2$ portions of ANCA are capable of low-level neutrophil and monocyte activation.¹³⁰⁶ The Fc portion of the molecule almost certainly causes leukocyte activation once the $F(ab')_2$ portion of the immunoglobulin has interacted with the antigen, either on the cell surface or in the microenvironment.¹²⁹¹ The signal transduction pathways of $F(ab')_2$ and Fc receptor activation through a specific p21ras (Kirsten-ras) pathway have also been elucidated.¹³¹⁶

Given the pathogenicity of ANCA, there has been considerable current effort directed toward identifying specific epitope(s) on MPO that are recognized by ANCAs in an effort to begin to identify therapeutic avenues to eliminate this interaction in vivo. Using a highly sensitive epitope excision and mass spectrometry approach, investigators from the United States, the Netherlands, and Australia have identified autoantibodies to specific epitopes of MPO in sera from patients with active disease that differed from those in remission. Furthermore, this same study reported what may be a seminal finding in that pathogenic MPO-ANCA were found in patients with ANCA-negative disease. These were detected after IgG purification that eliminated ceruloplasmin, the natural inhibitor of MPO, contamination from serum.¹³¹⁷

The role of T cells in the pathogenesis of pauci-immune necrotizing small-vessel vasculitis or glomerulonephritis, although suspected,^{1318,1319} is somewhat less well defined. Such a role is suggested by the presence of CD4⁺ T cells in granulomatous¹³²⁰ and active vasculitic lesions^{1321–1325} and by some correlation of the levels of soluble markers of T cell activation with disease activity,^{1320,1326} specifically, soluble IL-2 receptor and sCD3.^{1327,1328} Much is known about T cell responsiveness in ANCA disease, including the recognition of PR3 and MPO by T cells.^{1329,1330} The proportion of regulatory T cells in ANCA patients increases, although these regulatory T cells seem defective in their inability to suppress proliferation of effector cells in cytokine production. In addition, the percentage of T cells secreting IL-17 increases in the periphery, and serum levels of T_H17-associated cytokine IL-23 correlate with the propensity for disease activity.¹³³¹ Although yet to be replicated and validated, gene expression profiling of purified CD8⁺ T cells from patients with ANCA-associated vasculitis has identified a signature associated with poor prognosis and that included an expanded CD8⁺ memory T cell population.¹³³² A separate study examining purified CD4⁺ T cells from patients with ANCA-associated vasculitis has found an increased frequency of regulatory T cells with decreased suppressive function in patients with active disease, as well as a second T cell population that was resistant to regulatory T cell suppression.¹³³³

It has long been proposed that patients with ANCA-associated vasculitis and glomerulonephritis have a genetic

predisposition for disease. The first ever GWAS was performed using DNA samples from 1233 patients in the United Kingdom, with ANCA-associated vasculitis and 5884 controls, and a replication cohort of 1454 Northern European case patients and 1666 controls.¹³³⁴ Interestingly, genetic associations were made most strongly with respect to ANCA serotype rather than disease phenotype in that patients with PR3-ANCA had significant associations with HLA-DP and genes encoding α_1 -antitrypsin (SERPINA1, the endogenous inhibitor of PR3) and PR3 (PRTN3) itself ($P = 6.2 \times 10^{-89}$, $P = 5.6 \times 10^{-12}$ and $P = 2.6 \times 10^{-7}$, respectively). Patients with MPO-ANCA had a significant genome-wide association with HLA-DQ ($P = 2.1 \times 10^{-8}$). Of note, ANCA-associated vasculitis is notably rare in African Americans; however, an association of the HLA-DRB1*15 alleles with PR3-ANCA-positive disease has been found, conferring a 73.3-fold higher risk in African American patients than in community-based controls.¹³³⁵ Interestingly, the DRB1*1501 allelic variant, which is of Caucasian descent, was found in 50% of African American patients, whereas the DRB1*1503, of African descent, was underrepresented in this group. A recent GWAS of 1986 patients with GPA or MPA in North America identified risk alleles associated with HLA-DPB1, SERPINA1, PTPN22, and PRTN3 loci.¹³³⁶

Further establishment of a pathogenetic link between ANCA and the development of pauci-immune necrotizing glomerulonephritis and small-vessel vasculitis has greatly benefited from the development of animal models of this disease. Early models of disease were based on the finding of circulating anti-MPO antibodies in 20% of female MRL/lpr mice¹³³⁷ and in an inbred strain of mice, SCG/Kj, derived from the MRL/lpr mice and BXSB strains that develop a severe form of crescentic glomerulonephritis and systemic necrotizing vasculitis.¹³³⁸ Anti-MPO antibodies have been isolated from these strains of mice. Treatment of rats with mercuric chloride has led to the development of widespread inflammation, including necrotizing vasculitis in the presence of anti-MPO antibodies and anti-GBM antibodies.¹³³⁹ A more convincing model has indicated a pathogenetic role for ANCA. Aggravation of a mild, anti-GBM-mediated glomerulonephritis in rats, when the animals were previously immunized with MPO,¹²⁰⁵ suggests that minor proinflammatory events could be driven to severe necrotizing processes in the presence of ANCA.

More compelling models for ANCA small-vessel vasculitis now exist. MPO-deficient mice were immunized with murine MPO, and splenocytes from these mice were transferred to immunoincompetent recombination-activating gene (Rag2)-deficient mice.¹³⁴⁰ This resulted in the development of anti-MPO autoantibodies, severe necrotizing and crescentic glomerulonephritis and, in some animals, vasculitis in the lung and other organ systems. In a separate but similar set of experiments, anti-MPO antibodies alone were transferred into Rag2^{-/-} mice and induced pauci-immune necrotizing and crescentic glomerulonephritis.¹³⁴⁰ These studies indicate that anti-MPO antibodies cause pauci-immune necrotizing disease. The glomerulonephritis induced by anti-MPO antibodies is aggravated by the administration of lipopolysaccharide (LPS).¹³⁴¹ Conversely, the disease is abrogated when the neutrophils of anti-MPO-recipient mice are depleted by a selective antineutrophil monoclonal antibody.¹³⁴² In experiments to assess the role of T cells using this animal model,

the transfer of T cell-enriched splenocytes (>99% T cells) did not cause glomerular crescent formation or vascular necrosis. These data do not support a pathogenic role for anti-MPO T cells in the induction of acute injury.¹³⁴³ Furthermore, the role of genetic predisposition was investigated by inducing disease in 13 inbred mouse strains from the Collaborative Cross; however, a dominant quantitative trait locus was not identified, suggesting that differences in severity are likely polygenic in nature and possibly related to environmental milieu as well.¹³⁴⁴

Using the same model described previously, a previously unsuspected role of complement activation was demonstrated. Glomerulonephritis and vasculitis were abolished with the administration of cobra venom factor and failed to develop in mice deficient in complement factors C5 and B, whereas C4-deficient mice developed disease comparable with that in wild-type mice.¹³⁴⁵ These results indicate that the alternative complement pathway is required for disease induction, but not the classic or lectin pathways. Using this same mouse model, glomerulonephritis was completely abolished or markedly ameliorated by treating the mice with a C5-inhibiting monoclonal antibody either 8 hours before or 1 day after disease induction with anti-MPO IgG and lipopolysaccharide.¹³⁴⁶ Thus, anti-C5 had a dramatic therapeutic effect on this mouse model of ANCA vasculitis. These results have been corroborated by *in vitro* experiments demonstrating that blockade of the C5a receptor on human neutrophils abrogated their stimulation.¹³⁴⁷

More recent work has confirmed the immunopathogenetic importance of the alternative complement pathway in that blockade of C5a receptor (C5aR) activity protects against disease development. Mice expressing human C5aR were protected from anti-MPO autoantibody-induced disease when given an oral small-molecule antagonist of human C5aR called "CCX168."¹³⁴⁸ In contrast, using the same mouse model, mice deficient in complement factor 6 were not protected from disease, thereby supporting the concept that formation of the membrane attack complex is not necessary for disease development.¹³⁴⁸

In aggregate, these results suggest an important role for complement activation in the pathogenesis of ANCA vasculitis and have implications for possible future therapeutic interventions using blockers of the complement cascade. Although yet to be confirmed, there is also preliminary evidence for this in humans, because abnormal levels of C3a, C5a, and soluble C5b-9 in plasma and urine have been identified in patients with active disease.^{1349,1350} A randomized, placebo-controlled clinical trial, recruited adults with newly diagnosed or relapsing ANCA-associated vasculitis treated with cyclophosphamide or rituximab, as well as placebo plus corticosteroids, avacopan (oral anti-C5a receptor antagonist, 30 mg orally bid) with reduced prednisone (20 mg orally daily) or avacopan (30 mg orally bid) without steroids.¹³⁵¹ The primary outcome was the proportion of patients receiving a more than 50% reduction in the Birmingham Vasculitis Activity Score (BVAS) by week 12 and no worsening in any body system; 67 patients were enrolled. Clinical response at week 12 was achieved in 14 of 20 (70%) of the placebo group, 19 of 20 (86.4%) of the avacopan plus reduced-dose steroids group, and 17 of 21 (81%) in the avacopan without prednisone group (difference from control 11.0%; two-sided 90% CI, -11.0% to 32.9%; $P = .01$ for noninferiority). Adverse events were similar

across groups.¹³⁵¹ In addition, the phase 3 ADVOCATE clinical trial is underway to assess the efficacy of CCX168 (anti-C5a receptor antagonist) in a large group of patients with ANCA-associated glomerulonephritis (www.clinicaltrials.gov, NCT02994927). It is also important to note that all evidence for the role of the alternative complement pathway emanates from a model of anti-MPO autoantibody-mediated disease, and there is no direct evidence in mice or humans that this also applies to the anti-PR3 autoantibody-mediated disease.

The pathogenic role of anti-MPO antibodies has been documented in a second animal model, in which rats immunized with human MPO developed antirat-MPO antibodies and necrotizing and crescentic glomerulonephritis, as well as pulmonary capillaritis.¹³⁵² Using intravital microscopy, elegant studies have shown that anti-MPO-activated neutrophils undergo margination and diapedesis along the vascular wall.^{1352,1353} These two animal models have documented that anti-MPO antibodies are capable of causing necrotizing and crescentic glomerulonephritis and a widespread systemic vasculitis.

A model of anti-PR3-induced vascular injury was developed in PR3-neutrophil elastase-deficient mice in which the passive transfer of murine antimouse PR3 was associated with a stronger localized cutaneous inflammation, and perivascular infiltrates were observed around cutaneous vessels at the sites of intradermal injection of TNF- α .^{1343,1354} In summary, these animal studies have documented that both anti-MPO and PR3 antibodies are capable of causing disease.

As is true for most autoimmune responses, the inciting events in the breakdown of tolerance and the generation of anti-MPO or anti-PR3 antibodies are not known. Although genetic predispositions¹³⁵⁵ and environmental exposure to foreign pathogens,¹³⁵⁶ notably to silica,^{1357,1358} have been implicated, no direct link between these exposures and the formation of ANCAs has been established. A serendipitous finding in ANCA vasculitis has led to a theory of autoantigen complementarity.^{1218,1359} This theory rests on evidence that proteins transcribed and translated from the sense strand of DNA bind to proteins that are transcribed and translated from the antisense strand of DNA.¹³⁶⁰ It has been demonstrated that some patients with PR3-ANCA harbor antibodies to an antigen complementary to the middle portion of PR3.¹³³⁰ These anticomplementary PR3 antibodies form an antiidiotypic pair with PR3-ANCA. Moreover, cloned complementary PR3 proteins bind to PR3 and function as a serine proteinase inhibitor. Preliminary data have suggested that the complementary PR3 antigens are found on a variety of microbes, some of which have been associated with ANCA vasculitis and have also been found in the genome of some patients with both PR3-ANCA and MPO-ANCA.¹²¹⁸ Although these studies need to be confirmed and expanded to determine the source of the complementary PR3 antigen and their role (if any) in inducing vasculitis, these observations may provide a promising avenue for the detection of the proximate cause of the ANCA autoimmune response.

CLINICAL FEATURES AND NATURAL HISTORY

Most patients with pauci-immune necrotizing crescentic glomerulonephritis and ANCA have glomerular disease as part of a systemic small-vessel vasculitis. The disease is clinically limited to the kidney in about one-third of patients.¹³⁶¹ When

both renal-limited and vasculitis-associated pauci-immune crescentic glomerulonephritis are considered, this category of crescentic glomerulonephritis is the most common cause of RPGN in adults.^{1127,1133,1179,1361,1362} When the disorder is part of a systemic vasculitis, patients have pulmonary-renal, dermal-renal, or a multisystem disease. Frequent sites of involvement are the eyes, ears, sinuses, upper airways, lungs, gastrointestinal tract, skin, peripheral nerves, joints, and central nervous system. The three major ANCA-associated syndromes are MPA, GPA, and EGPA.^{1270,1363,1364} Even when patients have no clinical evidence of extrarenal manifestations of active vasculitis, systemic symptoms consisting of fever, fatigue, myalgias, and arthralgias are common.

Most patients with ANCA-associated pauci-immune necrotizing glomerulonephritis have RPGN with rapid loss of kidney function associated with hematuria, proteinuria, and hypertension. However, some patients follow a more indolent course of slow decline in function and less active urine sediment. In the latter group of patients, episodes of focal necrosis and hematuria resolve with focal glomerular scarring. Subsequent relapses result in cumulative damage to glomeruli.

Note that patients who have only pauci-immune crescentic glomerulonephritis at presentation may later develop signs and symptoms of systemic disease, with involvement of extrarenal organ systems.¹³⁶⁵ An autopsy study was conducted in deceased patients with ANCA-associated vasculitis. This study revealed the widespread presence of glomerulonephritis, but also demonstrated the finding of clinically silent extrarenal vasculitis. It was found that 8% of patients died from septic infections or progressive recurrent vasculitis.¹³⁶⁵

No studies currently available specifically examine the prognostic factors of pauci-immune crescentic glomerulonephritis in the absence of extrarenal manifestations of disease. In studies addressing the question of prognosis of patients with ANCA-related small-vessel vasculitis in general,^{1273,1365,1366} the presence of pulmonary hemorrhage was the most important determinant of patient survival. With respect to the risk of ESKD, the most important predictor of outcome is the entry serum creatinine level at the time of initiation of treatment.¹³⁶⁶ This parameter remained the most important predictive factor of renal outcome in a multivariate analysis that corrected for variables such as the presence or absence of extrarenal disease. Treatment resistance and progression to ESKD is also predicted by longer disease duration and vascular sclerosis on kidney biopsy specimens—presence of glomerular sclerosis, interstitial infiltrates, tubular necrosis, and atrophy¹³⁶⁷—and the presence of clinical markers of chronic disease, including cumulative organ damage (measured by the vasculitis damage index).¹³⁶⁸ A finding of vascular sclerosis on the biopsy was also found to be an independent predictor of treatment resistance¹³⁶⁹ and may be a reflection of chronic kidney damage due to hypertension or other atherosclerotic processes, with ANCA-associated nephritis providing an additional insult.

The impact of kidney damage as a predictor of resistance emphasizes the importance of early diagnosis and prompt institution of therapy. It is important to note that although the entry serum creatinine level is the most important predictor of renal outcome, there is no threshold of kidney dysfunction beyond which treatment is deemed futile, because more than 50% of patients who have a GFR less than 10 mL/min

at presentation reach remission and experience a substantial improvement in kidney function.¹³⁷⁰ Therefore, aggressive immunosuppressive therapy is warranted in all patients with newly diagnosed disease.¹³⁶⁹ However, the risk of progression to ESKD is also determined by the change in GFR within the first 4 months of treatment. In the absence of other disease manifestations, the decision to continue immunosuppressive therapy in patients with a sharply declining GFR should be weighed against the diminishing chance of renal recovery.¹³⁶⁹

Relapses of ANCA small-vessel vasculitis occur in up to 40% of patients. Based on a large cohort study, the risk of relapse appears to be predicted by the presence of PR3-ANCA (as opposed to MPO-ANCA) and the presence of upper respiratory tract or lung involvement.¹³⁶⁹ Patients with glomerulonephritis alone who have predominantly MPO-ANCA belong to the subgroup of patients with a relatively low risk of relapse, with a rate of relapse rate of about 25% at a median of 62 months.

Pauci-immune necrotizing glomerulonephritis and small-vessel vasculitis may recur after kidney transplantation.^{1371,1372} The rate of recurrence for ANCA small-vessel vasculitis in general, including pauci-immune necrotizing glomerulonephritis alone, is about 20%.¹³⁷³ The rate of recurrence in the subset of patients who have pauci-immune necrotizing glomerulonephritis alone, without systemic vasculitis is unknown, but may be lower than 20%. A positive ANCA test result at the time of transplantation does not seem to be associated with an increased risk of recurrent disease.

LABORATORY FINDINGS

Approximately 80% to 90% of patients with pauci-immune necrotizing and crescentic glomerulonephritis have circulating ANCA.^{365,1273,1276,1363,1374–1376} On indirect IF microscopy of alcohol-fixed neutrophils, ANCAs cause two patterns of staining, perinuclear (P-ANCA) and cytoplasmic (C-ANCA).^{1276,1376} The two major antigen specificities for ANCA are MPO and PR3.^{1269,1376–1379} Both proteins are found in the primary granules of neutrophils and the lysosomes of monocytes. With rare exceptions, anti-MPO autoantibodies produce a P-ANCA pattern of staining on indirect IF microscopy, whereas anti-PR3 autoantibodies produce a C-ANCA pattern of staining. About two-thirds of patients with pauci-immune necrotizing crescentic glomerulonephritis, without clinical evidence of systemic vasculitis, will have MPO-ANCA or P-ANCA, and approximately 30% have PR3-ANCA or C-ANCA.^{1270,1380} The relative frequency of MPO-ANCA to PR3-ANCA is higher in patients with renal-limited disease than in patients with MPA or GPA.¹²⁷⁰ A small percentage of patients will harbor both MPO- and PR3-ANCA; however, this likely represents primarily patients who have been exposed to levamisole-adulterated cocaine.^{1381,1382}

As mentioned previously, about one-third of patients with anti-GBM disease and approximately 25% of patients with idiopathic immune complex crescent glomerulonephritis test positive for ANCAs; therefore, ANCA positivity is not completely specific for pauci-immune crescentic glomerulonephritis.³⁶⁵ Maximal sensitivity and specificity with ANCA testing is achieved when both IF and antigen-specific assays are performed. Antigen-specific assays may be ELISA or radioimmunoassays. A variety of commercial tests are now available, and their diagnostic specificity ranges from 70% to 90% and

sensitivity from 81% to 91%.^{365,1383} However, tests still do not provide the necessary sensitivity, specificity, and predictive power to allow their use as the basis for initiating or altering cytotoxic therapy.

The positive predictive value (PPV) of a positive ANCA test result (i.e., the percentage of patients with a positive result who have pauci-immune crescentic glomerulonephritis) depends on the signs and symptoms of disease in the patient who is being tested. The signs and symptoms indicate the pretest likelihood of pauci-immune crescentic glomerulonephritis (predicted prevalence), which greatly influences predictive value. The PPV of a positive ANCA result in a patient with classic features of RPGN is 95%.³⁶⁵ In patients with hematuria and proteinuria, the PPV of a positive ANCA result is 84% if the serum creatinine level is more than 3 mg/dL, 60% if the serum creatinine level is 1.5 to 3.0 mg/dL, and only 29% if the serum creatinine level is less than 1 mg/dL.¹³⁸⁴ Although the PPV is not good in this last setting, the negative predictive value is greater than 95%, and thus a negative result can allay any concerns that the patient has early or mild pauci-immune necrotizing glomerulonephritis.

Urinalysis findings in pauci-immune crescentic glomerulonephritis include hematuria with dysmorphic red blood cells, with or without red cell casts, and proteinuria. The proteinuria ranges from 1 g of protein/24 hours to as much as 16 g of protein/24 hours.^{1365,1385} The serum creatinine concentration usually is elevated at the time of diagnosis and rising, although a minority of patients have relatively indolent disease. The erythrocyte sedimentation rate and C-reactive protein level are elevated during active disease. Serum complement component levels are typically within normal limits.

Whether a kidney biopsy is essential for the management of ANCA-associated pauci-immune glomerulonephritis depends on a number of factors, including the diagnostic accuracy of ANCA testing, pretest probability of finding pauci-immune glomerulonephritis, value of knowing the activity and chronicity of the renal lesions, and risk associated with immunotherapy for ANCA-associated pauci-immune necrotizing glomerulonephritis. Based on a study of 1000 patients with proliferative and/or necrotizing glomerulonephritis and a positive test for PR3-ANCA or MPO-ANCA, the PPV of ANCA testing was found to be 86%, with a false-positive rate of 14% and a false-negative rate of 16%. Considering the serious risks inherent in treatment with high-dose corticosteroids and cytotoxic agents, it is prudent to confirm the diagnosis and characterize the activity and chronicity of ANCA-associated pauci-immune crescentic glomerulonephritis by kidney biopsy, unless the patient is too ill to tolerate the procedure.¹³⁸⁴

TREATMENT

Data on the treatment of ANCA-positive pauci-immune necrotizing and crescentic glomerulonephritis have been derived from studies of ANCA-associated vasculitis, including GPA and MPA. There are scant data specifically addressing the treatment of patients with renal-limited pauci-immune necrotizing glomerulonephritis. The treatment of pauci-immune crescentic glomerulonephritis (with or without systemic vasculitis) is still based primarily on varying regimens of corticosteroids and cyclophosphamide.^{1366,1386,1387}

In view of the potential explosive and fulminant nature of this disease, induction therapy should be instituted using pulse methylprednisolone at a dose of 7 mg/kg/day for 3 consecutive days in an attempt to halt the aggressive, destructive, inflammatory process. This is followed by the institution of daily oral prednisone, as well as cyclophosphamide, either orally or intravenously. Prednisone is usually started at a dosage of 1 mg/kg/day for the first month, tapered to an alternate-day regimen, and then discontinued by the end of the fourth to fifth month. When a regimen of monthly intravenous doses of cyclophosphamide is used, the starting dose should be about 0.5 g/m² and should be adjusted upward to 1 g/m² based on the 2-week leukocyte count nadir.^{1387,1388} A regimen based on daily oral cyclophosphamide should begin at a dose of 2 mg/kg/day¹³⁸⁶ and should be adjusted downward, as needed, to keep a nadir leukocyte count above 3000 cells/mm³.

The optimal form of cyclophosphamide therapy (daily oral vs. intravenous pulse) has been the subject of investigation. In general, the intravenous regimen allows for an approximately twofold lower cumulative dose of cyclophosphamide than the oral regimen and is associated with a significant decrease in the rate of clinically significant neutropenia and other complications. In a metaanalysis of three RCTs, the rate of relapse associated with pulse cyclophosphamide was not statistically higher than the rate of relapse with a daily oral regimen, but the intravenous pulse regimen was associated with a statistically higher rate of remission and lower rates of leucopenia and infections.¹³⁸⁹ The final outcomes (death or ESKD) were no different for the two dosing regimens.

A large RCT (CYCLOPS) of pulse versus daily oral cyclophosphamide for the induction of remission was conducted that included 149 patients with newly diagnosed generalized ANCA vasculitis with kidney involvement.¹³⁹⁰ Patients were randomly assigned to receive pulse cyclophosphamide, 15 mg/kg every 2 weeks times 3, then every 3 weeks, or daily oral cyclophosphamide, 2 mg/kg/day. Cyclophosphamide therapy was continued for 3 months beyond the time of remission. All patients were then switched to azathioprine (2 mg/kg/day orally) until month 18. All patients received prednisolone, starting at 1 mg/kg orally, followed by a taper. Patients with a serum creatinine level more than 500 μmol/L (5.7 mg/dL) were excluded from the study; 79% of patients achieved remission by 9 months (median time to remission was 3 months for both groups). The two treatment groups did not differ in time to remission or proportion of patients who achieved remission at 9 months (88.1% in the pulse group vs. 87.7% in the daily oral group). The GFR did not differ between the two groups at any time point. By 18 months, 13 patients in the pulse group and 6 in the daily oral group had experienced a relapse (HR, 2.01; CI, 0.77–5.30). Absolute cumulative cyclophosphamide dose in the daily oral group was almost twice that in the pulse group (15.9 vs. 8.2 g, respectively; $P < .001$). The pulse group had a lower rate of leukopenia (HR, 0.41; CI, 0.23–0.71), but the frequency of serious infections was not statistically different between the two treatment groups. The long-term results of this trial were reported with a median duration of 4.3 years follow-up and data from 90% of patients in the original trial.¹³⁹¹ There was no difference in survival between the two groups; however, risk of relapse was significantly lower in the daily oral

cyclophosphamide arm (HR, 0.50; CI, 0.26–0.93; $P = .029$) although kidney function was similar by the end of the study ($P = .82$), as were adverse events.

This RCT confirms that the two cyclophosphamide regimens are associated with similar remission induction rates and time to remission induction, with the pulse cyclophosphamide regimen resulting in about 50% of the cumulative medication dose of the oral regimen and a significantly lower rate of leukopenia. The long-term results would suggest that the daily oral cyclophosphamide regimen portends less relapse risk, and there was a trend toward this in the original study. At this point, clinicians must weigh the risks and benefits of either regimen to determine which is most appropriate, and this decision may likely be based more heavily now on the level of patient compliance.

The length of cyclophosphamide therapy has changed significantly, largely based on the results of a large controlled trial in which patients who attained complete remission with cyclophosphamide after 3 months of therapy were randomly assigned to switch to azathioprine or to continue taking cyclophosphamide for a total of 12 months. After 12 months, both groups received azathioprine maintenance therapy for an additional year.¹²¹¹ Changing to azathioprine after 3 months of cyclophosphamide treatment appeared to be as effective as receiving oral cyclophosphamide for 12 months followed by 12 months of azathioprine, based on kidney function and frequency of relapse. It is noteworthy that patients whose PR3-ANCA titers remained positive at the time of the switch had about a twofold increased risk of subsequent relapse compared with patients whose ANCA titers had reverted to negative.¹³⁹²

In three relatively small RCTs addressing the role of plasmapheresis in the treatment of ANCA-associated vasculitis and glomerulonephritis,^{1393–1395} plasmapheresis was not found to provide any added benefit over immunosuppressive treatment alone in patients with renal-limited disease or patients with mild to moderate kidney dysfunction. However, the use of plasmapheresis in addition to immunosuppressive therapy appears to be beneficial in the subset of patients who require dialysis at the time of presentation.^{1395,1396} In a study performed by the European vasculitis study group (MEPEX trial) of 137 patients with a new diagnosis of severe biopsy-confirmed ANCA-associated glomerulonephritis, the use of plasma exchange was found to be superior to pulse methylprednisolone in producing recovery of kidney function in patients with severe kidney dysfunction at the time of entry into the study (serum creatinine level >5.8 mg/dL).¹³⁹⁷ Long-term follow-up of these patients did not show a significant difference in the proportion of patients free of ESKD or death; however, the small number of patients limited the power to detect differences.¹³⁹⁸ Because of the clinically observed increased risk of severe bone marrow suppression with the use of cyclophosphamide in patients receiving dialysis, such treatment should be pursued with extreme caution.

Patients who eventually are able to discontinue dialysis usually do so within 3 to 4 months of initiation of therapy.^{1370,1388} For this reason, continuing immunosuppressive therapy beyond 4 months in patients who are still undergoing dialysis is unlikely to be of added benefit (unless they continue to have extrarenal manifestations of vasculitis). In a retrospective analysis of 523 patients with ANCA vasculitis followed over a median of 40 months, 136 patients reached ESKD.¹³⁹⁹

Relapse rates of vasculitis were significantly lower in patients on long-term dialysis (0.08 episodes/person-year) than in the same patients before they reached ESKD (0.20 episodes/person-year) and in patients with preserved kidney function (0.16 episodes/person year). Infections were almost twice as frequent among patients with ESKD receiving maintenance immunosuppressants and were an important cause of death. Given the lower risk of relapse with hemodialysis and the higher risk of infection and death with long-term immunosuppression, the risk-benefit ratio does not support the routine use of maintenance immunosuppression therapy in patients with ANCA small-vessel vasculitis who are on long-term dialysis.

Clinical Relevance

Immune Suppression in Dialysis Patients

The risk-benefit ratio of maintenance immunosuppression therapy in patients with ANCA small-vessel vasculitis who are on long-term dialysis must be carefully considered because the relapse risk is lower on dialysis and infection risk is high.

Although high-dose intravenous pooled immunoglobulin has been used in the treatment of systemic vasculitis resistant to usual immunosuppressive treatment,^{1400–1404} there have been no published reports of its use in patients with pauci-immune crescentic glomerulonephritis alone, without systemic involvement.

Induction therapy with methotrexate has been compared with cyclophosphamide treatment in patients with early limited GPA and mild kidney disease.^{1405–1409} The rate of remission at 6 months was comparable in the two treatment groups.¹⁴⁰⁷ However, the onset of remission was delayed in methotrexate-treated patients with more extensive disease or pulmonary involvement. Methotrexate was also associated with a significantly higher rate of relapse than cyclophosphamide (69.5% vs. 46.5%, respectively), and 45% of relapses occurred while patients were receiving methotrexate. The dose of methotrexate must be reduced in patients whose creatinine clearance is less than 80 mL/min, and its use is contraindicated when creatinine clearances is less than 10 mL/min. Moreover, in the authors' experience, there are patients taking methotrexate who have progressive glomerulonephritis. Methotrexate is therefore unlikely to have any role in the treatment of pauci-immune crescentic glomerulonephritis alone.

Whether the use of cyclophosphamide can be reduced or avoided completely by the use of rituximab has been the subject of two RCTs. In the RITUXVAS trial, 44 patients with newly diagnosed ANCA vasculitis were randomly assigned in a ratio of 3:1 to receive rituximab (375 mg/m² weekly × 4) in addition to cyclophosphamide (15 mg/kg intravenously × 2; 2 weeks apart), or to cyclophosphamide (15 mg/kg intravenously every 2 weeks × 3, then every 3 weeks, for a maximum total of 10 doses) alone.¹⁴⁰⁹ Both groups received the same intravenous and oral prednisolone regimen. Patients in the rituximab group did not receive maintenance therapy, whereas those in the cyclophosphamide group were switched to azathioprine until the end of the trial. Minimum follow-up was 12 months. The rate of sustained remission was similar in the two treatment groups (76% in the rituximab group vs. 82% in the cyclophosphamide group; $P = .67$ for risk difference). Severe adverse events were common in both

groups, affecting 45% of patients in the rituximab group and 36% in the cyclophosphamide group ($P = .60$). This study has suggested that a combination of rituximab and reduced-dose cyclophosphamide may be no less effective than a traditional cyclophosphamide regimen, but it did not demonstrate a safety benefit of a rituximab-based approach. This study was not powered to establish equivalence or noninferiority.

In a large controlled trial designed to assess the noninferiority of rituximab compared with cyclophosphamide, 197 patients were randomly assigned to treatment with rituximab (375 mg/m² infusions once weekly × 4) or cyclophosphamide (2 mg/kg/day orally) for months 1 to 3, followed by azathioprine (2 mg/kg/day orally) for months 4 to 6. All patients received methylprednisolone (1 g/day intravenously for up to 3 days), followed by prednisone (1 mg/kg/day, tapered off completely by 6 months). The induction phase of this trial revealed similar rates between the two treatment groups in complete remission at 6 months (64% in the rituximab group vs. 55% in the cyclophosphamide group; $P = .21$).¹⁴¹⁰ No differences in relapse rates were observed between the groups. The 18-month efficacy of this single course of rituximab, as compared with cyclophosphamide followed by azathioprine, revealed persistent noninferiority; however, remission rates at the 18-month mark were 39% and 33% ($P = .32$), respectively, signifying the critical and persistent need for more effective remission induction and maintenance strategies.¹⁴¹¹ Although these studies suggest that substitution of cyclophosphamide with rituximab may be effective, rituximab has not been formally evaluated in patients with severe renal failure requiring dialysis.

Studies pertaining to maintenance immunosuppression for relapse prevention are primarily geared to patients with GPA or MPA. Current data have suggested that patients with pauci-immune glomerulonephritis alone and MPO-ANCA are at a relatively low risk of relapse.¹³⁶⁹ The value of prolonged maintenance immunosuppression in this group of patients is unknown, and any benefit in preventing a relapse would have to be weighed against the potential toxicity of immunosuppressive agents and the risks associated with their use.

MAINTENANCE TREATMENT

Following successful induction remission therapy, most patients with ANCA-associated vasculitis should remain on a maintenance immunosuppressive regimen to prevent relapses and limit end-organ damage. Patients with PR3-associated vasculitis or lung and upper respiratory tract involvement are more likely to relapse than others, with an HR of 1.7 for each risk factor.¹³⁶⁹

Cardiovascular involvement may be associated with a higher risk of relapse, whereas poor renal function is associated with a lower risk of relapse.¹⁴¹² Patients with drug-associated ANCA vasculitis who achieve remission after initial induction therapy may be followed closely, without the need for maintenance therapy. Whether all patients with the lower risks of relapse (MPO-associated vasculitis without lung or upper respiratory tract involvement) need maintenance therapy is debatable, but patients should be assessed on a case by case basis.

Prolonged treatment (>12 months) with cyclophosphamide (CYC) and glucocorticoids (GCs) has been the standard of care for a long time. Although the mortality rate has dramatically improved, both CYC and GCs have been associated with

significant morbidity and organ damage.¹³⁶⁸ This led to the search for a lesser toxic therapeutic regimen for maintenance therapy. Several major clinical trials have investigated different maintenance immunosuppressive regimens. These trials include CYClophosphamide versus AZAthioprine for early REMission phase of vasculitis (CYCAZAREM), WEGner Granulomatosis-ENTretein (WEGENT, comparing azathioprine versus methotrexate), MAINtenance of remission using RITuximab in Systemic ANCA-associated vasculitis (MAINRITSAN), MMF Protocol to Reduce Outbreaks of Vasculitides (IMPROVE), and prolonged REMission–MAINtenance therapy is systemic vasculitis (REMAIN).

In the CYCAZAREM study, newly diagnosed patients with ANCA-associated vasculitis (57% PR3-positive, 37% MPO-positive) underwent induction treatment with oral cyclophosphamide (2 mg/kg/day) and prednisone.¹⁴¹³ Most enrolled patients had renal involvement (mean eGFR, 49 mL/min; 95% CI, 43.7–54.6). Patients with a serum creatinine level greater than 5.7 mg/dL were excluded. After induction of remission, patients were randomized to azathioprine (2 mg/kg/day) or continued CYC (1.5 mg/kg/day) with prednisone, 10 mg/day. At month 12, both groups were treated with azathioprine (1.5 mg/kg/day) and prednisone, 7.5 mg daily. At 18 months, the relapse rate was not significantly different (~15%), and neither were the rates of serious adverse effects in the two groups; there were seven deaths during the induction-remission phase. This study has suggested that a prolonged course of CYC may be avoided with the substitution of azathioprine after achieving remission. Also, there was a similar degree of improvement in renal function in both groups from study entry to 18 months (17.5 mL/min and 23.5 mL/min in both the azathioprine and CYC groups, respectively).¹⁴¹³

In the WEGENT study, a multicenter open-label randomized control study, patients with a new diagnosis of Wegener granulomatosis (55%) and microscopic polyangiitis (45%) who achieved induction remission following treatment with pulse CYC and GCs underwent randomization to azathioprine (2 mg/kg/day) or methotrexate (0.3 mg/kg/week, progressively increased to 25 mg/week) for 12 months, followed by taper over 3 months.¹⁴¹⁴ The mean serum creatinine level at randomization was 1.46 mg/dL. The incidence of adverse events leading to discontinuation of the study drug or death was 11% in the azathioprine group and 12% in the methotrexate group, respectively. The HR was 1.65 (95% CI, 0.65–4.18) for the risk of an adverse event with methotrexate versus azathioprine. Also, grade 3 or 4 toxicity was higher with methotrexate (18%) compared with 8% with azathioprine ($P = .11$). The relapse rate was similar at 36% and 33% for azathioprine and methotrexate, respectively. Although, both drugs appeared similar in achieving remission, there was a trend toward more severe adverse events with methotrexate, probably due to use of higher dosing in the setting of impaired renal function. Moreover, the relapse rates were quite high in both groups.¹⁴¹⁴

In the IMPROVE study, patients with newly diagnosed GPA or MPA underwent induction-remission protocol of oral or intravenous CYC and GCs. Maintenance therapy constituted azathioprine (2 mg/kg/day) or MMF (2 g/day).¹⁴¹⁵ At month 12, the dose was reduced to 1.5 mg/kg/day for azathioprine and 1.5 g/day for MMF up to month 18, and then further tapered to 1.0 mg/kg/day for azathioprine and 1 g/day

for MMF until month 42. Both groups received a tapering course of prednisone for 24 months. The median follow-up was 39 months (interquartile range, 0.66–53.6 months). The median serum creatinine level was 2.7 mg/dL in the azathioprine group and 2.9 mg/dL in the MMF group. Relapses were more common in the MMF group, with an HR of 1.80 (95% CI, 1.10–2.93; $P = .02$) after adjustment for other factors.¹⁴¹⁵

In the MAINRITSAN trial, the superiority of rituximab over azathioprine for relapse prevention of ANCA-associated vasculitis was reported.¹⁴¹⁶ Among 115 patients, most patients had GPA (76%), whereas 20% of the cases had MPA. Most of the patients were newly diagnosed cases. All patients were randomized at 1:1 to rituximab or azathioprine after achieving remission following induction therapy with corticosteroids and pulse CYC. Patients received rituximab on days 0 and 14 after randomization and then at 6, 12 and 18 months thereafter. Azathioprine was dosed at 2 mg/kg/day for the first 12 months, followed by 1.5 mg/kg/day for the next 6 months and then at 1 mg/kg for 4 months. At month 28, there were 17 relapses in the azathioprine group (29%) compared with only 3 in the rituximab group (5%). In the rituximab group, 1 patient relapsed at month 8 and 1 each at months 22 and 24—that is, within 6 months of the last dose of rituximab. In contrast, 8 patients relapsed within the first 12 months, and 7 patients relapsed after azathioprine was stopped between months 24 and 28. Both groups had similar rates of adverse effects. Although the trial showed the superiority of rituximab over azathioprine for relapse prevention, it has drawn criticism. A major criticism is that most patients had GPA and only a small group of patients had MPO and renal-limited ANCA vasculitis. Thus, the superiority of rituximab may not hold true for both groups. Also, the dose of azathioprine was tapered after 1 year of standard dosing, but the dose of rituximab was kept the same. It begs the question whether the relapse rate would have been different if the dose of azathioprine had been uniform for the entire study period. All patients underwent induction therapy with CYC, so these conclusions cannot be extrapolated to patients who receive rituximab as induction therapy.^{1416–1418}

The REMAIN study investigated patients with ANCA-associated vasculitis in remission after a CYC-based regimen for induction remission.¹⁴¹⁹ In this study, patients were randomized into an azathioprine-prednisolone group for 48 months from diagnosis (continuation group) or to withdraw azathioprine/prednisolone at 24 months (withdrawal group). This prospective, randomized controlled study was designed to study the risk of relapse with prolonged duration of maintenance therapy. The study demonstrated that continuation of maintenance therapy for a longer duration (>2 years) results in a threefold reduction in relapse risk. Most relapses in the withdrawal group occurred after stopping azathioprine. Although the safety profile was not statistically different between the two groups, the study was underpowered to study the difference. Moreover, cytopenias and cardiovascular events were much more common in the continuation group. There were five deaths, with two of them due to cardiovascular events in the continuation group. There are many limitations of this study including an unknown cumulative dose of GCs and CYC, exclusion of patients with ESRKD, and prior severe relapse of disease. Although prolonged immunosuppression therapy

may reduce the risk of relapse, this study did not address the risks associated with this approach.^{1360,1419}

Based on the previous results, one could consider azathioprine, rituximab, or methotrexate for maintenance therapy. The patient's clinical and serologic characteristics, renal function, concomitant medications (azathioprine is best avoided with allopurinol), cost, and other factors should be considered when determining the best and most appropriate maintenance therapy. The duration of maintenance therapy remains undefined, and it must be weighed against the risks of relapse and effects of long-term immunosuppression (e.g., risks of infections, cardiovascular morbidity, malignancy). Open discussions between the clinician and patient are essential in determining the best maintenance therapy, duration, and if or when to stop therapy. Patients should be educated about the signs and symptoms of disease relapse when off therapy, and periodic urine dipstick measurement of hematuria in patients with kidney involvement should be encouraged.

Methotrexate should be avoided in patients with renal insufficiency or in those at high risk for kidney injury. Rituximab may lead to HBV reactivation, so screening for HBV infection is essential before the initiation of therapy. The optimal duration and dosing of azathioprine still needs to be determined, but because most relapses occur when the drug is stopped and/or tapered after 12 to 15 months, one could consider continuing azathioprine for a longer time. The duration of maintenance therapy also depends on the risks of relapse, tolerance of the drug, infections, adverse effects, and other factors. With rituximab, one may consider the dosing regimen used in the MAINRITSAN trial until the results of MAINRITSAN 2 (NCT01731561) and RITAZAREM (NCT01697267) trials are available. The MAINRITSAN 2 study will evaluate the dosing regimen of rituximab based on ANCA titers and CD19 lymphocytes versus infusions at a fixed interval in systemic ANCA-associated vasculitis. The RITAZAREM trial aims to compare the infusion of rituximab, 1000 mg, every 4 months for 2 years versus azathioprine at 2 mg/kg/day for patients with relapsing ANCA-associated vasculitis after an induction remission with rituximab and glucocorticoids.

Serum IgG levels should be monitored in patients receiving repeated courses of rituximab, especially if they previously received cytotoxic therapy. Because circulatory mature plasma cells, the source of most IgG, do not express CD20, a single dose of rituximab may not deplete IgG levels. Repeated dosing and depletion of B cell precursors may result in hypogammaglobulinemia. In a retrospective analysis following rituximab infusion, 34% of patients were found to have a low IgG level (<6 g/L) for at least 3 months, of whom only 4% had a severely low IgG level (<3 g/L). Although the median IgG level trended toward the hypogammaglobulinemic range in patients who received more than 6 g, this was not statistically significant.¹⁴²⁰ In a single-center cohort study of 29 GPA patients who received a median cumulative dose of 17 g of CYC one quarter of GPA patients developed hypogammaglobulinemia following rituximab infusion.¹⁴²¹ Male gender, total cyclophosphamide exposure, and baseline serum immunoglobulin levels were independent risks factors, whereas the total cumulative rituximab dose did not influence the risk.¹⁴²¹

In another series, 4% of patients with systemic vasculitis had moderate to severe hypogammaglobulinemia with

recurrent infections after rituximab infusion. Most of these patients had previously received other immunosuppressive medications. The median dose of CYC was 6 g. IgG was replaced intravenously at a dose of 0.4 g/kg monthly, titrated to achieve a goal of 8 to 10 g/L and a decrease in infection rate. This regimen resulted in a decreased incidence and severity of infections, and many patients were able to continue rituximab infusion along with IgG replacement.¹⁴²²

Following significant response to induction therapy, GCs should be tapered over 6 months. Although data are inconsistent on the duration of therapy, a metaanalysis has suggested that low-dose GCs for longer than 12 months may result in fewer relapses. However, the authors acknowledged that the nature of relapses (major or minor) and an accurate assessment of adverse effects were lacking in their analysis.¹⁴²³ Moreover, long-term, low-dose GC use may increase the risks of infections and probably new-onset diabetes mellitus.¹⁴²⁴

Sulfamethoxazole-trimethoprim (S-T) has been suggested for the maintenance treatment of ANCA-associated vasculitis.^{1425,1426} It has been proposed that S-T may act as an anti-inflammatory agent by interfering with cytotoxic oxygen metabolites from activated neutrophils and prevent tissue damage.¹⁴²⁷ In a prospective, randomized, placebo-controlled study of 81 patients with generalized GPA, cotrimoxazole (800 mg of sulfamethoxazole plus 160 mg of trimethoprim) or placebo was given twice daily during remission following treatment with CYC and prednisolone. The mean serum creatinine clearance was more than 60 mL/min in both groups, and there was no significant difference among groups in dosing of prednisolone and CYC. At month 24, there were fewer respiratory and non-respiratory tract infections; 82% of patients in the S-T group and only 60% in the placebo group remained in remission (relative risk of relapse, 0.4; 95% CI, 0.17–0.98). In 20% of patients, the drug was stopped due to side effects, including nausea, vomiting, rash, and a single case of possible interstitial nephritis and asymptomatic hepatotoxic effects. All adverse effects resolved with cessation of drug therapy.¹⁴²⁵

In a separate study of generalized GPA following induction remission with CYC and prednisone, patients were randomized to receive methotrexate or daily oral S-T (160 mg trimethoprim plus 800 mg sulfamethoxazole), with or without prednisone for maintenance therapy. In contrast to the previous study, remission (complete or partial) was maintained in 86% of patients on methotrexate but only 58% in the S-T group. Also, all patients on S-T and on low-dose prednisone relapsed after a median duration of 14 months. The discrepancy in the remission rates between the two studies could be partially explained by the use of different clinical tools for assessment of disease activity. In addition, the risk factors for relapse could have been different among different groups.¹⁴²⁸ Although S-T could be used in conjunction with standard maintenance immunosuppressive therapy to prevent infections and possibly reduce the risk of relapse, its use alone is not recommended to prevent relapses, especially in patients with pauci-immune glomerulonephritis.

Both PR3 ANCA-associated vasculitis (AAV) and MPO-AAV have been categorized together in most clinical trials. Multiple studies have suggested that PR3- and MPO-AAV have significant differences, including their genetic basis, histologic findings, and clinical manifestations.^{1318,1334,1429} Moreover, the risk of relapse is higher with PR3-AAV than with MPO-AAV,

and renal survival is worse with MPO-AAV compared with PR3-AAV and ANCA-negative vasculitis (HR, 2.1; 95% CI, 1.11–3.80; $P = .01$). Patients with PR3-AAV with pauci-immune glomerulonephritis have more active disease compared with those with MPO-AAV who may have more chronic lesions on renal histology, suggesting a smoldering disease with a delay in diagnosis and treatment.^{1318,1430} Based on a post hoc clinical trial analysis, patients with PR3-AAV are more likely to achieve and maintain remission with rituximab compared with CYC ($n = 131$; 65% vs. 48%; $P = .04$).¹⁴³¹

MPO ANCA patients have been underrepresented in many clinical trials. Future clinical studies for induction and maintenance therapy should be designed by separating the MPO and PR3 ANCA serotypes.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. A 25-year-old Caucasian man presents to clinic with 3 weeks of extreme fatigue, worsening lower extremity edema, and frothy urine. He takes no medications. On physical examination, he is afebrile, blood pressure is 150/90 mm Hg, heart rate is 70 beats per minute, and there is 2+ lower extremity pitting edema to the knees. Laboratory data show hemoglobin, 12.0 g/dL, creatinine, 1.2 mg/dL (baseline unknown), albumin, 3.0 g/dL, low-density-lipoprotein (LDL) cholesterol, 1180 mg/dL, and proteinuria of 8 g/24 hours. Urine sediment review reveals four or five dysmorphic red blood cells per high power field, with no evidence of cellular casts or crystals.

A native kidney biopsy is performed, which reveals normal-appearing glomeruli by light microscopy, no immunoglobulin or complement positivity by immunofluorescence (IF) microscopy, and podocyte foot process effacement accompanied by microvillous transformation by electron microscopy consistent with minimal change glomerulopathy. Of note, no focal areas of interstitial fibrosis and tubular atrophy are apparent. Which of the following is the most appropriate next treatment step for this patient?

- Supportive care with prompt initiation of an angiotensin inhibitor and an LDL-lowering agent
- Oral corticosteroids**
- Calcineurin inhibitor
- Cyclophosphamide
- Rituximab

Answer: b

Rationale: Corticosteroids remain the mainstay of therapy for minimal change disease because the response, with improvement in proteinuria and edema, is prompt. It may take longer to achieve complete remission in adults than in children.

2. A 30-year-old African-American man is seen in the emergency department and admitted to the hospital with anasarca. He takes venlafaxine for depression. On physical examination, he is afebrile, blood pressure is 170/100 mm Hg, heart rate is 60 beats per minute, and there is 3+ lower extremity pitting edema that extends to the low back. Laboratory data show hemoglobin, 10.0 g/dL, creatinine, 1.5 mg/dL (baseline unknown), albumin, 1.5 g/dL, and proteinuria, 12 g/24 hours.

Urine sediment review reveals numerous lipid droplets and hyalofatty casts, with no evidence of dysmorphic red blood cells, cellular casts, or crystals. A native kidney biopsy is performed, which reveals focal segmental glomerulosclerosis (FSGS). Which of the following is the clinically aggressive variant of FSGS?

- Tip lesion FSGS
- FSGS not otherwise specified
- Collapsing FSGS**
- Perihilar FSGS

Answer: c

Rationale: The collapsing variety of FSGS may present with severe nephrotic syndrome and carries a poor prognosis.

3. Which of the following is the most accurate statement regarding the epidemiology of IgA nephropathy in the United States?

- IgA nephropathy is more common in women in the 5th and 6th decades of life.
- IgA nephropathy is more common in men in the 5th and 6th decades of life.
- IgA nephropathy is more common in women in the 2nd and 3rd decades of life.
- IgA nephropathy is more common in men in the 2nd and 3rd decades of life.**
- IgA nephropathy occurs equally in men and women.

Answer: d

Rationale: IgA nephropathy can occur in individuals at all ages but is more common in men in their second and third decades of life.

4. A 20-year-old Asian woman presents to clinic with three episodes of macroscopic hematuria that have occurred over the past 6 months around the time of recurrent upper respiratory tract infections. The physical examination is unremarkable except for a blood pressure of 140/90 mm Hg. Laboratory data show hemoglobin, 11.0 g/dL, creatinine, 0.9 mg/dL (unchanged from baseline), and proteinuria of 3 g/24 hours. A native kidney biopsy is performed, which reveals diffuse mesangioproliferative glomerulonephritis and mild endocapillary hypercellularity consistent with IgA nephropathy and associated mild interstitial fibrosis and tubular atrophy. The patient is started on lisinopril, which is uptitrated over 6 months to a maximally tolerated dose; however, proteinuria of 3 g/24 hours persists. Which of the following is the most appropriate next treatment step for this patient?

- Fish oil containing omega-3 fatty acids plus tonsillectomy
- Intravenous plus oral corticosteroids**
- Oral corticosteroids and intravenous cyclophosphamide followed by azathioprine
- Oral corticosteroids, intravenous cyclophosphamide and plasma exchange
- Oral corticosteroids and rituximab

Answer: b

Rationale: Glucocorticoid therapy may be considered in patients with IgA nephropathy with preserved renal function and have persistent significant proteinuria despite maximum RAAS blockade.